

SMP Year 3 – CAH – Study Notes

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By Louisa Leone, MD3 2025

1. ENT

Ear

When you perform otoscopy, make sure to pull the pinna up and back to visualise the tympanic membrane.

Foreign Body in Ear

- May remove via direct visualisation or with microscopy. Only batteries and organic matter (that may expand with moisture) require urgent removal.
- Direct vision: instruments (e.g. right angle hook or wax loop/curette) can manoeuvre behind the foreign body then pull forward.
- Live insects: kill first by instilling oil (e.g. sesame or olive oil)
- Irrigation can be used (clean water + Betadine, aiming postero-superiorly), but ensure tympanic membrane is intact first!

Otitis Externa (skin infection!)

- Clinical features: tenderness of external auditory meatus skin, tragus, draining node on mastoid tip; decreased hearing **late!**. "Soothing" dressings (e.g. warm oil) will hurt.
 - Otagia is one of the first symptoms. Watch out for **pain when pulling the pinna on examination**.
 - *Note: this is not mastoiditis (which has pain on the top and back of ear)*
- Prevention of otitis externa: dry ears out after swimming with 'acidified alcohol' drops e.g. Aquaeear, Ear Clear drops for swimmers.
- Management: take a swab of discharge and commence topical antibiotics; give 1-2 doses of oral steroid if severe pain or swelling; place an otowick in the canal if there is severe oedema (stents open the ear canal to allow delivery of topical antibiotics)
 - The antibiotic drops listed below are combined with steroid.
 - Do **not** syringe the ear canal to clean out debris – may carefully suction; keep ears **dry** until infection has resolved.

	Acute Otitis Externa		
	Diffuse	Focal	Itchy
Pathogen	Pseudomonas (commonest) Staphylococcus spp.	Infected hair follicle (a 'boil') Staph aureus	Fungal/yeast; Aspergillus niger or Candida
Treatment	Mild-moderate pain: analgesics, antibiotic drops with steroid (Sofradex/ Ciproxin HC) Severe pain (i.e. needing opioid): drops + oral ciproxin, IV gentamicin, 1-2 doses oral steroid (e.g. prednisone) + clean after 48h. 	Antibiotic drops with steroid (Sofradex/ Ciproxin HC) Systemic (oral) anti-staph antibodies Analgesia	Swab/mop out then locacorten-vioform drops 

"Wax" Impaction: Skin + sebum + debris. Clean with solvent – sodium bicarbonate ear drops then hydrogen peroxide, or Ear Clear Drops (clean water is also good, but **not olive oil!**).

- Ear wax problems are seldom due to over-production; more commonly under-removal. Skin problems e.g. eczema, dermatitis, and occasionally psoriasis, can contribute. Prevent with frequent mild steroid cream.

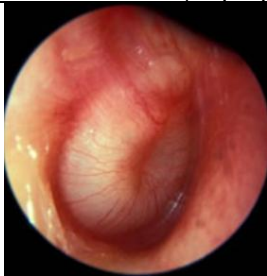
Exostoses (from cold water exposure): osteochondroma; outgrowth of bones of ear canal. Suspect in surfers or kids who swim a lot with wax impaction and/or excess otitis externa.

Acute Otitis Media (AOM)

- Infection of the middle ear cleft: Eustachian tube, middle ear, and mastoid air cells (lined by *respiratory mucosa*).
- Epidemiology: common, peak age **6 – 18 months**, 75% have **at least one AOM episode by school age**.



- Impacts hearing, speech, language acquisition, and learning.
- **Aetiology: AOM is usually viral.** The most common bacterial causes are *Streptococcus pneumoniae* (*most common cause of acute complications*), *Haemophilus influenzae*, *Moraxella catarrhalis* (in that order).
 - *Pseudomonas* may be seen in chronic otitis media or otitis externa.
- Particularly significant in **Aboriginal and Torres Strait Islander children**, prevalence up to 90%.
- Diagnosis: requires acute onset of signs and symptoms e.g. otalgia and fever, of acute middle ear infection.
 - Infants and toddlers won't be able to describe **symptoms**. Beware of **the infant who is tugging at their ear** (this may be the only sign of ear pain).
 - **Signs:** otalgia (including *ear pulling*), ear discharge (otorrhoea, purulent and blood-stained, following perforation), fever, irritability, vomiting, diarrhoea; **tympanic membrane "full"/bulging, opaque, reduced mobility.**
- The most accurate method of diagnosis is **tympanocentesis** but is seldom performed.
- Investigations: no role for routine diagnostic investigations; consider CT/MRI if intracranial complications suspected.

Intact tympanic membrane	Acute Otitis Media (AOM)	Otitis media with effusion (OME); 'Glue Ear'	Perforated tympanic membrane with otorrhoea
Not AOM, often seen with fever, eustachian tube obstruction, or viral URTI	Bulging and red tympanic membrane, loss of landmarks (especially handle of malleus), opaque	TM retracted, prominent handle of malleus; bulging or air-fluid level; yellow/amber appearance	Otorrhoea and relief of pain.
			

- **Watchful waiting** is recommended to make the diagnosis of treatable (bacterial) AOM.
 - Most AOM spontaneously resolves.
 - Treat pain with adequate, simple **analgesia** – topical analgesia (e.g. 2% lignocaine, 1-2 drops Benzocaine + Phenazone) may be considered as an adjunct provided **the tympanic membrane is intact**.
 - Decongestants, antihistamines, and corticosteroids are useless.
- **When to give antibiotics for AOM?**
 - Immediately for 'at risk': < 6 months, < 2 years + bilateral AOM, ear discharge, craniofacial anomalies e.g. Down syndrome, reduced immune function, cochlear implant, Aboriginal or Torres Strait Islander.
 - Also consider if **systemically unwell e.g. lethargic, pale, very irritable** – fever alone is not sufficient.
 - Most effective for bilateral AOM or discharge (NNT = 4 and 3 respectively).
 - Watchful waiting for 48 hours, then if not improving **amoxicillin 40 – 90 mg/kg/24 hours for 5 days.**
- **Complications of AOM:** subperiosteal abscess, facial palsy, sigmoid sinus thrombophlebitis, intracranial abscess, meningitis, labyrinthitis.

One feared complication of AOM is **acute mastoiditis**. It is the most common suppurative complication of AOM (though rare in absolute terms) and may be associated with intracranial complications. Recall the bone is thinnest between the skin and mastoid air cells.

- Presents in <3 years old. Swelling is **always** postero-superior to the pinna; therefore, **it pushes the pinna down and forward.**
- Usually **strep pneumoniae**!
- The diagnosis of AM is based on post auricular inflammatory signs (erythema, oedema, tenderness or fluctuance), protruding auricle, external auditory canal oedema, AOM.
 - Clinical diagnosis
 - Imaging is only indicated if there are clinical features of extra- or intracranial complications (e.g. hearing loss, tinnitus, vertigo, CN signs), severe illness/toxic appearance, or AOM not responding to Abx



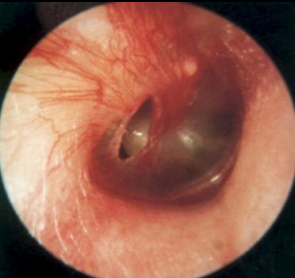
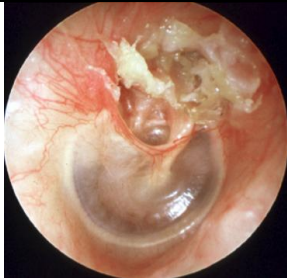
- **Management:** IV antibiotics (E.g. 3rd generation cephalosporin), surgical drainage (skin and tympanic membrane); cortical mastoidectomy.
 - Patients will require a grommet to relieve middle ear pressure and ventilate the middle ear.
 - Cortical mastoidectomy drains the developing abscess.

Otitis Media with Effusion (OME; Glue Ear)

- Fluid in middle ear with no signs or symptoms of inflammation. Fluid description varies - no predictive value.
- **Sequelae:** blocked ear, reduced hearing (conductive hearing loss), imbalance.
- Bubbling and air-fluid levels can signify resolution.
- Antibiotics and ENT referral are **not** routinely required for OME; most cases occur after an episode of AOM and resolve spontaneously with no long-term effects on language, literacy, or cognitive development.
- **Grommets** "break the suction" of negative middle ear pressure. Shepard is small (6 months); Reuter-Bobbin is regular (12 months); Sheehy and T-type are large (18 months and 2 years, respectively).
 - Adding adenoidectomy to grommet insertion reduces reinsertion rate by **20%**.
 - Large adenoids can obstruct the Eustachian tube orifice; the adenoid surface can also act as a reservoir for resistant bacteria (biofilm) → AOM
 - Remember: kids must avoid dirty water; pools need to be high chlorine and low pH (pseudomonas is sensitive to acid); beach and showers okay.
 - Recurrent infection or grommets can lead to **tympanosclerosis**.



Chronic Suppurative Otitis Media ("Open" Otitis Media).

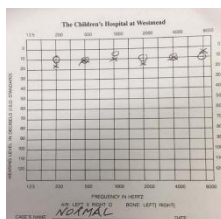
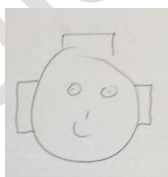
Safe	Unsafe
Pars tensa (central)	Adjacent to margin; skin (squamous epithelium) can grow medial to tympanic membrane (cholesteatoma).
Repaired via myringoplasty which uses connective tissue as a scaffold, e.g. temporalis fascia, or tragus perichondrium +/- cartilage.	Surgically treated, requires mastoidectomy .
	

Hearing Assessment in Paediatrics

Audiology: Round symbol = Right ear. Threshold < 20 dB normal.

Hearing loss:

Mild: 20 – 40 dB
Moderate: 40 – 60 dB
Severe: 60 – 80 dB
Profound: > 80 dB



Bone conduction: helps differentiate conductive hearing loss from SNHL. Represented with a bracket system. [= right ear,] = left ear; other ear is masked. Bone conduction > air conduction = conductive hearing loss.

Tympanometry

Measures the compliance of the tympanic membrane by reflecting sound off it.

- Peak should be at 0 (atmospheric pressure).
- Size of peak reflects compliance; location of peak (should be at 0) reflects middle ear pressure.
- A normal tympanogram is **Type A**.
- **Type B (flat):** middle ear effusion (i.e. glue ear) → middle ear fluid resists changes from external ear pressure or TM perforation (no longer mobile)
- **Type C:** negative pressure in middle ear → eustachian tube dysfunction or "congestion"

Nose

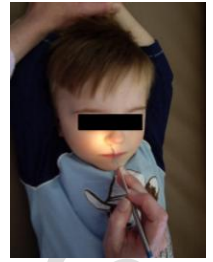
Nasal examination: have a light source. Examine outside, then lift the nasal tip with your finger and use an otoscope with speculum. You should be able to see the adenoids, inferior/middle turbinates, and the middle meatus.

Nasal Foreign Body: usually history; unilateral nasal discharge with excoriation or epistaxis. Button batteries require urgent removal (can → septal perforation in hours).

- Remove with a metal ear curette. May require decongestant or topical anaesthetic.

Septal perforation: managed with saline, intermittent topical antibiotics, and Vaseline.

Epistaxis: usually anterior and > 2 years old. Commonest causes are digital trauma, winter, allergies, and sprays.



- Beware of the juvenile angiofibroma in the **adolescent male** with **recurrent, unilateral, severe nosebleeds!**
- Blood dyscrasias and foreign bodies are also causes of epistaxis to rule out.
- **Treatment:** topical anaesthetic (helps bleeding vessel stand out), silver nitrate cautery.
 - Remember, only cauterise one side at a time (otherwise you can cause a septal perforation).
 - If crusting and digital trauma, suspect **low-grade vestibulitis or infection** → Bactroban or Nasalate.

Neonatal/infant nasal obstruction: can be an emergency because infants are obligate nasal breathers until 8-16 weeks!

- Choanal atresia is the most common cause, usually unilateral. May be syndromic e.g. **CHARGE**.
- Investigations: mirror misting test, passage of nasal catheter, nasoendoscope, CT scan.

Older Children Nasal Obstruction

- Anterior nose = allergy; posterior nose = adenoids.
- Examination: adenoid facies, allergic salute, examine tonsils and throat.
- Investigations: lateral airway X-ray (For adenoids); CT scan (rare); polysomnogram (if OSA suspected); allergy tests (skin tests, RAST testing).
- 1. Allergy: usually FHx. Managed via avoidance, steroid spray, antihistamines, desensitisation; surgery as a last resort.
- 2. Adenoids: history of snoring, nasal discharge/ nasal-sounding speech, adenoid facies, mouth breathing, glue ear (due to blockade of Eustachian tube). You can see the adenoid pad with otoscope +/- lateral airway X-ray. An adenoidectomy may be indicated.

Paediatric Sinusitis

- Acute (6 – 8 weeks) vs chronic (> 3 months).
- Multifactorial aetiology: allergy, viral URTI, daycare, reflux, air pollution.
- Acute inflammation e.g. viral URTI followed by secondary bacterial infection. The commonest pathogens are **the same as AOM** → *Streptococcus pneumoniae*, *Haemophilus influenza*, anaerobes, Staph (more chronic).
- Symptoms: nasal obstruction and congestion; facial pressure and pain; thick nasal discharge; toothache; fever; cough.
- Treatment: antibiotics, decongestants, saline nasal spray.
- Surgical management (adenoidectomy or conservative endoscopic sinus surgery) may be required to maintain patency of ostia, reduce tissue oedema, and prevent chronic sinusitis.

Septal Haematoma

- Kids have a cartilaginous nasal septum and nasal vault that can bend even with minor displacement → septal haematoma. No external deformity.
- **Clinical features:** trauma followed by nasal obstruction and new onset snoring.
- **Diagnosis:** palpate with cotton bud (feels spongy). *Don't confuse with a turbinate, which originates from the lateral wall instead!*
- **Management:** urgent drainage! Fractures must be reduced within 7 – 10 days.
- **Complications (if untreated):** septal perforation or abscess



Throat

Brodsky Grading Scale of Tonsil Size: Reflects the percentage of the oropharynx width covered by tonsils.



Viral Pharyngitis: the 'common cold'/URTI is the most common cause of sore throat in children!

- **Aetiology:** rhinovirus, influenza, parainfluenza, adenovirus, coxsackievirus, echovirus, Epstein-Barr virus, respiratory syncytial virus (RSV).
- **Clinical features:** mild-moderate sore throat, pharyngeal erythema, low-grade fever.
 - **Adenovirus** is more painful – higher fevers and conjunctivitis.

Virus	Clinical Features and Management
Adenovirus	Exquisitely painful, higher fevers, and conjunctivitis.
Epstein Barr Virus (Glandular Fever or Infectious Mononucleosis)	Fever, general malaise, headache, pharyngitis, dysphagia, odynophagia. Normal-sized or hypertrophic tonsils with grey exudate; palatal petechiae; large, tender cervical lymphadenopathy. • Very high fevers should raise your suspicion of a viral infection! • Remember to gently palpate the liver and spleen to assess for enlargement • Malaise is often prolonged and can last for weeks Diagnosis: $\uparrow$ WCC ($\uparrow\uparrow$ lymphocytes and atypical lymphocytes), $\uparrow$ ALT/AST, ALP, bilirubin • $\uparrow$ LFTs with hepatomegaly. • Heterophile agglutination test (Monospot test) for anti-EBV antibodies: rapid and specific, but lacks sensitivity (and children < 4 years may not develop antibody at all). $\uparrow$ in 40-60% in first month, remains $\uparrow$ for a month. • EBV-specific serology is being increasingly used, but takes longer. Management: symptomatic – avoid amoxicillin (causes a severe rash); NSAIDs; corticosteroid for airway obstruction. Spread through droplets; therefore, do not share eating utensils/drink bottles with others or swap saliva for at least 2 months!
Herpangina	Vesicles which become ulcers and spread over the anterior tonsillar pillars, palate, and posterior pharynx +/- cutaneous rash. Caused by coxsackie virus. Coxsackie A16 specifically causes hand, foot, and mouth disease. Highfever, oral and throat pain due to vesicular eruption in the mouth, rash or vesicles on the palms, soles, and buttocks. HSV causes cold sores and gingivostomatitis in younger children.

Tonsillitis

- Most common bacterial cause: Group A β -haemolytic streptococcus (GABHS)
- Peak incidence: 5 – 6 years (rare in kids < 3 years old)
- Symptoms: fever, sudden throat pain, odynophagia, malaise, dysphagia, otalgia, headache, nausea, and abdominal pain.
 - Remember to always check the throat in a kid with nausea and abdominal pain!
- Signs: oropharyngeal erythema, enlarged tonsils +/- exudate, swollen uvula, palatal petechiae, enlarged, tender cervical lymphadenopathy
- **Management:** antibiotics (penicillin or amoxicillin typically first-line; macrolides and cephalosporin may be more effective)
- **Tonsillectomy for recurrent tonsillitis:** 7+ episodes in 1 year, 5+ episodes a year for 2 years, 3+ episodes each year for 3 years.
 - Also consider severity, abscess, response to treatment, school absence.
 - Numerous techniques, all the same.

- Perform following **complete** resolution of acute infection; “**hot tonsillectomy**” very rare – often for peritonsillar abscess.
 - The most serious complication is **bleeding**; secondary bleeding occurs around Day 10.
 - **Pain**: avoid codeine post-tonsillectomy in those with OSA (hypermetabolisers); Nurofen remains controversial due to its GI problems.
 - Other complications of tonsillectomy include velopharyngeal insufficiency (if adenoids were large) and injury to the lips, tongue, or teeth.
- **Scarlet Fever** is a variation of GABHS pharyngitis with descending rash, from the neck to the trunk and extremities, that develops **12 to 24 hours after the fever**.
 - Patients interestingly have a **strawberry tongue** (with enlarged papillae) +/- white membrane.

Since a rash and strawberry tongue are present in both *Scarlet Fever* and *Kawasaki Disease*, they may easily be confused. Please consult the table below.

Feature	Scarlet Fever	Kawasaki Disease
Aetiology	Group A Streptococcus (S. pyogenes) toxin	Unknown; immune-mediated vasculitis
Typical Age Group	5–15 years	<5 years (mostly infants/toddlers)
Fever Characteristics	Abrupt onset, responds to antipyretics	Persistent (>5 days), minimally responsive
Rash	Fine, sandpaper texture; starts neck/chest, then generalizes; blanching; Pastia's lines (<i>erythema in skin folds with linear petechial character</i>) 	Polymorphous, diffuse; often trunk/groin/genitalia; sheet-like desquamation
Strawberry Tongue	Red with accentuated papillae 	Red, but not white tongue, cracked lips 
Mucosal Involvement	Pharyngitis, conjunctivitis, circumoral pallor , no	Dry, cracked lips; bilateral conjunctivitis (non-exudative, limbal-sparing) 
Extremity Involvement	Flaking/desquamation, diffuse 	Red/swollen hands and feet, periungual sheet-like desquamation 
Lymphadenopathy	Tender cervical adenopathy	Painless, unilateral cervical adenopathy
Complications	Rheumatic fever, PSGN	Coronary artery aneurysm, myocarditis
Treatment	Penicillin/antibiotics	IVIG, aspirin

Random fun point: diphtheria is another cause of tonsillitis. Patients have a grey or whitish, sharply demarcated diphtheritic pseudomembrane that adheres to underlying tissues and can bleed if touched. Hoarseness can be a sign of laryngeal diphtheria.



Also see the sections on [Obstructive Sleep Apnoea](#) and [Foreign Body Ingestion/Aspiration](#).

2. General Paediatrics

Growth and Nutrition

Determinants of Growth: genetics (mid-parental height), nutrition, environment, health, ethnicity.

- A breastfed infant grows rapidly in the first 2 months and less rapidly from 3-12 months.

Growth Shortcuts	
Mid-parental height: Target Height Female = $(\text{Height of Father} + \text{Height of Mother} - 13)/2$ Target Height Male = $(\text{Height of Father} + \text{Height of Mother} + 13)/2$	Weight estimate = $(\text{Age in years} + 4) \times 2$

Feeding:

Energy requirements depend on age, basal metabolic rate, physical activity, growth, and illness.

Age	Energy Requirement
Neonate	100 – 120 kcal/kg/day
Older infant (6 – 12 months)	90 – 100 kcal/kg/day
Young children (1 – 3 years)	80 – 90 kcal/kg/day

Breast milk provides the complete nutritional profile in infants up to 6 months. Solids are introduced around 6 months of age, where developmental readiness is present (e.g. head control, sitting, loss of tongue thrust reflex). Parents are urged to begin with small amounts of iron-rich pureed foods (e.g. meats, fortified cereals) and gradually increase variety and texture.

- **Breast milk is insufficient in iron, Vitamin D, zinc, and Vitamin B12.** Iron and Vitamin D deficiency are particularly notable for infants older than 6 months who are exclusively breastfed.
- Neonates should not drink water due to the risk of water intoxication and electrolyte imbalances.
- Infants can begin drinking water at 6 months; by 12 months, they are encouraged to transition to water and cow's milk as the primary fluids.

Recall the neonatal 50th percentiles (the 3's and 5's): BW 3.5 kg, HC 35 cm, L 53 cm.

Assessing Growth: weight, height/length, head circumference, plotted on the appropriate growth chart, serially.

	Birth – 2 years	> 2 years
Head circumference	Tape around widest part of head (above eyebrows to occipital prominence); nearest 0.1 cm	-
Weight	Pan scale; record to the nearest 10g	Regular digital scales; to the nearest 0.1 kg
Length	Two-person job; ensure legs are straight and flat (remove shoes and socks); nearest 0.1 cm	Standometer

So cute!



Standards:

- **WHO charts** are used for children **aged 0 to 2 years** (whether breastfed or formula fed, and for all ethnic groups). They were originally based on healthy breastfed infants from 6 countries living in optimal conditions.
 - Correct for prematurity for the first 2 years!
- The **CDC Growth Charts** are used for ages 2 – 18 years (weight, height, BMI).
- There are disease-specific charts e.g. Down Syndrome, Turner Syndrome, Achondroplasia etc.
- There are also many ethnic-specific growth charts.

Healthy Growth	Serial measurements “track” along percentiles; weight, length, and height are in proportion.
Poor Growth	Weight and/or length tracking downwards on percentiles.
‘Flags’	Weight or length < 3 rd percentile or > 97 th percentile, BMI > 85 th percentile
Further investigate	Unexplained weight loss, weight not regained following acute illness, weight or length plateau, increasing or decreasing centiles.

Unhealthy Excess Weight

BMI is the standard tool to assess unhealthy weight in >2-year-olds. It depends on age and stage of growth.

Overweight ≥ 85th percentile for age on BMI chart; obese ≥ 95th percentile.

Antecedents of Adult Disease

- The first 2000 days are critical for brain development; people develop over **90% of their total head circumference**.
- Adverse childhood events (ACE) predict many adverse outcomes in adulthood.
 - School performance, involvement in criminal justice system, substance abuse, obesity, learning difficulties.
 - 3 y/o childhood self-control measures predict: general health, substance dependence, income/SES, financial problems, criminal conviction, single-parent child rearing, welfare receipt, life satisfaction.
- Protective factors can improve outcomes: love, reading and speaking, music; skin-to-skin

Child Development

Development: skills across a number of domains – gross motor, visual-motor, language (expressive, receptive, social pragmatic), personal social/ self-care.

- Progressive acquisition of skills, complex with spurts and plateaus.
- Influenced by child-related factors and environmental factors e.g. family
- Development in different domains can occur at different rates

History (GRADING)

Genetics and relatives	Consanguinity, pedigrees
Reading/ writing, recreation	Level of schooling/ pre-school, sport, social skills, bullying
Antenatal/ perinatal	Pregnancy: obstetric problems, abnormal antenatal scans and screening tests, medications during pregnancy. Gestational age at delivery. Labour and delivery including antibiotic coverage. Neonatal: did the child require admission to a special care unit, and if so, why?
Developmental	Hearing and vision; gross motor; fine motor; language; social. Screening: - When did the child sit unaided? (~6 months) - When did they talk? (~12 months) - When did they walk? (~12 months) - When were they toilet-trained? (~2-3 years)
Immunisation History	Is the child up to date with their immunisations?
Nutrition History	Infant: which, when, how much, how often; breast vs bottle; solids (which, when weaned). Child/adolescent: breakfast, lunch, dinner, snacks
Growth	Growth charts: WHO if < 2 years (correct for gestational age), CDC for 2-18

Examination:

- Observe the child’s interactions during the consultation
- Plot parameters on growth charts
- Reflexes
- Hearing and vision
- Gross and fine motor assessment
- Appearance
- Interaction (parent and others)
- Behaviour and play
- Motor skills (gross and fine)
- Communication

The **Ages and Stages Questionnaire (ASQ)** is a parent-completed screening tool used to monitor and assess developmental progress in children from 1 month to 5 ½ years of age. Parents/ caregivers complete the questionnaire by observing their child performing specified activities, or by answering based on recent everyday observations if the child already does the skill.

- Score in white area = normal; grey area = monitor; black area = refer
- If language difficulties or concern parent can't complete questionnaire, there is a kit with items that clinician can undertake.

Normal Developmental Milestones

For major milestones, recall "*Heads, shoulders, knees and toes; eyes, ears, mouth, and nose (knows)*".

Age	Milestone
3 months	Holds head upright
6 months	Rolls over (shoulders)
9 months	Crawls (knees)
12 months	Walks (toes)
12 months	Eyes (peekaboo)
18 months	Ears (follows one-step commands)
2 years	Mouth (forms phrases)
3 years	Knows (their age)

Comprehensive Table:

Age	Gross Motor	Visual-Motor	Language	Social/Adaptive
1 month	Raises head from prone	Visually fixes, follows to midline, tight grasp	Alerts to sound	Regards face
2 months	Holds head in midline, lifts chest off table	Follows past midline, fist not as tight	Smiles socially	Recognises parent
3 months	Supports on forearms, holds head up steadily	Hands open at rest, follows in circle, reacts to visual threat	Coos	Reaches for objects, anticipates feed
4 months	Rolls over, supports on wrists, shifts weight	Reaches with both arms, hands to midline	Laughs, orients to voice	Enjoys looking around
6 months	Sits unsupported , puts feet in mouth	Unilateral reach, raking grasp, transfers objects	Babbles, lateral orientation to bell	Recognises stranger
9 months	Pivots with sitting, crawls, pulls to stand, cruises	Immature pincer, probes with forefinger, holds bottle, throws	Mum, Dad indiscriminant, waves, understands "no"	Explores environment, gesture games, object permanence
12 months	Walks alone	Mature pincer , voluntary release	Uses 2 words , 1 step command with gesture	Imitates actions, comes when called, scared of strangers
15 months	Creeps up stairs, walks backward	Scribbles in imitation, tower of 2 blocks	4-6 words, 1 step commands without gesture	Uses spoon and cup
18 months	Runs, throws objects from standing without falling	Scribbles spontaneously, tower 3 blocks, turns 2-3 pages	7-10 words, knows 5 body parts	Copies parents, plays with other kids
24 months	Walks up and down steps, jumps	Imitates pencil stroke , tower 7 blocks, removes shoes, pants	2 step command, 50 words, 2 words together	Parallel play
3 years	Alternates feet up steps	Copies O, undresses completely, partly dresses, unbuttons	Minimum 250 words, 3 word sentences	Group play, shares, takes turns
4 years	Hops, skips, alternate feet going down steps, pedals tricycle	Copies square, buttons, dresses self, catches ball	Knows colours, asks questions, repeats song	Tells "tell tales", plays cooperatively
5 years	Skips alternating feet, jumps over obstacles	Copies triangle, ties shoes, spreads with knife	Prints name, asks word meaning, labels feelings	Competitive games, helps with tasks

Putting it together, you can remember that we start solids in a **6 month old** because this is when they can sit unsupported, when they grow their first tooth, and when they can rake at their food.

Genetic Syndromes: Pattern Recognition

Fragile X Syndrome	Commonest neurodevelopmental condition; intellectual disability, behavioural challenges (e.g. autism spectrum features, macroorchidism, long face, long ears, hypermobility).
CHARGE Syndrome	Coloboma, heart defects, choanal atresia, growth retardation, genital anomalies, ear abnormalities.
Foetal alcohol syndrome	Smooth philtrum, thin upper lip, growth retardation, neurodevelopmental impairments (intellectual and behavioural difficulties)
Neurofibromatosis	NF1 commonest; café-au-lait spots, neurofibromas, skeletal dysplasias, learning disabilities
Tuberous sclerosis	Autosomal dominant neurocutaneous syndrome, with benign tumours in the brain, skin (ashleaf spots/ shagreen patches) and other organs; neurodevelopmentally can manifest as intellectual disability, autism, or seizure disorders.

Red flags at all ages: regression, lack of response to stimuli (social, visual, auditory), absent or limited eye contact, asymmetry of strength/tone/movement.

Developmental Delay

- Not developing skills according to the expected timeframe (< 5 years). May result from:
 - Biological factor e.g. chromosomal disorder
 - Environmental e.g. family mental health, neglect, vulnerability
 - Unknown (most common)
- **Language** difficulties are usually the first manifestation of developmental delay.
- **Specific developmental delay:** confined to 1 area of development
- **Global developmental delay (DSM-5):** fails to meet milestones in ≥ 2 categories (< 5 years old **only**)
 - This is a clinical diagnosis. Children are too young to participate in standardised testing. Reassess once old enough to participate in standardised testing
- Early identification of developmental delay is critical to the wellbeing of children and their families.

Terminology: Surveillance (recognising those *at risk* of developmental disorders), Screening (standardised tools), Referral and further evaluation.

Developmental Surveillance: a flexible, continuous process with broad scope. Every visit with a clinician is seen as an opportunity for **developmental surveillance**.

Developmental Screening: brief assessment procedure designed to identify children who should receive more intensive formal assessment. Simple, reproducible, readily available to identify children with high probability of a developmental disorder.

Screening and Surveillance Tools: history from parent, informal observation of the child, Blue Book (NSW 2021; 6, 12, 18 mo, 2, 3, 4 years), Ages and Stages Questionnaire, Mary Sheridan 'From Birth to 5 Years' (timeless reference).

Development should be assessed at every consult opportunistically. Parental concern is very reliable for delay. Also look out for red flags (above).

Other Investigations

- **Hearing and vision tests** with speech pathology referral.
 - **ALWAYS TEST HEARING AND VISION FOR DEVELOPMENTAL CONCERNS!**
 - **Hearing if language delay.**
- **Bloods:** FBC and blood film, EUC, LFT, TFT, iron studies, Vitamin D, Vitamin B12/folate, CK, TORCH titres
 - **Not usually for isolated delay**
- Chromosome microarray and genetic testing (e.g. FISH probe)
- Fragile X testing
- Urine metabolic screen
- Very occasionally we'd get neurology involved: neurologist referral, neuro-imaging, EEG.

Genetic testing has variable yield. No cause is found in 40% with moderate-severe intellectual disability, and 70% with mild intellectual disability. The most common result is a **variant of uncertain significance (VOUS)**.

Management

- Refer to allied health, e.g. speech, physio, OT (community or private)
- **Early Childhood Early Intervention (NDIS):** kids under 7 don't need formal diagnosis

- If over 7, need diagnosis – usually intellectual disability or ASD
- Need to be Australian Citizen or permanent resident
- Refer to paediatrician; if isolated gross motor skills delay → paediatric neurologist

Formalised Developmental Assessment

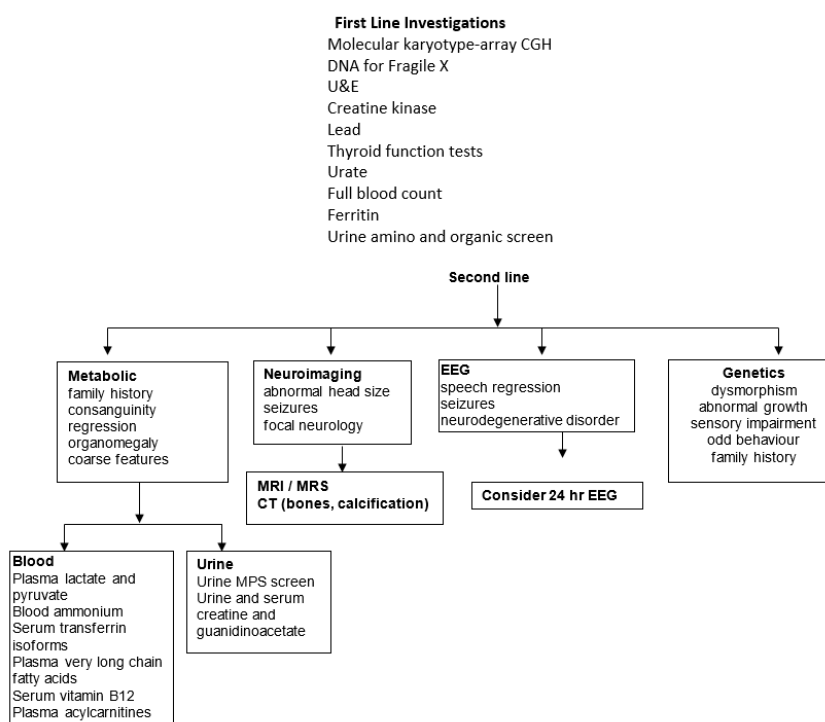
These are for children with established developmental concerns or intellectual difficulties.

Tool	Age	Description
Griffiths III	0 – 6 years	Gross and fine motor skills, foundations of learning/non-verbal cognitive abilities, and overall picture.
Bayleys Scale of Infant and Toddler Development	16 days – 42 months (3.5 years)	Often done in neonatal follow-up clinics. Language scale separated into expressive and receptive.
Autism Diagnostic Observation Schedule (ADOS)		In addition to developmental assessment; not always necessary for Dx of ASD.
Intellectual Assessment	4+ years	Performed by clinical psychologists: Wechsler Preschool and Primary Scales of Intelligence 3rd Edition (WPPSI IV) Stanford-Binet Intelligence Scales 5th Edition Wechsler Intelligence Scales for Children 5th Edition (WISC V) Wechsler Adults Intelligence Scales 4th Edition (WAIS – IV) We get concerned when scores fall < 2 SD below the mean (2.5th percentile) → intellectual disability. Verbally-based, so not suitable for children who are nonverbal or who have language limitations.
Adaptive Function		ABAS, Vineland (parent interview questionnaires) Note: to be diagnosed with an intellectual disability, you need to demonstrate difficulties in adaptive behaviour.

Flow-Chart from Recommended Readings

Recommended approach to global developmental delay of unclear aetiology in preschool children: (flow diagram adapted and modified from McDonald et al)

- Thorough history and physical examination
- Formal developmental assessment
- Formal hearing assessment
- Formal vision assessment
- If diagnosis not apparent consider following investigations:



Faltering Growth (Failure to Thrive)

Growth occurs when there is sufficient supply and proper use of energy. Faltering growth/ failure to thrive (FTT) occurs when this balanced is perturbed.

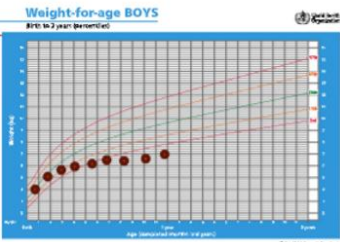
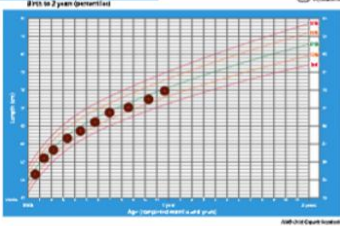
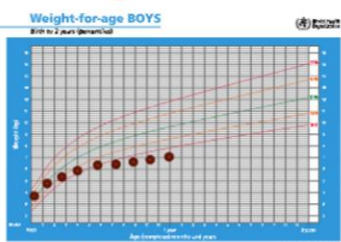
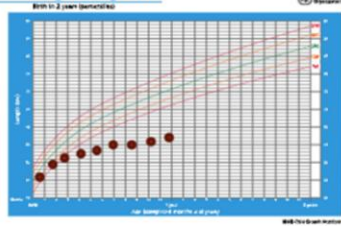
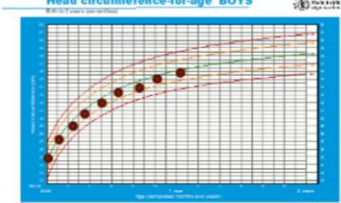
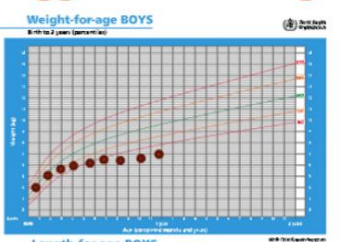
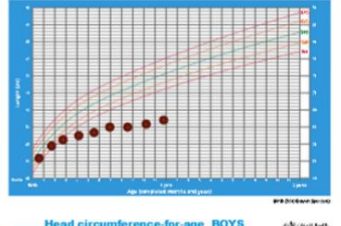
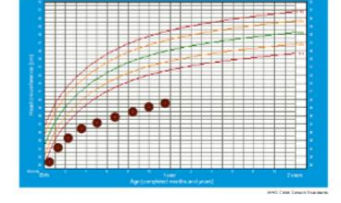
Normal growth: **Peak 0 – 3 months (150 – 200 g/wk)**, 2 – 5 months (100 – 150 g/wk), 5 – 12 months (70 – 80 g/wk).

Rate of growth	Attained growth	Depressed weight gain
Fall off from previously established growth curve. Downward crossing of 2+ major percentiles on growth chart.	Weight < 3 rd percentile on growth chart Weight for height < 5 th percentile on growth chart Weight 20% or more below ideal height for weight (wasting)	< 20 g/d from 0 to 3 months of age (<150 g/week) < 15 g/day from 3 to 6 months of age (< 100 g/week)

Five mechanisms for faltering growth

Inadequate intake	Decreased absorption	Excessive losses	Abnormal use	Increased need
Diet e.g. breast milk, formula reconstitution, parent education, poverty Oromotor or feeding dysfunction e.g. cleft palate, poor suck or swallow Other e.g. oral aversion, anorexia, food refusal	Small intestine: coeliac disease, cow milk protein intolerance, Crohn's Colonic: entamoeba histolytica, short-segment Hirschsprung's Other: CF (pancreas), chronic cholestatic disorders (liver), biliary atresia	Vomiting: GORD, GI obstruction, raised ICP, metabolic problems Diarrhoea: IBD, infection, immunodeficiency, coeliac, CF Renal: polydipsia, polyuria (DM), RTA	Endocrine: DM, hypopituitarism Inborn errors of metabolism: mitochondrial disorders Other: chromosomal, prenatal insult	Chronic disease e.g. congenital heart disease, chronic respiratory disease, chronic or recurrent infection Hyperthyroidism Neoplasms

The pattern of growth parameters can provide insight into the aetiology of FTT. In children without genetic conditions, the head circumference is always spared (i.e. maintained). A child with a brain insult around birth may have a normal HC to start, but then it crosses centiles.

↓ Weight	↓ Weight and height, normal HC	↓ Weight, height, HC
Caloric insufficiency, decreased intake, increased loss (e.g. diarrhoea, malabsorption), hypermetabolic state	Structural dystrophies, endocrine causes, constitutional growth delay, genetically short.	Intrauterine insult and genetic abnormality, e.g. congenital CMV
  	  	  

- In the first few months of life, a healthy baby who is gaining weight may **cross and track along a lower centile than their birthweight**, particularly if they were LGA. This is known as ‘**catch down**’ growth.
- **Constitutional growth delay** is benign; growth is slower, but parents may be assured their child will “catch up”. **Bone age (L wrist)** shows skeletal immaturity.
- Whereas in **familial short stature** the bone age is normal (comparable to chronological age)

Remember for evaluation of faltering growth, the **blue book** is vital in the first 2 years, as are **growth charts**.

History: food intake, output (including urine output and other losses), food behaviour and dietary practices, past Hx (chronic and current illness, recurrent infection), FHx (pattern of weight gain and growth in other family members), psychosocial assessment.

Examination:

- General observation: in proportion and well, or unwell?
- Hydration
- Signs of underlying systemic diagnosis
- Mid-parental height
- Muscle bulk (buttocks), subcutaneous fat stores (thighs), skin, hair, gums, eyes, nails
- Developmental level, caregiver-child interactions
- Feed if able

Also assess psychosocial causes: mother (depressed, external demands, care of others); poverty; drug and alcohol abuse; lack of support regarding parenting; physical absence of the father/DV; parental intellectual disability.

Investigations

- Urine: urinalysis, microscopy and culture (especially infants < 12 months).
- Bloods: FBC, ferritin (inflammation), EUC, TSH, glucose, LFT, **coeliac serology + total IgA** (if on gluten-containing diet).
- Micronutrients, especially **active B12** (for malabsorption/ suspected restricted dietary intake).
- Stool: microscopy, fat globules (fat maldigestion e.g. biliary/ pancreatic insufficiency), fatty acid crystals (fat malabsorption, e.g. intestinal mucosal damage).
 - If > 12 months, we may consider **ESR** and **faecal calprotectin** for IBD.
- Bone age if suspecting constitutional growth delay (e.g. growth plates in L wrist)
- Karyotype to rule out Turner Syndrome in girls
- Consider GH stimulation test if suspecting pituitary GH deficiency (delayed skeletal maturation, young/ “cherubic” appearance, increased body fat, muscle weakness, delayed tooth eruption, delayed/ absent puberty).
 - Neonatal signs can be severe: hypoglycaemia, prolonged jaundice, micropenis (males), midline defects e.g. cleft palate

Children with isolated <3rd centile weight-for-age, with **normal neurodevelopmental progress** and **no red flags** may be within normal limits for growth – just **monitor**.

Management

- Adequate calories, appropriate diet for “catch up growth”
- Supplement with extra calories e.g. added fats to food, Polyjoule, Duocal
 - SCHN has a ‘High energy eating for children’ factsheet
- Treat underlying acute and chronic disease
- Needs and rights of the child and family: protect the child if they are not safe, parental support (antenatal, early child support services)
- Monitor growth in all children!

Example clinical cases:

5 months, big baby with vomiting, no distress or haematemesis → **overfeeding**

7 months, poor weight gain since 4 months old with increasing vomiting, flecks of blood, weak cry, hoarse voice, head tilt; head circumference has been increasing centiles → **posterior fossa tumour**

2-year-old refusing to eat solid food; drinks 2L cow’s milk per day; otherwise well but hypochromic microcytic anaemia → **iron deficiency (confirmed on iron studies)**

18 months, plateaued weight since 6 months old, gaseous abdomen distension and buttock wasting → **coeliac disease (needs serology, endoscopy + duodenal biopsy, gluten free diet)**

- The dead giveaway is that the child is completely okay until solids are introduced.

Please see **Adolescent Health** for notes on [Puberty](#) and [Obesity](#).

Recommended Reading: Assessment of Childhood Short Stature (for GP)

- The prevalence of short stature is linked to **poverty**.

Assessment of childhood short stature: a GP guide

Physiological Causes of Childhood Short Stature	
Familial short stature	Within 3 centiles of MPH, no red flags.
Constitutional delay of growth and puberty.	Short stature for age and sex, normal growth rate, delayed puberty and pubertal growth spurt. M:F 2:1; delayed skeletal maturation (bone age) seen.
Pathological Causes	
Primary	Secondary
Secondary to being SGA with no catch-up in growth (consider in kids > 4 years with height < 2 nd centile and BW or L < 2 nd centile)	Nutritional insufficiency e.g. malabsorption, poor intake, feeding problems
Skeletal dysplasia (usually associated with body disproportion)	Systemic diseases/ inflammation e.g. Crohn's disease
Chromosomal disorders e.g. Turner Syndrome in girls	Psychosocial nutritional deprivation
	Endocrine disorders e.g. hypothyroidism, GH deficiency

Determinants of Childhood Growth

Infancy: nutrition, rapid height velocity (up to 25 cm/year)

Childhood (from 3 years old): growth hormone and thyroid hormone, slower 4-8 cm/year.

Puberty: growth spurt, growth hormone and sex hormones, dramatic increase in height velocity (8 – 12 cm/year).

Normal growth and pubertal development

Remember 4 concerning scenarios: child with Sx chronic illness, height < 2nd centile, height > 3 centiles below parents' heights, growth failure.

Know the importance of a targeted history, examination (screening for *secondary causes* e.g. chronic illness), accurate growth measurement, and parental height assessment.

Basic investigations in primary care: FBC and ESR (inflammation and anaemia), EUC, LFT, CMP, coeliac serology, TFTs (TSH, FT4).

Obesity

Overweight: BMI 85 – 95 centile for age and sex

Obese: BMI ≥ 95th centile for age and sex

Currently, **1 in 4 Australian children/ teenagers aged 5 – 16 years are overweight or obese (8% obese)**

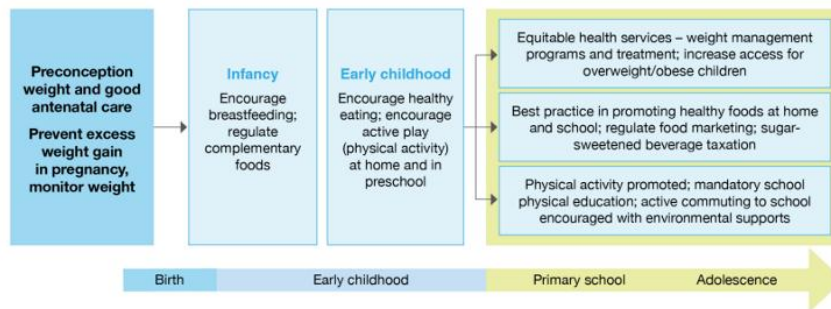
- Increasing prevalence worldwide since 1980s, but has recently plateaued in Australia, except for disadvantaged communities. Small island nations most affected.
- Preschool-aged children are now affected in numerous countries.
- Many bio-psycho-social impacts: physical health and metabolic syndrome, lower self-esteem, increased bullying, reduced school attendance, poorer employment
- The Harvard Growth Study (NEJM 1992) showed that for males, adolescent weight status predicted later adult health risks.

Drivers of obesity epidemic:

- **Genetic contribution 40 – 70%**; polygenic. Implicated genes include appetite/satiety signalling, adipocyte signalling, adrenal action, energy homeostasis, insulin sensitivity, nutrient partitioning etc.
 - Role of in utero and early environments
- Environment: screen time, decreased activity, changed diet, shortened sleep duration
 - The environment ultimately determines whether the genetic predisposition is manifest.

Prevention of obesity:

Environmental change (**upstream population health approaches**) is more important than individual change. The National Obesity Strategy launched in March 2022 has some great plans (but implementation TBD) involving school/child care, the food environment (food marketing and access to healthy foods), physical activity infrastructure, workplaces, and tackling the drivers of inequity.



Assessment of Obesity

- History: general, weight, complications, family history
- Measure height and weight, and plot on BMI for age chart.
 - Note: weight **maintenance** can be a good strategy for children who are still growing!
- Central fat distribution: waist:height ratio > 0.5 (if >6 years old) associated with increased cardio-metabolic risk

Findings requiring further investigation: short stature, abnormal physical stigmata (syndromic cause), developmental disability, visual disturbance or headache (idiopathic intracranial hypertension).

When to investigate overweight/obesity:

- Age (adolescents > young children)
- Higher levels of BMI and central obesity
- Clinical suggestion of complications e.g. acanthosis nigricans
- Higher-risk ethnic groups or high risk family history

Investigations

Initial investigations	Other
Glucose, LFTs, lipids, TFTs, insulin	HbA1c, OGTT, liver ultrasound, reproductive hormones, FBC, micronutrient deficiencies, sleep assessment

Insulin:glucose (pmol/mmol) ratio > 15 consistent with insulin resistance. Note, **insulin pmol/L ~ insulin mIU/L x 6**

Basic treatment goals:

- Reduction in weight and weight-related outcomes
- Change in weight-gain trajectory
- Improvement in obesity-related complications
- Change in markers of future health, psychological, and social complications

Obesity Management in Children and Adolescents

Remember to manage **obesity-associated complications** e.g. OSA, gastrointestinal issues, orthopaedic etc.

Behaviour change: typically, small reduction in body weight status, more effective at older age. Multicomponent interventions may be effective for kids ≤ 6 years old.

- This data comes from studies; recall, there are barriers in real life.
- **Principles:** family involvement, long-term, developmentally appropriate, plan for long-term weight maintenance

Practical recommendations

- Changes in food intake
- Increased physical activity
- Decreased screen time
- Goal-setting
- Stimulus control
- Self-monitoring
- **Role modelling and parental involvement**
- Regular sleep routines
- Involvement of health professionals e.g. dietitian, exercise professional

NSW Health specifically promotes 8 healthy habits to support children's well-being: drink water, aim for 5+ serves of vegetables and 2+ serves fruit daily, start each day with a healthy breakfast, be mindful of portions and when you feel full, choose healthier snacks, limit screens, be active for at least 1 hour a day, get enough sleep.

Severe Obesity

- Bariatric surgery improves weight and cardiometabolic outcomes in adolescents with severe obesity (**30% weight loss in 3 years, Roux-en-Y bypass or sleeve gastrectomy**)
- Obesity pharmacotherapy: only **orlistat** is approved in Australia (GI lipase inhibitor)
 - Not FDA-approved but off-label: metformin, topiramate, exenatide (GLP-1 agonist), lisdexamfetamine
 - STEP TEENS Study**: RCT on **semaglutide**. Can reduce body weight and improve cardiometabolic risk factors. But, not on PBS so need to pay out-of-pocket.

Access to effective treatment is challenging. And, remember to tackle weight stigma!

Prevention approaches require multi-point and multi-level interventions. Must extend beyond the immediate family and school environments to up-stream drivers of food and physical activity environments.

Brief Resolved Unexplained Episode (BRUE)

Brief	Resolved	Unexplained	Episode
Usually < 1 minute in child < 12 months old	Complete return to baseline	Cannot be explained by medical cause	1+ of marked change in breathing, tone, colour, or responsiveness

- Incidence**: 1 case per 1000 births
- Aetiology**: most commonly, exaggerated physiological response to feed, vomit, or secretions
 - It is common for no specific diagnosis to be made after evaluation and a period of observation
- There is a minimal risk of SUDI after BRUE; higher risk if maternal smoking or non-accidental injury.
- History** is vital, characterizing the event, plus environmental risks and any family history (unexplained death, arrhythmia, genetic conditions).
 - Red flags for non-accidental injury**: unexplained bruising, delays seeking medical attention, story inconsistent with child's age, changing version of story, domestic violence warning, suspicious bruising patterns or burns/nail marks/ scratches.
- By definition, a child with a BRUE does **not** require ongoing acute treatment! If discharged, it is recommended the infants have early medical follow-up. In practice, many with a low-risk BRUE are admitted to hospital for observation for reassurance

	High-risk BRUE	Low-risk BRUE
Factors	< 2 months old Prematurity (<32 weeks) More than 1 event	Age > 60 days Born > 32 weeks No CPR by trained healthcare professional First event Duration < 1 minute Normal exams and no concerns on Hx
Investigations	FBC, EUC, CRP, septic W/U Glucose Nasopharyngeal aspirates Blood gas ECG Further Ix based on history, e.g. EEG, VFSS (swallowing), milk scan (aspiration)	No investigations required Blood glucose Urinalysis
Disposition	Admit for observation, oximetry, +/- telemetry Consultant review/ surgery/ neuro/ CPU depending on concern.	May be discharged if parents confident Parental reassurance Early medical follow-up.

Commonest underlying Dx (no longer a BRUE).

Gastro-esophageal reflux	Results in laryngospasm or exaggerated physiological response
Lower respiratory tract infection	Can cause apnoea more common with prematurity
Seizures	Consider non-accidental injury

Differentials:

Category	Differentials	Features
Physiological	Exaggerated response to secretions or refluxate	Gagging, choking
	Periodic breathing	Pauses in breathing < 10 seconds
Infection	Sepsis	Fever, floppy, non-blanching rash, poor feeding, lethargy
	Viral respiratory infection, pertussis, RSV	Apnoea, cough, infectious contact, secretions
	Meningitis	Bulging fontanelle, abnormal posturing (opisthotonos), irritability, poor feeding, lethargy
Airway obstruction	Structural anomaly of the airway (stenosis, cyst), vascular ring	Stridor, appearing not to be able to get air despite effort, recession, head bobbing
	Croup	Stridor, barking cough, fever
Metabolic	Hypoglycaemia	Drowsing, history of vomiting and reduced intake
	Hypocalcaemia	Seizure, twitching, maternal Vit D deficiency
	Hypokalaemia	Cardiac arrhythmia, abnormal ECG, history of vomiting or diarrhoea
	Inborn error of metabolism	Family history, consanguinity, seizure, drowsiness
Toxin/drug	Accidental or non-accidental	Parental drug use, vaping liquid, parental methadone use
Cardiac	Arrhythmia	Abnormal ECG, sudden onset pallor, sweating
	Congenital heart disease	Murmur, reduced or absent femoral pulses or differential UL/LL BP (coarctation), cyanosis
	Prolonged QT	Family history of sudden death, long QT on ECG
Neurological	Head injury	Trauma or fall from height, bruising, leaking of CSF from ear or nose (base of skull #), bruising around eyes (raccoon eyes)
	Seizure	Jerking movements, eyes deviating, not responsive to voice or touch, sleepy after movement stops
	Infection	Fever, seizure (encephalitis), reduced level of consciousness, abnormal extensor posturing
Surgical	Intussusception	Screaming and in pain, vomiting, 'red currant jelly' stool, can appear to have reduced level of consciousness
	Strangulated hernia	Hernia in groin with overlying erythema
	Testicular torsion	Painful testis, sits higher and horizontal on affected side
NAI	Shaken baby	Irritable, seizure, reduced LOC, retinal haemorrhages, fractures, bruising – finger pattern
	Drug overdose	Drowsiness, parental drug abuse

Risk factors for SUDI: teenage pregnancy, alcohol use in pregnancy, premature birth, LBW, tobacco use in pregnancy, assigned male at birth.

Constipation

Definition: delay or difficulty passing stool for >2 weeks. The exception is in **breastfed infants** where it is not uncommon to have days-weeks between bowel motions; consider stool **consistency** instead (Type 1 or 2 Bristol).

Aetiology: functional constipation is the commonest, caused by environmental factors with no underlying pathology (e.g. toilet training, changes in routine or diet, stress, illness, unavailability of toilets or too busy).

Rome III Diagnostic Criteria for Functional Constipation: no organic pathology; 2+ of; <2 bowel motions/week, 1+ episode incontinence per week after being toilet trained; Hx stool retention, painful or hard bowel movements; large faecal mass in rectum; large diameter stools that may obstruct toilet.

- <4 years need criteria for 1 month
- >4 years need criteria for at least once per week for 2 months

Red flags: onset < 1 month old, **delayed passage of meconium > 48h**

(Hirschsprung's Disease!), fever, vomiting, bloody diarrhoea, failure to thrive, anal stenosis (may have tight, empty rectum), impaction, abdominal distension, sacral dimple/hair on spine/gluteal cleft deviation (tethered cord or occult spina bifida); reduced tone/ reflexes/ strength in lower limbs (spinal cord issue).

Bristol Stool Chart

Type 1		Separate hard lumps, like nuts (hard to pass)
Type 2		Sausage-shaped but lumpy
Type 3		Like a sausage but with cracks on the surface
Type 4		Like a sausage or snake, smooth and soft
Type 5		Soft blobs with clear-cut edges
Type 6		Fluffy pieces with ragged edges, a mushy stool
Type 7		Watery, no solid pieces. Entirely Liquid

Management of functional constipation:

- Education (that soiling is not intentional, just overflow around a hard faecal mass)
- Acute disimpaction:
 - > 1 year: lactulose or mineral oil + suppository/enema
 - < 1 year: lactulose + suppository/enema (mineral oil has an aspiration risk).
- Maintenance therapy: diet, behaviour, laxatives if necessary (lactulose is first-line).
 - Give laxatives for at least 2 months then wean if free of constipation for 1 month

A child with intractable constipation **may** have cow's milk protein allergy; a 2-4-week trial of cow's milk protein avoidance may be indicated.

Lactose Intolerance

Lactose, a disaccharide, must be broken down by **lactase** in the small intestine brush border. Inability to break down lactose leads to lactose intolerance.

Clinical features: abdominal pain, diarrhoea, nausea, flatulence, or bloating after ingestion of lactose-containing food.

Primary lactase deficiency (lactase nonpersistence)	Secondary lactase deficiency	Congenital lactase deficiency	Developmental lactase deficiency
Most common form Genetically programmed gradual decline in lactase activity that begins after weaning i.e. not in <2 y/o More common in some ethnic groups e.g. Asian, Hispanic	Due to small bowel injury e.g. gastro, bacterial overgrowth, chemotherapy More common in infancy	Extremely rare autosomal recessive genetic disorder due to mutations in the lactase (LCT) gene Severe diarrhoea and FTT Would not survive without lactose-free human milk substitute	Relative lactase deficiency due to immature development of the intestinal mucosa A disease of premature infants; usually resolves shortly after birth

Diagnosis:

- 2-week trial of lactose-free diet leading to resolution of symptoms (then recurrence after lactose re-introduced)
- If subtle, hydrogen breath test can be used. Colonic flora ferment undigested lactose, producing lactic acid and H₂ gas as a byproduct.
- Stool can be tested for pH (low due to lactic acid) and faecal reducing substances (of which lactose is one).
 - Note, most disaccharidase deficiencies will have elevated reducing substances, except for **sucrase**, because sucrose is not a reducing substance.
- Intestinal biopsies for lactase activity in the gut.
- Stool specimen for Giardia or cryptosporidium
- Antibodies for coeliac disease (TTG, total IgA): coeliac disease can cause lactose intolerance

Management: lactose-free formula in non-breastfed; in breastfed infant, continue breastfeeding. Kids should avoid milk and dairy; milk substitutes e.g. rice or soy milk are good (though yoghurt and aged cheese may be tolerated).

- Also ensure the kid's diet contains enough Ca!

Frothy stools in a breastfed infant may indicate a relative lactose malabsorption secondary to an oversupply of breastmilk. Recall that the foremilk is CHO rich while the hindmilk is fat-rich and slows down the absorption of CHO. Therefore, we can advise mothers to **allow the baby to fully feed from one breast** before swapping.

Congenital lactose intolerance is exceptionally rare, and therefore changes to lactose-free formula are usually not indicated!

Also see: [Non-IgE-Mediated Food Allergy](#) for info on **Cow's Milk Protein Intolerance**.

Gastroesophageal Reflux

Regurgitation is **very common**: 67% parents report regurgitation at 4 months. Most children outgrow by 1 year old.

Possetting: common, particularly when the stomach is full after a feed. Rarely causes problems for the child.

Pathological Reflux

Gastro-oesophageal reflux (GOR)	Gastro-oesophageal reflux disease (GORD)
Doesn't usually need investigation or treatment	May require investigation and/or treatment
Begins in the first 2 months Can be frequent Improves as the child gets older, often resolving by 1 year Growth is normal	Haematemesis Melaena Dysphagia Ongoing regurgitation beyond 1 year Poor growth associated with regurgitation Retrosternal or epigastric pain not responding to therapy Sandifer's syndrome (abnormal posturing of the neck) or persistent back arching with feeding



Risk factors for GORD: premature infant, parental history of GORD, obesity, hiatus hernia, congenital diaphragmatic hernia, repaired oesophageal atresia, neurodisability.

Vomiting Red Flags

Symptom	Implication	Investigation
Forceful, projectile vomiting if < 2 months	Pyloric stenosis	Abdominal ultrasound (if < 2 months)
Bile-stained vomiting and/or abdominal distension	Bowel obstruction e.g. malrotation volvulus	Urgent plain AXR (supine and erect) then upper GI contrast study
Haematemesis	Serious bleeding from GIT	ABCs, bloods, NGT aspiration, upper GI endoscopy
Onset after 6 months or persisting after 1 year	Consider other Dx e.g. reflux, cerebral space-occupying lesion	
Systemically unwell	Infection e.g. UTI	Septic screen
Dysuria	UTI	Septic screen
Rapidly increasing head circumference and/or headache with morning vomiting	Raised ICP caused by hydrocephalus or brain tumour	Neuroimaging – CT/MRI
Acutely bulging fontanelle +/- altered level of consciousness	Raised ICP e.g. meningitis	Neuroimaging + lumbar puncture if safe

Investigations for GORD:

- Oesophageal pH study +/- impedance monitoring
- Upper GI endoscopy

Management

- Conservative: no positional changes (↑ risk of SIDS), lactation consultant if breastfed, if formula fed → ↓ volume if getting too much, smaller & more frequent feeds, trial of thickened feeds
- Consider **4 week PPI trial (esomeprazole or omeprazole)** in those with features of GORD or at risk.
 - If PPI ineffective, consider 2-4 week trial of cow milk and soy-free diet (maternal avoidance if breastfed; partially or fully hydrolysed formula in formula fed)
- Prokinetic agents such as **metoclopramide**, **domperidone** and **erythromycin** are not recommended and in specialist territory.
- If severe
 - Tube feeding: NG, or jejunal (transpyloric) if not tolerated – but need to be continuous (↓ risk of dumping syndrome and osmotically overloading the small bowel).
 - Fundoplication (severe, intractable GORD)

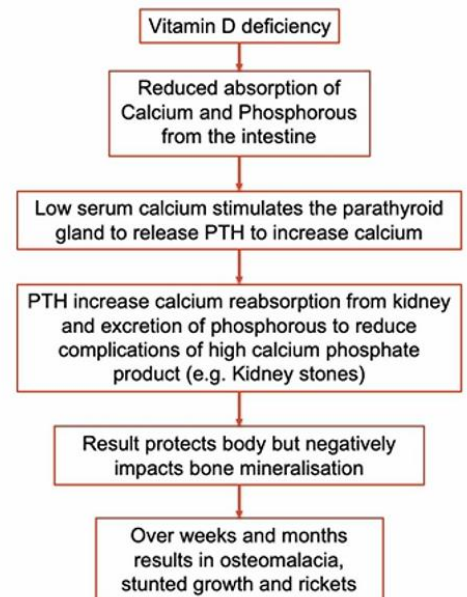
Vitamin D Deficiency

Vitamin D is essential for bone and muscle health, as it regulates Ca concentration in the body. It ↑ intestinal absorption of calcium, ↓ bone reabsorption, and ↓ renal Ca and PO₄ excretion. It is especially important in **children** since they are growing.

- Severe deficiency: rickets, hypocalcaemic seizures, limb pain, fractures
- In Australia, only margarine has **mandatory** fortification with Vitamin D, but dairy and formula may also be fortified.

Sources of Vitamin D: sunlight is the most important (>90%), as it promotes the conversion of 7-dehydrocholesterol in the skin into cholecalciferol (Vitamin D3)

- Recall that Vitamin D2 or D3 undergo hydroxylation in the liver to make 25-hydroxy Vitamin D (calcidiol), the storage form.
 - Calcidiol is stored in the liver and adipose tissue.
- Then, the kidney converts calcidiol into 1,25-dihydroxy Vitamin D (calcitriol).
 - Calcitriol is an **active hormone that is used almost immediately after it is produced**.
- Calcitriol synthesis is mediated by PTH, serum PO₄, and growth hormone, and can also occur in non-renal sites (e.g. lung alveoli and osteoblasts).



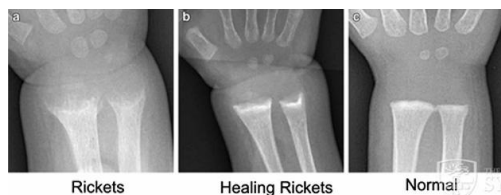
Risk factors for low Vitamin D: dark skin, skin coverings, medical conditions (obesity, liver and renal failure, severe malabsorption, medications – corticosteroids, antiepileptics e.g. valproate).

Importantly, **breastmilk contains very little Vitamin D** so exclusively breastfed infants are at risk of Vitamin D deficiency.

History: diet (breastfed, calcium), previous Vitamin D treatment, bone pain, fatigue with exercise, growth, motor delay, symptoms of hypocalcaemia (irritability, muscle cramps, tetany, stridor, seizures if < 6 months old).

Examination: growth, delayed dentition (no teeth by 9 months), rickets (see below), hypocalcaemia (**Chvostek sign**: twitch of lip when tapping facial nerve).

- **Rickets** are caused by severe Vitamin D deficiency that leads to issues with bone calcification.
- **Clinical features** include long bone bowing (if weight-bearing), metaphyseal widening/ splaying/ fraying/ cupping at the wrist and ankle; rachitic rosary (expansion of costochondral cartilage), fracture, wide fontanelle



Investigations:

- Vitamin D: Normal > 50 nmol/L; mild 30 – 49 nmol/L; moderate/severe < 30 nmol/L.
- PTH if Ca intake low or symptomatic
- ALP can be elevated as a measure of bone activity
- CMP
- X-ray if clinical features of rickets (usually left wrist)

According to the flow chart (next page), those with at least 1 risk factor and features of rickets require the full kit and kaboodle of investigations. For those who are at risk of Vitamin D deficiency but do not have features of rickets, the focus is obviously on **prevention**.

High PTH can be a sign of impending hypercalcaemia even if Ca is normal ... increase Ca intake or supplement!

Management: Obviously just Vitamin D supplementation; treatment is based on age and severity.

Children with low Vitamin D should be treated to restore their levels to the normal range (50+ nmol/L). Ensure adequate calcium intake.

Terminology: Note, both Vitamin D2 and Vitamin D3 can make calcidiol and calcitriol.

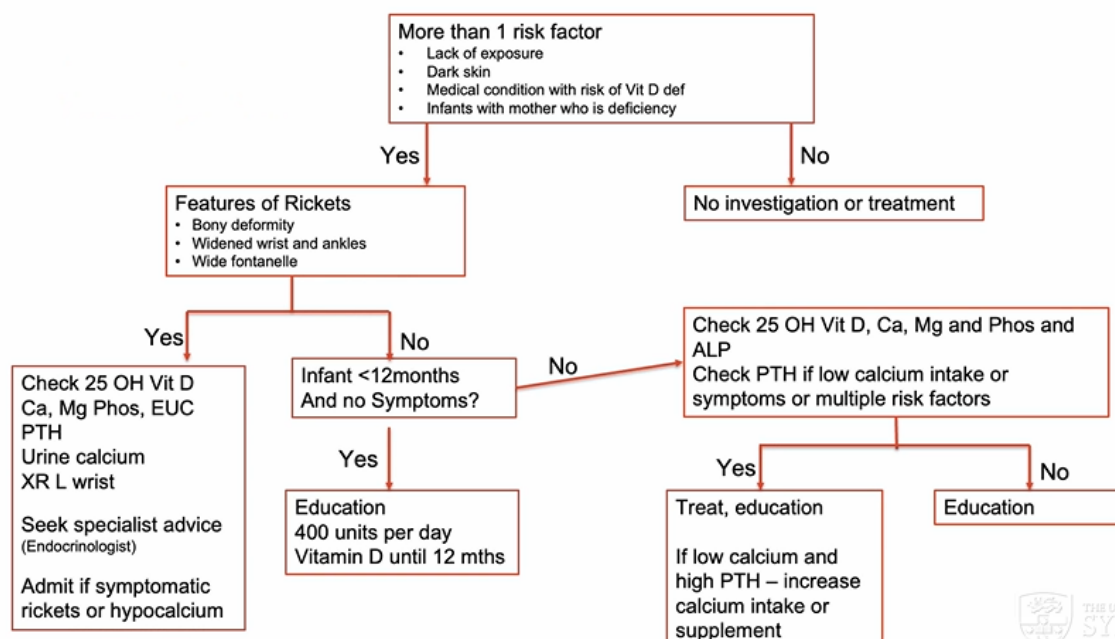
Vitamin D2	Vitamin D3	25-OH Vitamin D	1,25-OH Vitamin D
Ergocalciferol	Cholecalciferol	Calcidiol	Calcitriol

Age	Treatment		Maintenance
	Mild	Moderate/ Severe	
Pre-term	200 IU/kg/day, max 400 IU	800 IU/day, review 1 mth	200 IU/kg, daily max 400 IU
< 3 months	400 IU daily for 3 months	1000 IU daily for 3 months	400 IU daily
3 – 12 months	400 IU daily for 3 months	1000 IU daily for 3 months or 50 000 IU stat and review 1 month	400 IU daily
1 – 18 years	1000 – 2000 IU/day for 3 months or 150 000 IU stat	1000 – 2000 IU daily for 6 months or 3000 – 4000 IU daily for 3 months or 150 000 IU stat, repeat 6 weeks	400 – 600 IU daily or 3000 – 4000 IU weekly or 150 000 IU at the beginning of Autumn

Vitamin D supplementation is also recommended in **at-risk children:** 400 IU daily in 0 – 12 months (e.g. if exclusively breastfed); 500 – 600 IU daily if 1 – 18 years.

Ongoing monitoring

- Follow-up bloods measure **25 OH Vitamin D (Calcidiol)**
 - Infants are followed up after **1 month** and children > 1 year **after 3 months**.
 - Those with mild deficiency do not require follow-up.
- The recommended level is **> 50 nmol/L regardless of age**.
- The preferred supplement is **Vitamin D3 (cholecalciferol)**, especially in the oral form. Compared to Vitamin D2, it has a longer half-life, better oral bioavailability, and higher potency.
- Calcitriol is not recommended as a supplement itself due to its short half-life, the inability to increase Vitamin D stores, and the risk of adverse effects e.g. hypercalcaemia and hypercalciuria.
 - However, it is useful in CKD-associated secondary hyperparathyroidism, because the kidneys lose the ability to perform the 1 α -hydroxylation step that converts 25-hydroxyvitamin D to calcitriol.



Nocturnal Enuresis

Very common, affects 10% of 5-year-olds, M>F, FHx common.

Primary nocturnal enuresis: never dry for more than a few months.

Secondary: dry for >6 months then wets again.

Pathophysiology: impaired cortical arousal + reduced bladder capacity +/- nocturnal polyuria.

History: fluid intake, daytime voiding (frequency, volume, any urgency or incontinence), identify contributing factors (constipation, toilet phobia, fear of dark).

- We ought to consider underlying diagnoses: UTI, DM, and constipation.

Assessment: time and volume chart

Management: most grow out of it!

- Treat underlying factors e.g. constipation
- Behavioural modification: avoid caffeinated drinks, ensure adequate fluids, encourage child to pass urine regularly during the day
 - Counterintuitively, restricting fluids before bed and waking during the night do not work!
- Children who **bed wet > 6 years old need professional help**
 - Bed wetting alarm first-line (2/3 dry)
 - Desmopressin (reduces the number of wet nights, but effect is not sustained). Children must drink no more than 240 mL fluid 1 hour before and up to 8 hours after desmopressin to avoid water intoxication. It might be recommended when the child is going to a sleep over or camp.

Sudden Infant Death Syndrome (SIDS)

SIDS is the most common cause of death in the first year of life. Its peak incidence is **3 months old**. Following a cot death, siblings should be screened for potential sepsis and inborn errors of metabolism.

Major risk factors	Other risk factors	Protective factors
Putting the baby to sleep prone Parental smoking Prematurity Bed sharing Hyperthermia or head covering	Male sex Multiple births Low SES Maternal drug use Winter season	Breastfeeding Room sharing (but not bed sharing!) The use of dummies

3. Respiratory

3.1 Acute Respiratory Problems

Infants and young children are more predisposed to respiratory failure than adults:

- Smaller diameter airways → increased resistance
- Compliant chest wall → less support for maintenance of lung volume
- Fewer alveoli → potential for VQ mismatch
- Increased proportion of foetal Hb until 6 months → less O₂ offload to tissues

Anatomy of Infant and Child Airway: Compared to an adult: (i) smaller in diameter and shorter in length, (ii) narrowest at cricoid cartilage (cf glottic opening in adults), (iii) epiglottis longer, floppier, and narrower, (iv) large tongue, (v) large occiput (can flex neck when lying flat).

See Emergency Medicine Section for: [Foreign Body Aspiration](#), [Anaphylaxis](#), [Basic Life Support](#)

Respiratory Tract Infections

Usually acute viral infections, most prevalent among young children and in the winter.

- Risk factors: daycare/ preschool, overcrowded living, exposure to cigarette smoke
- Common cold treatments and remedies don't reduce disease duration; GPs usually prescribe to manage parents' expectations.

Upper Respiratory Tract	Lower Respiratory Tract
Rhinitis/ Pharyngitis/ Tonsillitis Otitis Media Sinusitis	Bronchiolitis Viral-Induced Wheeze Pneumonia

3.1.1 Upper Airway

Acute Upper Airway Obstruction: [Clinical Practice Guidelines : Acute upper airway obstruction](#)

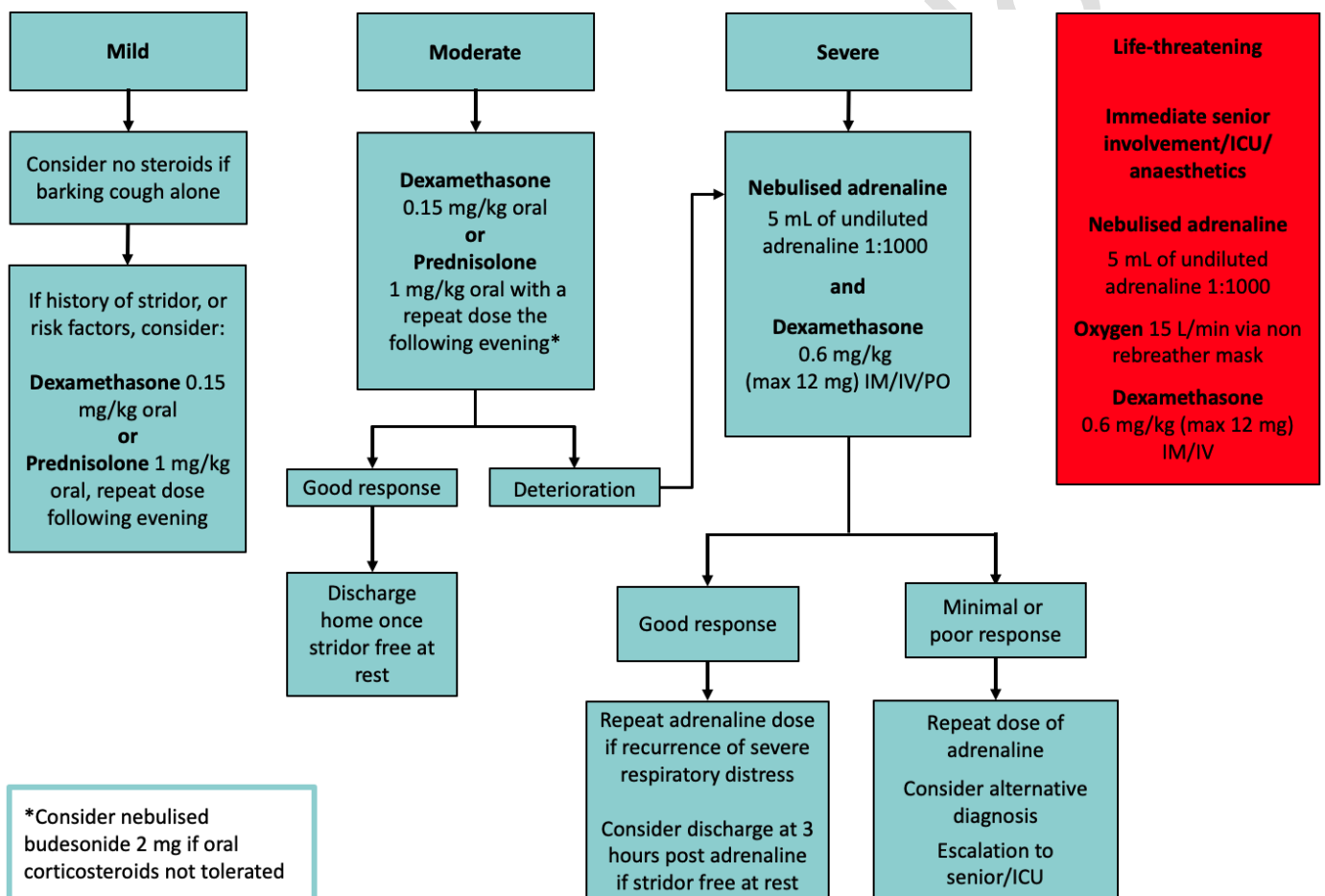
- Allow children with acute upper airway obstruction to **adopt a position of their choice**.
- **Nebulised adrenaline** may provide temporary relief while awaiting definitive measures.

Differential	Features
Croup	Young child (rare < 3 months); rapid onset harsh <u>barking cough</u> ; hoarse voice/ cry; <u>stridor</u> ; may be febrile and miserable but otherwise systemically well.
Anaphylaxis	Swelling of face and tongue; wheeze; urticarial rash; allergen exposure; haemodynamic compromise
Foreign Body Inhalation	Young child (or developmentally similar); abrupt onset; coughing, choking, vomiting episode (may not be witnessed); <u>unilateral chest findings</u> ; wheeze
Reduced pharyngeal tone or size	Reduced consciousness after ingestion, recent seizure or head injury; pre-existing narrow or floppy upper airway
Retropharyngeal abscess	Sore throat, fever, neck pain and stiffness or torticollis, fullness and redness of posterior pharyngeal wall, dysphagia and drooling.
Peritonsillar abscess (quinsy)	Severe sore throat (often unilateral); hot potato/ muffled voice; trismus; swollen posterior palate and tonsil, medial tonsillar displacement, uvula deviation
Epiglottitis	<u>Inadequate Hib immunisation or immunocompromised</u> ; high fever and systemically unwell (<u>"toxic-appearing"</u>); muffled voice; hyperextension of neck; dysphagia; pooling of secretions, drooling; absent cough; low pitched expiratory stridor or stertor.
Bacterial tracheitis	Systemically unwell, prior URTI, <u>Staphylococcus aureus is the commonest cause</u> ; fever, stridor, cough, and a preference to lay flat. Markedly tender trachea. Diphtheria can present similarly (sore throat, malaise, low-grade fever); exceptionally rare with immunisation.
Ludwig angina	Infection of the sublingual and submandibular spaces. Swollen, tender floor of mouth and under tongue; facial laceration or dental abscess; submandibular swelling.
Airway burns	Burns elsewhere (especially facial), singed nasal hair, sooty sputum.
Trauma	Bruising and swelling of neck, subcutaneous emphysema, may → pneumothorax or pneumomediastinum.

Management: emergency airway management with anaesthetic or ENT support. If awaiting other measures, can give O₂, consider **nebulised adrenaline** (for life-threatening upper airway obstruction), **dexamethasone 0.6 mg/kg (max 16 mg) IM/IV/PO** to reduce swelling.

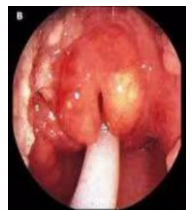
Croup (Laryngotracheobronchitis)

- Inflammation of the upper airway, larynx, and trachea, usually triggered by a virus (**commonest = parainfluenzae Type 1, 2, 3**).
- Clinical features:** Low-grade fever, night-time increase in oedema with stridor, "seal bark" cough, hoarse voice or cry, increased WOB +/- associated widespread wheeze.
- Croup is a **clinical diagnosis**.
 - Alarm features** for severe croup: agitation, drowsiness, persistent stridor at rest, markedly abnormal RR, marked chest wall retraction, **hypoxia only if life-threatening**.
- Treatment (DNO):** minimise distress to avoid worsening symptoms; supplemental O₂ not usually required.
 - Oral dexamethasone** to all: 0.15 mg/kg if mild/moderate, 0.6 mg/kg if severe.
 - Nebulised adrenaline** if severe: 0.5 mL/kg to 5 mL of 1:1000
 - Oxygen** if life-threatening.
- Discharge home if no stridor; observe for 3 hours if nebulised adrenaline given.



Epiglottitis

- Rapid onset, severe distress in hours.
- Usually *Haemophilus influenzae type b (Hib)* so vaccine decreasing incidence.
- Clinical features:** high fever, intense sore throat, difficulty swallowing, drooling, stridor; child sits up, leans forward, extends neck. "Toxic-appearing"; anxious, restless, irritable.
 - No cough!**
- Do not examine throat** (can trigger spasm of epiglottitis): "cherry red epiglottitis".
- Keep child calm, intubate via gaseous induction by experienced anaesthetist.



Bacterial Tracheitis

- URTI symptoms for 1-3 days prior to stridor and dyspnoea
- Most often caused by *Staphylococcus aureus*.
- Clinical suspicion requires direct laryngoscopy.

Pertussis/Whooping Cough

- **Aetiology:** Bordetella Pertussis, a gram-negative intracellular coccobacillus.
- Recurs in 2-3 year outbreaks. Most prevalent in those > 5 years old, but most devastating in kids < 5 years.
 - 1 in 125 babies under 6 months with whooping cough will die from pneumonia or brain injury.
- Significant in unvaccinated babies, and spread via infected family members.
 - Note: supposed to receive pertussis-containing vaccine in a 5-dose schedule – 2, 4, 6 and 18 months, and 4 years of age (see Immunisation).
 - Immunisation of **pregnant women** prevents early neonatal disease; antibodies last until the 2 month vaccines.
 - Those who are vaccinated are **unlikely to develop whooping cough (disease)** but are still **contagious**.
- 4 stages:

Incubation	Catarrhal/Prodromal	Paroxysmal	Convalescent
7 – 10 days	1 – 2 weeks Rhinorrhoea, intermittent, dry cough, <u>most contagious period</u> .	1 – 6 weeks Severe coughing fits (paroxysms), inspiratory “whoop”, post-tussive vomiting, cyanosis, apnoea, attacks <u>worse at night</u> .	Gradual recovery; cough may persist for months.

- Treatment/contact prophylaxis: intracellular penetrating antibiotic (e.g. azithromycin)

3.1.2 Lower Airway

Pneumonia

- Fever, tachypnoea, cough, pleuritic chest pain. Signs may include respiratory distress, dullness to percussion, reduced breath sounds and bronchial breathing over involved area (consolidation).
- Focal chest signs; **viral most common** but can be atypical e.g. mycoplasma.
- Streptococcal pneumonia is the most common cause of bacterial pneumonia.
- Mycoplasma is common in older children, causing “**walking pneumonia**” (non-specific, protracted) → requires azithromycin.

Parapneumonic Effusion

A pleural effusion is a collection of fluid in the pleural space. A parapneumonic effusion is a pleural effusion associated with pneumonia. It may be:

1. Simple: non-loculated, sterile transudate
2. Complicated: loculated or empyema, due to bacterial invasion of the pleural space

Most common infective causes of empyema: Streptococcus pneumoniae, Staphylococcus aureus, Group A Streptococcus.

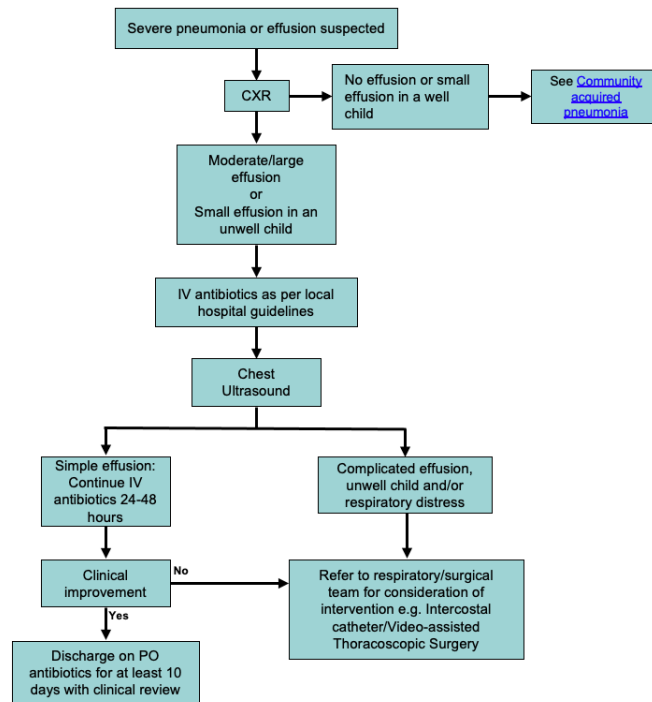
Suspect empyema in **children with a fever that persists after 48 hours of appropriate treatment of pneumonia!**

Investigations

- Bloods: FBC, EUC, CRP, blood culture, LFTs and coagulation studies if severely unwell (consider **toxic shock**)
- Viral swabs
- Testing for TB
- If inserting a drain: pleural fluid gram stain, MCS, pneumococcal PCR
- Chest imaging: AP CXR, chest ultrasound (to differentiate simple vs complicated effusion)

Treatment

- Small simple effusion and no respiratory distress → outpatient management with **oral antibiotics**.
- Moderate/large effusion or unwell → admit for **IV antibiotics**
- Supportive: supplemental O₂, limit maintenance fluids to 2/3 child's fluid requirement (to avoid fluid overload), analgesia, encourage mobilisation



Viral-Induced Wheeze

- Very common, especially in the first 2 years of life. Wheeze + RTI signs.
- Nil specific investigations. Can do nasal swab as RSV in infants suggests **bronchiolitis**.
- Most common cause in school-aged children is **rhinovirus**.
- Management is mainly supportive, but may trial salbutamol in > 1 years old.

Bronchiolitis vs Croup

	Bronchiolitis	Croup
Symptoms	Initial cough and coryza Progress to tachypnoea, ↑ WOB, wheeze Usually associated with fever	"Barking" cough Progress to stridor and ↑ WOB
Peak age (Range)	3 – 6 months (0 – 12 months)	2 years (6 months – 6 years)
Sex	M > F (3:2)	M > F (3:2)
Season	Winter peak	Autumn peak
Most common pathogen	Respiratory syncytial virus (RSV)	Parainfluenzae Type 1, 2, 3
Risk factors	Cigarette smoke, no breast feeding, crowded living conditions, preschool, cardiorespiratory compromise	Family history
Disease burden	1/3 children by 12 months 90% children by 2 years	2% preschool-aged children every year
Prognosis	10 – 14-day natural history 50% recover by 14 days; 90% by 21 days; 10% prolonged cough or wheeze	2-3 days of symptoms Frequently have recurrent episodes
Management	Supportive: just oxygen and hydration (NG, IV routes acceptable in admitted patients) Antibiotics, bronchodilators, and corticosteroids have no demonstrable benefit.	Corticosteroids → nebulised adrenaline → oxygen (in that order: <i>Croup Never Oxygenates</i>)

Why do bronchiolitis and croup have a male preponderance? Male infants have smaller airways than females; testosterone may also delay lung maturation and surfactant production. Female infants also have a stronger innate and adaptive immune response to viral infections.

Bronchiolitis

- The first episode of viral-induced lower respiratory illness with wheeze and/or crackles in infants under 12 months.
 - Disease of infants with **small-calibre airways**; viral infection → inflammation, oedema, and mucus plugging of terminal bronchioles.

- After 12 months, airway diameter and smooth muscle development increase, and the pathology of wheeze begins to resemble that of asthma or viral-induced wheeze.

Initial Severity Assessment			
Symptoms	Mild	Moderate	Severe & Life-Threatening
Appearance	Well	Moderate	Unwell
Work of breathing	Normal	Mild-moderate	Moderate-severe grunting
Cyanosis	No	No	Yes (or pale)
Oxygen saturation	> 92% in room air	≤ 92% in room air, or needs < 1 L/kg/min for WOB	≤ 90% in room air, needs > 1 L/kg/min for WOB
Heart rate	Normal	Mild tachycardia	Tachycardia > 180
Feeding	Normal or slightly decreased (>75% normal feed)	Taking > 50% normal feed	Taking < 50% normal feed
Management			
Oxygen	No	Nasal prongs to maintain SaO ₂ > 92% and/or improve WOB. Consider hiflow if not responding.	Maintain SaO ₂ > 92%; consider HFNP 2L/kg/min.
Hydration	Smaller, frequent feeds	Smaller, more frequent feeds +/- NG feeds	IV fluids at 2/3 maintenance, NBM*
Investigations	Nil	Nil (maybe NPA)	Consider CXR, blood gas, NPA

*Risk of water retention (resp distress can → ADH secretion) and pulmonary oedema with full maintenance fluids

3.2 Chronic Airway Problems

Excellent RACGP algorithm on wheeze: [RACGP - The wheezing child: an algorithm](#)

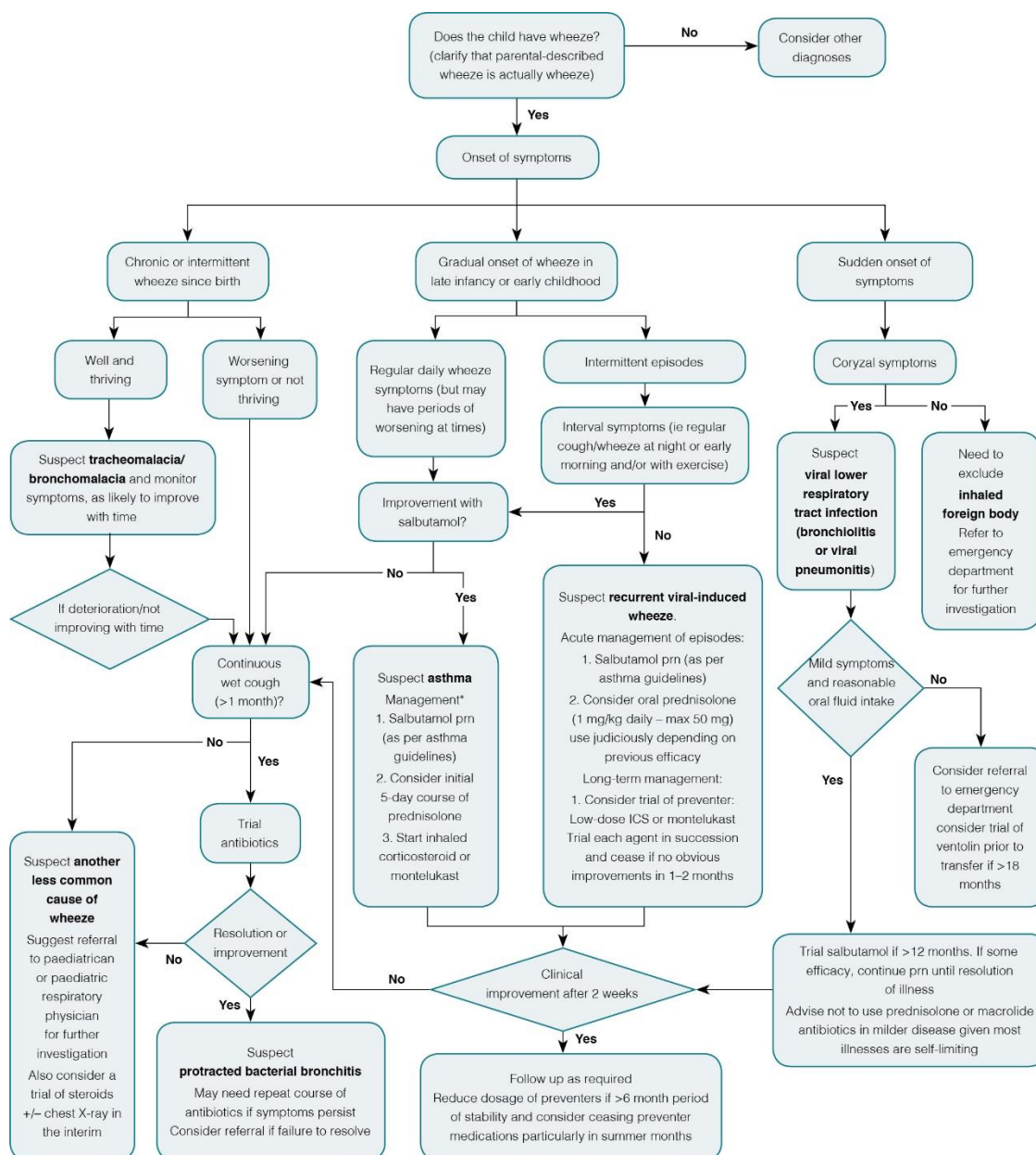


Table from the same article:

Condition	Incidence	Clinical Signs	Investigation	Expected Course	Management
Viral wheeze	Very common, esp first 2 years 50% kids have at least 1 wheezing episode	Wheeze assoc w RTI Singular or recurrent Bronchiolitis (<2 years): fine crackles +/- wheeze on auscultation	None specific May send NPA but does not change Mx; positive for RSV may support Dx of bronchiolitis	60% outgrow by 6 years 15% acquire wheezing after 6 years After 7-8 years, only 1 in 5 outgrow	Trial salbutamol only if > 1 year old Supportive care, including monitoring fluid intake (>50% usual intake adequate) and for signs of resp distress
Asthma	15 – 20% paediatric population	Regular wheeze Some have persistent/ interval Sx	Spirometry with BD response if 5+ years old	Usually expected to be lifelong, but clinical course can vary	Exacerbations: regular salbutamol and consider oral

		between episodes of viral wheeze, e.g. cough and/or wheeze at night or with exercise			prednisolone for up to 5 days Regular preventer usually indicated
Airway malacia (tracheo- or broncho-malacia)	1 in 2100	Neonate with wheeze, stridor, cough, and rattling Appears well; 'happy wheezer'	Bronchoscopy usually diagnostic but often not necessary	Most outgrow by 2 years Secondary PBB can occur (poor cough clearance)	Treatment rarely needed Specialist referral if worsening Sx or FTT
Protracted bacterial bronchitis (PBB)	Probably uncommon but incidence unknown	Chronic wet cough, >4 weeks Concurrent wheeze and/or rattly breathing uncommon	Bronchoscopy may assist with Dx but usually unnecessary CXR normal or non-specific	Most resolve with 1-2 courses Abx	2-6 week courses Abx, typically amoxicillin/ clavulanic acid (Augmentin)

Q: What are the most likely organisms in protracted bacterial bronchitis?

The same as pneumonia – Haemophilus influenzae, Streptococcus pneumoniae, and Moraxella catarrhalis

Q: How do I distinguish PBB from recurrent viral infections?

Viral RTI: average duration of Sx is 4 weeks, with an improving dry cough at the end. Coryzal symptoms wax and wane.

3.2.1 Asthma

Definition: recurrent, reversible airway hyperresponsiveness of bronchial smooth muscle in an obstructive lung disease pattern often triggered by allergens, viral URTIs, stress, and associated with atopy. Intermittent airway narrowing due to bronchoconstriction, mucosal oedema, bronchial secretions (mucous plug).

- Typical exacerbation trigger is a **respiratory viral infection**.
- Viruses can be detected (on PCR) in 80-90% of children with asthma exacerbations, 2/3 are **rhinovirus**
- The frequency of asthma exacerbations coincides with peak periods of rhinovirus – late autumn to early Spring

Paediatric vs. adult asthma: preschool wheeze that doesn't become asthma; don't have spirometry to prove reversibility; capacity to deteriorate quickly; different signs to decide assessment of acute exacerbations (tachypnoea, accessory muscle use, "quiet chest", lethargy, hypoxia, poor feeding); issue of overuse of steroids (e.g. bone health, growth, behavioural change).

Diagnosis:

Infants (age 0 – 12 mo)	Preschool wheeze (1 – 5 y)	Children aged 6 – 11 y	Adolescents (as adults)
Do not diagnose asthma. Wheezing is usually due to acute viral bronchiolitis or small and/or floppy airways.	Recurrent respiratory Sx with no red flags, and rapid clinical response to salbutamol . Can trial reliever/ preventer.	Spirometry showing bronchodilator response (FEV1 ↑ >10% from baseline); trial of therapy with clinical improvement. OR FeNO ≥ 25 ppb ¹	Spirometry showing bronchodilator response (FEV1 ↑ >12% and ≥ 200 mL at 10-15 minutes) OR FeNO ≥ 40 ppb ¹

¹FeNO = Fractional exhaled nitric oxide. Provides an index of eosinophilic activity in the respiratory tract.

50% of children will wheeze, but a large proportion will not have persisting wheezing episodes at age 6y.

- Consider atopy and FHx of atopy (asthma, eczema, hayfever); triggers (e.g. viral infection, exercise, smoke, weather, animals, dust).
- Viral-induced wheeze: **only** triggered by URTI
 - Asthma: chronic disease, multiple triggers, atopic history, interval symptoms.

DDx Wheeze	DDx Chronic Cough	DDx Dyspnoea
Other infections e.g. RSV (bronchiolitis?)	Inhaled foreign body	Hyperventilation/ anxiety Poor cardiopulmonary fitness

Inhaled foreign body Anaphylaxis Tracheomalacia and other structural airway issues	CF and other suppurative lung disease Pertussis/protracted bronchitis (responds to prolonged oral Abx) Habit cough (exacerbated by stress, settles in sleep) Drugs e.g. ACEi	
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Is my airway sound really wheeze?

Wheeze	High-pitched, continuous, most prominent on expiration but can be inspiratory.
Stridor	Harsh, loud, high-pitched, monophonic; best heard over the neck; typically on inspiration.
Stertor	Snoring from pharyngeal or nasal blockage, low-pitched (think adenoids, tonsillar hypertrophy)
Secretions	Coarse crackles/rhonchi, bubbly or wet sounds, associated with infection or retained fluid

Asymmetry on auscultation is often found due to mucus plugging, but persistent asymmetry may indicate other causes e.g. inhaled foreign body or pneumothorax.

Reversibility:

- Spirometry (pre- and post-bronchodilator response): school age only.
- Mannitol challenge: triggers asthma response
- Eucapnic voluntary hypopnoea: exercise-induced asthma
- Clinical response: Ventolin or preventer response with SABA +/- ICS trial.

Classification of Asthma

Intermittent: no symptoms between exacerbations

- Infrequent: exacerbations every 6 weeks or less
- Frequent: exacerbations > once every 6 weeks

Persistent: between exacerbations, either (i) daytime symptoms (>1/week), (ii) nocturnal symptoms (>2/month), (iii) symptoms restrict activity or sleep.

Then, for persistent asthma, classify severity:

Mild	Moderate	Severe
FEV1 $\geq$ 80% predicted and at least one of • Daytime symptoms > once per week but not every day • Night-time symptoms > 2x per month but not every week	Any of • FEV1 <80% predicted • Daily daytime symptoms • Night-time symptoms > once per week • Symptoms sometimes restrict activity or sleep	• FEV1 $\leq$ 60% predicted • Daytime symptoms continual • Night-time symptoms frequent • Flare-ups frequent • Symptoms frequently restrict activity or sleep

Severity	Symptoms (Daytime)	Night Symptoms	Reliever Use	Activity Limitation	Lung Function (if measured)
Intermittent, infrequent	<6 episodes/year AND well between	None	$\leq$ 2 times/month	None	Normal
Intermittent, frequent	$\geq$ 6 episodes/year, but well between	<1/month (may have some)	>2 times/month	None	Normal
Persistent, mild	>1 episode/month or symptoms >2 days/ week (but not daily)	>1/night/month	>2 per week	Minor	Normal or FEV1 $\geq$ 80% predicted
Persistent, moderate	Daily	>1/week	Daily	Some	FEV1 60–80% predicted
Persistent, severe	Throughout the day, most days	Frequent/nightly	Frequent	Marked, limited play/school	FEV1 <60% predicted

Asthma Action Plan

- Always **use a spacer**. Contains 4 sections: when well, when not well, if symptoms worsen, and emergency.
- **3 considerations:** pattern of asthma, age, and weight.

Reliever

- **Always specify** what reliever, how many puffs (or how often/ delivery device), and divide reliever into 0-5 years vs. 6+ years.
 - **In children, the reliever is almost always Ventolin (salbutamol).**
 - **Always refer to medications using their trade name.**

	0-5 years	6+ year.
When well	2 puffs as needed	4 puffs as needed
When not well	4 puffs every 6 – 8 hours	6 puffs every 6 – 8 hours
Acute flare up	6 puffs every 3 – 4 hours	12 puffs every 3 – 4 hours
Exercise symptoms	2 puffs 10-15 minutes before exercise	2-4 puffs 10-15 minutes before exercise

Preventer

- The preventer remains the same throughout each section of the asthma action plan.
- There are two common medications. **Fluticasone** is first-line; montelukast was previously more frequent with allergic-type asthma, now reserved to specific cases (seems to be very good with allergy and exercise)
 - Usually, montelukast is recommended as an add-on for persistent Sx despite regular inhaled steroids
 - Beware: montelukast can also have **behavioural** and **neuropsychiatric effects**

Generic name	Brand name	Mechanism	Indications	Usual dose
Montelukast	Singulair	Oral (chewable) tablet Leukotriene receptor antagonist	Frequent intermittent, mild persistent asthma Only in >2 yrs if unable to use pMDI + spacer/mask; <u>significant allergic rhinitis</u> ; patient preference.	1 tablet daily 2-5 yrs: 4 mg 6+ years: 5 mg
Fluticasone	Flixotide Junior (50 mcg) Flixotide (150 mcg) Flixotide (250 mcg)	Inhaled steroid	First-line if frequent Sx, restrict sleep, or Hx moderate or severe flare-ups.	100 – 200 mcg/day 250 mcg/day 250-500 mcg/day

Side-effects of inhaled steroids: voice changes, sore throat, oral thrush. Rinse your mouth out or have a drink.

Most kids aged 1-8 are on Flixotide Jr; if needing higher dose than 100 mcg bd, might switch to Flixotide.

Acute flare up

Generic name	Brand name	Mechanism	Indications	Usual dose
Oral prednisone	Redipred/ Predmix (5 mg/mL) 5 mg tablet or 25 mg tablet.	Corticosteroid	Acute exacerbation	1 – 2 mg/kg/day (depends on severity), max 50 mg Parent initiated OCS at 1 mg/kg.

Risk factors for severe asthma exacerbations in children: high probability of respiratory viral infection, confirmed food allergy or history of anaphylaxis, poor asthma symptom control, ED/hospital admission for asthma in past 12 months, Hx severe exacerbation, over-use of salbutamol, poor adherence or inhaler technique, psychological or socioeconomic problems, exposure (smoke, allergens, pollution), obesity, high eosinophil count.

Asthma Management

- Aim for good control of symptoms to limit exacerbations.
- Identify triggers and manage comorbidities.
- Monitor regularly and adjust treatment regimen, e.g. aim for minimum dose of ICS.
- Education e.g. technique, compliance, asthma action plan, immunisations, limit exposure to triggers, avoid tobacco.

We **do not use any long-acting medications (LAMA or LABA)** in children under 12. Adolescents are treated like adults – and we *do not like using SABA alone in adolescents* (maybe SABA PRN in very, very mild asthma).

	Children 1-5 years old	Children 6-11 years old	Adolescents
Indications for preventer	Mild Sx <u>every 4-6 weeks</u> or more frequently. Any Hx of moderate-severe flare ups (requiring ED care or oral CS) or life-threatening flare ups (hospitalisation or PICU)	Severe or life-threatening flare-ups, regardless of frequency. Flare-ups more than once every 6 weeks Persistence (daytime symptoms >1/week, night-time symptoms >2/month, restrict activity or sleep)	Symptoms >2/month, waking due to asthma, flare requiring oral CS in past 12 months, Hx severe exacerbation.
Stepped approach	1. SABA PRN, then add: 2. Preventer: low-dose ICS or montelukast 3. Stepped-up preventer: high-dose ICS or low-dose ICS + montelukast	1. SABA PRN, then add 2. Preventer (ICS low dose or montelukast) 3. Stepped-up preventer: high-dose ICS, low-dose ICS/LABA, low-dose ICS + montelukast	1. Regular ICS + PRN SABA or low-dose budesonide-formoterol (ICS/LABA; Symbicort) 2. Medium-high dose ICS + PRN SABA or low-dose ICS-LABA + PRN SABA or ICS-formoterol maintenance and reliever therapy (SMART; on private prescription)

Remember to review the child 4-6 weeks after commencing preventer medication; sooner if concerned.

SMART = Single Maintenance and Reliever Therapy; combines an inhaled corticosteroid + LABA.

Asthma Devices



ASTHMA & COPD MEDICATIONS

2025

BIOLOGIC THERAPIES FOR SEVERE ASTHMA

Fasenra
benralizumab 30mg/mL

Dupixent
dupilumab
200mg/1.14mL • 300mg/2mL

Nucala
mepolizumab 100mg/mL

Xolair
omalizumab
150mg • 75mg/0.5mL • 150mg/mL

All the above are section 100 (Highly Specialised Drugs)

RESOURCES

TREATMENT GUIDELINES

Australian Asthma Handbook:
astmahandbook.org.au

COPD-X Plan:
copdx.org.au

COPD Inhaler Device Chart Poster:
lungfoundation.com.au/resources/copd-inhaler-device-chart-poster/

INHALER TECHNIQUE

How-to videos, patient and practitioner information
nationalasthma.org.au

pMDIs should be used with a spacer (and face mask if needed)

HOW-TO VIDEOS

SABA RELIEVERS

Ventolin Inhaler †
salbutamol 100mcg
Additional brands: Asmol, Zempren

Airomir Autohaler †
salbutamol 100mcg

Bricanyl Turbuhaler †
terbutaline 500mcg

SAMA MEDICATION

Atrovent Metered Aerosol †
ipratropium 21mcg

NON STEROIDAL PREVENTER

Montelukast Tablet
montelukast
4mg • 5mg • 10mg
Multiple generic brands

LABA MEDICATIONS

Oxis Turbuhaler †
formoterol 6mcg • 12mcg

Serevent Accuhaler †
salmeterol 50mcg

Onbrez Breezhaler †
indacaterol 150mcg • 300mcg

ICS PREVENTERS

Fixotide Inhaler †
fluticasone propionate
50mcg • 125mcg • 250mcg
*Fixotide Junior †

Fluticasone Cipla Inhaler †
fluticasone propionate
125mcg • 250mcg

QVAR Inhaler †
beclomethasone
50mcg • 100mcg

Alvesco Inhaler †
ciclesonide
80mcg • 160mcg

Axotide Inhaler †
fluticasone propionate
50mcg • 125mcg • 250mcg
*Axotide Junior †

Fixotide Accuhaler †
fluticasone propionate
100mcg • 250mcg • 500mcg
*Fixotide Junior

Pulmicort Turbuhaler †
budesonide
100mcg • 200mcg • 400mcg

QVAR Autohaler †
beclomethasone
50mcg • 100mcg

Arnuity Ellipta †
fluticasone furoate
100mcg • 200mcg

Axotide Accuhaler †
fluticasone propionate
100mcg • 250mcg
*Axotide Junior

LAMA MEDICATIONS

Spiriva Respimat †
tiotropium 2.5mcg

Spiriva Handihaler †
tiotropium 18mcg

Bratrus Zonda †
tiotropium 13mcg

Seebri Breezhaler †
glycopyrronium 50mcg

Incruse Ellipta †
umeclidinium 62.5mcg

LAMA-LABA COMBINATIONS

Spiolto Respimat †
tiotropium-olodaterol
2.5/2.5mcg

Brimica Genuair †
acridinium-formoterol
340/12mcg

Ultibro Breezhaler †
indacaterol-glycopyrronium
110/50mcg

ICS-LAMA-LABA COMBINATIONS

Trelegy Ellipta †
fluticasone furoate-umeclidinium-vilanterol
100/62.5/25 • 200/62.5/25 • 200/114/114

Enerzair Breezhaler †
mometasone-glycopyrronium-indacaterol
68/48/114 • 136/48/114

Trimbow Inhaler †
beclomethasone-glycopyrronium-formoterol
100/10/6 • 200/10/6

Breztri Aerosphere †
budesonide-glycopyrronium-formoterol
160/7.2/5
all units in mcg

ICS-LABA COMBINATIONS

Seretide Inhaler †
fluticasone propionate-salmeterol
50/25 • 125/25 • 250/25 • 500/50
Additional brands: Pavide, Fluticasone + Salmeterol Cipla, SalplusF, Evocair

Symbicort Turbuhaler †
budesonide-formoterol
100/6 • 200/6 • 400/12 • 800/12
Additional brand: Rilast Turbuhaler

DuoResp Spiromax †
budesonide-formoterol
200/6 • 400/12

Flutiform Inhaler †
fluticasone propionate-formoterol
50/5 • 125/5 • 250/10

Salflumix Easyhaler †
fluticasone propionate-salmeterol
250/50 • 500/50

Fostair Inhaler †
beclomethasone-formoterol
100/6 • 200/6

Bufomix Easyhaler †
budesonide-formoterol
200/6 • 400/12

Breo Ellipta †
fluticasone furoate-vilanterol
100/25 • 200/25

Aetectura Breezhaler †
mometasone-indacaterol
62.5/125 • 127.5/125 • 260/125
all units in mcg

This chart was developed independently by the National Asthma Council Australia with support from AstraZeneca Australia, Chiesi Australia, GlaxoSmithKline (GSK Australia) and Orion Pharma.

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PBS PRESCRIBERS † Asthma unrestricted benefit ‡ Asthma restricted benefit * Asthma authority required * COPD unrestricted benefit # COPD restricted benefit ° COPD authority required
 ‡ Anti-inflammatory reliever (AIR) therapy † Maintenance-and-reliever therapy (MART) ° Severe asthma * Check TGA and PBS for current age and condition criteria.
 Improving asthma control can also reduce the environmental impact of asthma. For more information refer to the Reducing the environmental impact of asthma treatment information paper at nationalasthma.org.au.

ICS: inhaled corticosteroid; LABA: long-acting beta₂ agonist; LAMA: long-acting muscarinic antagonist; SAMA: short-acting beta₂ agonist; SAMA: short-acting muscarinic antagonist

- The most common device is a **puffer (pressurised metered dose inhaler pMDI) with spacer**.
 - Under school age → small-volume spacer + face mask for seal.
 - School age → small or large volume spacer
 - Clean in warm soapy water and air dry.
- Puffer without spacer:** > 12 yrs (Can consider from age 8 yrs).
- Spacers are just as effective as nebulisers as long as they are well-maintained and used properly. Nebulisers might be needed in the emergency setting.



	Advantages	Disadvantages
Puffer without spacer	Placebo trainer Portable Can wash in warm soapy water	Coordination difficult Post-inhalation cough Not all have dose counters
Spacer (vs nebuliser)	Placebo trainer As effective as nebuliser but much cheaper and more portable	Large volume spacer is big Static, incorrect cleaning Face mask can absorb medication, need good seal

Other devices:

Device	Age	Advantages	Disadvantages
Puffer			
Autohaler e.g. Airomir (salbutamol)	> 7 years	Breath activated Less coordination than puffer	Lever can be hard to pull into position for young kids Might be unable to take big enough breath especially during flare-up No dose counter More expensive
Dry powder inhalers (the limitations are all the same).			
Accuhaler (e.g. Flixotide, Seretide)	> 7 yrs (popular with teens!)	Placebo trainer Size Dose counter Easy to clean (wipe mouthpiece)	Post-inhalation cough Medication on back of mouth and throat
Turbuhaler e.g. Symbicort	> 12 yrs	Size/portable Minimal taste Dose indication window/counter Easy to clean (wipe mouthpiece)	Device must be upright for loading May be difficult to use in asthma flare-up Post-inhalation cough
Ellipta e.g. Arunity, Breo	> 12 yrs	Placebo trainer Once daily Size Dose counter Easy to clean (wipe mouthpiece)	Post-inhalation cough Medication on back of mouth and throat

Acute Asthma

iWOB: RR, accessory muscle use, tracheal tug, suprasternal or subcostal recession, abdominal breathing. Other respiratory signs include SaO₂, air entry, wheeze, nasal flaring, hyperinflation.

If a child has features in different severity categories. **always choose the highest.** We can rapidly assess the severity of the acute asthma episode by **observation** and **pulse oximetry**.

“Burst therapy” refers to giving frequent doses of a reliever medication (usually salbutamol via MDI or nebuliser) for a short period during an acute exacerbation. Typically, 4–6 puffs if 1-5 years old, and up to 12 puffs if > 6 years old, every 20 minutes for 1 hour under supervision.

	Mild/moderate	Severe	Life-threatening
Definition	Can walk, speak in phrases, moderate iWOB SpO ₂ > 94% RA	Can't speak in sentences, significant iWOB SpO ₂ ≤ 94% RA	Reduced consciousness/collapse, cyanosis, poor respiratory effort SpO ₂ < 90% RA
Treatment: 1-5 years old			
Immediate	Salbutamol 2 – 6 puffs via pMDI + spacer +/- mask	Salbutamol 6 puffs via pMDI + spacer +/- mask Start O ₂ targeting SpO ₂ 92 – 95%	Transfer to ICU Salbutamol 2 x 2.5 mg nebulises via continuous nebulisation driven by O ₂ Maintain SpO ₂ to target 92-96%.
Continued	Repeat salbutamol 4 – 6 actuations every 20 – 30 mins for the first hour, if needed	Transfer to acute care Salbutamol 6 puffs every 20 minutes for the first hour and as needed	
- Start systemic corticosteroids if no good response to salbutamol within 30 minutes: prednisone/prednisolone 1 mg/kg PO mane for 3 days. - If the bronchodilator does not relieve dyspnoea/iWOB or the condition deteriorates, you may add: Ipratropium bromide: 4 actuations via pMDI (21 mcg/actuation) every 20 minutes for the first hour, then every 4-6 hours for 24 hours. IV MgSO₄ (not for <2 years old): 0.1 – 0.2 mmol/kg (max 8 mmol) diluted in a compatible solution as an IV infusion over 20 minutes.			
Treatment: > 6 years old			
Immediate	4-12 puffs salbutamol via pMDI + spacer.	12 puffs salbutamol via pMDI + spacer Ipratropium 8 puffs via pMDI + spacer every 20 mins for first hour. Repeat 4-6 hourly for 24 hours. O ₂ supplementation targeting SpO ₂ 92 – 96%.	Higher-level care. Salbutamol 2 x 5 mg nebulises via continuous nebulisation driven by O ₂ . Ipratropium 500 mcg added to nebulised solution every 20 minutes for the first hour. Repeat bronchodilators 4-6 hourly. O ₂ supplementation targeting SpO ₂ 92 – 96%.
Systemic corticosteroids within 1 hour: - Prednisone/prednisolone 1 mg/kg PO mane for 3 days - Oral dexamethasone 0.6 mg/kg as single dose - If not tolerating orals, IV hydrocortisone 4 mg/kg 6 hrly or IV methylprednisolone 1 mg/kg 6 hrly Inhaled ipratropium bromide and IV magnesium sulfate infusion are again indicated if the child does not improve or deteriorates.			

Salbutamol is 100 mcg per actuation.

Once symptoms have resolved, **observe the patient for at least 4 hours**. At discharge, write an interim asthma action plan and ensure the child has appropriate asthma treatment that includes an adequate maintenance dose of ICS. Send them home with a **Ventolin weaning plan too (over 3 days, decrease frequency of puffers)**. Arrange follow-up with the child's GP within 3 days.

- The child can be discharged if **they do not require oxygen** and **Ventolin requirement is every 3 hours or less frequent**.
- Only children over 6 can cooperate with spirometry; they always need repeat spirometry before discharge.

Emergency first aid for a severe/life-threatening asthma flare up:

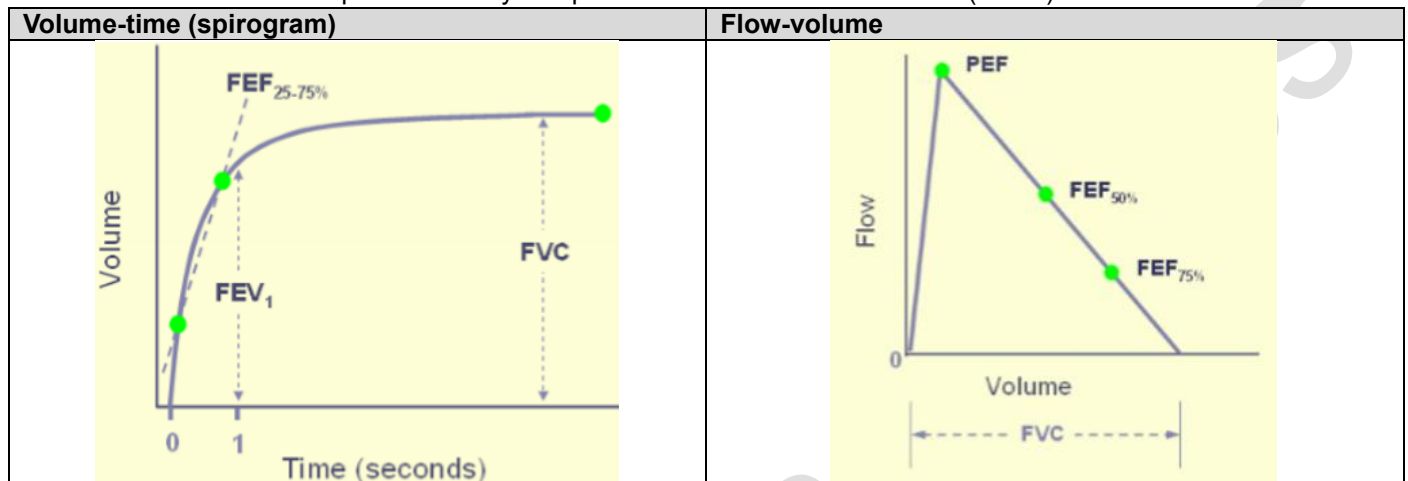
- Place child in a seated upright position; **call 000!**
- 4 by 4 by 4**: 4 puffs of reliever with spacer (e.g. Ventolin); allow child to take 4 breaths through spacer; repeat every 4 minutes.

Asthma Spirometry

Recall that spirometry measures **dynamic lung volumes** from a maximal expiration (volume and flow). Expiration is above the x-axis, inspiration is below the x-axis.

Forced vital capacity (FVC)	Maximum amount of air exhaled in a single breath
Forced expiratory volume in 1 second (FEV1)	Amount of air exhaled in first second
FEV1/FVC	Ratio
FEF 25 – 75	Mid-range forced expiratory flow
PEF	Peak expiratory flow

All volumes and flows are reported at Body Temperature and Pressure Saturated (BTPS)



There are strict criteria for spirometry to be **acceptable** e.g. no glottic closure, no cough, no obstruction, no leak. Must achieve one of three criteria: expiratory plateau, expiratory time at least 15 seconds (*uncommon in paediatrics*), FVC within repeatability tolerance.

- Need to have a good strong peak.
- Repeatability is everything – obtain at least 3 technically acceptable blows

Common problems with spirometry: submaximal effort, incomplete inhalation or exhalation, not closing lips on mouthpiece, posture, cough, glottic closure.

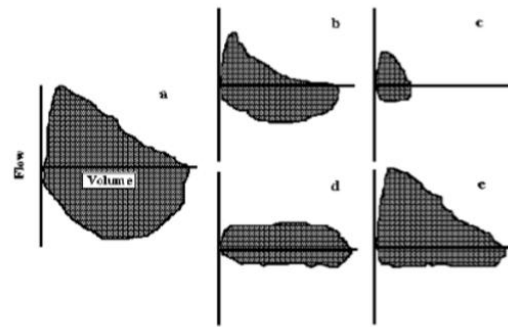
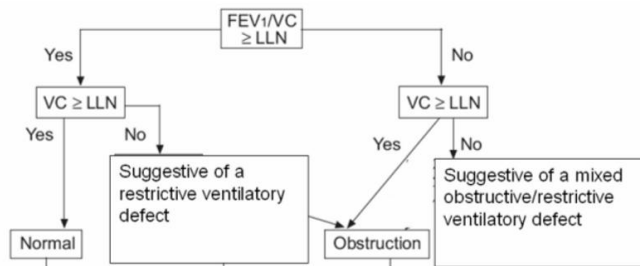
Assessment of reversibility

- Perform pre-BD spirometry
- Administer BD, e.g. 4 puffs salbutamol with pMDI/spacer
- Wait 15 minutes and repeat spirometry
- **FEV1** is most commonly used to quantify reversibility, however **reversibility may be an increase in FEV1 or FVC > 10%**
 - $100 \times (\text{FEV1 post BD} - \text{FEV1 pre BD}) / \text{predicted FEV1}$

Recall patterns might be obstructive (e.g. asthma), restrictive (e.g. scoliosis, DMD), or mixed

Interpretation

- Patient results compared to reference values: age, height, biological sex, ethnic origin.
- Fixed percentages are not ideal
- Z-scores (FEV1 % predicted) are used to quantify **severity**
 - FVC Z-score can help determine if there is a mixed picture
- Strategy: graphs → FEV1/FVC → FVC → FEV1 Z-score



Flow-Volume Loops

A is normal, B is obstruction (concave), C is restriction (narrow), D is fixed obstruction (e.g. subglottic stenosis), E is extrathoracic obstruction (inspiratory flow problem only).

Tip: a clue that spirometry has been performed well is that pre and post-BD curves are superimposed.

3.2.2 Cystic Fibrosis

Most common childhood onset life threatening genetic condition in Australia.

- **Carrier frequency:** 1 in 20 Caucasians

Prevalence in Northern European Descent: 1 in 3200

Genetics: pathogenic CFTR variant; commonest mutation worldwide is the p.Phe508del variant (85% of Caucasian CF patients).

Pathophysiology: CFTR is a cAMP-dependent protein involved in the transport of Cl⁻, Na⁺, and HCO₃⁻ across the cell membrane. Note the function of CFTR depends on its anatomic location.

- Lungs and GIT: Cl⁻ out
- Skin: Cl⁻ in
- *Therefore, people with CF get thick, sticky mucus but salty sweat.*

Normal physiology requires **10% CFTR functioning**; CF develops at 5% function, and pancreatic insufficiency when <1%.

CFTR mutations are classified as I – VI. The commonest mutation, F508del, is a Class II mutation.

	I	II	III	IV	V	VI
Molecular defect	No synthesis	Block in processing	Block in regulation	Reduced conductance	Reduced synthesis	Reduced half-life
Functional anomaly	Protein not synthesised	Folding defect, doesn't reach membrane	Channel opening defect	Ion transport defect	Decreased protein synthesis	Decreased protein half-life
Effective Modulator Drugs	Ineffective (no protein made)	Correctors	Potentiators	Potentiators	Potentiators +/- correctors	Corrector + potentiator

Diagnosis

1. Newborn screening
 - Clinical suspicion e.g. meconium ileus
 - Newborn bloodspot screening for **immunoreactive trypsinogen (IRT)**: high levels are a biomarker of CF-related exocrine pancreatic disease.
 - This is because pancreatic ductal obstruction leads to a backflow of trypsinogen into the bloodstream.
 - Elevated IRT → **CFTR mutation analysis**. Positive NBS (↑ IRT + 1-2 common CFTR mutations, or ↑↑ IRT) → refer to CF centre for sweat chloride test +/- repeat genetic testing.

- Sweat chloride test (**gold-standard!**)
 - Detects the amount of Na⁺ and Cl⁻ in sweat (↑ in CF)
 - Collect at least 75 mg sweat, child must be > 4 kg for test to be conclusive.
 - False negatives in skin oedema and malnutrition.
 - False positive in metabolic conditions or hypothyroidism.

Salt level	Interpretation
> 60 mmol/L	CF
29 – 60 mmol/L	Possible CF, repeat
< 29 mmol/L	Normal

Clinical Manifestations: multisystem!

- Bronchiectasis and progressive respiratory impairment.
- Epithelial dysfunction: pancreas (malabsorption), liver (biliary cirrhosis), sweat glands (heat shock), absence of the vas deferens (infertility).
- Other: sinusitis, nasal polyps, constipation or distal intestinal obstruction.

Respiratory Pathogens in CF

- Staphylococcus aureus is the most common chronic respiratory infection in early childhood (*children with CF require **Staphylococcus prophylaxis for the first year of life***).
- Pseudomonas becomes the most common coloniser over 25 years of age.
- People with CF can grow different pathogens at different times ... look at recent sputum samples on admission to hospital and/or before physio etc.

Management of CF

4 pillars:

Airway Clearance	Antibiotics	Nutrition	Emerging Therapies
Chest physio (with 3% or 6% hypertonic saline) Exercise Mucolytics (nebulised pulmozyme; hydrolyses DNA in sputum) Bronchodilators	Increased cough above baseline or new productive cough Oral, inhaled, or IV Early Augmentin, <u>or</u> tailored to specific colonisers	Pancreatic enzymes (Creon) Supplemental fat-soluble vitamins	CFTR correctors and potentiators Gene therapy Personalised medicine

Any patient with CF and a significant worsening of lung function requires hospital admission for IV antibiotics or nebulised antibiotics.

CFTR Modulators: Correctors (ellexacaftor, tezacaftor, lumacaftor) correct flaws in the CFTR protein. Potentiators (ivacaftor) bind to the defective CFTR protein at the cell surface and hold the channel open.

- Ivacaftor, a potentiator, is good for Class III mutations because they involve a channel opening defect.
- Those with the F508 mutation (Class II) require a corrector, because there is a processing defect.

Kalydeco	Orkambi	Symdeko	Trikafta
Ivacaftor	Lumacaftor + ivacaftor	Tezacaftor + ivacaftor	Ellexacaftor, tezacaftor, ivacaftor
>12 mo with 1 Class III mutation	> 2 years, F508 homozygous	> 12 years, F508 homozygous or 1 approved mutation	> 6 years, 1x copy F508

Side-effects of CFTR modulators: LFT derangements, rashes, small risks of cataracts (**require annual screens**).

Life expectancy: 38.1 years with basic supportive care, rises to 45-48 years on Orkambi/Symdeko and 71.6 years on Trikafta.

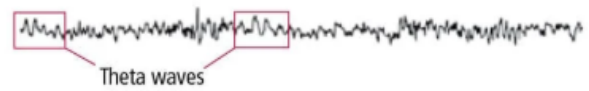
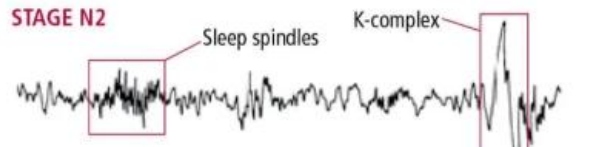
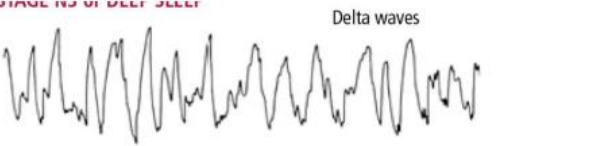
Remember to address psychosocial aspects too.

3.3 Sleep in Infancy and Children

Sleep is a reversible behavioural state of decreased responsiveness and interaction with the environment. It is required to restore body and mind function and is a period of considerable neurologic and physiologic activity. Overall sleep duration decreases with age, though **nighttime sleep specifically** peaks at about 2 years old.

Newborns	6 months old	Toddlers	Adolescents
18 – 20 hours	11 hours	10 hours (2 yrs), no daytime sleep by 4 yrs	7.5 to 8 hours

Sleep Stages

Awake		
Alert, active	Eyes open: beta waves (high frequency, low amplitude). Eyes closed: alpha waves	Awake (mental activity)   Beta (13-30 Hz) Awake (resting)   Alpha (8-12 Hz)
Non-REM Sleep: primarily controlled by sleep pressure. Occurs early in the sleep period. SWS for recovery and recuperation.		
Stage N1	Light sleep, theta waves (mixed frequency).	
Stage (SWS) N2	Deeper sleep with sleep spindles and K complexes .	
Stage (SWS) N3	Deepest non-REM sleep; delta waves with low frequency (0.5 – 2 Hz) and high amplitude. Physiologically restorative.	
REM Sleep: Controlled by the circadian rhythm. Occurs late in the sleep period.		
Mix between stage 1 and awake.	Low voltage, mixed frequency, rapid eye movements. Loss of motor tone (muscle paralysis), memory processing, vivid dreaming. Unstable autonomic activity, irregular breathing. OSA manifests here (skeletal muscle atonia)	Beta waves.

The duration of REM sleep reduces from 50% in newborns to 25% in adolescents, while the duration of NREM sleep remains relatively stable.

Symptoms of Sleep Disorder

Duration	Disorders of Timing	Sleep-Related Behaviours	Sleep-Disordered Breathing
Excessive daytime sleeping Insomnia	Delayed sleep phase disorder Abnormal circadian rhythm due to blindness or neurological disease	Movement disorders e.g. head banging Parasomnias Sleep related seizure disorder	Primary snoring. OSA

Snoring

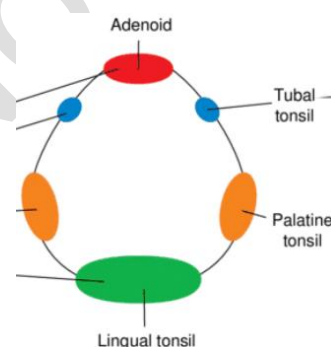
- Common symptom of sleep-disordered breathing; prevalence is **10%**.
- Primary snoring: without apnoeas, arousals, or gas exchange abnormalities → 2/3 resolve over 2 years.

Obstructive Sleep Apnoea (OSA)

Disruption of normal ventilation during sleep: prolonged partial upper airway obstruction and/or intermittent complete obstruction.

- **Prevalence:** 2 – 5%
- **Peak prevalence:** 3 – 5 years old.
 - Coincides with age of maximal growth of adenoids (2 – 6 years)
- **Commonest cause:** enlarged adenoids and tonsils.
 - **Recall, the lingual tonsil, palatine tonsils, tubal tonsils, and adenoids form the Waldeyer Ring.**
- Other causes: craniofacial (micrognathia, maxillary hypoplasia), laryngomalacia, hypotonia, obesity.
- **Pathophysiology:** enlarged adenoids and tonsils, structure of airway, loss of neuromuscular tone in sleep.
 - This is why OSA is worse in REM sleep.
 - Nose breathing is important because nasal receptors maintain pharyngeal tone and produce NO (pulmonary vasodilator) → adenoid hypertrophy predisposes to OSA.
- Several complex illnesses increase the risk of OSA: craniofacial anomalies e.g. Crouzons, genetic syndromes e.g. Downs, cerebral palsy, obesity, metabolic e.g. Pompe's, neuromuscular e.g. muscular dystrophy, laryngomalacia.
- Characterised by **apnoeic episodes:** intermittent cessation of airflow lasting 10 seconds or greater.
- **Sequelae:**
 - **Increased WOB:** failure to thrive → decreased appetite, growth restriction from altered IGF-1/GH pathway, increased caloric expenditure.
 - **Intermittent hypoxia:** neurobehavioural and neurocognitive decline, pulmonary hypertension (more with continuous hypoxia), systemic hypertension (sympathetic tone).
 - **Sleep fragmentation:** poor concentration, executive function, aggressive outburst. Enuresis may also result.
 - **The commonest complication of OSA is sleep and memory decrements.**
- **Physical examination:**
 - Mallampati score (oropharynx): Class 3 or 4
 - Tonsillar grade (1 – 4)
- **Investigations:**
 - Lateral airway X-ray for enlarged adenoids (60 – 70% sensitive for OSA)
 - Sleep diary
 - Pulse oximetry and morning venous gas (CO₂ and bicarbonate)
 - Actigraphy, video, or audio tapes
 - **Polysomnography:** ↓ airflow despite effort; ↑ apnoea/ hypopnea index (AHI)
 - AHI > 10 = severe OSA.
 - Can also measure SaO₂ (min < 92%) and end-tidal CO₂ (>53 mmHg)
- **Diagnosis:** clinical symptoms have **50 – 70% reliability**; polysomnogram (PSG) is the **gold-standard** but does not *have* to be performed.
 - Usually, a careful history of sleep disturbed breathing (snoring, restless sleep, apnoeic episodes +/- enuresis, poor concentration during the day) + physical exam (adenotonsillar hypertrophy) consistent with OSA is sufficient to make a surgical decision for a child.
 - When in doubt, order a **formal sleep study** e.g. syndromic cases, craniofacial abnormalities, or neuromuscular dysfunction.
- **Management:**
 - Observe and reassess 6-monthly.
 - Nasal steroids and leukotriene receptor antagonists.
 - Weight management.
 - CPAP (may be offered before surgery in children at high risk of OSA e.g. syndromes or CP).
 - Adenotonsillectomy.

Clinical Identification of Moderate-Severe OSA	
Nocturnal	Daytime
Snoring Difficulty breathing Mouth breathing Sweating Throat clearing Chest recession Paradoxical breathing Observable apnoea Nocturnal enuresis Frequent awakenings Abnormal sleeping position (e.g. head extended)	Morning headache Difficulty waking up Poor school performance Poor attention span Hyperactivity Intrusive naps Frequent sore throat/ dry mouth Failure to thrive Daytime somnolence Hypertension



- Tracheostomy rare.
- Note, **adenotonsillectomy** cures 60 – 80% of children with OSA. Other surgical modalities include lingual tonsillectomy, uvulopalatopharyngoplasty (UPPP), tongue reduction, mandibular distraction, bariatric surgery, tracheostomy.

Parasomnias

Undesirable motor, autonomic, or behavioural phenomena that occur exclusively or predominantly during sleeping.

Parasomnia	Aetiology	Key Features	Onset	Duration	Management
Night Terror (slow wave arousal)	Begins ~18 mo, 6% healthy children; usually benign. Genetic factors Sleep disordered breathing Acute triggers e.g. fever, stress, sleep deprivation.	Often occurs in clusters (several months); sudden onset; bolts upright, stare, diaphoretic (autonomic), irregular breathing; inconsolable; no recall .	1 st third of sleep (NREM III and IV)	Seconds to 10 minutes	Parent reassurance (outgrow by school years), safe environment, do not awaken or discuss with child, review sleep hygiene. Refer to specialist: new onset in school-age child, frequency >3/week, lasting >30 min, affecting daytime performance. Resistant: scheduled awakenings, medications (e.g. BZD – rare).
Partial confused arousal	Early childhood	Child wakes up; confused, agitated or dazed; brain divided between wakefulness and sleep	Transition from SWS to REM	5 – 15 mins	Reassurance, sleep log or video +/- PSG if another sleep disorder suspected. Outgrow by school years, do not awaken child or discuss in morning.
Head banging	8 months to 4 years M:F 3: 1	Preceded by rhythmic habits e.g. body rocking; head banging at bedtime; no abnormal behaviour at daytime	Variable	Variable	Reassure unless causing injury
Nightmares	Onset 3-6 years Prevalence 10-50%	Child cries out, visibly upset, easily consoled, can recall vividly	REM arousal, last third of night	<1-2 minutes	Caregiver comfort at time of episode; minimize night-time discussion; decrease exposure to identifiable stressor; rule out medications

Tip: You REMember your nightmares!

Adolescence: Delayed Sleep Phase Disorder

- Circadian rhythm disturbance; affected individuals have an unusually long circadian rhythm.
- **Presentation:** typically adolescents and young adults, reported prevalence 7 – 16%. Late sleep onset during the week, may interfere with daily function e.g. concentration at school. 'Catch up' sleep-ins in the holidays or on the weekend.
 - Up to 40% have a positive family history.
- **Management:**
 - Behavioural: reschedule sleep time (move bedtime ½ hour earlier each week), improve sleep hygiene, morning routine, regular exercise, relaxation techniques.
 - Melatonin if refractory.

4. Gastroenterology

Gastroenteritis

Covered in [18.1 Recognising and Managing the Sick Child \(Vomiting\)](#) and [18.5 Managing Fluids in Young Children and Infants](#).

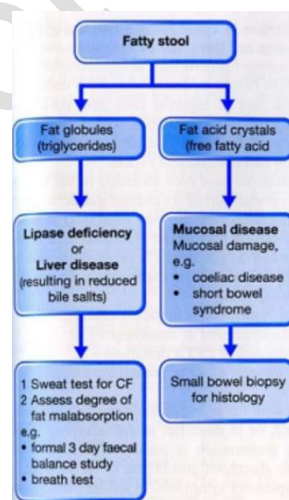
Chronic Gastrointestinal Conditions in Children

Diarrhoea and Malabsorption

Malabsorption is not a diagnosis per se. It presents as diarrhoea and is investigated with stool assessment. Recall that diarrhoea may be fatty (steatorrhoea), watery, or bloody.

Steatorrhoea

- Fat absorption requires (i) gastric acid (hydrolysis), (ii) bile salts (solubilise fat), (iii) pancreatic enzymes (digest fat) and (iv) absorption (across intact intestinal mucosa).
- Lipids are typically absorbed through the lymphatic system as chylomicrons; one exception is small and medium-chain TAG which can enter the portal venous system directly.
 - This makes **short-chain triglycerides/short chain fatty acids** an excellent supplement for those with malabsorption/maldigestion.
- Steatorrhoea in childhood is typically caused by diseases of the **pancreas and small intestine**.
- When assessing steatorrhoea, we ought to differentiate between **fat globules** vs **fatty acid crystals** in the stool:
 - Fat globules implies lipase deficiency or liver disease (+ jaundice) → investigate with a faecal elastase breath test (may lack sensitivity), formal 3 day faecal fat study, or CF sweat test.
 - Fatty acid crystals implies OK digestion but poor uptake → small bowel biopsy



Suspect cystic fibrosis in the kid with meconium ileus, chronic steatorrhoea, failure to thrive, and respiratory illnesses!

Chronic liver disease can lead to steatorrhoea for many reasons. Suspect in a kid with jaundice + hepatosplenomegaly.

- Extrahepatic biliary atresia (presents with **cholestatic jaundice**) → warrants biliary tree imaging e.g. cholangiogram.
- Cirrhosis (decreased bile salts)

Watery Diarrhoea (Secretory or Osmotic)

Recall that osmotic diarrhoea stops on fasting; secretory diarrhoea persists on fasting.

Osmotic	Abnormal water and electrolyte transport	Infection	Mucosal disease	Bacterial overgrowth syndromes	Bile salt malabsorption
Disaccharidase deficiency: lactase, sucrase/ isomaltase Glucose-galactose malabsorption Excessive intake: sorbitol, fructose	Congenital chloride diarrhoea or sodium diarrhoea		Coeliac disease Milk protein intolerance Inflammatory bowel diseases Immunodeficiency disorders Autoimmune enteropathy	Gastrointestinal motility disorders Anatomical (blind loop)	Congenital Ileal resection

CHO Physiology Recap:

Sucrose → glucose + fructose via **sucrase**

Lactose → glucose + galactose via **lactase**

Glucose and galactose absorbed actively via SGLT

Fructose absorbed via facilitated diffusion by GLUT-5

Assessment of watery stools: requires a fresh stool sample

- Reducing substances: diagnoses carbohydrate malabsorption - glucose, galactose, fructose, lactose, and maltose can act as reducing agents by donating electrons in chemical reactions.
 - Note: in breastfed infants, up to 0.5% reducing substances is normal → abundant lactose, rapid intestinal transit, immature gut enzyme systems
 - The one exception is **sucrose** which is a non-reducing sugar (note: it may act as a reducing agent if hydrolysed by boiling or acid).
- **Stool osmotic gap:** $280 - 2 \times (\text{Stool Na} + \text{Stool K})$, normal 50 – 100. **Osmotic gap >100 = osmotic diarrhoea.**

Clinical Features of Osmotic Diarrhoea:

- Stops when feeds stop
- Mild **hyperchloraemic acidosis**
 - Unabsorbed CHO in colon fermented to SCFA e.g. butyrate → **absorbed in exchange for HCO_3^-** → decreased Cl^- excretion to maintain electrochemical neutrality
- SCFA effect outweighs alkalinising effect of HCO_3^- loss → **stool acidity and perianal excoriation**

Dietary history is **critical** for determining the type of CHO malabsorption. For instance, the onset of osmotic diarrhoea seen with the introduction of fruit in the diet may suggest a diagnosis of **congenital sucrase-isomaltase deficiency (a rare autosomal recessive condition)**.

- **Disaccharide malabsorption** may occur secondary to any condition that causes damage to the intestinal brush border: viral gastroenteritis, coeliac disease, chronic giardiasis, milk protein enteropathy, small bowel bacterial overgrowth, immunodeficiency, autoimmune enteropathy
- **Monosaccharide malabsorption** is **life-threatening** because it can lead to massive fluid loss and dehydration.
 - Monosaccharide transporters are deeper in the brush border; thus, you need severe enteropathy to cause monosaccharide malabsorption.
 - Otherwise, can result from congenital transporter defects e.g. **glucose/galactose malabsorption (autosomal recessive SGLT1 mutation)**
 - The only treatment is to **remove the offending carbohydrate** e.g. substitute fructose for glucose/galactose; and **treat the underlying disease where present**.

Secretory Diarrhoea (Osmotic Gap < 50)

- Most commonly due to mucosal damage or inflammation (**e.g. infection**) but may be due to inherited abnormal fluid and electrolyte transport.
- Bile salt malabsorption can also cause secretory diarrhoea (bile acids irritate the colonic mucosa).

Bloody Diarrhoea

Infection	Bacterial, parasite
Inflammatory bowel disease	Crohn's disease, ulcerative colitis, milk protein intolerance

- Blood is not always obvious in the stool – you must do microscopy.
 - The **colonic mucosa** is vulnerable to friability and ulceration during inflammation → **RBC and leukocytes on stool microscopy suggest colitis**.
 - Enteritis usually does not result in bleeding, with the exception of some bacteria e.g. Salmonella, EHEC, or Shigella.
- The reason patients with colitis get **diarrhoea** is because there is malabsorption of fluid and electrolytes by the inflamed colonic mucosa.
- **Protein-losing enteropathy**: excessive blood loss + protein loss from inflamed intestinal mucosa → **iron deficiency anaemia** and **hypoproteinaemic oedema**
 - Measure with **stool alpha 1 antitrypsin**
- Recall that **faecal calprotectin** (released by neutrophils during intestinal inflammation) can differentiate IBD from IBS.

Specific GI Conditions

Coeliac Disease

- Immune-mediated inflammatory disease of the small intestine caused by sensitivity to dietary gluten and related proteins in genetically predisposed individuals

- **Epidemiology:** 1.4% (about 1 in 100)
- **Genetics:** HLA DR3-DQ2 (90%) or HLA DR4-DQ8 (10%). Good negative predictive value if absent.
- **Clinical presentation:** **diarrhoea/steatorrhoea** or constipation, failure to thrive/ poor weight gain, abdominal distension
 - **Extraintestinal manifestations:** fatigue, lethargy, iron-deficiency anaemia, bone/ muscle/ joint pain, dental defects, neurologic symptoms, mouth ulcers (aphthous ulcers), dermatitis herpetiformis, short stature and delayed puberty, dental defects, menstrual disturbance
- **Test for coeliac in kids with the following:** iron deficiency anaemia (if Fe-resistant or unexplained), IgA deficiency, Down Syndrome, Turner Syndrome, Williams Syndrome, those with first-degree relatives with Coeliac Disease.
- **Diagnosis:** made on gluten-containing diet, 3g gluten/day (if on gluten-free diet, challenge with 10g/day gluten for 8 weeks then re-test).
- Coeliac Serology:
 - **Anti-tissue transglutaminase IgA:** first-line, most sensitive
 - If high index of suspicion but low anti-TTG, suspect **IgA deficiency** → order **deamidated gliadin peptide IgG (DGP-IgG)**
 - Endomysial antibody is sometimes used as a confirmatory test

Coeliac Criteria	Description
Serological	Normal IgA + anti-TTG IgA > 10x ULN, AND positive endomysial antibody (EMA) on separate blood test. <i>These are diagnostic of coeliac disease according to the ESPGHAN 2020 guidelines (i.e. no biopsy needed).</i>
Small bowel biopsy	Marsh criteria for intraepithelial lymphocytosis (plus crypt hyperplasia and villous atrophy). Coeliac disease diagnosed if Marsh 2 or above: >40 IEL per 100 enterocytes in jejunum; > 30 in duodenum; crypt hyperplasia (<i>villous atrophy need not be present</i>)

- **Treatment:** lifelong gluten-free diet. This will normalise growth, reverse symptoms, minimise the risk of GI cancers and GI Non-Hodgkin Lymphoma.
 - Children with severe coeliac disease may also benefit from 6 weeks of a lactose-free diet (recall lactose digestion requires lactase activity at the intestinal brush border)
- **Monitoring:** TTG is useful for monitoring treatment response, dietary adherence. Performed at diagnosis, 6 months, and then annually.

Inflammatory Bowel Disease

Genetic, environmental, and microbiome predisposition leading to autoimmune inflammation of the bowel.

Crohn's Disease

- "From mouth to anus"
- Small bowel Crohn's may present insidiously: non-specific abdominal pain, loose stool, growth failure
- **Diagnosis:**
 - Faecal calprotectin (neutrophil degradation product) for IBD > IBS; best in >2 years old
 - Gold-standard: endoscopy and biopsy showing chronic inflammation +/- granulomas
- **Management:** immunosuppression → induction of remission (exclusive enteral nutrition, steroids, or monoclonal antibodies), maintenance
- **Complications:** fistulas, segmental stenosis, bowel obstruction, malabsorption, osteoporosis

Acute Severe Colitis

- Acute presentation of IBD; these patients have a higher chance of having refractory disease and a higher risk of colectomy.

1-2% with ASC will develop **toxic megacolon**.

- Additional risk factors include infection (CMV or C difficile), electrolyte disturbances (hypokalaemia, hypomagnesaemia), drugs (anticholinergics, antidepressants, loperamide, opioids)
- **Presentation of toxic megacolon:** colonic dilatation ≥ 56 mm, fever, tachycardia, dehydration, electrolyte disturbance.
- **Investigations:** abdominal x-ray and surgical consult

- **Management:** steroids, antibiotics (+ oral vancomycin if *C. difficile* involved), IV ganciclovir if CMV involved, gut rest +/- immunosuppressants (e.g. infliximab, cyclophosphamide, tacrolimus) + early surgical consult.

Jaundice in Children

Recall the bilirubin excretion pathway:

- Splenic macrophages break down RBC, releasing **heme** which is converted into **biliverdin** by heme oxygenase.
- Biliverdin reductase then converts biliverdin into unconjugated bilirubin, which circulates bound to **albumin**.
- Bilirubin is conjugated with glucuronic acid in the liver via **UDP-glucuronyltransferase (UGT)**.
- It is excreted into the small intestine where it may be removed in faeces, or taken up via enterohepatic circulation and excreted in urine.

Cholestatic Jaundice

Conjugated hyperbilirubinaemia is defined as hyperbilirubinaemia with conjugated fraction $\geq 20\%$ total.

In a neonate, conjugated jaundice is always abnormal. Recall the most important thing to ask about is **stool colour**: acholic stool represents a failure of bile to enter the intestine.

Biliary atresia is the leading cause of cholestatic jaundice in paediatrics, and the top indication for a paediatric liver transplant in Australia.

- **Presentation:** typically 2-6 weeks of life, with conjugated hyperbilirubinaemia, dark urine, and pale stools
 - **Note, there is usually a progressive inflammatory process that blocks the bile ducts** → stools become progressively acholic.
- **Pathophysiology:** in utero/idiopathic biliary ductal inflammation and damage → scarring and little-no bile flow
 - **There is a syndromic version called biliary atresia and splenic malformation (BASM) where patients will also present with asplenia or polysplenia.**
 - Biliary atresia can also be associated with heterotaxy syndromes (check where the heart/abdo organs are)
- **LFTs:** ↑ AST, ALT, ALP, GGT (*cholestasis and hepatocellular injury*)
- **Abdominal ultrasound** (can lack sensitivity i.e. a normal USS does not exclude biliary atresia)
- **Management:** **Kasai portoenterostomy** where part of the small intestine is connected to the porta hepatis, allowing drainage to occur.
 - The earlier the better (<60 days old)
 - **Rule of 1/3's:** 1/3 good drainage, 1/3 good then fail (need transplant), 1/3 fail immediately.
- Cholangitis and chronic liver disease are major complications.

Paediatric Liver Failure

Acute liver failure	Chronic liver failure
Moderate coagulopathy (INR > 1.5) with encephalopathy OR severe coagulopathy (INR > 2) only The INR is not correctable with Vitamin K	Can be compensated or decompensated
Jaundice, encephalopathy, ascites, hepatomegaly, oedema	Clubbing, spider naevi, splenomegaly, jaundice, encephalopathy, varices, ascites ad oedema; liver usually small and cirrhotic

Common LFT Patterns in Paediatrics

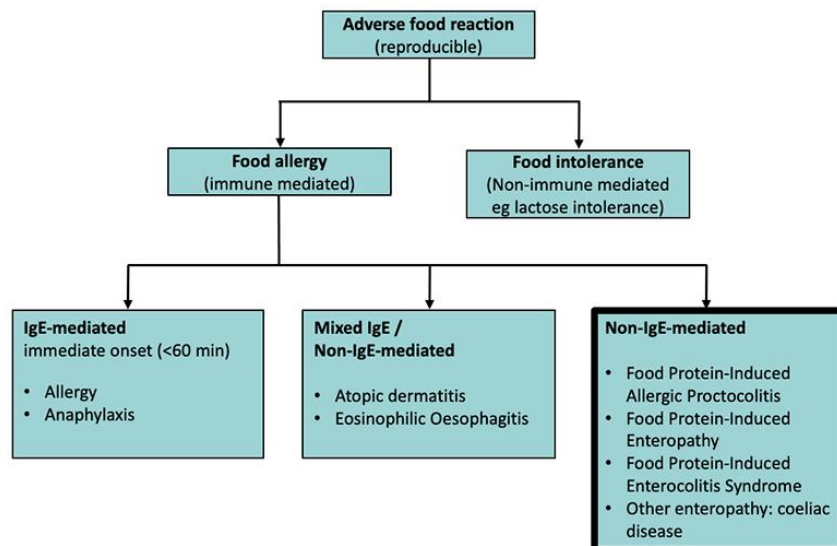
Condition	HSV Hepatitis	Wilson's Disease	Low GGT Cholestasis	High GGT Cholestasis	Not the liver!
AST and ALT	↑↑↑	↑	↑	↑	
GGT	↑	↑	LOW < 15	↑↑	Normal
ALP	↑	Low cf others		↑	Normal
Other	Consider paracetamol toxicity	ALP is Zn dependent; Zn is displaced by excess Cu	Congenital defects in hepatocyte transporters affecting bile secretion into canaliculi*; bile salt synthesis disorders	Multiple; biliary atresia, choledochal cyst, Alagille syndrome, metabolic causes	AST and ALT found in skeletal muscle too; screen for myopathies!

*Low GGT cholestasis suggests the problem is in bile formation or transport – not bile duct obstruction or inflammation. Genetic causes of low GGT cholestasis include **Progressive Familial Intrahepatic Cholestasis 1 and 2 (PFIC)** and **ARC (Arthrogryposis, Renal dysfunction, Cholestasis) Syndrome**, plus bile salt synthesis disorders (MYO5, TJP2, SLOS, TALDO)

5. Immunology

Food Allergy

IgE-Mediated	Non-IgE-Mediated
Stereotypical signs and symptoms: urticaria, angioedema, airway compromise.	Mainly GI symptoms e.g. diarrhoea, vomiting
Rapid, within 60 minutes of ingestion	Delayed, usually hours to days
Skin prick testing for IgE-mediated sensitisation	No good tests; provisional clinical diagnosis then limination and re-challenge where indicated.



IgE-Mediated Food Allergy

Prevalence: 1 in 10 children. Overall, on the rise, but may have plateaued.

Mechanism:

- Sensitisation: macrophage presents allergen to Th2 cell, which stimulates B cell to produce IgE.
- Re-exposure: IgE bound to mast cells recognises allergen → cross-linking of IgE → mast cell degranulation and release of histamine and mast cell mediators.

Clinical Presentation

IgE-mediated food allergy is characterised by its short time course (onset < 2 hours, resolution generally < 24 hours). A prior exposure history is not always present.

	Mild-Moderate	Anaphylaxis
Clinical Features	Swelling of lips, face, eyes Hives or welts Tingling mouth Abdominal pain, vomiting (these are signs of anaphylaxis for insect allergy)	Difficult or noisy breathing Swelling of tongue Swelling or tightness in throat Wheeze or persistent cough Difficulty talking or hoarse voice Persistent dizziness or collapse Pale and floppy (young children)
Management	Oral non-sedating antihistamine e.g. cetirizine, loratadine, fexofenadine	See section on Anaphylaxis too. Intramuscular adrenaline: 0.01 mg/kg 1:1000. Can use an EpiPen autoinjector: • EpiPen Jr (0.15 mg) for kids 1 – 5 years (7.5 – 20 kg) • EpiPen for kids > 5 years (> 20 kg) Neffy is a new intranasal adrenaline device.

Severity of Food Reactions

Unpredictable, and can vary with identical exposures in the same person! Risk factors for severe allergic reaction: older child, higher dose of allergen, poorly controlled asthma, nature of allergen (nuts, shellfish, fish), cofactors.

1. Cofactors can reduce the reaction threshold or increase severity, and include poorly controlled asthma, exercise, NSAIDs, alcohol, sleep deprivation, oestrogen, ACE-I, beta-blockers.

Investigations

- Skin prick testing (reported as wheal size)
- Serum specific IgE (uses ELISA): detects antibodies that are specific to certain allergens
 - Variable utility, validated for egg/milk/peanut, poor NPV in low-intermediate risk subjects.
- Component-resolved diagnostics (**high specificity but low sensitivity**)
 - Detects a specific **single component of interest** e.g. Omega-5-gliadin for wheat-dependent exercise-induced anaphylaxis; Ara h 2 = peanut; Ana o 3 = cashew; tropomyosin = shrimp
 - Ovomucoid (Gal d 1) for extensively heated egg products (**helps distinguish allergy to raw vs cooked egg**)
- Provocation tests: the gold-standard is double-blind, placebo-controlled food challenge but this is mainly confined to the research sphere.
 - Open oral food challenge: establishes tolerance, de-labels food allergies, can be used for positive skin test with sketchy history (e.g. no clinical exposure or inconclusive exposure history).

None of these tools really help determine the severity of clinical food allergy.

Importantly, skin prick testing alone does not diagnose allergy. 30 – 50% of children have sensitivity on SPT without clinical allergy.

- It has a good **NPV if > 2 years** and **fair PPV when combined with positive history**.
- Wheal size is an indication of the likelihood of clinical reaction, not severity.
- SPT is good for monitoring tolerance to an allergen.

Allergy testing is not recommended for kids with no history of an allergic reaction.

You also should not test allergen mixes, for IgG to foods (it's IgE!), and order random pseudoscience tests.

Natural History of Allergies

Note: allergic reactions typically do not get worse over time or with accidental exposure.

High propensity for outgrowing	Low propensity for outgrowing
Milk, egg, soy, wheat	Peanut, tree nut, fish, shellfish

Cross-Reactive Foods

Note that SPT and sIgE cross-reactivity is **much higher** than clinical cross-reactivity.

- Tree nuts: cashew and pistachio, walnut and pecan
- Cow's milk has high cross-reactivity with goat and sheep milk
- Within fish, crustaceans, and molluscs groups (*though uncommon between groups*)
- **Low cross-reactivity**: wheat and other grains; peanuts with other legumes, soy and tree nuts.

Prevention and Management of Food Allergy

The 2015 LEAP Study changed our understanding of preventing peanut allergy in children considered high-risk of developing food allergy. It found that early oral introduction and sustained consumption of peanuts can prevent allergy in high-risk sensitised infants and non-sensitised infants.

- See previous section for management of **acute allergic reaction**.
- **Food ladders (egg and milk)**: pragmatic tool for gradually reintroducing allergens into the diet for young children with a history of IgE-mediated symptoms.
- **Food allergen treatment options**:
 - Daily oral immunotherapy (e.g. daily ingestion of peanut protein), epicutaneous patch; **window of opportunity under age 4 years!**
 - Biologics: omalizumab (anti-IgE antibody) subcutaneous injection every 2-4 weeks; can treat multiple allergies at once.

The mainstay of treatment of allergy is aimed at allergen avoidance, education, and provision of emergency medication for accidental exposure.

Mixed IgE and Non-IgE-Mediated Food Allergy

Eosinophilic Oesophagitis (EoE)

Children with EoE are likely to present with difficulty feeding, failure to thrive, or vomiting. M > F.

Diagnosis requires **upper GIT endoscopy** and **oesophageal biopsy**. Your suspicion of EoE should be raised with (i) endoscopic findings (rings, furrows, exudates, oedema, stricture, narrowing, crepe paper mucosa), (ii) atopic conditions.

Diagnostic criteria:

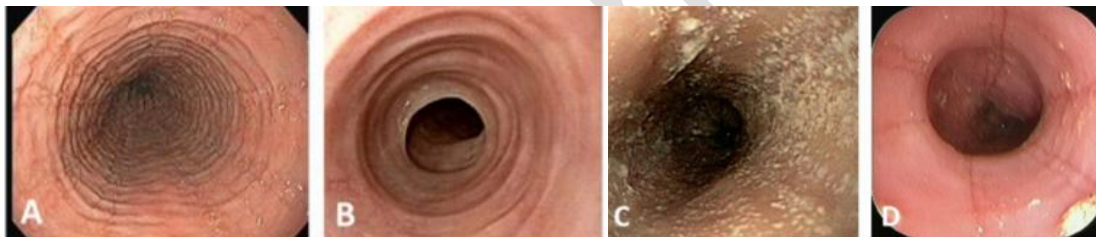
1. Symptoms of oesophageal dysfunction: food refusal, slow eating/ sipping liquids, vomiting, dysphagia/ choking/ gagging on food, abdominal pain, food impaction/ food bolus obstruction, chest pain when eating (adolescent and adult).
2. ≥ 15 eos/hpf on oesophageal biopsy (*eosinophilic infiltration should be isolated to the oesophagus*).

Treatment

- Allergy referral
- PPI (though effective in a minority of patients)
- Dietary therapy (6 – 8 weeks):
 - Elemental diet for young children (tastes gross!), amino acids consumed directly to avoid antigenicity
 - Directed elimination diets are not effective!
 - Empiric elimination: "2-food" (milk and wheat), or "4-food" (milk, wheat, egg, soy).
- Corticosteroids: budesonide oral disintegrating tablet, or swallowed inhaled steroids (budesonide slurry)
- Dupilumab (anti-IL4/IL-13)

There is essentially no role for oesophageal dilatation in children, and it also carries the risk of perforation.

From L to R: oesophageal rings, rings with focal stricture, white exudates, longitudinal furrows.



Non-IgE-Mediated Food Allergy

Cow's Milk Protein Intolerance: umbrella term encompassing the non-IgE-mediated allergic conditions.

	Food Protein-Induced Allergic Proctocolitis (FPIAP)	Food Protein-Induced Enteropathy (FPE)	Food Protein-Induced Enterocolitis Syndrome (FPIES)
Age of presentation	First few months of life	Early infancy	6-12 months old, following introduction of solid foods
Vomiting	Usually not prominent	May be present	Profuse +++
Stools	Blood, mucous usually present	Mucous +/- blood may be present	May have loose stools
Pallor	No	Possible	Common
Weight gain	Not affected	Can be affected	Rarely affected
Timing of reaction after ingestion	> few hours-days	> few hours-days	Average 2-4 hours
Behaviour	Normal	Unsettled	Lethargic
Improvement of symptoms	Over a few days-weeks after eliminating trigger food	Over a few days-weeks after eliminating trigger food	Once vomiting ceases and fluids tolerated, improvement in a few hours
Associated features		Secondary lactose malabsorption with bloating and perianal excoriation	Neutrophilia, thrombocytosis +/- acidosis
Common triggers	Cow's milk, soy	Cow's milk, soy	Rice, cow's milk, soy, oats
Treatment	Elimination	Elimination	Treat vomiting Ondansetron Fluid resuscitation

Outcomes			Usually outgrown by 2-4 years; rice and grain FPIES improve earliest.
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Investigations

- Not routine, diagnosis is clinical. Remember, **no allergy testing for non-IgE-mediated allergies**.
- If **petechiae**, order an urgent FBC for thrombocytopaenia (+ FBC for anaemia if persistent blood in stools).
- The biggest differential is **infectious gastroenteritis** (or, if just vomiting, infection more broadly). You may consider stool testing for MCS and viral PCR to exclude infection.

Management: for proctocolitis and enteropathy, food elimination. Note that improvement may not be seen for 2 weeks. This includes maternal elimination if the child is breast-fed. If a breastfeeding mother needs to eliminate cow milk and soy, suggest calcium supplements.

If formula is required, go for a 2-week trial of **extensively hydrolysed formula** or **rice formula**.

We may consider reintroducing trigger foods at around 12 months. A graded approach is advised.

Immunodeficiency

Suspect immunodeficiency in children with: unusual or opportunistic infections; disseminated infections; recurrent serious infections (sinopulmonary, fungal), infections in association with failure to thrive, chronic diarrhoea, unusual skin rashes, or persistent oral Candida.

Humoral Deficiencies (Antibody Deficiencies – IgG/M/A/E)

Characterised by **recurrent bacterial infections**, particularly **encapsulated organisms**: Streptococcus pneumoniae (invasive pneumococcal disease), Neisseria meningitidis (meningococcal disease) or Haemophilus influenzae.

- Often ears and sinopulmonary infections
- Subtypes: X-linked agammaglobulinaemia (XLA), IgA deficiency, specific antibody deficiency, common variable immunodeficiency (CVID).
 - Note: CVID is characterised by low IgG, IgM, and IgA.

Severe Combined Immunodeficiency (SCID) (T, B, and NK cell)

- Can be X-linked (defects in gamma chain for receptors on T and NK cells), or autosomal recessive (JAK 3, ADA, RAG1/2).
- Lymphopaenia: mostly T cells, with variable B and NK cell deficiency depending on subtype.
- **CXR shows absent thymus**.
- Clinical presentation:
 - Unusual and opportunistic infections (PCP, fungal, overwhelming viral – EBV, CMV)
 - Failure to thrive, diarrhoea
 - Rash: **Omenn-like** (diffuse erythroderma; scaly, peeling skin) in those with specific RAG1/2 mutation, due to autoreactive T cells.
- SCID can be diagnosed on **Newborn Bloodspot Screening** via absent T-cell receptor excision circles (TRECS).
 - TRECS are usually generated during T-cell maturation in the thymus.
 - Infants with SCID have markedly low or absent TRECS in peripheral blood due to diminished or absent thymic production of T cells.



Phagocytic Cell Disorders

- Defects in **neutrophils**.
- Chronic Granulomatous Disease (CGD): X-linked, autosomal recessive.
 - Neutrophils fail to undergo oxidative burst.
 - Characterised by **recurrent bacterial and fungal infections** particularly involving skin, soft tissues, mucosal surfaces, lungs, and liver.
 - Chronic mucocutaneous ulcers and gingivitis, formation of granulomas, and delayed wound healing.
- **Leukocyte adhesion disorders:** high neutrophil count with inability to form pus.
 - Interestingly, an early sign is **delayed separation of the umbilical cord** (recall, this usually relies on phagocytic activity) +/- omphalitis (infection of the cord stump).

Complement Deficiencies

- Terminal pathway deficiencies (C5-C9) lead to susceptibility to Neisserial infections. Recall, the membrane attack complex (MAC) is **critical** for lysing Neisseria bacteria.
- C2 deficiency**: recurrent bacterial infections resembling antibody deficiency, autoimmune disease.
 - C2 ordinarily helps clear immune complexes and apoptotic debris.

Summary of Type of Infection and Immune Deficiency

Type of Infection	Suspected Immunodeficiency
Recurrent bacterial infection	Humoral deficiencies (IgG, M, A, E) or complement deficiencies
Recurrent Neisseria infection	Complement deficiency
Opportunistic or fungal infections	T cell/SCID
Fungal, staphylococcal, pseudomonal, nocardia	Neutrophil (CGD)

Simple Investigations

- FBC + differential (note lymphocyte count)
- IgG, IgA, IgM
- Functional antibodies (to assess vaccine response): diphtheria, tetanus, pneumococcal, salmonella typhi
- T, B, NK cell subsets (flow cytometry)
- Next Generation Sequencing for specific genetic variants leading to immunodeficiency

6. Infectious Disease

Infectious Disease in Childhood: Global and Regional Perspective

Child deaths are declining in Australia, largely attributed to a **decrease in infant deaths**. This is owing to neonatal healthcare, sanitation, vaccination, and community awareness.

Neonatal Sepsis

Recall that **neonates are relatively immunosuppressed**. They have a 10 – 30% risk of bacterial sepsis.

Early signs of neonatal sepsis are non-specific and include lethargy, poor feeding, apnoea, and temperature instability.

Early-Onset Neonatal Sepsis (< 7 days)	Late-Onset Neonatal Sepsis (> 7 days)
Rapidly progresses to fulminant multi-system disease with pneumonia.	Slowly progressive focal infection, most often meningitis.
Group B streptococcus and E. coli	Coagulase-negative staphylococcus (linked to prolonged IV access/ventilation).

Group B Streptococcus Infection (e.g. Streptococcus agalactiae)

- Septicaemia +/- meningitis (30%), pneumonia and focal infection e.g. osteomyelitis, septic arthritis, and cellulitis.
 - Almost 50% of survivors of GBS sepsis have long-term neurological sequelae.
- Beta-haemolysis on blood agar (complete lysis of RBCs, zone of clearance).
- GBS is a coloniser of the GIT, vagina, and urethra, present in ~20% of mothers. Up to 70% of babies are colonised if their mother has GBS.
- Incidence of invasive GBS disease**: 0.43 in 1000 live births
- Risk factors for infection**: pre-term birth, prolonged rupture of membranes, intrapartum fever.
- Intrapartum antibiotics** can reduce the risk of EOGBS but not LOS. Screening > 35 weeks may be protective.

Neonatal Herpes Simplex Virus (HSV)

- Subtypes**: localised skin, eyes, mouth; pneumonitis; involving CNS; disseminated
 - Classically, we'd consider HSV if a vesicular rash + abnormal CSF, especially in the setting of seizures.
- Incidence**: 6 in 100 000. Usually presents within 2 weeks of birth. May be HSV 1 or HSV 2.
- Neonates have the greatest risk of acquiring HSV if the mother had primary genital herpes late in pregnancy. However, most (3/4) mothers are unaware and shed the HSV asymptomatically.
- It is common for infants to have recurrent skin lesions.
- Treatment**: IV aciclovir for 3 weeks

Also recall your **congenital infections**: CMV, toxoplasmosis, rubella, varicella, and syphilis.

Infants/Children

Respiratory tract infections are common in children. Most are viral. See the [Respiratory](#) section (Pertussis, Bronchiolitis, Viral-Induced Wheeze), and [ENT section \(Otitis Media\)](#).

Children under 5 have 5-7 URTIs per year on average.

Dental Disease

- Highest cause of morbidity among Australian children.
- Interplay between fermentable carbohydrates, saliva, and oral plaque-forming bacteria (*Streptococcus mutans*).
- Periodontal infection is polymicrobial – Porphyromonas, Treponema, Actinobacillus (Periodontal Tooth Abscess!)
- **Sequelae:** feeding problems, nutrition deficiency.
- **Treatment:** may need antibiotics, tooth removal, and abscess drainage.

Measles

- Enveloped, RNA paramyxovirus virus.
- Highly contagious, spread by respiratory aerosols/direct contact with respiratory secretions.
- **Incubation period:** 10 – 12 days (to prodromal symptoms); 14 days to rash.
 - Patients become contagious 24 hours prior to the onset of flu-like prodromal symptoms.
- Prodromal symptoms: fatigue, cough, coryza, conjunctivitis.
- **Diagnosis:** on the day of measles rash onset, the optimal diagnostic tool is **PCR or IF on urine or nasopharyngeal samples**. Measles IgM develops after 3-7 days.
- Followed by fever and maculopapular rash (spreads from **head to toe**, may have confluent areas).
- Vaccine-preventable (live vaccine), just one dose is usually enough!



Slapped Cheek Disease (Parvovirus B19, aka Fifth Disease)

- Droplet spread
- Mild preceding febrile illness. Rash occurs in two stages:
 - Cheeks appear red and flushed.
 - 1-2 weeks later, lacy maculopapular rash (mainly arms and legs); fluctuate, exacerbated by hot baths.
- Major risks: pregnancy (bilateral, symmetrical arthralgia; foetal anaemia and hydrops); bone marrow suppression and anaemia.



Roseola Infantum: Human Herpesvirus 6 (HHV-6)

- High fever for 3-4 days, then rash onsets as temperature falls.
 - Widespread maculopapular rash mainly on trunk, **discrete (cf measles which is confluent, and has rash with fever)**.
- May be associated febrile convulsions
- Encephalitis is a risk in immunocompromised patients; otherwise self-limiting.



Enterovirus

- Include polioviruses, Coxsackie viruses, echoviruses and other.
- **Transmission:** respiratory and faecal-oral
- **Presentation:** respiratory; hand, foot, and mouth disease; stomatitis; gastroenteritis; pericarditis/ myocarditis; neurological (e.g. meningitis, transverse myelitis).



Impetigo

- **Extremely** contagious.
- Can be caused by Group A Streptococcus (Streptococcus pyogenes, GAS), or Staphylococcus Aureus
- **Presentation:** red macule that becomes vesicular; vesicles burst to leave honey-coloured crust. Classically around the mouth, but can occur elsewhere.
- Trigger of **rheumatic fever/ rheumatic heart disease** in Indigenous populations.
- Treated with antibiotics; no longer contagious 24 hours after starting antibiotics.



Meningococcal Disease

- Two major clinical presentations: meningitis or septicaemia (some will have both)
 - Rare manifestations: septic arthritis, pericarditis.

Stage	Duration	Key Features
Incubation	1 – 7 days (rarely up to 10)	Initial asymptomatic phase
Early/prodromal phase	Hours to 1-2 days	Non-specific flu-like symptoms, fever, headache, malaise, myalgia, sore throat, nausea, vomiting, diarrhoea Maculopapular rash that may be blanching.
Septicaemic/ fulminant phase	Rapid progression, hours to days	High fevers, chills, cold/mottled extremities, severe fatigue, limb pain, <u>petechial or purpuric rash (non-blanching)*</u> , hypotension, shock, altered consciousness, multi-organ failure.



*Bacteria can be isolated from these lesions.

- Management: prompt initiation of antibiotics, notification of public health unit (even on suspicion)
- Primary prevention of meningococcal disease is achieved through vaccination (Meningococcal ACWY); secondary prevention via antibiotic prophylaxis and vaccination for close contacts.

Adolescents: don't forget STIs (chlamydia, gonorrhoea, syphilis, HIV). On the rise, especially in Indigenous youth.

Extra: Group A Streptococcus (e.g. Streptococcus pyogenes)

Seems to be a pretty nasty bug with a bunch of associated diseases and post-infectious sequelae. Below is a summary of all GAS-associated conditions.

- GAS is so bad because it has a variety of virulence factors (exotoxins e.g. streptolysins; surface proteins e.g. M protein; enzymes e.g. streptokinase) and superantigens, broad tissue tropism (can infect throat, skin, soft tissues) and can undergo molecular mimicry (autoimmune disease → ARF, PSGN).

Note that **antibiotic treatment** of **Group A Streptococcus**, particularly **pharyngitis** prevents acute rheumatic fever (ARF) and its long-term consequence rheumatic heart disease, as well as suppurative complications e.g. peritonsillar abscess and invasive GAS infection. Antibiotics do not reliably prevent PSGN.

Disease	Type	Pathophysiology	Clinical Features	Timing	Diagnosis	Treatment	Notes
Pharyngitis	Direct infection	GAS infects pharyngeal mucosa	Sore throat, fever, tonsillar exudate, tender cervical lymphadenopathy, absence of cough	Acute infection	Throat swab culture or rapid antigen detection test	10-day course of penicillin or amoxicillin; symptomatic relief	Prevents ARF and reduces transmission
Scarlet fever	Toxin-mediated	Erythrogenic exotoxin producing diffuse rash	Sandpaper rash, strawberry tongue, circumoral pallor, fever	During or shortly after pharyngitis	Clinical rash and positive GAS test	Penicillin or amoxicillin	Rash desquamates as illness resolves
Impetigo (non-bullous)	Direct infection	Localized skin infection	Honey-crusted lesions on face, limbs	Active skin infection	Clinical diagnosis; culture if needed	Topical antibiotics, oral if severe or widespread	Highly contagious
Cellulitis/ erysipelas	Direct infection	Invasive skin and soft tissue infection	Warm, tender erythema, swelling	Acute infection	Clinical diagnosis; culture if abscess	Oral/IV beta-lactams; surgical drainage if abscess	Erysipelas involves upper dermis; sharply demarcated

Necrotizing fasciitis	Invasive infection	Rapid soft tissue necrosis by bacterial toxins	Severe pain, rapidly spreading erythema, systemic toxicity	Acute	Imaging and surgical exploration definitive	Emergency surgery; broad-spectrum IV antibiotics	Surgical emergency; high mortality risk
Streptococcal toxic shock syndrome (STSS)	Toxin-mediated	Superantigen-induced systemic inflammation	Fever, hypotension, multi-organ failure	Acute	Positive blood culture; clinical shock	ICU support; IV clindamycin + penicillin; consider IVIG	High mortality without prompt treatment
Acute rheumatic fever (ARF)	Immune-mediated	Autoimmune cross-reactivity after GAS pharyngitis	Migratory polyarthritis, carditis, chorea, erythema marginatum, subcutaneous nodules	2–4 weeks post-pharyngitis	Elevated ASO or anti-DNase B titres; clinical Jones criteria	IM benzathine penicillin G; anti-inflammatory therapy; secondary prophylaxis	Major cause of acquired heart disease in children
Post-streptococcal glomerulonephritis (PSGN)	Immune complex-mediated	Immune complex deposition in glomeruli	Haematuria, oedema, hypertension, proteinuria	1–3 weeks post-pharyngitis or skin infection	Low complement C3, recent GAS infection history	Supportive care; treat hypertension; antibiotics if active infection	Usually self-limiting; watch for complications
Rheumatic Heart Disease (RHD)	Chronic auto-immune sequela	Autoimmune cross-reactivity (molecular mimicry) causes inflammation and scarring of heart valves causing valvular stenosis or insufficiency	Mitral valve most commonly affected; symptoms of heart failure, murmur, arrhythmias; progressive valve damage	Due to repeated or severe ARF	Echo-cardiogram detecting valve disease	Long-term benzathine penicillin prophylaxis	Leading cause of acquired CVD worldwide; especially in Indigenous and low-resource settings
PANDAS (Paediatric autoimmune neuropsychiatric disorders associated with streptococcal infections)	Immune-mediated	Autoimmune CNS effects after GAS infection	Sudden onset OCD, motor/vocal tics	Weeks after infection	Clinical diagnosis; elevated streptococcal antibodies	Antibiotics to eradicate GAS; symptomatic neuropsychiatric management	Controversial; multi-disciplinary approach

More on Acute Rheumatic Fever

It is a nonsuppurative sequela of Group A Streptococcus (**S. pyogenes**) pharyngitis (or impetigo) that occurs 2-4 weeks after infection.

Epidemiology:

- Common in developing countries + Indigenous children in Australia, New Zealand, Canada, and the United States
- Risk factors for S. Pyogenes: poor living conditions, e.g. overcrowding, limited access to washing facilities
- Predominantly children aged 5-14 years
- 60% ARF (untreated) progresses to RHD
- RHD prevalence peaks in 3rd/ 4th decade

Pathophysiology: autoimmune, exact causal pathway unknown

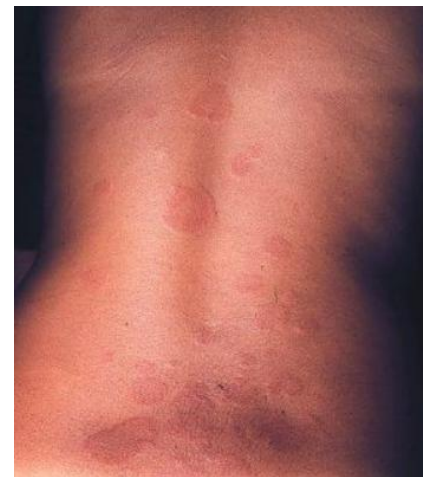
Diagnosis: Evidence of streptococcal infection plus 2 major or 1 major + 2 minor criteria

- **Evidence of streptococcal infection:** elevated or rising anti-Streptolysin O titre (ASOT), anti-DNase B
- ASOT peaks after 3-6 weeks while DNase B peaks a bit later (6-8 weeks); ASOT can remain high for 6-12 months while DNase B can persist for > 1 year. **A before B!**
- Other forms of evidence: positive throat culture (not sensitive), positive GAS RAT

Duckett-Jones Diagnostic Criteria

Major (JONES): **J**oints, **O**[carditis; think of the O as a heart shape], **N**odules (subcutaneous), **E**rythema marginatum, **S**ydenham's chorea

- Arthritis is usually the earliest symptom: inflammation affecting several joints in quick succession with overlapping onset
- Erythema marginatum is an evanescent, pink or faintly red nonpruritic rash involving the trunk and sometimes the limbs but not the face. The margins are erythematous while the skin in the centre returns to a normal appearance.
- Evanescent means it appears rapidly and fades without leaving a mark



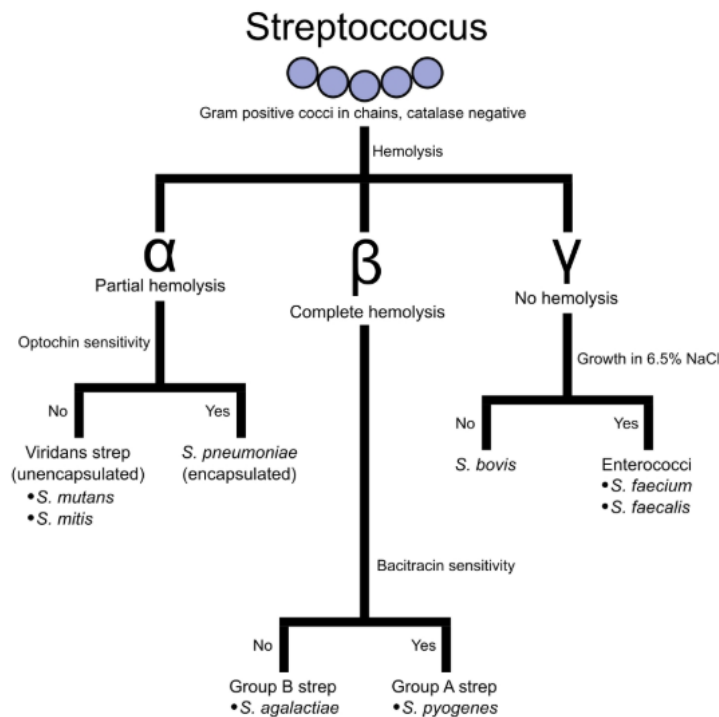
Recall that the most common sequela of ARF is rheumatic heart disease, 10-20 years after the original illness.

Treatment of ARF: IM benzathine benzylpenicillin (*also for secondary prophylaxis*). For arthritis or severe arthralgia, give naproxen, ibuprofen or aspirin PO.

Prevention of recurrent ARF:

- Benzathine benzylpenicillin IM for at least 10 years; may decrease GAS colonisation

Random Streptococcus Diagram (because there are so many species!)



Acute Sepsis in Children

Please refer to [Recognising and Managing the Sick Child](#).

Immunisation

Vaccination introduces the body to a non-infectious antigen to elicit a primary response. This means that if someone encounters the real pathogen, a secondary immune response is initiated (which is faster and more effective).

Vaccines can either be **live attenuated** or **inactivated** (whole, subunit, toxoid, polysaccharide, conjugate, recombinant, viral vector, mRNA).

	Live attenuated vaccine	Inactivated vaccine
Description	Weakened version of the live virus while retaining ability to replicate	Bacterium/virus inactivated with heat or chemical; mostly humoral response
Advantage	Mimics natural infection, long-lasting protection	Cannot cause disease; safe for immunocompromised
Disadvantages	Some pathogenicity; requires good cold chain; not for immunocompromised; affected by circulating antibodies	No amplification/ replication; antibody titres decrease with time; multiple doses needed
Examples	MMR, Varicella/Zoster, Yellow Fever, BCG, oral polio (outside Australia)	See below

Note: live vaccines must be administered **at least 4 weeks apart**. If a child is sick with a mild illness (e.g. common cold), vaccinate once afebrile and well. If moderate/severe febrile illness, you may need to wait 3-4 weeks.

Subtypes of inactivated vaccines:

Type	Description	Examples
Whole or fractional (subunit) bacterium/ virus	Note, the old whole-cell pertussis vaccine has been replaced by the acellular pertussis vaccine.	Pertussis, inactivated polio, influenza, Hepatitis B
Toxoid	Inactivated toxin that causes the disease	Tetanus, diphtheria
Recombinant	Engineered; made by inserting DNA encoding an antigen into mammalian cells to produce an antigen and then purifying them	Hepatitis B, HPV
Polysaccharide	Sugar capsule of bacteria, T cell independent, induces mostly IgM. Poor immunogenicity in children under 2 years old.	Pneumococcal (Pneumovax) 23 valent for older children/adults at risk Older polysaccharide meningococcal vaccine no longer available in Australia.
Conjugate/ Conjugated	Antigen is linked to a highly immunogenic protein, eliciting a T-cell-dependent response. Unlike polysaccharide vaccines, it is immunogenic in under 2 y/o.	Used for encapsulated bacteria (sort of an alternative to polysaccharide vax): Pneumococcal (Prevenar) 13 valent; Meningococcal ACWY (Nimenrix); Meningococcal B (Bexsero); Haemophilus influenza B
Messenger RNA (mRNA)	Short sequence of synthetic mRNA encodes antigen from pathogen; when injected, translated into protein which is expressed on the cell surface.	SARS CoV-2 (Pfizer and Moderna)
Viral vector	Unrelated harmless virus is modified so it delivers the genetic material of the target pathogen; human cells transcribe and translate the gene sequence into a viral protein. May be replicating or non-replicating.	Ebola, SARS CoV-2 (Astra Zeneca Vaxzevria)

Impact of Maternal Antibodies

- Maternal IgG antibodies persist for up to 6-12 months in infant circulation.
- They can interfere with the infant's immune response to vaccination.
- One notable example is **measles antibodies**; the MMR vaccine is **not recommended before 9 months** because the maternal antibodies mop up the live attenuated virus.

Enhancing Immunogenicity of Vaccines

- Repeat doses with appropriate spacing (*too close can → insufficient time for memory development*)
- Adjuvants e.g. aluminium: depot for antigen, produces particulate antigens to present to APC, activates complement and stimulates macrophages.

Adverse Events Following Immunisation (AEFI)

Any untoward medical occurrence which follows immunisation, and which does not necessarily have a causal relationship with the usage of the vaccine.

WHO classifies AEFI into 5 categories: (i) vaccine product related, (ii) vaccine quality defect related, (iii) immunisation error related, (iv) immunisation anxiety related, (v) coincidental.

Common AEFI:

- Mild fever that does not last long, < 38.5 °C:** fluids, do not overdress, paracetamol
- Injection site reaction:** tender initially, pain settles when redness appears. No systemic toxicity, movement not affected, nontender lymphadenopathy (**unlike cellulitis**).
 - May last 1-2 days; small hard lump may occur, lasting a few weeks
- Rash:** early onset is most concerning, ensure you screen for anaphylactic symptoms.
 - Delayed rash may be unrelated e.g. intercurrent virus.
- Fainting:** in adolescents and adults; typically occurs within 5 minutes (up to 30 minutes).
 - Infants and children rarely faint**, so assume any young child who suddenly loses consciousness is having an **anaphylactic reaction!**
 - This is a **vasovagal episode**. Patients should have a strong central pulse (e.g. carotid).



- Ensure the patient is safe – lay on their back, elevate legs, loosen tight clothing.

Following vaccination, infants may be grumpy, unsettled, unhappy, and sleepy. Teenagers may specifically report myalgia.

Simple (home) management: avoid too many clothes; breast feed more frequently and/or give more fluids; cold wet cloth on sore injection site; paracetamol for fever or pain.

AEFI specific to vaccines:

Vaccine	AEFI
BCG	Ulcer at injection site (2-6 weeks)
MMR	7 – 10 days after vaccine: fever (can be > 39.4 °C) lasting 2-3 days; faint red rash; coryza; drowsiness or tiredness; swelling of salivary glands; thrombocytopaenia.
Varicella	Mild pustular rash at injection site, 5 – 26 days after vaccination
DTPa	Extensive limb swelling
Hepatitis B	Lump at injection site
Rotavirus	Can cause intussusception ; highest risk 6-12 months, especially following first vaccine. This is why the rotavirus vaccine is given earlier (2, 4, and 6 months)

Serious AEFI: anything life-threatening or resulting in death, or a congenital anomaly. Examples include anaphylaxis, intussusception, febrile seizures, GBS, disseminated disease in immunocompromised patients receiving live attenuated vaccine, thrombosis/ thrombocytopaenia, myocarditis.

Contraindications to Vaccination

- Anaphylaxis to a previous dose of a vaccine or vaccine component
- For live vaccines: significantly immunocompromised; pregnant (*women should avoid becoming pregnant within 28 days of receiving a live vaccine*)

Food allergies (e.g. egg allergy) usually okay!

Reporting of AEFI: In NSW, it is a legislative requirement to report any AEFI. Report to the Public Health Unit (PHU) – 1300 066 066 or via national (TGA) form.

Children with AEFIs, complex medical conditions or immunisation needs, should be referred to a **Specialist Immunisation Clinic**.

National Immunisation Schedule and Immunisation for Special Populations

For those who might need extra vaccination, recall the HALO Principle: Health, Age, Lifestyle, Occupation

Recommended sites: <12 months old → lateral thigh; 12+ months old → deltoid muscle

Don't: prime the administration needle, mix vaccines in the same vial, mix with local anaesthetic

Groups with special vaccine requirements:

- Indigenous Australian:
 - Pneumococcal 13vPCV at 6 months and 23vPPV at 4-5 years
 - Hepatitis A vaccine at 12 and 18 months
 - Influenza annually for all aged 6+ months
 - Meningococcal B at 2, 4, and 12 months
 - BCG if living in high-risk areas in NT, QLD, SA
 - Japanese encephalitis for outer Torres Strait Islanders
- Pregnancy: influenza, pertussis (20 – 32 weeks), RSV (Abrysvo) from 27 weeks, **no live vaccines!**
 - **Note, the only contraindicated vaccine in breastfeeding is the yellow fever vaccine.**
- Pre-term babies: < 28 weeks or CLD → pneumococcal; < 32 weeks or < 2000g → HBV at 12 months (additional dose).
 - Vaccines are given at **chronological age**.
 - The pneumococcal vaccination is given on the basis of **gestational age**, and the Hepatitis B vaccine is given based on **birthweight**
 - **Increased risk of apnoea (usually self-limiting):** monitor hospitalised pre-term infants up to 48 hours post-vaccination e.g. with oxygen sats monitoring

- Impaired immunity: influenza vaccine, extra pneumococcal and HPV, **no live vaccines (exception: asplenia, those on low-dose corticosteroids and safe dose of methotrexate)**
 - Those with **asplenia** specifically require pneumococcal and meningococcal vaccines.
 - We might pre-emptively vaccinate people with anticipated immunodeficiency e.g. chemotherapy
- See the table below for specific immunosuppressive conditions:

Asplenia	Encapsulated bacteria: Pneumococcal (PCV13 and 23vPPV), Meningococcal (MenB and MenACWY), Hib. Note, live vaccines are safe for asplenic patients.
HIV	The degree of immunosuppression is important: CD4+ count, viral load, AIDS-defining illnesses Always contraindicated: BCG Consider if asymptomatic and CD4+ $\geq 200/\text{mL}$: yellow fever, MMR, VZV Recommended vaccines: influenza, pneumococcal, MenACWY/MenB, HAV, HBV (double dose), HPV
Solid organ transplant	Must be up to date with vaccines at least 4 weeks before transplant: <u>check MMR and varicella serology</u> After transplant: inactivated vaccines best given 6-12+ months after transplant Influenza vaccine: 2 doses in first year after transplant, then one dose annually Additional vaccines: 23vPPV, consider 9vHPV, Hepatitis A vaccine (liver transplant, chronic liver disease) +/- travel vaccines
Haematopoietic stem cell transplant	No live vaccines until 24 months after transplant (provided no ongoing immunosuppression) Revaccinate, and order Hepatitis B and MMR serology to determine need for further doses Consider travel vaccines
Blood product recipients	If receiving a blood product that contains antibodies (e.g. Ig), you must <u>wait 8 months</u> before having a live viral vaccine (to ensure no interference)
Coagulopathy	There is a risk of <u>haematoma</u> at IM injection site – defer IM injection if INR > 3.0 or anti-Xa > 0.5 units/mL at 4h post dose Haemophilia: consider SC route instead of IM, firm pressure for 5-10 minutes, ice/immobilise if haematoma develops.

In 2025, the **influenza vaccine** was NIP-funded for 6 months to < 5 years; pregnant; Indigenous; people aged 6+ months with medical conditions that make them vulnerable to influenza.

Help! Which RSV vaccine?

- **Arexyv**: older adults 60+ years only (X, V and Y are at the end of the alphabet)
- **Abrysvo**: sounds like embryo; used in pregnancy.
- **Beyfortus** (nirsevimab): for **babies**; single IM dose per season; if mother did not receive the RSV vaccine during pregnancy.

National Immunisation Program Schedule Summary

Age	Everyone	Aboriginal and Torres Strait Islander	Specified Medical Risk Conditions
Birth	Hepatitis B		
2 months	Diphtheria, tetanus, pertussis, hepatitis B, polio, Haemophilus influenzae type B Rotavirus Pneumococcal	Meningococcal B	
4 months	Diphtheria, tetanus, pertussis, hepatitis B, polio, Haemophilus influenzae type B Rotavirus Pneumococcal	Meningococcal B	
6 months	Diphtheria, tetanus, pertussis, hepatitis B, polio, Haemophilus influenzae type B (Hib)	Pneumococcal Meningococcal B	Pneumococcal
6 months to < 5 years (annually)	Influenza		
12 months	Meningococcal ACWY	Meningococcal B	

	Measles, mumps, rubella Pneumococcal		
18 months	Haemophilus influenzae type B (Hib) Measles, mumps, rubella, varicella Diphtheria, tetanus, pertussis	Hepatitis A	
4 years	Diphtheria, tetanus, pertussis, polio	Hepatitis A	Pneumococcal
5+ years		Influenza	Influenza
12-13 years	Human papillomavirus Diphtheria, tetanus, pertussis		
14-16 years	Meningococcal ACWY		
All ages		Influenza	Influenza Pneumococcal

Influenza vaccine: in children aged 6 months to 9 years on first administration, give 2 doses a minimum of 4 weeks apart, then one dose annually for subsequent vaccines.

“Catch-Up Schedules”

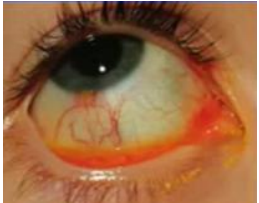
- Three groups of patients who are eligible for National Immunisation Program (NIP) vaccines can catch up on free vaccines missed in childhood:
 - All people under 20 years old
 - People under 26 years old who have missed HPV vaccination
 - Refugees and humanitarian entrants at any age

7. Ophthalmology

Conjunctivitis in Children

Conjunctiva: mucus membrane that lines outer eye, stops at edge of cornea, extends to upper/lower fornix. The bulbar conjunctiva lines the sclera, while the tarsal conjunctiva lines the eyelids

Conjunctivitis is inflammation of the conjunctiva, and may be infective, allergic, chemical or a special subtype (neonatal, associated with systemic disease)

	Bacterial	Viral	Allergic
Most common organism	Haemophilus influenzae, then Streptococcus pneumoniae, Staphylococcus aureus, Moraxella catarrhalis	Adenovirus	-
Seasonality	Autumn, winter, early spring	Summer	Spring, may be recurrent
Systemic symptoms	May be associated with acute otitis media	Viral prodrome e.g. adenopathy, fever, pharyngitis, or URTI; preauricular lymphadenopathy Pharyngoconjunctival fever: sore throat + fever + conjunctivitis	Nasal congestion, sneezing, wheezing
Laterality	Usually unilateral, except chlamydial conjunctivitis which is usually bilateral	Usually bilateral	Usually bilateral
Itching	None	Limited-none; mainly grittiness, burning, irritation	Primary complaint; may also have grittiness, burning, or irritation
Ocular discharge	Mucopurulent; may be yellow, white, or green. Purulent discharge seen in gonococcal conjunctivitis	Watery with strands of mucus	Watery
Conjunctival appearance	Pink or red 	Pink or red; tarsal conjunctiva may have a follicular (bumpy) appearance	Pink; bulbar conjunctiva may be puffy, tarsal conjunctiva may have follicular appearance Phlyctus (raised area) at limbus of eye 
Management	Swab not needed Topical chloramphenicol or tobramycin Eye drops 2-4 hourly during the day, eye ointment at night Infection control measures	Conjunctival swab + PCR for adenovirus (doesn't influence management) Self-limiting condition Lubricant eye drops Infection control measures Steroid drops are sketchy: can mask infection, glaucoma risk	Systemic antihistamines Topical olopatadine HCl (0.1%) (Patanol) for recurrent presentations: mast cell stabiliser and antihistamine

			<u>No steroids (unless chronic/vernal – see below)</u>
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* A good rule of thumb is that **topical steroids are not prescribed for children.**

Exam

- Visual acuity: normal unless keratitis or mucous
- Conjunctival changes e.g. follicles → viral
- Fluorescein for corneal disturbance
- Skin rashes, e.g. Steven-Johnson Syndrome
- Lymphadenopathy (preauricular, submandibular) in viral conjunctivitis

Laboratory Investigations: usually not necessary in primary care. Conjunctival swab/ scraping indicated in (i) neonatal conjunctivitis, (ii) chronic or recurrent infection, (iii) suspected HSV 1 or 2, (iv) atypical reaction.

Neonatal Conjunctivitis

Pathogen by onset:

Day 1	Day 4 – 6	Day 8 – 12	Day 7 – 14
Gonococcal	Bacterial (Strep, Staph, Haemophilus, etc)	Chlamydia	Herpes simplex

The online video gave an example of **chlamydial conjunctivitis** presenting in a 10-day-old with bilateral mucopurulent discharge, red and swollen eyelids. He was investigated with **conjunctival swab/scraping culture, Gram stain, Chlamydia culture and PCR, and HSV PCR.** He was treated with **IV erythromycin 50 mg/kg/day.**

- **Remember to treat the parents!**

HSV Conjunctivitis

- Presentation: red, swollen eyelid with **vesicles**, injected conjunctiva, watery discharge.
- You **must** examine the cornea for a **dendritic ulcer**.
- Management: oral aciclovir/famvir (reduces recurrence), topical aciclovir ointment. Do **not** use steroids (can cause viral particles to enter the corneal stroma → antigen-antibody reaction → scarring and vision loss).
 - If a patient **only** has keratitis, we may use ointment 4x/day for 5-7 days

Vernal Conjunctivitis

- Chronic allergic inflammation of the conjunctiva, typically triggered by seasonal allergens.
- Features: subtarsal conjunctival papillae, limbal papillae (trantas dots), corneal involvement (plaque, shield), ropy/mucoid discharge
 - Corneal involvement can lead to vision loss
- Management: identify and avoid affecting allergen, cold compress, topical eye drops (Patanol, steroid – FML, Maxidex), lubricant
- **Vernal keratoconjunctivitis** is chronic, recurrent, and debilitating. It may lead to visual loss, corneal scarring, and secondary glaucoma (steroid-related).

Congenital Nasolacrimal Duct Obstruction

- Occurs in 6% of infants
- Presents with epiphora (watery eye) followed by infection
- Conservative management until 12 months (by which it usually resolves); subsequently warrants probing irrigation of the nasolacrimal duct under GA.

Stevens-Johnson Syndrome

- Severe immune-mediated hypersensitivity reaction where CD8+ T cells attack skin and mucosal keratinocytes
- Typically drug-induced (1-3 weeks after drug exposure) and preceded by **flu-like symptoms including sore throat.**
- Bullous erythematous skin lesions, bilateral red eyes, mucoid discharge, conjunctival vesicles or bullae.
- **Emergency! Requires hospitalisation.**
- Ocular treatment: lubricants, topical antibiotics for secondary infection, daily symblepharonolysis with glass rod.



Chemical Conjunctivitis

- Alkali: penetrate the cornea, **worse** (household cleaning agents, caustic soda, cement dust)
- Acid burns: superficial damage only, heals with local antibiotics
- Management: tap water at home, primary care (local anaesthetic, irrigate with normal saline for 30 minutes, remove particles of lime, instil antibiotic eye drops/ointment, cycloplegia with Homatropine 2%)
 - Cycloplegia is useful because some chemicals cause ciliary spasm → pain
 - Requires **urgent referral to the ED! Time is eye!**

Strabismus

Prevalence: 2% of children (same as amblyopia)

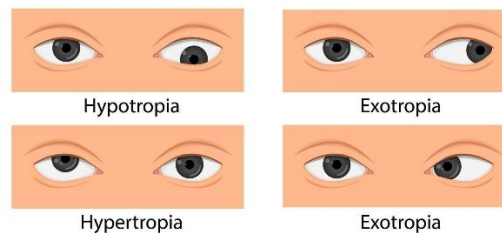
Misalignment of the two eyes, where only one eye is directed towards the object of regard.

Risk factors: FHx, prematurity/LBW, drug or alcohol use during pregnancy, refractive error, intraocular pathology (e.g. cataract, retinoblastoma), neurological disorder, craniofacial disorder.

Beware of **pseudostrabismus**! Some children, particularly those with broad flat noses and prominent epicanthal folds may appear to have strabismus, particularly esotropia (eye inwards), but they do not. The trick is **to assess the symmetry of the corneal reflections**.



Strabismus types



True strabismus:

True strabismus may be intermittent, constant, non-paralytic, or paralytic.

Type of Strabismus	Onset	Description	Outcomes
Infantile esotropia 	First few months of life	Constant strabismus; cross-fixation	Good outcome if referred for surgical correction within first 12 months.
Accommodative esotropia 	2 – 3 years old	Intermittent at first, often precipitated by illness, associated with hyperopia (farsightedness; eyes work too hard to focus on near objects)	Corrects with glasses; surgery with poor response; good prognosis if treated early.
Intermittent exotropia 	Mostly < 5 years	Eye drifts outward intermittently, often during fatigue and daydreaming. Child may close eye in sunlight (looking outwards → eyes drift → diplopia → uncomfortable)	Non-surgical first (minus lenses, orthoptics, prisms); often good prognosis if managed
Constant exotropia 	Infancy/childhood	Persistent outward deviation, often associated with underlying disease	Often requires surgical correction; often has poor vision in one eye e.g. optic atrophy; binocular outcomes vary.

Paralytic strabismus¹	Any age	Sudden onset diplopia, abnormal head posture, extraocular muscle paresis/paralysis May be congenital, viral, trauma, raised ICP (e.g. cavernous sinus tumour or aneurysm → CN III palsy)	Patient should <u>always</u> get an MRI due to the possibility for intracranial pathology!
Restrictive strabismus²	Any age	May be genetic (e.g. Duane syndrome, Brown Syndrome) or acquired e.g. trauma/ thyroid disease	Treat underlying cause; surgical release if severe. Prognosis depends on aetiology and amount of damage.
Associated with intra-ocular pathology	Any	Retinoblastoma Congenital cataract Toxoplasmosis	

1: More on cranial nerve palsies

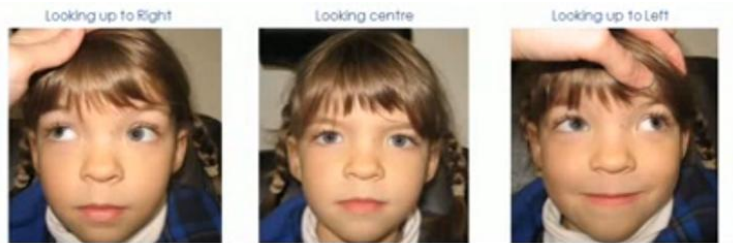
CN VI Abducens Nerve	CN IV Trochlear Nerve	CN III Oculomotor Nerve
Lateral rectus	Superior oblique	Superior, medial, inferior rectus, inferior oblique Levator palpebrae superioris Sphincter pupillae + ciliary muscles
Abduction of eye	Intorsion, depression of adducted eye (looking to your nose)	
Limited abduction of affected eye with ipsilateral esotropia; horizontal diplopia worse when looking to affected side	Diplopia worse when looking down and in (e.g. reading or walking down stairs) Eye drifts up and out → head tilt away from side of lesion Eye may be slightly elevated, limited depression when adducted	Eye "down and out" (unopposed lateral rectus and superior oblique) Ptosis, dilated pupil , loss of accommodation and pupillary light reflex Diplopia, difficulty with eye movements except abduction and intorsion
Looking Right Looking centre Looking Left  This child has a left abducens palsy.	Looking down to the right Looking up to the right 	Looking Right Looking Left 

2: Two causes of restrictive strabismus

- **Duane Syndrome (Duane Retraction Syndrome):** abnormal development and innervation of the LR muscle by CN VI with anomalous innervation of LR by branches of CN III.
 - There is simultaneous contraction of the medial and lateral recti during attempted adduction → **globe retraction and eyelid closing**. This makes it different from a CN VI palsy.
 - Compensatory face turn.



- **Brown's syndrome:** mechanical limitation of elevation during adduction, classically due to abnormality in superior oblique tendon or sheath, especially at trochlea.
 - The eye can't elevate when it is adducted because it is mechanically tethered by the superior oblique



Test for Strabismus

- General observation: obvious strabismus, abnormal head posture
- Visual acuity
- Corneal light reflections test (Hirschberg test):
 - Shine light 50 cm away and observe where corneal reflection appears
 - Normally centred and symmetric in both pupils
- Cover/uncover test: cover the **fixing eye**; the deviated eye will move to take up fixation.
 - Test with child looking near (1/3 m) first, then distance (6 m)
- Ophthalmoscopy: red reflex, optic disc/retina



Strabismus red flags: concern about vision, constant from any age, sudden-onset, restricted eye movements, unequal red reflexes, suspect underlying eye or intracranial pathology, child with risk factors for strabismus.

The management of strabismus overlaps with the management of amblyopia below. Essentially, correct refractive error first (**glasses**). After 6-8 weeks, if vision is still reduced, **occlusion** treatment for amblyopia. If the strabismus is uncosmetic, **surgery** might be performed.

Amblyopia

Reduction in visual acuity that occurs due to interruption of normal visual development during the “critical period”. Amblyopia during the sensitive period is potentially reversible

- Babies should fix well by 4 months.
- Vision develops rapidly by 2-2.5 years then slowly till 7-8 years; the first 2-3 years is most **sensitive to deprivation**.

There are two types of amblyopia:

Unilateral	Bilateral
Form deprivation e.g. ptosis, cataract Strabismus Anisometropia (refractive difference between eyes)	Form deprivation (bilateral ptosis, cataracts) Ametropia (high refractive error in both eyes)

Testing Vision:

- Pre-verbal children: forced preferential looking tests, e.g. Teller acuity card or Cardiff acuity card (looks at stripes/ picture respectively until they can no longer differentiate the two).
 - We can also test vision **qualitatively** e.g. following objects, eye contact.
 - Check a child's response to **occlusion** – they might get upset if their good eye is occluded.
- **Verbal vision tests:** Kay picture test (name the pictures), Sheridan Gardiner (matching letters), Snellen (only if whole alphabet known).

Management:

Prognosis is most favourable if treatment is commenced during the sensitive period of visual development (0 – 8 years). Note that children **don't just grow out of strabismus!**

- Early diagnosis
- Assess visual acuity in all preschool children (StEPS IN nsw)
- Correct refractive error
 - Glasses first-line
- Correct impediment to visual development
- **Occlusion/patching:** stimulates visual development in amblyopic eye
 - Occlude the good eye!
 - Mild-moderate: 2 h/day patching
 - Severe: 4 h/day patching
- Alternative: atropine eye drops (*last resort*) to blur vision in good eye.

Leukocoria

Leukocoria may be unilateral (in which case it is amblyogenic) or bilateral.

It is typically picked up via an abnormal red reflex (e.g. neonatal examination), using an ophthalmoscope of 10D held at 1/3m from the eye.

Differential Diagnoses:

- Leukocoria is **retinoblastoma** until proven otherwise!
 - Distinct golden appearance; accompanied by strabismus.
- Cataract
- Persistent foveal vasculature (persistent hyperplastic primary vitreous)
- Coats' disease (Exudative retinal condition)
- Endophthalmitis (infective).

Retinoblastoma (**Remember, beware of underlying intraocular pathology as the cause of strabismus!**)

Prevalence: 1 in 20 000 births

Rule of 75's: 75% unilateral, 75% present under age 3, 75% due to somatic mutation (non-hereditary)

Retinoblastoma can spread to the brain via the optic nerve, or haematogenously to the rest of the body.

Management: chemotherapy then radiation or laser; enucleation of eye if large.



Congenital Cataracts

Bilateral leukocoria with decreased vision (amblyopia) and nystagmus.

Aetiology: **50% idiopathic**, 25% familial, 25% other (intrauterine infection in first trimester; metabolic e.g. galactosaemia or aminoaciduria; syndromic e.g. T21).

Investigations: paediatric consultation, genetic studies, specific investigations for aetiology (e.g. TORCH titres, urine metabolic screen)

Surgery: if **vision significantly affected**. Unilateral cataracts require surgery in first 1-4 weeks of life (lest the child becomes extremely amblyopic); bilateral in first 3 months of life.



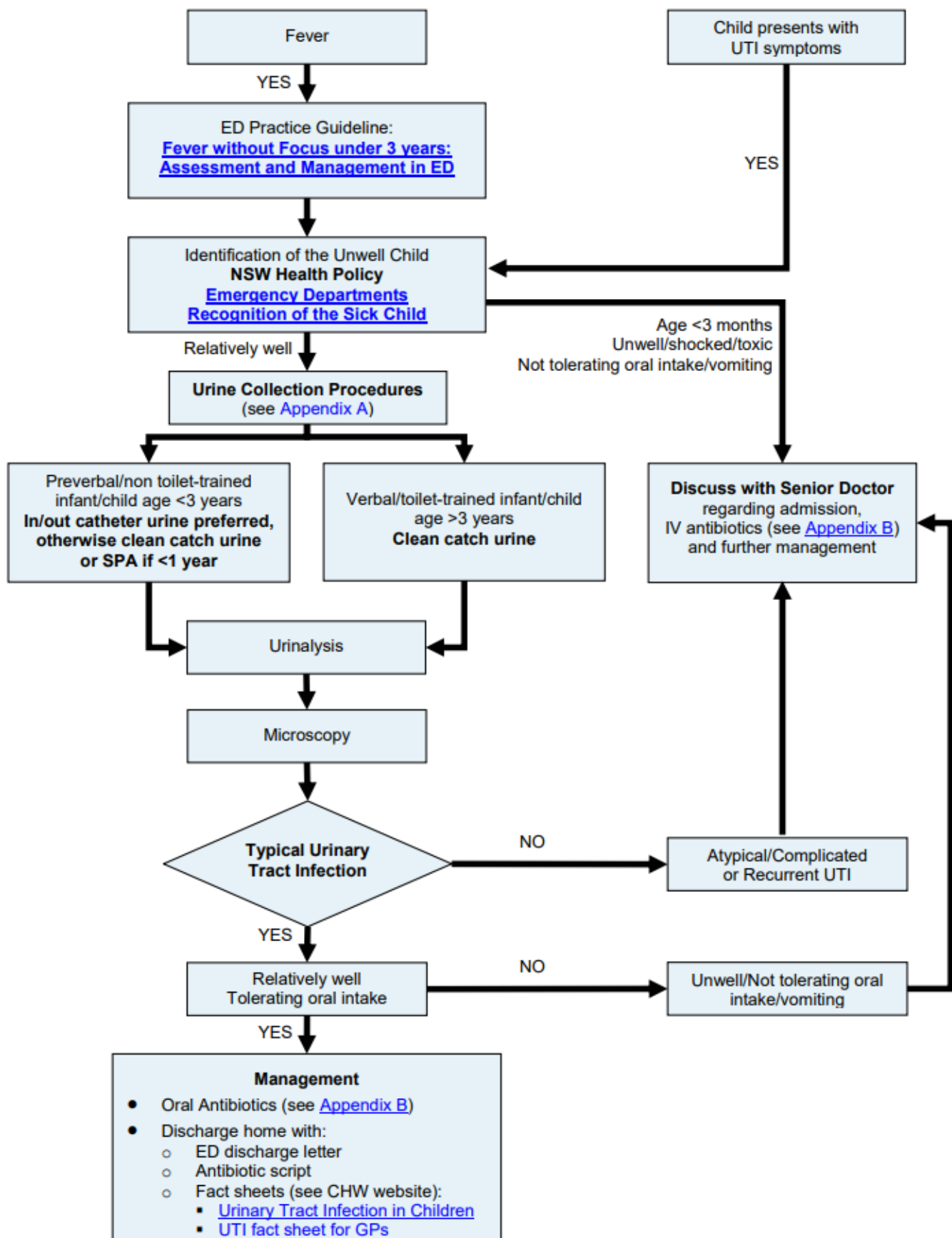
Correction of aphakia (difficulty focussing due to absent lens) after cataract surgery:

- **No intraocular lens before age 2** – big risk of complications requiring surgical intervention!
- Age 0-2: contact lenses (extended wear lenses), glasses
- Age > 2: intraocular lenses, extended wear contact lenses, glasses

8. Renal

Urinary Tract Infection (UTI)

See [NSW Health Policy on UTI](#)



Epidemiology: UTI is extremely common in kids. 5% of febrile children have a UTI.

- M>F until 12 months, then F>M
- 30% recurrence, usually within 1 year

Commonest pathogen: >80% E. coli. 10-15% other gram negatives e.g. Klebsiella, Enterobacter, proteus, pseudomonas.

- **Beware** a urinary tract anomaly in a child with a Pseudomonas UTI

Aetiology: 30% have vesico-ureteric reflux (VUR) → ureteric dilation and clubbing of the calyces

Clinical Features

Infants	Children
Poor feeding, vaguely unwell Fever of unknown origin Neonatal jaundice Vomiting	Dysuria Abdominal/ loin pain Incontinence Haematuria (Proteus) Fever

UTI Red Flags

- Infants < 3 months: all require full septic screen and IV antibiotics
- Shock: immediate resuscitation
- Toxic/vomiting: admit, IV fluids, antibiotics
- Atypical/complicated, e.g. poor urine flow, catheter in situ, pre-existing urinary tract disease, abdominal or bladder mass, spinal cord lesion, high BP, immunosuppression, or non-responsive to treatment
 - Discuss with senior clinician

Recurrent UTI: defined as 2+ pyelonephritis, 1 pyelonephritis with 1+ cystitis, or 3+ cystitis.

Diagnosis

Recall the preferred urine specimens

Preverbal, not toilet-trained: in-out catheter

Verbal or toilet trained: clean catch specimen

Urinalysis: dipstick for **leukocytes** and **nitrites**. ≥97% chance of UTI if both are positive; if one is positive, rely on the clinical picture. If both are negative, consider other causes.

Microscopy and culture: **all urine specimens for kids <3 y/o should be sent for culture.**

- WCC > 100 x 10⁶/L or single organism ≥ 10⁷ (catheter) or ≥ 10⁸ (voided)

Treatment of UTI

Afebrile implies cystitis; febrile is pyelonephritis.

Clinical Features	Antibiotics	Route	Duration
Afebrile (cystitis)	Cephalexin Bactrim	Oral	3 days
Febrile (pyelonephritis)	Cephalexin Bactrim	Oral	7 days
< 1 month old or very unwell	Ampicillin and gentamicin	IV	Until appropriate to switch to oral (48 h afebrile, then 7 day oral)

Follow-up of UTI:

	First, typical UTI	Recurrent UTI
<6 months old	Ultrasound at 6 weeks	Ultrasound during acute infection DMSA 4-6 months later MCUG
>6 months old	No follow-up	Ultrasound within 6 weeks DMSA 4-6 months later

There is no role for MCUG in infants and children > 6 months – it is invasive and does not change management.

Preventative Measures

Voiding history: a child under 10 should void 5-6 times per day!

- Regular fluids and voiding
- Avoid/treat constipation
- Avoid caffeinated drinks
- Cranberry juice (only effective by virtue of being a fluid)
- Correct wiping front to back
- Probiotics sketchy
- Circumcision in boys (2+ infections or reflux and ≥ 1 UTI)
- Surgery exceedingly rare (pyeloplasty or deflux ureteric opening)

Dysfunctional voiding: abnormal pattern of urination in children where the LUT muscles fail to relax appropriately during bladder emptying, in the absence of anatomical or neurological causes.

- Common in school-aged children, especially after toilet training.
- **Symptoms:** daytime wetting/incontinence, urinary urgency and frequency, hesitancy and straining to void, weak/intermittent/spraying urinary stream, dribbling or post-void leakage
- **Diagnosis:** toilet habits (withholding urine, rushing toilet routine, straining or hesitating), symptoms, and "holding manoeuvres".
- **Management:** increase fluids, bladder re-training, complete bladder emptying, double voiding, avoiding irritants

Nephrotic Syndrome

Clinical Practice Guidelines : Nephrotic syndrome

Triad of proteinuria, hypoalbuminaemia and oedema.

- Oedema, particularly peripheral, can worsen as the day progresses.
- Patients may also have **hypercholesterolaemia** as the liver compensates for albumin loss by upregulating lipoprotein synthesis.

Epidemiology: 2 – 7 cases per 100 000 children.

Remember the **80% rule for nephrotic syndrome! 80% of children...**

- Present under 6 years old.
- Have minimal change disease
- Steroid responsive (higher in MCD)
- Relapse (often triggered by mild illnesses)

Diagnosis: heavy proteinuria (dipstick $\geq 3+$ or spot protein/creatinine ratio >200 mg/mmol); hypoalbuminaemia (albumin < 25 g/L); oedema.

Examination may include a fluid status assessment, blood pressure (confirmed **manually**), and assessment of extra-renal features e.g. joint involvement.

Complications of nephrotic syndrome:

- Hypovolaemia
- Severe oedema
- Infection/sepsis (urinary loss of Ig and complement $\rightarrow$ encapsulated bacterial infection)
- Pneumococcus can cause spontaneous bacterial peritonitis
- Coagulopathy (urine loss of anticoagulant proteins, $\uparrow$ synthesis of prothrombotic factors, haemoconcentration)
- Severe hypertension

Red Flags that Nephrotic Syndrome is not minimal change disease: renal impairment, significant HTN, age < 1 year or > 10 -12 years, macroscopic haematuria or persistent microscopic haematuria, RBC casts in urine, FHx nephrotic syndrome, "syndromic" child, steroid-resistant.

Standard investigations: urine MCS, urine protein/creatinine ratio, FBC, EUC, albumin.

- Nephrotic syndrome *may* be associated with renal impairment due to intravascular hypovolaemia, but it should not be persistent.
- **Renal biopsy** might be considered in children whose presentation is atypical/steroid non-responsive.

Treatment

Recall that minimal change disease is exquisitely steroid-sensitive!

First episode	Relapse
Oral prednisolone 60 mg/m² for 4-6 weeks then wean (treated for 2-3 months overall) Teach parents to test urine No added salt diet; may need fluid restriction if grossly oedematous (1 L/day for older kids, 500-700 mL/day if younger) Pneumococcal vaccine Annual flu vaccine (since virus can precipitate nephrotic syndrome)	Shorter course of prednisone (2 months overall, but weaning commenced on Day 3 of admission) Steroid-sparing agents e.g. cyclosporine, cyclophosphamide, MMF, tacrolimus, rituximab.

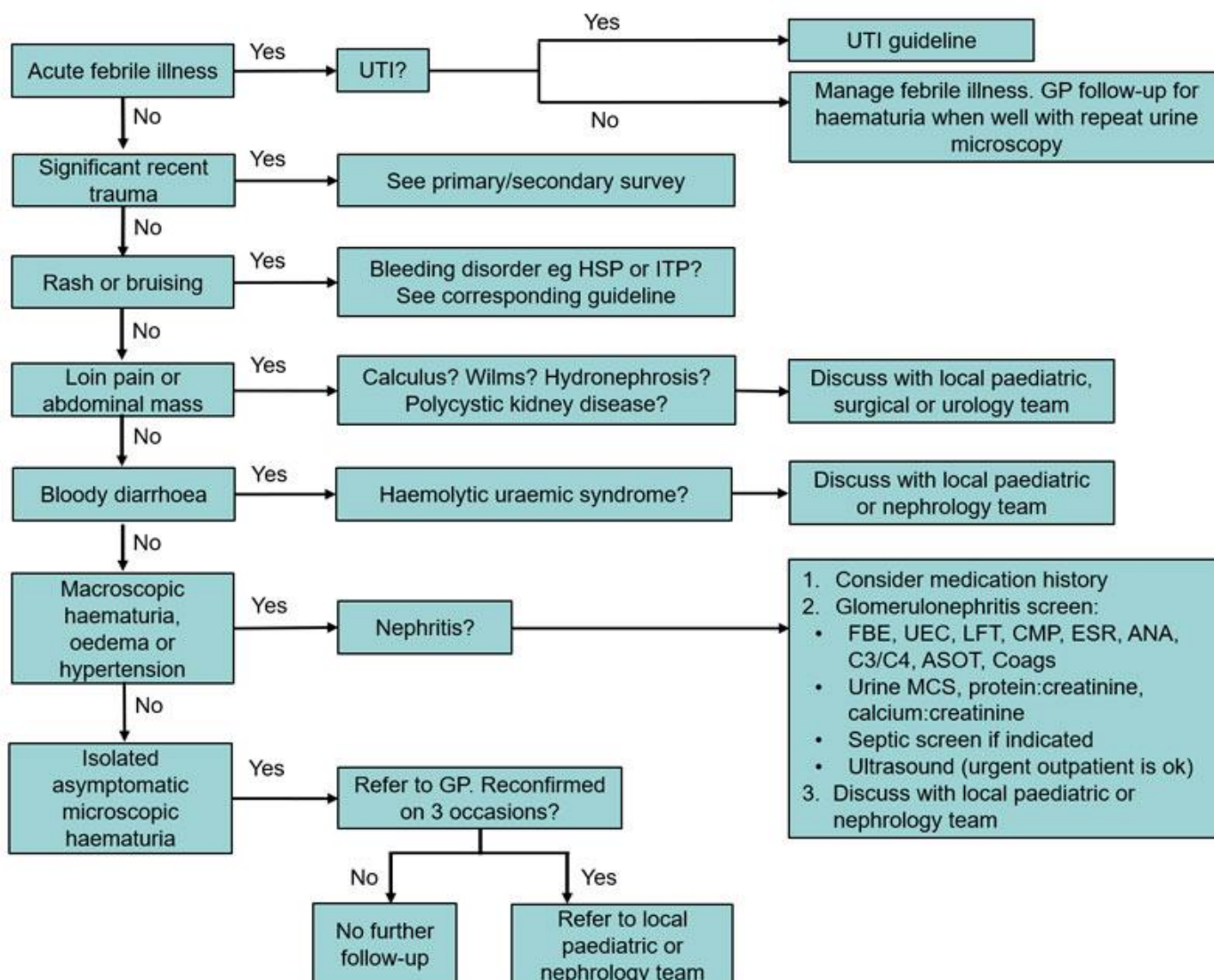
Children might also receive **intermittent concentrated albumin infusions (5 mL/kg of 20% albumin, or 1g/kg)** if they have intravascular volume depletion and/or severe or symptomatic oedema.

It was mentioned that 80% who get nephrotic syndrome will relapse. Unfortunately, half of these people will become frequent relapsers or steroid-dependent.

Nephritic Syndrome and Haematuria

[Clinical Practice Guidelines : Haematuria](#)

Microscopic haematuria (> 10 RBC/microlitre) can only be detected by urinalysis (urine colour normal). **Macroscopic haematuria** is when blood is present in the urine without microscopy.



Note, urate crystals are commonly present in the urine of neonates, and they can produce a red discolouration of the nappy which is sometimes mistaken for blood. They can be more prominent when the baby is dehydrated.



Nephritic Syndrome: results from glomerulonephritis. The classic features are haematuria (may be microscopic; sometimes visible as red or cola-coloured urine), hypertension, mild-moderate (non-nephrotic) proteinuria.

- Other features: oliguria, oedema

Differentials for acute nephritis:

- **Post-infectious glomerulonephritis** is the most common; most will not have significant renal impairment
- IgA vasculitis nephritis (Henoch Schonlein Purpura)
- Membranoproliferative glomerulonephritis
- Rapidly progressive GN

Spotlight: Post-Streptococcal Glomerulonephritis

Generally mild, self-limiting condition caused by nephritogenic strains of Group A Streptococcus (*pyogenes*).

- Much greater incidence in Aboriginal and Torres Strait Islander peoples.
- Usually present 2-4 weeks after GAS pharyngitis or impetigo with a triad of **hypertension, haematuria, and oedema**.
- C3 is low due to activation of the alternative complement pathway; complement C4 is usually normal or only slightly low.

Management is mostly supportive

- Strict fluid balance with daily weights (same time, same scales) and monitoring of urine output. Management is directed to fluid retention
- If fluid-overloaded, furosemide (consider IV if gut oedema suspected), low or no salt-added diet, fluid restrict (750 mL/m²/day)
- Antibiotics as per local guidance if active Streptococcal infection
- Hypertension can mostly be treated by fluid management; nifedipine if needed

Course: macroscopic haematuria should resolve in 1-2 weeks, proteinuria should resolve within 6 weeks, microscopic haematuria may persist for 2 years, low complement should normalise within 5-8 weeks.

Nephritis Red Flags

- Previous history of rashes, swollen joints, or other systemic symptoms → lupus nephritis or IgA vasculitis nephritis (HSP). Puzzlingly, HSP can occur post-streptococcal infection.
- The pattern of complement deficiency can also tell you a lot, for example:
 - Low C3 → complement activation, consistent with **PSGN**
 - Low C3/C4 → complement sequestration e.g. in SLE
- Previous episodes of macro- or microhaematuria may indicate IgA nephropathy or (if accompanied by hearing loss) Alport syndrome. This is especially the case if in the presence of family history of kidney disease at a young age.
- Creatinine rapidly rising or clinically deteriorating: consider rapidly progressive GN

Investigations

- EUC, CMP, LFT
- FBC + film
- Anti Streptolysin O titre (ASOT) + anti-DNase B
- C3/4
- Urinalysis, microscopy
- Renal ultrasound (usually swollen and mildly hyperechoic kidneys)
- +/- ANA (SLE), ANCA, anti-GBM (pulmonary renal syndrome), CK
- +/- renal biopsy

Management: usually conservatives; diuretics for HTN; immunosuppression (after renal biopsy) depending on cause. If acute renal failure → fluid restriction, manage acute hyperkalaemia, dietary intervention (low salt, low-moderate protein)

Hypertension in Childhood

Measurement of BP:

- The cuff bladder should be > 40% of the mid-upper arm width, and 80-100% of its circumference in length.
- Positioning: legs uncrossed, cuff at heart level. Young kids can lie down.
- **Hypertension:** SBP or DBP > 95th centile by age and height

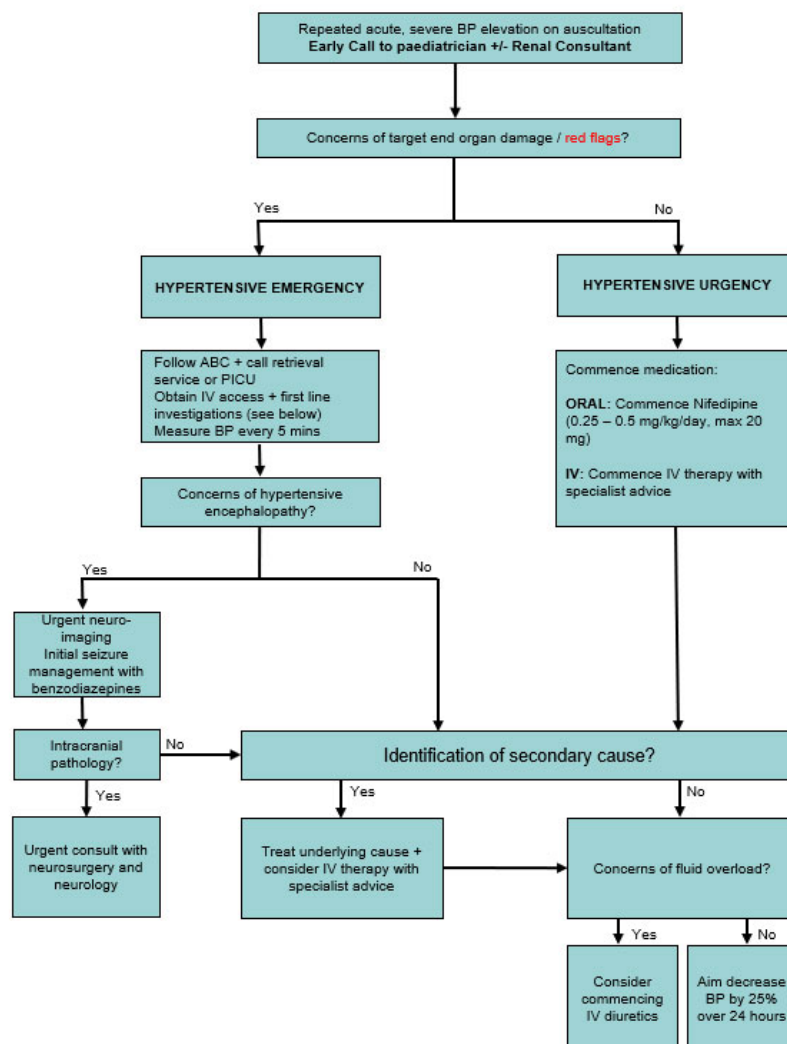
If a child has an abnormal BP reading: first re-do (you may need a bigger cuff). Afterwards, consider a 24 hourly ambulatory BP monitor (for a child > 5-6 years old).

Causes of Hypertension in Childhood

Primary (essential) hypertension accounts for most hypertension in kids > 6 years old and is usually associated with obesity or a family history of HTN. **Secondary hypertension** is more common in younger kids (<6 years old) with renal disease being the most prevalent cause. Secondary HTN is more likely than primary HTN to lead to emergencies.

Infants	Young children	Older children	Adolescents
Renal vein or artery thrombosis Coarctation of aorta Congenital renal abnormalities Bronchopulmonary dysplasia*	Tumours e.g. Wilm's or neuroblastoma Renal parenchymal disease Vascular e.g. renal artery stenosis or aortic coarctation	Renal parenchymal disease Renovascular disease Endocrine e.g. adrenal, thyroid, parathyroid, pituitary Essential hypertension	Metabolic syndrome Essential hypertension Renal parenchymal disease Endocrine

*Bronchopulmonary dysplasia is classically associated with **pulmonary hypertension** as it leads to inflammation and impairment in alveolar and vascular development. **Chronic hypoxia and oxidative stress** may lead to endothelial dysfunction and fluid overload, contributing to hypertension, and this is exacerbated by steroid use.



Examination

- Vitals: tachycardia, four limb BP
- Height and weight

Signs of end-organ damage

- Fundoscopy
- Cardiovascular
- Palpable kidneys in CRF
- Focal neurology

Signs of cause

- General appearance e.g. Cushingoid, proptosis, goitre, webbed neck (Turner syndrome), elfin facies (William syndrome)
- Skin lesions, rash, birthmarks
- Cardiovascular: murmurs +/- radiation, apical heave, ↓ femoral pulse, oedema, hepatomegaly
- Abdomen: masses, palpable kidneys, bruits
- Genitourinary: ambiguous genitalia in CAH

Investigations of HTN

Tailored to the individual.

- EUC, FBC (glucose/ lipids)
- Renin, aldosterone
- Urine (and/or plasma) catecholamines
- Urinalysis
- Renal Doppler/ abdominal US (plus other imaging as indicated)
- Angiogram
- Echo (LVH) and ophthalmology review

Treatment

First-line is lifestyle: weight loss, low salt diet, Mediterranean diet, physical activity (40 minutes moderate-vigorous exercise 3-5 days/week).

Medical management: if conservative Mx failed, symptomatic HTN, severe with no modifiable risk factors, CKD or diabetes

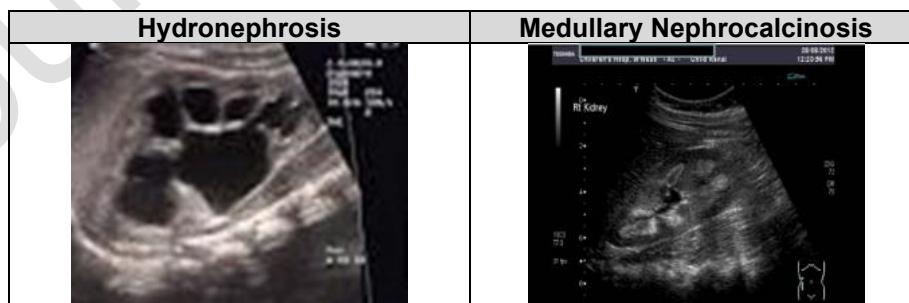
- First-line: long-acting calcium channel blockers e.g. amlodipine
- Other medications:

Class	Caution
ACEi/ARB e.g. lisinopril, losartan	Fetopathic, cough, K ⁺
Calcium channel blocker e.g. amlodipine or nifedipine	Fluid retention, flushing
Beta blocker e.g. atenolol	Asthma
Alpha blocker e.g. doxazosin	Dry mouth, blurred vision
Diuretics	
Centrally acting agents e.g. clonidine, moxonidine	Sleep disorder

- ACEi/ARB, beta blockers, alpha blockers, diuretics, centrally acting (e.g. clonidine)

Renal Tract Imaging

Renal Tract Ultrasound: structural anomalies (dilatation, cysts, masses, stones), size/shape/position of kidneys, bladder outline (pre- and post-void volumes).

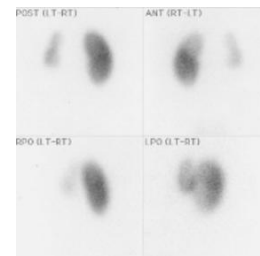


Micturating Cystourethrogram (MCUG): invasive investigation where a catheter is placed into the bladder with contrast inserted. Only performed in children < 1 years old. The gold-standard for vesicoureteric reflux.

- Directly visualises vesicoureteric reflux and bladder anatomy during voiding
- Catheterisation is straightforward in younger infants; limited radiation exposure too.
- In older children, we can use **nuclear medicine cystograms (MAG3)** instead.

DMSA (Dimercaptosuccinic acid) Scan: Static scan that provides information on the differential function of each kidney. A scarred kidney may also appear small on DMSA.

- Maybe remember S for scarring/static!
- DMSA binds to the renal cortex and so is good for detecting cortical defects and scarring e.g. post-pyelonephritis. It is **more sensitive than a renal ultrasound**.
- It does not assess renal function/ drainage.



Renal Functioning Scan (MAG-3/DTPA)

Feature	Tc-99m MAG3	Tc-99m DTPA
Clearance Mechanism	Tubular secretion (~90%)	Glomerular filtration (~100%)
Best Clinical Use	Infants, impaired renal function, obstruction assessment	Older children/adults, GFR measurement
Imaging Quality	High extraction, better image quality, faster clearance	Lower extraction, less clear images
Radiation Dose	Lower due to higher extraction	Higher due to lower extraction
Functional Parameter Measured	Effective Renal Plasma Flow (ERPF)	Glomerular Filtration Rate (GFR)
Age Preference Reason	Better for immature kidneys and impaired function	Better for mature kidneys, accurate GFR

CT Scan: Rarely needed. May be good for complex anatomy or in planning of complex urological surgery. It is also the gold-standard for suspected kidney stones.

I think in summary a **renal ultrasound** is the first-line investigation. MCUG will diagnose vesicoureteric reflux in an infant, MAG-3 is better for older children. DMSA for renal scarring after UTI (not obstruction or drainage – it is S for Static!). And DTPA can assess GFR in older kids.

Modality	Main Indication	Typical Clinical Scenario/Use
Ultrasound	First-line for anatomy, screening	Initial UTI evaluation, hydronephrosis, masses, congenital anomalies; pregnancy; stone screen
MCUG	Assess vesicoureteric reflux, lower tract anatomy	Infants <1yr with UTI, abnormal US, suspected reflux, posterior urethral valves; pre/post surgical evaluation
DMSA	Detect renal cortical defects/scarring	Post-UTI, rule out/confirm scarring or pyelonephritis, differential renal function
MAG-3	Renal drainage/function, obstruction	Hydronephrosis, suspected obstruction, differential renal function, poor renal function (infants/kids)
DTPA	Measure GFR, differential function	Older children/adults, suspected parenchymal disease; assessing renal transplantation
CT	Anatomy, stones, tumors, trauma, hematuria	Persistent hematuria, stone evaluation, mass characterization, trauma, pre-op planning, complicated UTI

9. Cardiology

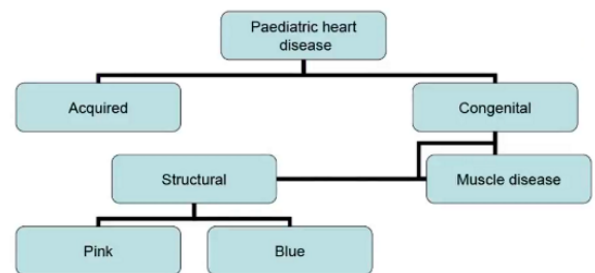
Recall the foetal circulation:

- Oxygenated blood is supplied from the placenta via the **umbilical vein**, which joins the IVC via the **ductus venosus (DV)**.
- In the foetus, pulmonary vascular resistance is high. The **ductus arteriosus (DA)** and **foramen ovale (FO)** allow oxygenated blood to bypass the lungs and enter the systemic circulation.
- Most blood shunts from RA to LA via the **foramen ovale**; only a small amount crosses the tricuspid valve into the RV. The foramen ovale forms a one-way valve.
- The transition after birth occurs **with breathing**: ↓ PVR, ↑ SVR, closure of FO, DA, DV

Foetal shunt	Foramen Ovale (FO)	Ductus Arteriosus (DA)	Ductus Venosus (DV)
Description	Opening between RA and LA	Connects pulmonary trunk to aortic arch	Connects umbilical vein to IVC
Timing of closure	Closes functionally at birth; anatomically in first few months (may remain patent)	Functional: 10 – 15 hours Permanent: 2 – 3 weeks	Commences shortly after birth; complete by 3-7 days
Stimulus for closure	↑ Pulmonary venous return + pulmonary blood flow → ↑ LA pressure ↓ RA pressure due to ↓ venous return (placental circulation removed)	↑ PaO ₂ , removal of prostaglandin Permanent closure: thrombosis, intimal proliferation, fibrosis	Passive, ↓ flow and pressure

Neonates have **RV dominance**. This diminishes slowly throughout childhood, so that as an adult, the LV becomes dominant instead.

There are some conditions where **fetal shunts** are reopened as a temporising measure before surgery, e.g. PGE1 administration to open ductus arteriosus in duct-dependent conditions like aortic coarctation



Routine checks are critical, particularly the 6-week exam!

The newborn exam might miss duct-dependent cardiac disease e.g. aortic coarctation.

Recall PVR continues to drop over the first 6 weeks of life. A VSD murmur won't be apparent until PVR falls and there is a pressure differential between LV and RV.

Acquired Heart Disease in Children

Condition	Description
Acute rheumatic fever	10 days – 6 weeks after Group A Streptococcal infection (pharyngitis or pyoderma); nonsuppurative inflammation of heart (plus joints, subcutaneous tissue, and CNS). High-risk groups are Indigenous communities living in rural or remote areas or disadvantaged suburban areas.
Rheumatic heart disease	Caused by recurrent acute rheumatic fever; antibiotic (penicillin) prophylaxis essential for prevention.
Kawasaki disease	Can cause coronary artery aneurysm; treatment is with IVIg
Endocarditis	Most commonly normal mouth commensals that seed in damaged endocardium. More virulent organisms (e.g. Staphylococcus) can cause endocarditis in a normal heart. Prophylactic Abx needed for anyone with prosthetic material in their circulation Prolonged IV antibiotics (6 weeks)
Some cardiomyopathies	Chemotherapy-induced (e.g. anthracyclines), infectious, metabolic, etc.

Congenital Heart Disease

Incidence: 6 – 8 in 1000 (remember **1 in 100**).

Acyanotic	Cyanotic (5 T's)
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Ventricular septal defect (the most common CHD) Atrial septal defect Persistent ductus arteriosus Obstructive Conditions Aortic stenosis Pulmonary stenosis Aortic coarctation	Tetralogy of Fallot (TOF) (commonest cyanotic CHD) Transposition of the Great Arteries (TGA) Truncus Arteriosus Total Anomalous Pulmonary Venous Return (TAPVR) Tricuspid Valve Abnormalities and Hypoplastic Right Heart Syndrome
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Aetiology: most have no clear association

- Chromosomal association: Down syndrome (**AVSD**), T13, T18, Turner Syndrome
- Genetic syndromes: Velo-Cardio-Facial, Marfan syndrome, Williams syndrome, CHARGE and VACTERL associations
- Toxins: alcohol, maternal diabetes, warfarin
- Maternal SL, foetal rubella

History

- Cyanosis
- Respiratory distress (tachypnoea, recessions, grunting, nasal flaring)
- Sweating (catecholaminergic response)
- Feeding problems
- Growth problems

Exam

- End-of-bed
- Dysmorphic features
- Growth
- Vital signs: HR, BP, RR
- Clubbing
- Pulses (*radio-femoral delay in aortic coarctation*)
- Apex beat
 - < 7 y/o: left 4th IC space MCL
- Heaves and thrills
 - **Parasternal heave in RVH**
 - Thrill in VSD and aortic stenosis
- Heart sounds, murmurs
- Chest auscultation
- Hepatomegaly, splenomegaly
 - Kids don't get peripheral oedema like adults!

Murmurs and Heart Sounds

Recall S1 marks the start of ventricular systole and S2 diastole. In children, S2 usually splits during inspiration (A2 then P2) because increased venous return delays P2.

- S3 is a low-frequency sound heard in early diastole, reflecting rapid ventricular filling. It is usually physiological.
- **Anything that increases blood on the right side will → split S2**
 - Atrial septal defect (fixed, wide split S2)
 - Pulmonary hypertension
 - Pulmonary stenosis
 - Large VSD

Grades of murmurs:

Grade	Intensity	Thrill
Grade 1	Scarcely audible	No
Grade 2	Soft	No
Grade 3	Loud	No
Grade 4	Loud	Faint
Grade 5	Very loud	Easily felt
Grade 6	Audible without stethoscope	

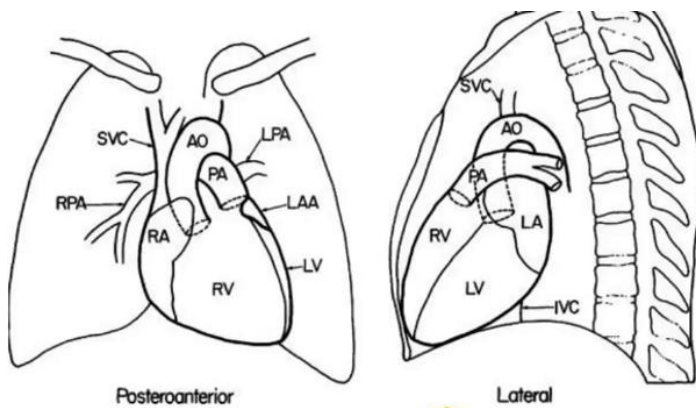
Character of murmurs:

Murmur Type	Timing	Pathophysiology	Paediatric Examples
Crescendo	Increasing intensity	Blood flow accelerates through narrowing or obstruction	Pulmonary stenosis, aortic stenosis, hypertrophic cardiomyopathy
Decrescendo	Decreasing intensity	Rapid initial regurgitation with slowing backflow as pressures equalize	Aortic/pulmonary regurgitation

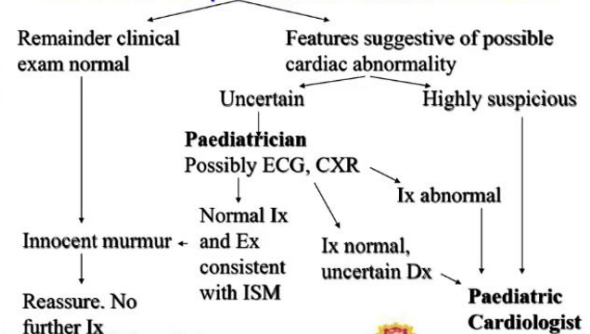
Continuous	Throughout cycle	Persistent pressure gradient causing continuous flow between vascular compartments	Patent ductus arteriosus (PDA), AV fistulas
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Investigations: CXR, ECG, then gold-standard investigations (Echocardiogram, CT, MRI)

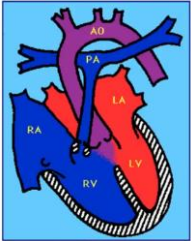
- CXR: Pay attention to position, contour, and pulmonary vascularity.
 - Patterns: “**Boot-shaped heart**” = TOF, “**Egg-on-a-string**” = TGA, “**Snowman shaped heart**” = TAPVR
- ECG: right dominant in neonate; becomes left dominant by 1 – 5 years old.



Management of a child with a murmur or possible heart disease



Common Congenital Heart Defects

Condition	Description	Murmur	CXR	Clinical Features
Ventricular septal defect ¹	Membranous-type commonest L→R shunt at ventricles	Continuous harsh pansystolic murmur	Cardiomegaly (fills out posteriorly); ↑ pulmonary vascular markings	Presents at 4-8 weeks
Atrial septal defect	L→R shunt at atria, often asymptomatic early Common type is ostium secundum	Wide fixed split S2, systolic flow murmur at left upper sternal border	RA and RV enlarge (anterior , parasternal clear space filled on lateral CXR)	May present with recurrent resp infection (↑ pulmonary blood flow → congestion) IR can manage
Atrioventricular septal defect	Large atrial and ventricular septal defect with common AV valve	Variable systolic murmur, wide fixed split S2	Cardiomegaly, increased pulmonary markings	Presents early infancy (first 2-3 months), often in Down Syndrome, severe heart failure
Patent ductus arteriosus	L→R shunt from aorta to pulmonary artery	Continuous “machinery” murmur at left infraclavicular area	Cardiomegaly, increased pulmonary vascular markings	Bounding pulses, wide pulse pressure
Tetralogy of Fallot 	Tetrad of: Right ventricular outflow obstruction Right ventricular hypertrophy Ventricular septal defect Aortic override Caused by cranial displacement of the infundibular septum into the RVOT (usually meets apical septum)	Harsh systolic ejection murmur at left upper sternal border	Boot-shaped heart due to RV hypertrophy; avascular lung fields; absent aortopulmonary window	Early infancy or neonatal period; cyanosis, “tet spells” (hypercyanotic episodes) ² , failure to thrive Many acyanotic at birth; classic is progressive cyanosis by 5-6 months as PVR falls and R-sided pressure (RVOT obstruction)

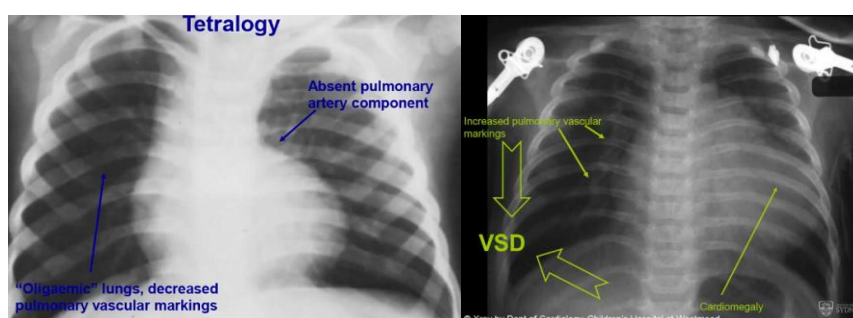
	Acyanotic type if RVOT obstruction mild			and progressive infundibular stenosis) increases Surgery by 6 mon.
Transposition of the Great Arteries (TGA)	Aorta arises from RV and pulmonary artery from LV	Single loud S2, no murmur	"Egg on string" shaped heart, increased pulmonary vascular markings	Severe cyanosis in early neonatal period, needs PDA/ ASD to maintain mixing
Total anomalous pulmonary venous return (TAPVR)	Pulmonary veins drain abnormally into systemic circulation; requires interatrial communication for survival e.g. ASD or PFO	No specific murmur unless associated with increased pulmonary flow or defect e.g. ASD	Cardiomegaly and increased pulmonary vascular markings; "snowman" appearance due to dilated 'vertical' vein (combination of pulmonary veins) draining into SVC	Cyanosis in the neonatal period, respiratory distress and shock if obstructed. Otherwise features of CCF (tachypnoea, poor feeding, hepatomegaly).
Ebstein Anomaly	Malformation of tricuspid valve → 'atrialisation' of RV and tricuspid regurgitation	Holosystolic murmur of tricuspid regurgitation, split S1/S2	Massive cardiomegaly	Cyanosis, HF, arrhythmias
Aortic Coarctation	Narrowing in aorta, typically just distal to insertion of ductus arteriosus	Systolic murmur on back or left infraclavicular area	Rib notching (older; collateral circulation through intercostal arteries), cardiomegaly if HF	Deterioration 3-7 days to 2 weeks after birth (when DA closes), weak/ absent lower limb pulses
Hypoplastic Left Heart Syndrome (HLHS)	Underdeveloped left heart structures → systemic outflow obstruction	No murmur or soft ejection murmur	Cardiomegaly, pulmonary plethora if PDA open	Severe cyanosis and HF as ductus closes; emergency management needed Surgery needed where RV becomes systemic ventricle

1: Eisenmenger Phenomenon is a complication of a long-standing left-to-right shunt e.g. large uncorrected VSD. Over time, increased pulmonary arterial flow → pulmonary arterial hypertension → exceeds systemic pressure → shunt reverses direction (R → L).

- **Associated features:** clubbing, polycythaemia, gout (RBC turnover), gallstones (pigment stones), scoliosis

2: A Tet spell is caused by a sudden increase in right-to-left shunting of deoxygenated blood through the ventricular septal defect due to worsening right ventricular outflow tract obstruction.

- Triggered by events that decrease systemic vascular resistance e.g. crying, defaecation, exertion, hypovolaemia (i.e. decrease left-to-right flow of blood)
- Children may squat instinctively to increase SVR and reduce shunting
- Catecholamine release paradoxically increases RV contractility, worsening obstruction
- Vicious cycle of hypoxia, cyanosis, hyperpnoea, acidosis, and shunting
- **Acute management:** knees to chest/ sling baby over carer's shoulder, oxygen (decreases PVR), propranolol (reduces infundibular spasm and myocardial contractility); morphine/sedation to decrease agitation



Other complications of R-L shunts: hypoxia, myocardial and somatic tissue dysfunction, paradoxical embolus, polycythaemia, neurological sequelae (stroke, plegia, abscess), exercise intolerance, bleeding.

Prognosis

- Small VSDs and ASDs close off themselves.
- Valvular regurgitation usually worsens with life.
- Valvular stenosis: some get worse, some stay the same, some get better.

General Management

- Growth and nutrition
- Immunisation (for all unless immunocompromised)
- Dental hygiene (endocarditis prophylaxis especially if cyanotic defect, prosthetic material in situ, or residual defect post-repair)
- General infection
- Exercise

Innocent Cardiac Murmurs

- Normal history (incidental finding) and normal peripheral examination
- **Typical features:** systolic, ejection, musical/vibratory, 4th left intercostal space, Grade 1-3/6 (no thrill), change with body position, varies with review/ augmented by illness, might worsen with exercise (hyperdynamic)
 - **S's:** Soft, Systolic, Short, Symptomless, Standing/Sitting (vary with position)
- **Exclusions:** symptomatic, diastolic component (**exception: venous hum**), loud/thrill/radiates, cyanosis, abnormal pulses/ heart sounds (e.g. clicks) / CXR / ECG

Innocent Murmur	Typical Age	Location	Timing and Quality	Pathophysiology	Clinical Features
Still's	2 – 7 years	Left lower sternal border	Early-mid systolic, musical/ vibratory	Turbulent flow across LVOT or near chordae tendineae	Loudest supine, softer sitting/ standing; no radiation
Pulmonary flow murmur	Children and adolescents	Upper left sternal border	Mid-systolic crescendo-decrescendo	Increased flow velocity across pulmonary valve	Louder with fever or exercise
Carotid bruit	Any age (rare in kids)	Over carotid artery in neck	Systolic, harsh	Turbulent flow through narrow carotid artery or increased flow	Can be benign in kids
Venous hum	Mostly children	Right upper chest or neck	Continuous, low-pitched	Turbulent flow in jugular veins due to large venous return	Loudest sitting, abolished by jugular vein compression

Arrhythmias

- Ectopic beats are very common and of no importance
- Wolff-Parkinson-White Syndrome due to accessory pathway conduction: delta waves (short PR interval, pre-excited ECG)
 - Diagnosis of the syndrome per se requires an abnormal ECG + symptoms of tachyarrhythmia (palpitations, syncope, or documented SVT)

Supplemental: Paediatric Auscultation

Remember to listen to the heart sounds first, then the murmur second!

Ejection Systolic vs Pansystolic: ejection systolic implies an outflow tract issue (aortic stenosis, pulmonary stenosis, ASD, HCM). A pansystolic murmur implies flow for the entire duration of systole, e.g. VSD.

What causes the murmur in ASD? It's actually turbulent flow across the pulmonary valve due to increased blood flow on the right side of the heart - not flow through the defect itself. Note that there is insufficient pressure gradient across the atria for a murmur to be heard. The murmur is an **ejection systolic murmur**.

You might also hear a diastolic murmur due to increased flow across the tricuspid valve during diastole.

ASD also presents with fixed split S2 because of this increased blood flow through the RVOT that does not change with respiration.

What causes the murmur in Tetralogy of Fallot? It's actually the pulmonary stenosis producing a harsh **ejection systolic murmur**. Although TOF includes a large VSD, it typically produces no murmur because the pressure between the left and right ventricles equalizes, so there is little turbulence across the defect.

What happens to the murmur in a hypercyanotic "tet" spell? It disappears, because the RVOT becomes completely obstructed and flow through the stenotic pulmonary valve diminishes entirely.

Basic Paediatric Auscultation

	Auscultation		Diagnosis	Differentials	Symptoms
PSM (or holosystolic murmur)	S1 soft/absent Pansystolic murmur S2 soft/absent		VSD	MR TR	FTT Vomiting Tachypnoea
ESM (or crescendo-decrescendo murmur)	S1 normal Crescendo-decrescendo murmur S2 normal		AS	PS Innocent*	FTT Vomiting Tachypnoea
Fixed split S2	S1 normal No murmur S2 normal		ASD		None (Right heart failure in adulthood)
	S1 normal Crescendo-decrescendo murmur** S2 normal				
Respiratory split S2 (or inspiratory or physiological split)	S1 normal No murmur S2 normal		Innocent*		None
	on expiration=				
Continuous murmur (or machinery murmur)	S1 normal S2 normal Mechanical murmur throughout systole and diastole (with variable intensity)		PDA	Systemic to pulmonary collaterals	FTT Vomiting Tachypnoea

ESM = ejection systolic murmur, PSM = pansystolic murmur, FTT = failure to thrive, PDA = patent ductus arteriosus

Innocent*: for any finding to be innocent, there must be no symptoms and no chest x-ray changes and no ECG changes.

**The ESM in ASD arises if there is turbulent flow across the pulmonary valve (which is otherwise normal). The ESM is not required to make the diagnosis of ASD.

10. Community/Developmental

Please refer to [General Paediatrics: Growth and Nutrition](#) section.

Autism Spectrum Disorder (ASD)

Epidemiology: 0.5 – 1%, M>F, children = adolescents. Typically detected prior to school entry (i.e. **earlier** than ADHD).

- Problems usually extend into adulthood
- **33% with autism have intellectual disability**

Note that ASD and ADHD have divergent early neurobiologies. However, both ultimately result in faster cortical thinning.

ASD	ADHD
Overproliferation of cortical material	Delayed cortical maturation
Faster cortical thinning	Faster cortical thinning

Red flags for autism:

Impairment in social interaction	Impairment in communication	Restricted behavioural patterns	Regression of developmental skills
Poor eye contact No sharing of enjoyment No response to name	Lack of gesture No pointing Odd voice quality Accents Stereotyped phrases from TV shows Echolalia	Restricted patterns of behaviour, interests, and activities Spinning, twirling, peering, fixation on detail, odd body movements, licking, sniffing, stroking	

Some children may have islets of normal or supernormal ability (think, Rain Man).

Absolute indicators for referral of preschoolers for autism assessment:

Language	Social Communication	Restricted Repetitive Interests
No babbling No single words by 18 months No 2 word combinations by 2 years Any regression in language	No pointing or gestures by 12 months No sharing of interests	Twirling, flapping Obsession with same toy or activity Hypersensitivity to sound, food textures

Investigation and Assessments:

- Hearing and vision assessment
- Speech and OT
- Blood and urine tests: FBC, EUC, iron studies, Pb levels, CK, TFTs, Vitamin D, Fragile X screen, urine metabolic screen, CGH microarray (replaced karyotype)
- Preschool report/ feedback
- Developmental screens, e.g. ASQ, ARS
- Formal developmental assessment (**Griffiths III, <5 years old**) or **IQ test (>5 years old)**
 - The **ADOS 2** is an autism-specific tool that can be used when diagnosis is equivocal

The gold-standard for diagnosis of Autism Spectrum Disorder is the DSM 5.

- Persistent deficits in social communication and interaction across multiple contexts: reciprocity, non-verbal behaviours, and relationships
- Restricted repetitive patterns of behaviour, interests, or activities
- **Symptoms must be:** present early, cause impairment, not explained by intellectual disability or developmental delay

The key **distinguishing feature** of ASD vs intellectual disability or developmental delay is **social communication**.

Management

- Autism-specific early intervention
- Speech therapy
- Problem behaviour management
- Supportive communication, visual strategies (e.g. Picture Exchange Communication System; PECS)
- Medication does not target **core** symptoms but may help improve quality of life, e.g.
 - **Stimulants** for attention and hyperactivity
 - **Risperidone** for aggressive behaviour
 - **SSRI** for anxiety and obsessive compulsive behaviours
 - **Clonidine/melatonin** for sleep

Attention Deficit Hyperactivity Disorder (ADHD)

Epidemiology: 5 – 7%, M>F (5:1), children > adolescents. Typically detected **after school entry**.

- There is an uneven socioeconomic distribution due to comorbidities in vulnerable families.

Aetiology: highly heritable; 76% genetic predisposition based on twin studies. **Family history is the biggest risk factor.**

- Low birthweight and maternal smoking during pregnancy have strongest evidence as environmental factors.
- Epilepsy is also a strong risk factor (25 -40%)

Pathophysiology: primarily a disorder of executive function (frontal lobe). However functional imaging has also shown hypoactivation of the caudate nucleus and cerebellum (frequency and timing of events) and parietal cortex (biasing attention).

Clinical features: ADHD consists of two related but distinct symptom domains: inattention and hyperactivity-impulsivity.

- The DSM 5 TR Diagnosis requires 5+ inattentive and hyperactive/impulsive symptoms that are inconsistent with developmental level and interfere with function.
 - Present for at least 6 months, and present prior to 12 years of age, in 2+ settings.
 - As with anything DSM, they can't be accounted for by another disorder.

Inattention	Hyperactivity-impulsivity
Poor attention to detail	Fidgets and squirms
Cannot sustain attention	Often out of seat
Does not listen	Runs or climbs excessively
Does not follow instructions	Cannot play quietly
Poorly organized	Excessive talk
Avoids tasks requiring mental effort	Shouts out in class
Loses things	Cannot wait turn
Easily distracted	Interrupts or intrudes
Forgetful	Acts without thinking
	Impatient
	Uncomfortable doing things slowly, systematically
	Can't resist temptations or opportunities

Diagnosis can be problematic because it is subjective, and can be difficult incorporating reports from different sources e.g. family and school. Note that **girls can present differently too**.

Comorbidities

80% with ADHD have at least 1 comorbidity: learning disability (30 – 50%), disruptive behaviour disorders (e.g. ODD, conduct disorder) (40%), emotional or mood disturbances, autism, developmental delay.

Assessment:

- History is critical!
- Examination: height, weight, BP
- **Always** check hearing and vision
- Psychometric testing: cognitive, academic achievement (feedback from teachers)
- Behaviour rating scales in 2 settings
- Connors Parent and Teacher
- Vanderbilt
- DSM 5 TR specific checklists
- **Limited role for further investigations:** possibly baseline ECG due to (unsubstantiated) cardiac concern with stimulant medication

Treatment of ADHD

MTA Trial (1999 – 2001): medication alone is superior to behavioural treatment and routine community care for core ADHD symptoms; medication + behaviour management for non-core ADHD symptoms (e.g. anxiety, aggression)

Note, there is little evidence for diet or for alternative therapies for ADHD (*though moderate evidence for fish oil*).

- **Behavioural modification (governed by child psychologist):** counselling on consistency, routine, positive reinforcement for good behaviour, appropriate consequences for unacceptable behaviour.
- Classroom strategies including single desk seating, positive role modelling, fidget toys, individualized education plans, and learning support.

80% of children with ADHD respond to medication.

- Stimulants are first-line. They are **non-addictive** and have **minimal side effects**.

- Dose is individual, and weekends and school holidays are optional.
- They can be prescribed by paediatricians, child psychiatrists, and approved prescribers.
- In NSW, clinicians can prescribe ADHD medications for children **between 4-17 years old. Modal age 11 y/o (just before high school).**

	Methylphenidate (Ritalin)	Dexamphetamine
Short-acting	10 mg tabs (3 – 4 hours)	5 mg tabs (3 – 4 hours)
Long-acting	Ritalin LA (7 – 8 hours) Concerta (12 hours)	Lisdexamphetamine (Vyvanse) (10 – 12 hours)

The choice of drug depends on clinician preference and the child's response.

Other medications include melatonin (sleep), clonidine, atomoxetine (SNRI), guanfacine (Intuniv; long-acting α agonist).

- Clonidine/guanfacine might be good for aggression.

Side-effects of stimulant medication: the most common is a **loss of appetite**, especially for lunch.

- Less common: difficulty falling asleep (esp long-acting drugs), emotional lability (settles after 2-3 weeks), stomach ache
- Rare side-effects: mood lability, medication-induced psychotic symptoms, growth delay (not if drug holidays on weekends and school holidays)
 - There has been some association reported between ADHD medications and tics, but children with comorbid tic disorders seem to have a decreased tic frequency with stimulants.

Stimulant medication requires **6 monthly monitoring of height, weight, heart rate, and BP.**

Prognosis: 60 – 80% continue into adolescence; impulsivity and hyperactivity tend to ease while inattention persists.

- Predictors of adult outcome: severity of childhood ADHD symptoms, parent mental health problems
- Early intervention and additional support may assist in improving adult functioning

11. Dermatology

Birthmarks

Aetiology: too much or not enough of a component of the skin, may or may not be evident at birth.

Café au lait spot	Naevus flammeus	Ash leaf macule	Epidermal naevus
Increased melanocyte activity	Dilated capillaries	Seen in tuberous sclerosis, preceding developmental delay and seizures	Excess keratinocytes, linear pattern
			

Infantile Haemangioma (Haemangioma of Infancy, HOI)

- Most common benign soft tissue tumour of childhood.
- Incidence: 1 – 2% neonates
- **Not** usually present at birth, though “precursor” lesion noted in 30% of neonates.
- 3 phases: proliferative (3-6 months), plateau, involution (50% by 5 years, 90% by 9 years)
- May be superficial, deep (lump), or mixed.
- Deep haemangiomas *may* need ultrasound, though history usually gives the diagnosis away.
- Most haemangiomas will not cause complications, so **reassurance is okay**.
- **Complications:** ulceration is the most distressing (very painful, slow to heal).
 - Bad locations, e.g. ophthalmic → amblyopia; nasal tip → excess tissue (Cyrano de Bergerac deformity)
- Some haemangiomas can herald an underlying or regional problem, e.g. **sacral HOI** linked to **spinal and urogenital defects**; **mandibular HOI** linked to **laryngeal HOI**



Infantile haemangioma may also present as part of **PHACES Syndrome**

Posterior intracranial fossa anomalies	Haemangioma	Aortic arterial anomalies	and Cardiac anomalies	Eye anomalies	Sternal defects
Dandy-Walker and other cerebellar anomalies	Large facial and cervical	Coarctation, anomalous arteries, aneurysms	Valvular atresia, VSD, Cor triatriatum	Coloboma, glaucoma, cataracts	

- **Treatment:** not always needed, consider if ulceration, obstruction of function, long-term cosmetic outcome
 - Oral propranolol
 - Oral corticosteroids

Naevi: remember ABCDE; *children can still get melanoma*.

Eczema and Psoriasis

Atopic Dermatitis

- Chronic inflammatory disease characterised by epidermal barrier dysfunction (e.g. filaggrin deficiency) and immune dysregulation (TH2 imbalance).
- **Prevalence:** 10 – 20% children. 45% begin in first 6 months, 85% by 5 years old.
 - Children usually grow out of atopic dermatitis, prevalence only 1-3% adults.
- **Atopic association:** asthma 50%, allergic rhinitis 50-80%, food allergy 30%.

- **Appearance:** dry skin, fine scale, poorly demarcated margins. More severe eczema may present with papules and vesicles.
 - Patients may have excoriations which can lead to scarring.
 - Chronic scratching may lead to lichenification → stubborn to treat (topical steroids, moisturisers, immunosuppressants)
- **Distribution:** affects flexures. In babies and young children, seen on cheeks (Headlight Sign = prominence of cheeks but sparing of folds).
 - **The napkin area is typically spared due to moisture retention of nappies, although irritant contact napkin dermatitis can still develop**



- Avoid environmental irritants: drying agents (soap, shampoo, detergent), abrasive clothing (wool, nylon, acrylic), abrasive surfaces (carpets, woollen rugs, sandpits), water (frequent bathing, pools), heat, dry air, chemicals
- Secondary infection may occur with **Staphylococcus aureus** or **Streptococcus pyogenes**
 - **If there is no response to treatment, think of secondary infection!**
- Children with eczema are also more likely to be **colonised with Staphylococcus aureus**, which can exacerbate the inflammatory process via a superantigen effect.
 - Colonisation may be managed with bleach baths or anti-microbial washes +/- low-dose Bactrim
 - This, in turn, may help the eczema.



Management of atopic dermatitis: the aim is to **control, not cure!**

- Avoid environmental triggers
 - **Food allergies** may exacerbate atopic dermatitis but **avoidance does not help** (and can lead to malnutrition, which is bad)
- Regular emollients/ moisturisers. Continue long-term even if the skin looks and feels comfortable.
- **Topical steroids** are the gold-standard, remember to use an **appropriate strength** and **vehicle** (e.g. ointments are oily, lotions might be better for scalp and hair-bearing areas but are painful for broken skin).
 - Sigmacort (1% hydrocortisone) for face, aristocort (moderate) for trunk, diprasone (potent) for thickened areas on limbs
 - While steroid phobia is real, steroids are safe when used appropriately (1-2x per day until cleared)
 - **Do not be afraid to treat intensively with potent topical steroids!**
- Topical calcineurin inhibitors can be used in sensitive sites e.g. eyelids, skin folds, genital areas
- **Systemic treatment:** narrowband UVB phototherapy, immunosuppressives, dupilumab (biologic)

Note: **plant** and **contact** dermatitis require oral steroids specifically.

Psoriasis

- Immune-mediated disorder; triggers include trauma, strep infection, stress.
- **Prevalence:** 2-3% population, 30-45% of adults with psoriasis onset < 20 years.
- **Clinical features:** erythematous, well-demarcated plaques, thick silvery-white scale, symmetrical, "classical" locations (extensor knees, elbows, lumbosacral). Children may have a different presentation.

Guttate (raindrop)	Facial psoriasis	Flexural (inc. nappy)	Scalp psoriasis
			

The classic presentation of **guttate psoriasis** is sudden appearance of scaly lesions 2-3 weeks post- streptococcal infection. It typically starts on the trunk and proximal extremities and may spread to the face and scalp.

- Do not confuse with **viral exanthem** which usually spreads head to toe, presents as pink or red macules and papules, and lack the characteristic scaling.
- Ptyriasis amniantacea: excessive scaling of scalp that may be mistaken for tinea.
- Management:** address triggers (e.g. streptococcal infection, beta blockers, trauma), topical therapy (corticosteroids, tar, calcipotriol, dithranol), phototherapy (> 8 years old), systemic agents, biologics (etanercept)
 - Topical tars and Vitamin D analogues seem to be better than topical steroids in children

Dermatological Infections

Impetigo

Common, superficial, highly contagious bacterial skin infection characterised by pustules and honey-coloured crusted erosions. Remember, **impetigo** is **itchy** (whereas **herpes** **hurts**)!

- Most common causative pathogens: Staphylococcus aureus and Streptococcus aureus

Non-bullous (70%) – ‘school sores’	Bullous
Broken skin	Staphylococcus aureus produces exfoliative toxins → dissociation of epidermal cells → blistering of intact skin
Typically face or extremities, begins as erythematous macule which evolves into pustule or vesicle	Usually on face, trunk, extremities, buttocks, perineal regions; large thin roofed bullae which spontaneously rupture and ooze yellow fluid, leaving scaly rim
Typically otherwise well	More likely to have systemic symptoms
	

- Complications:** sepsis (septic arthritis, otitis media, lymphadenitis, pneumonia), autoimmune associations (guttate psoriasis, scarlet fever, post-streptococcal glomerulonephritis).
- Management:** general measures (clear crust with saline or warm water; hot washing of clothing and bedding), antibiotics (topical/ oral/ IV), treat underlying carriage (e.g. intranasal Bactroban).
 - We may also treat family members for underlying carriage.**

Staphylococcal scalded skin syndrome

- Typically a **child <5 years old** with haematogenous spread of staphylococcal toxin. **Extremely painful.**
 - Can occur in neonates, beware outbreaks in neonatal nurseries
- It can look like Steven Johnson Syndrome (SJS), but remember, **SSSS has no mucosal involvement!**
- Widespread exfoliation of the epidermis.
- Begins with localised infection in eyes, ears or throat. The site of infection is typically not the eroded skin.
- Management: dressings to control pain, antibiotics and fluids



Streptococcal Skin Infections

Streptococcus is the **great chameleon**. It produces **deeper infections (e.g. boils, abscesses)** than Staphylococcus and can have multiple presentations. It also has a bunch of autoimmune sequelae e.g. PSGN and rheumatic fever.

Erysipelas: a more superficial form of cellulitis, affecting the upper dermis and subcutaneous lymphatics. Abrupt onset, systemic symptoms (fevers, chills, shivering) with sharply demarcated border +/- ‘orange peel’ appearance.

Can test with **Antistreptolysin O titre (ASOT)** and **anti-DNAse B**.

Erysipelas	Perianal streptococcal dermatitis	Streptococcal dactylitis	Scarlet fever
			

Viral Skin Infections

Some viruses cause a classic picture. Urticaria in children is most commonly caused by a virus.

Recall the **herpetic lesions**. They are always **punched out, grouped**.

- Herpes Simplex Virus (HSV): primarily gingivostomatitis then cold sores.
 - We worry about child sexual abuse if lesions are in the anogenital region.
- Herpes Zoster Virus (HZV): affects >2 dermatomes; post-herpetic neuralgia is rare in children.

Molluscum contagiosum

- Caused by the **molluscum pox** virus; one of the most common viral skin infections.
- Highly contagious, spread by close contacts, usually in the context of **warm water** (baths, swimming pools)
- Skin-coloured papules with umbilicated centres
- Can develop eczema around lesions (**molluscum dermatitis**)
- Lesions can become painful, red, and discharge if an immune reaction develops. Otherwise, the virus can be persistent (lives in the **core** which is surrounded by a protective membrane).
- Treatment aims to irritate the core to stimulate the immune system
- **Management:** can resolve spontaneously e.g. if young; cryotherapy; derof lesion with needle.
 - Other treatments include topical imiquimod (stimulates immune system), cantharidin (derived from blister beetle, forms a blister on the lesion)



Warts

- Caused by **human papillomavirus (HPV)**, spread via skin-to-skin transmission or fomites
- 2/3 spontaneously resolve within 2 years, though more stubborn than molluscum lesions
- Types of warts

Common wart (verruca vulgaris)	Plane wart	Filliform wart	Periungual wart	Plantar wart	Extensive warts (immunodeficient)
					

Treatment of warts is similar to molluscum – aimed at **destruction** (e.g. cryotherapy, laser, wart paint with salicylic acid).

Fungal Infections

Tinea ('Ringworm')

- Caused by fungi known as 'dermatophytes': may be zoonotic, geophilic or anthropophilic.
- Can affect hair, skin, and nails
- Ring-like, scaling rash with incomplete central clearance (may also be serpiginous)
 - **Annular lesions are tinea until proven otherwise → diagnose with skin scraping**
- Beware **tinea incognito**: clinical appearance of tinea altered by inappropriate treatment, usually topical steroids. The tinea slowly extends.
 - The steroid cream reduces inflammation, so the condition feels less irritable.
 - When the cream is stopped, the itch worsens, so the cream is reused. Vicious cycle.



- **Tinea corporis** is the specific name given to tinea of any part of the body excluding the hands and feet, scalp, face and beard, groin, and nails.
- **Two main organisms:** Microsporum (from animals), Tricophyton (animals and humans)
- Beware the rash that does not respond to steroids.



- Tricophyton tonsurans can cause tinea capitis (scalp), spread by close contact and living quarters.
- **Kerion:** painful pustular hypersensitivity reaction against dermatophyte. Can be mistaken for bacterial infection but does not improve with antibiotics. Children are **systemically well (unlike, for example, a bacterial abscess)**!
 - **Skin scrapings and hair plucks may establish diagnosis.**
 - Treatment: oral antifungals + low dose prednisolone



Pytiasis Versicolor

- More common in hot, humid climates and in people who perspire heavily.
- Hypopigmented scaly eruption caused by overgrowth of commensal yeast.
- Affects trunk, neck, and/or arms; patches appear paler in darker skin, and may otherwise be coppery brown or pink.
- Treated with topical azole cream/shampoo e.g. ketoconazole; oral antifungals if topical agents have failed.



Scabies

- Caused by mite, *S. scabiei*
- Obligate human host; mites form burrows under the skin that can be seen on skin scrapings
- **Appearance:** papules, nodules, vesicles, and burrows. Bumps look like insect bites (hypersensitivity reaction to mites).
 - Papules and pustules on the palms/soles in < 1 year old is scabies until proven otherwise.
 - Dermographism <1 year old is scabies until proven otherwise.
- **Common locations:** genital area, feet (transmitted from carers)
- Intense **pruritus** (infants become irritable); “hard to treat eczema” → **topical tar** helps
- Can develop **secondary impetigo**.
- The whole family needs to be treated at the same time.



Other Important Conditions

Dermatitis artefacta: self-inflicted lesions, often linked to psychological distress. Treatment requires a sensitive and multidisciplinary approach.

Trichotillomania (hair pulling): well-circumscribed hair loss, often on dominant side, hair of various lengths. Habit tic.

Fun Extra Notes: Viral and Bacterial Exanthema

An exanthem is an eruptive widespread rash. Historically, there were six **viral/bacterial exanthema** named First through to Sixth Disease. You might recall that Parvovirus B19 is still called "Fifth Disease"; the rest have been phased out.

Let's go back to the early 20th Century...

First Disease	Second Disease	Third Disease	Fourth Disease	Fifth Disease	Sixth Disease
Measles	Scarlet Fever	Rubella	Duke's Disease	Parvovirus B19	Roseola Infantum
			Obsolete class, non-specific viral rash		
Blanching erythematous macular rash with some confluent areas, classically starting behind the ears before spreading to the rest of the body. Koplik spots (grey-white spots on the buccal mucosa) are pathognomonic 	Diffuse blanching erythema with numerous papular elevations, giving a "sandpaper" quality to the skin; flushed face with perioral pallor; strawberry tongue. The rash usually begins in the groin and armpits, then covers the trunk and extremities → desquamation Due to exotoxin produced by Group A Streptococcus (streptococcus pyogenes)	Often mild or asymptomatic. Milder erythematous macular rash. Like measles, it starts on the face then spreads to the rest of the body. Other salient clinical features: low-grade fevers, generalised lymphadenopathy, arthralgia/arthritis	Doesn't actually exist	"Slapped cheek" disease Diffuse bright red rash on both cheeks followed by a reticular mildly erythematous rash affecting the trunk and limbs (can be raised and itchy)	Caused by HHV-6 Classically, high fever that comes on suddenly, lasts for 3-5 days, then disappears suddenly. The rash appears when the fever settles: mild erythematous macular rash across the arms, legs, trunk, and face; not itchy
Measles PCR (throat, urine, or nasopharyngeal swab) Measles IgM	Throat swab for Group A Streptococcus culture or RAT	Rubella IgM serology, PCR		Parvovirus B19 IgM, PCR If high-risk e.g. immunocompromised, pregnant, or haematological conditions	Rarely needed, PCR for HHV-6 if diagnosis unclear
Otitis media Pneumonia Encephalitis	Suppurative: peritonsillar/retropharyngeal abscess, otitis, sinusitis, mastoiditis, invasive GAS Non-suppurative: ARF, PSGN	Congenital rubella syndrome in exposed fetuses Arthralgia/ arthritis Thrombocytopaenia Encephalitis		Transient aplastic crisis, especially haemolytic anaemia Hydrops foetalis in pregnancy Arthropathy	Febrile seizures (from high fever) Rarely encephalitis or myocarditis

12. Endocrinology

Puberty

Adolescence is a time of physical, cognitive, emotional, and psychosocial maturation and change. **Puberty** is specifically the physical transitions that occur during adolescence.

Recall the 6 **arches**. They are listed below in chronological order. **The typical duration of puberty is 2-5 years.**

Pubertal Event	Description	Mean age (Aus)	Typical Age Range	Clinical Features
Adrenarche	↑ androgen production from adrenal cortex.		6 – 8 years	Early ↑ DHEA/DHEAS, subtle physical signs
Pubarche	Response to adrenarche		8 – 13 years	Pubic/ axillary hair, acne, body odour
Gonadarche	Activation of gonads by FSH/LH: ovaries → oestrogen; testes → testosterone Stimulated by pulsatile secretion of GnRH .	Males: 12 years (testicular volume 4 mL)	8 – 14 years (girls) 9 – 15 years (boys) May overlap with adrenarche/pubarche	In girls, ovarian enlargement. In boys, testicular enlargement.
Thelarche	Appearance of breast tissue (girls) due to ovarian oestrogen	10.5 years	8 – 13 years (girls)	Breast bud formation
Menarche	First menstrual bleed (girls), endometrial shedding	12.25 years	10 – 15 years (girls);	Menstrual bleeding, ovulation follows.
Spermarche	First sperm production (boys); testicular activation by FSH/LH	13.5 – 14 years	11 – 15 years (boys)	Sperm in urine, nocturnal emissions

Menarche and spermarche are the last major events in female and male puberty, respectively.

Delayed and precocious puberty are defined by gonadarche.

Hormone Changes Throughout Life

In utero, infancy, and early childhood:

- The foetal adrenal gland produces large amounts of **DHEA/DHEA-S**.
- Males will get a **testosterone surge** in utero, driving masculinization of the reproductive tract and external genitalia as well as sexually dimorphic brain areas.
- There is a **mini puberty** in infancy (0 – 6 months), owing to the removal of negative feedback from maternal oestrogens.
- Levels of **oestradiol** and **testosterone** are then undetectable until **adrenarche**. Note, suppression of sex steroids is more profound in males than females.

Control of Puberty and the HPG Axis:

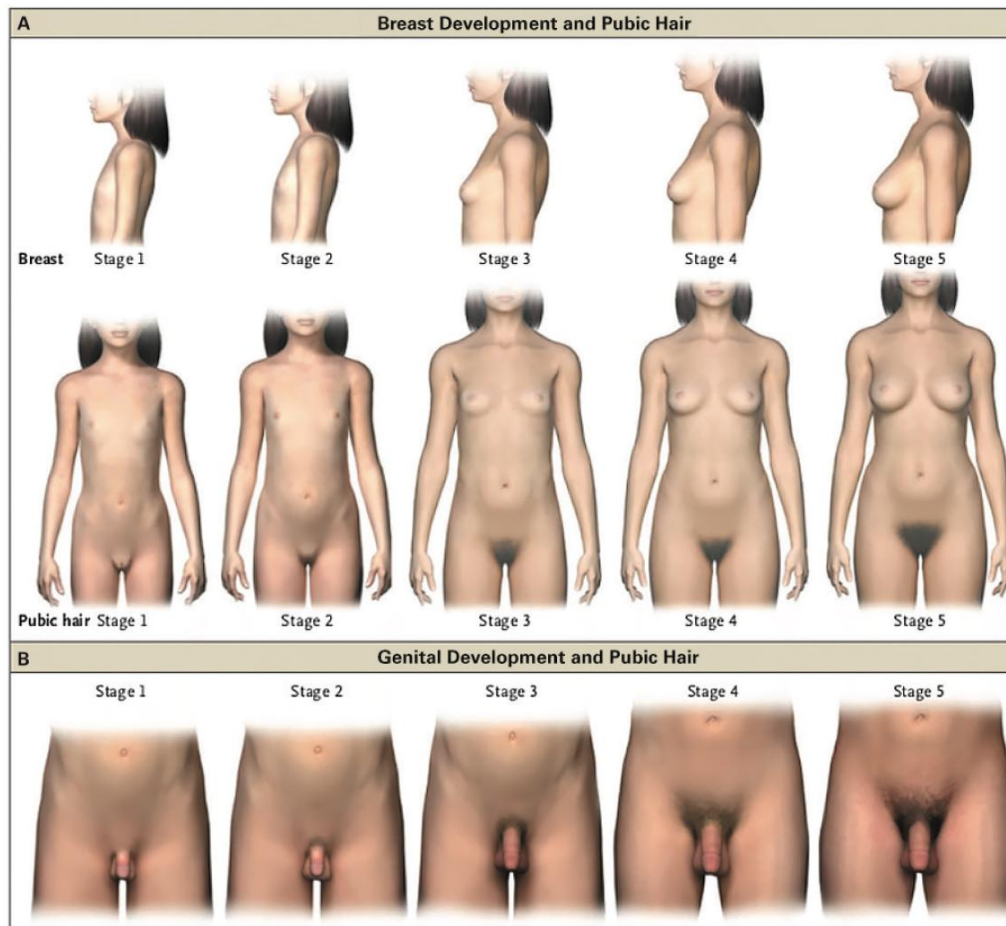
- **Higher centres:** nutrition, health, stress, exercise, genes (e.g. KISS-1/GPR54, GnRH, GnRHR, FGFR1, Leptin, LeptinR).
- GnRH pulses increase in amplitude from mid-childhood. At puberty, there is an ↑ in the **amplitude** and **pulsatility** of GnRH and, in turn, FSH/LH.
 - There is also a **circadian rhythm** e.g. **nocturnal LH surge**.

	Ovary	Testes
AMH	Produced by ovarian follicles; indicator of ovarian reserve.	Sertoli cell product that inhibits development of Mullerian structures.
Inhibin	From follicular cells; modifies FSH secretion.	From Sertoli cells; modifies FSH secretion
Insulin Like 3		Controls testicular descent

Assessing Physical Changes in Puberty

Once genitalia have become Tanner 4-5, and, in males, testicular volume is 15-20 mL, puberty is done!

Each change in puberty has an associated **Tanner staging system** (Stage 1 to 5). These can be plotted on puberty centiles. Note that testicular volume is usually assessed using a **Prader Orchidometer**.



	Girls	Boys
Methods of Assessment	Breast staging B1 – B5 Pubic hair staging P1 – P5	Genital staging G1 – G5 Pubic hair staging P1 – P5 Testicular volume (mL) or penis length
Timing of Peak Growth Spurt	11.5 years (early); after thelarche but before menses	13.5 years (late)
Peak growth velocity	6 – 10 cm per year	7 – 12 cm per year
First physical feature of puberty	Thelarche	Gonadarche (testicular volume 4 mL)
Menarche/ Spermarche	Menarche at 12 years Initial irregular periods, some anovulatory	Sperm production from 13.5 years
Other Changes	Psychological/emotional Somatic (breast, pubic hair) Skeletal (widening pelvis, carrying angle) Adipose tissue in female pattern Genital and uterine maturation Bone mineralization ¹	Psychological/ emotional Genital enlargement Increase in protein, decrease in fat Bone mineralization ¹ Later: continuing muscle development, facial hair, chest hair
Height and weight change	↑ 20 cm height, 20 kg weight 50% weight gain is in peri-menarcheal year ↑ lean body mass with relatively ↑ fat mass	↑ 30 cm height and 30 kg weight ↑ lean body mass and ↓ fat mass (testosterone) Maximum muscle gain is a late adolescent event

1: Close to peak bone mass (100 – 200% increase) is achieved by the end of adolescence. Inadequate increases in sex steroids (testosterone and oestrogen) plus any deficiency in GH, Ca, Vitamin D, and physical activity will inhibit optimal accumulation of bone mass. This can potentially ↑ the risk of osteoporosis and fractures.

Bone age can also be used to assess puberty stage. This involves an X-ray of the (non-dominant) hand/ wrist.

- Two grading systems where X-ray is compared with radiographic pictures: Greulich Pyle, Tanner Whitehouse

Growth Disorder in Adolescence

History	Examination
• Birth - growth in utero, birth weight, gestation, maternal factors (e.g. GDM, smoking, alcohol) • Neonatal period – floppiness, hypoglycaemia, feeding issues • Detailed past medical history – prolonged illness or hospitalisation, chronic or systemic illnesses, medications, developmental progress • Systems review to exclude systemic disease • Dietary history – malnutrition, anorexia nervosa • Family history – consanguineous parents, parental and sibling heights, pubertal timing in relatives	• General exam – include specific systems exams based on history, dysmorphism • Body proportions – arm span, upper limb and lower limb segments • Height • Weight on calibrated scales • Pubertal assessment – Tanner stage, Prader orchidometer for testicular volume ◦ Note: watch out for a buried penis in boys with high BMI. • Visual field examination, assessment of smell (assessing intracranial pathology)

Short Stature

RACGP - Growth disorders in adolescents

Height for age that is > 2 SD below the 50th centile for gender (i.e. $< 3^{\text{rd}}$ centile).

The most common causes are constitutional growth delay and familial short stature. In familial short stature, the child is short compared to their peers but appropriate according to the mid-parental height.

Constitutional delay of growth and puberty (CDGP), AKA constitutional growth delay

- **Definition:** absent breast development (Tanner Stage 2) in girls aged >13 years, and testicular volume <4 mL by age 14 in boys.
- **Aetiology:** often genetic; M:F 3:1 (but possibly referral bias).
- **Clinical features:** delayed bone age, normal gonadotrophins for bone age.
- **Management:** puberty usually occurs spontaneously; if experiencing emotional distress, may trial low-dose testosterone in boys or oestrogen in females (no clear guidelines in females).

Turner Syndrome (45 XO or Mosaic)

Common syndromic cause of short stature, incidence **1 in 2000**. Consider in any female with short stature or delayed/arrested puberty. May be addressed with GH therapy but outcomes vary.

Endocrine Causes

- Primary endocrine causes are uncommon in adolescence, usually $\uparrow$ weight fo height.
- **Acquired hypothyroidism** may occur secondary to autoimmune thyroiditis $\rightarrow$ lethargy, cold intolerance, constipation, poor school attention.
- **Cushing's syndrome** is usually exogenous (high-dose inhaled or systemic steroids)
- **Acquired growth hormone deficiency** should be considered in any child who has had an intracranial tumour, cranial radiation, or brain trauma.
 - Clinical features: proportional short stature, decline in growth velocity, intracranial Sx
 - Diagnosis: serum IGF-1 followed by provocation test
 - Treatment: recombinant human GH therapy

Non-Endocrine Causes

- Antenatal conditions e.g. extreme pre-term and IUGR/LBW have long-lasting effects on growth and final height
- Nutrition and extreme physical activity
- Chronic disease and/or inflammatory conditions e.g. IBD, CKD, coeliac
- Treatment side-effects e.g. oncology sequelae, steroids, recurrent blood-product transfusions, stimulant medications

Growth hormone treatment is tightly regulated on the PBS. The most common indication for GH is "short slow-growing". Note, if a young person has significant maturational delay, it might be more appropriate to induce puberty.

Delayed Puberty

Gonadal Causes (primary hypogonadism)	Pituitary causes (central, secondary)	Hypothalamic causes (central, tertiary)
Chromosomal (Turner 45 X0, Klinefelter 47 XXY) Trauma, torsion, infections e.g. mumps orchitis Radiation damage Chemotherapy damage Cryptorchidism ?COVID-19	Tumours (e.g. craniopharyngioma) Trauma Radiotherapy Genetic pituitary deficiency syndromes e.g. septo-optic dysplasia	Familial/constitutional delay (normal variant) Following sickness in childhood Malnutrition, excess exercise → negative energy balance Lesions interfering with GnRH (tumours, trauma, infections, radiotherapy etc) Genetic abnormalities of GnRH neurone migration Kallmann syndrome (+ anosmia)
High LH, FSH levels Low testosterone or oestradiol	Low LH, FSH, testosterone, oestradiol; does not respond to GnRH stimulation	Low FSH, LH, testosterone, oestradiol Rise in FSH and LH after GnRH stimulation

Investigations

- Bloods: FSH, LH, oestradiol or testosterone, TFT, coeliac serology, FBC, EUC, LFT
- Bone age x-ray
- Females with primary amenorrhoea:** may perform pelvic ultrasound.

Premature Thelarche

- Non-progressive breast growth with no other signs of puberty; exclude exposure to exogenous oestrogen e.g. mother's oestrogen gel.
- FSH, LH, oestradiol are normal for bone age.

Pubertal Gynaecomastia

- 80% of normal males in Tanner Genital Stage 3-4
- Clinical presentation:** tender breasts, unilateral or bilateral, varying size
- Pathophysiology:** changing aromatase activity with a local change in testosterone:oestrogen ratio, which promotes breast tissue development. *It's actual breast tissue.*
- Investigation:** USS shows breast tissue present
- Management:** surgery in 10%; usually resolves spontaneously

Note there are several causes of pathological gynaecomastia e.g. drugs (such as marijuana), sex-steroid secreting tumours, hypogonadism (e.g. Klinefelter syndrome).

Precocious Puberty and Other Changes

Recall the median age of onset of puberty in the developed world has fallen by 1 year over the last 50 years.

The mechanism is multifactorial and may include overweight & obesity; LBW + subsequent adequate nutrition; endocrine disruptor compounds (plasticisers); absent biological fathers with unrelated male in the household; overseas adoptions; blindness.

Premature Adrenarche

- Present before 8 (F) or 9 (M): pubic or axillary hair; acne, seborrhoea; increased body odour
- May have slight advancement of bone age, mild increase in DHEAS
 - Marked** discrepancy → consider **adrenal or gonadal pathology**.
- This is **not** precocious puberty! However, excess weight gain can drive early puberty.

Central Precocious Puberty

- Not early puberty, which is still within the normal range.
- Normal sequence of events, but much earlier.
- Aetiology:** usually idiopathic in girls (> 90%); CNS lesion in boys (75%)
- CPP has significant morbidity:** short stature in adulthood (early epiphyseal closure), psychosocial distress (varied evidence, but some examples are being out of step with peers, unable to cope with menstruation, etc).
- Evaluation:** confirm hormonal diagnosis; associated pathology; baseline for monitoring pubertal suppression.

- In boys with testicular volume > 4 mL before age 9, always order a brain MRI!

Treatment: highly individualized. Treat underlying cause if needed (e.g. CNS tumour); puberty suppression (GnRH agonists); peripheral hormone blockade.

Louisa Leone Notes

Diabetes in Children and Adolescents

Clinical Practice Guidelines : Diabetes Mellitus: new presentation

Remember to check blood glucose in most acutely unwell children, since diabetes can present non-specifically.

Classic symptoms: polydipsia, polyuria, new onset nocturnal enuresis, unexplained weight loss, unexplained fatigue.

- Other non-specific symptoms may indicate ketosis/ketoacidosis: vomiting, respiratory distress, dehydration, abdominal pain, altered consciousness

Most diabetes mellitus in children is **Type 1 diabetes mellitus**.

- It is typically diagnosed between age 4-6 years or between age 10-14 years
- Type 2 DM:** consider if obesity, acanthosis nigricans, strong family history. *On the rise.*
- Monogenic diabetes:** neonatal, autosomal dominant FHx of diabetes (MODY)
- DM may be associated with medications (e.g. glucocorticoids, tacrolimus) and other disease processes that affect the pancreas e.g. CF, hereditary haemochromatosis.

Diagnosis of diabetes: random BGL ≥ 11.1 mmol/L or fasting BGL ≥ 7.0 mmol/L **with symptoms**.

- If the child is **asymptomatic**, do not diagnose diabetes based on a single BGL reading.
- If the diagnosis is equivocal, we may order HbA1c, OGTT, ongoing monitoring of BGLs.

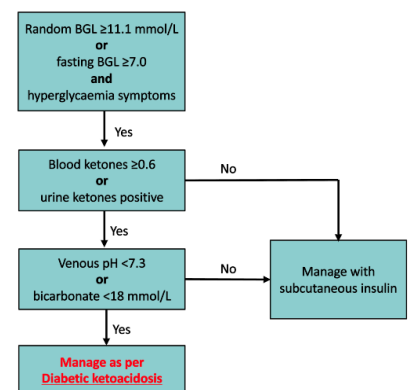
Examination: conscious state (AVPU/GCS), respiratory signs, assess hydration status, weight.

Investigations: formal serum BGL, point of care blood ketones if random BGL ≥ 11.1 mmol/L, VBG.

- Other investigations: EUC, diabetes autoantibodies (e.g. anti-GAD, IAA, anti-IA2, ZnT8), coeliac serology, TSH, FT4, insulin level, C-peptide, thyroid autoantibodies (thyroid peroxidase, thyroglobulin)

Subcutaneous Insulin (if not in DKA):

- The two most common regimens for insulin dosing are **fixed multiple daily injection (MDI)** and **flexible insulin dosing**.
 - In flexible insulin dosing, a basal insulin is given (evening), then rapid-acting insulin is calculated before each meal based on the meal composition and amount of CHO being consumed.



Diabetic Ketoacidosis

May be the initial presentation of T1DM.

Diagnosed with a triad of: serum glucose > 11 mmol/L, venous pH < 7.3 (or $\text{HCO}_3^- < 18$ mmol/L), ketonaemia (ketones ≥ 3 mmol/L) or ketonuria (ketones $\geq 2+$ on urinalysis)

- Euglycaemic ketoacidosis occurs when BGL < 14 mmol/L (*euglycaemic, obviously*). It can arise due to starvation, low carb diet, or off-label use of SGLT2 inhibitors
- Hyperglycaemic hyperosmolar state is more associated with T2DM – has **more severe hyperglycaemia** (BGL > 33.3 mmol/L), not acidotic/absent ketonaemia, **serum osmolality > 320 mOsm/kg**.

Precipitants: first presentation of T1DM, missed or inadequate insulin in known T1DM (e.g. pump failure), illness in known T1DM.

Clinical features: dehydration, tachypnoea (deep, Kussmaul respirations), nausea/ vomiting/ abdominal pain, confusion and drowsiness.

Assessment: assess severity, level of consciousness, precipitants, and complications

- Children with DKA are often dehydrated, but **acidosis can itself cause peripheral shutdown** $\rightarrow$ inaccuracy of dehydration estimate.
- The level of severity of DKA is determined by **venous pH** and **HCO_3^-**

Investigations

Insert IV cannula. Collect bloods.

- Serum glucose level
- VBG
- Point of care ketones
- EUC, CMP
- FBC
- Additional bloods for first presentation T1DM
 - Diabetes autoantibodies (anti-GAD, anti-IA2, zinc transporter 8)
 - Coeliac serology and IgA
 - TSH, FT4

Urinalysis: glucose, ketones, culture if clinical suspicion of UTI.

Management

There are four goals: (1) correct dehydration, (2) correct acidosis, ketosis, and BGL, (3) monitor for complications of DKA and its treatment (cerebral oedema, hypo/hyperkalaemia, hypoglycaemia), (4) identify and treat precipitant.

- ABC
- Nurse head up, monitor airway, consider NGT for aspiration risk
- Keep NBM until child is alert
- IV access x2
 - Commence rehydration fluids
 - Take bloods (above)
- Administer supplemental oxygen if severe circulatory impairment or shock
- Cardiac monitoring for ECG changes related to K^+
- Antibiotics if febrile and signs of infection
- Consider urinary catheter if unconscious for strict fluid balance

Mild DKA (pH 7.25 – 7.3, no altered sensorium, can tolerate oral fluids)	Oral rehydration, subcutaneous insulin, monitoring of electrolytes
Moderate-severe DKA	Fluid replacement (NaCl + KCl), insulin infusion, correction of electrolytes (IV potassium replacement), observation sheet (hourly BGL + blood ketones, 4 th hourly EUC/pH)

Correct Dehydration

- Bolus **10 mL/kg 0.9% NaCl** if tachycardic with delayed central capillary refill
- Use fluid containing **40 mmol/L potassium chloride** if the serum potassium ≤ 5.5 mmol/L and the child is passing urine.

Glucose	Potassium	Sodium	Phosphate
BGL decreases rapidly during initial rehydration and continues falling once insulin started. Once BGL ≤ 15 mmol/L, change fluids to 0.9% NaCl + 5% glucose + KCl Aim for BGL 5-10 mmol/L Increase glucose to 10% before adjusting insulin infusion rate	Total body K^+ deficit, but K may be normal, high, or low Insulin causes rapid intracellular shift of K^+ If hypokalaemic, replace K^+ before commencing insulin	Calculate corrected sodium (dilutional effect of hyperglycaemia) If extremes of corrected Na, discuss with senior clinician	Acidaemia → hypophosphataemia Severe hypophosphataemia is uncommon during treatment – if < 0.32 mmol/L, may consider IV PO ₄ replacement

We can see that **potassium is a real problem in DKA**. Vomiting leads to **K loss**. However, recall that insulin usually stimulates the Na^+/K^+ -ATPase to drive **Na^+ out and K^+ in**. Therefore, insulin deficiency results in **extracellular K^+ redistribution**. This is exacerbated by **acidosis** where H^+ and K^+ are exchanged, driving more K^+ out of the cell. Ultimately, K can be high, low, or normal.

Reverse ketosis and correct acidosis

- **Insulin** infusion **one hour** after rehydration.
- The insulin infusion consists of **rapid acting insulin added to 0.9% NaCl**

Monitor for complications of DKA and its treatment

Complications: cerebral injury, AKI, hypoglycaemia, hypo/hypernatraemia, hypo/hyperkalaemia, hypophosphataemia

Cerebral injury

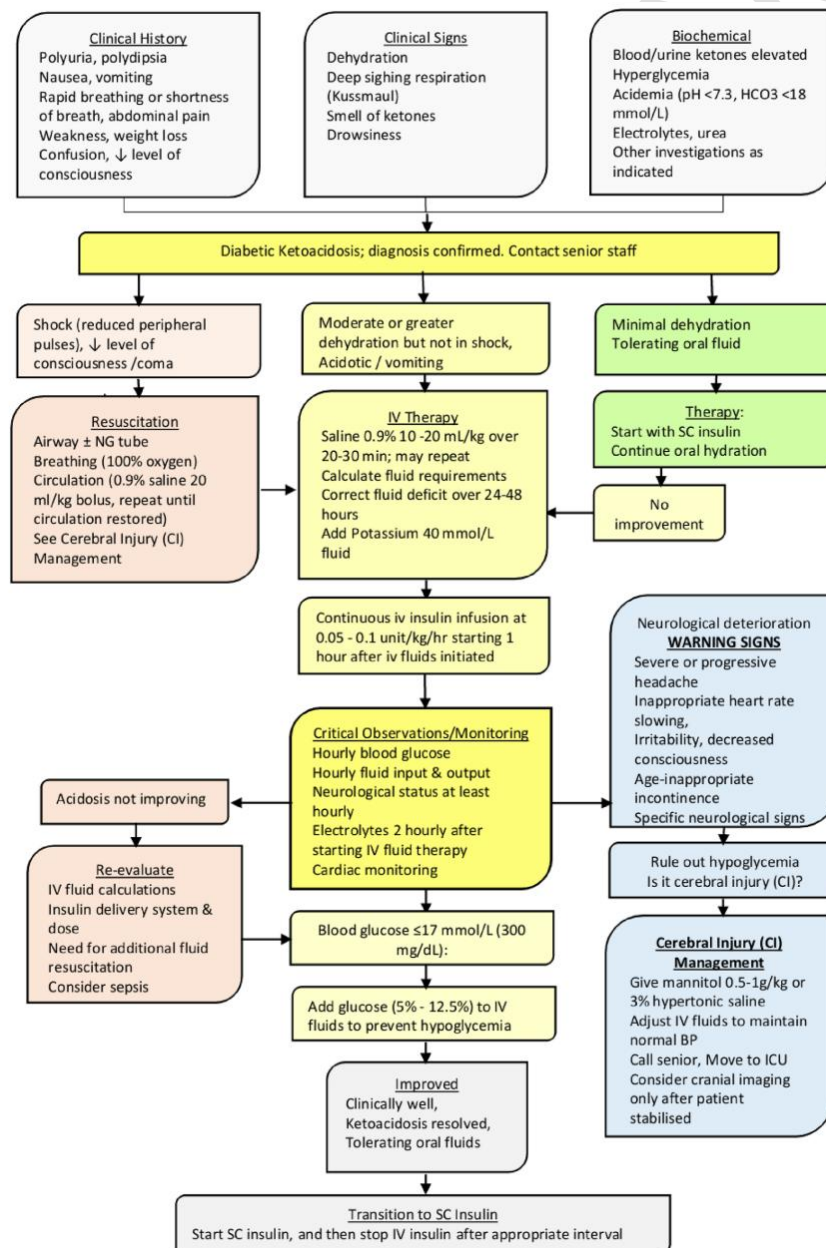
- **Pathophysiology:** Ischaemia-reperfusion injury - hyperventilation ($\downarrow$ CO₂) + dehydration $\rightarrow$ cerebral hypoperfusion and vasoconstriction
 - Osmotic shifts and oedema
- **Risk factors** include severe hypocapnia, severe acidaemia, increased urea
- **Clinical diagnosis:** headache, irritability, lethargy, worsening vomiting; depressed consciousness, incontinence, Cushing triad
- **Management:** Nurse head up, reduce infusion rate by 1/3, hyperosmolar therapy if clinically significant cerebral injury (mannitol or hypertonic saline)

Hypoglycaemia

If BGL < 4.0 mmol/L

- **10% glucose IV bolus 2 mL/kg**, and change rehydration fluids to include 10% glucose + KCl
- Do **not** discontinue insulin! Only decrease insulin if hypoglycaemia refractory to 10% glucose treatment.
- Oral treatment (high GI carbohydrates) if tolerated: 5g if < 5 years (<25 kg), 10g if > 5 years (or >25 kg)

Ongoing Monitoring: hourly fluid balance, vital signs, neurological observations. Perform hourly point of care ketones and hourly blood glucose measurements (alternating between VBG and BGL). 2-hourly EUC, CMP.



Diabetes Sick Day Management

ISPAD Chapter 13: Sick day management in children and adolescents with diabetes

Ketosis prevention:

- Never completely stop insulin!
- Monitor glucose and ketones every 1-2 hours
- Blood ketones preferred to urine ketones
- Adjust insulin doses based on glucose and ketone levels
- Maintain hydration

Vomiting and/or GI illness:

- Nausea and vomiting are signs of insulin deficiency and ketosis until proven otherwise.
- If hypoglycaemic but normal ketones, decrease insulin dose.
- Hypoglycaemia + hyperketonaemia – give insulin + carbohydrates.

Management where ketosis is suspected or confirmed:

- Liquids containing water and electrolytes every 5-10 minutes, plus carbohydrate-containing liquids if BGL < 14
- Frequent extra doses of short-acting insulin
- Medical review if signs of ketoacidosis developing / glucose continues to rise despite extra insulin

BGL can be monitored using CGM. Those with an insulin pump can keep using it, but it is preferable to switch to open loop or manual mode.

Adrenal Insufficiency

Clinical Practice Guidelines : Adrenal crisis and acute adrenal insufficiency

- An **adrenal crisis** usually presents in children with known adrenal insufficiency who develop an intercurrent illness or injury.
- Unexplained severe dehydration or shock may be a **new presentation** of adrenal insufficiency.
- **Biochemical triad:** low sodium, high potassium, low glucose.
- **Pillars of treatment:** fluid resuscitation, steroid replacement, management of glucose and potassium levels.

Classification

Primary adrenal insufficiency	Secondary adrenal insufficiency
Disease intrinsic to adrenal cortex	Lack of ACTH stimulation of cortisol production
Commonest: classic congenital adrenal hyperplasia	Hypopituitarism
Other: adrenal haemorrhage, autoimmunity, infection, peroxisomal defects (e.g. X-linked adrenoleukodystrophy), adrenal hypoplasia	Abrupt cessation of steroids following prolonged high-dose glucocorticoid therapy (at least 2-4 weeks in last 12 months)

Clinical Features of Adrenal Insufficiency

We may separately consider signs owing to glucocorticoid, mineralocorticoid, and adrenal androgen deficiency. Primary adrenal insufficiency is also associated with **hyperpigmentation**, particularly in high-pressure areas (e.g. palmar creases, axillae, gingival borders). This is because ACTH and melanocyte stimulating hormone (MSH) share a common precursor, POMC.

Note, patients with secondary (central) adrenal insufficiency do not have mineralocorticoid deficiency (because most aldosterone is released via RAAS) or ACTH-related hyperpigmentation.

Glucocorticoid deficiency	Mineralocorticoid deficiency	Adrenal androgen deficiency*
GI symptoms (nausea, vomiting) Fatigue, weakness Weight loss, anorexia Morning headache Fasting hypoglycaemia Decreased gastric acidity Decreased free water clearance (hyponatraemia) Decreased height velocity	Hypotension, dizziness Muscle weakness Lethargy Nausea, vomiting, anorexia Salt craving Weight loss Dehydration Hyponatraemia, hyperkalaemia, metabolic acidosis	Decreased pubic and axillary hair if pubertal +/- decreased libido Typically presents in females because in males, most androgens are produced by the testes, not the adrenal glands

*Note, in congenital adrenal hyperplasia, there is **excessive** production of adrenal androgens. This is because impaired steroidogenesis leads to excess ACTH signalling, and the steroidogenic pathway is shunted towards the production of adrenal androgens.

When do I consider an acute adrenal crisis?

1. Neonates with atypical genitalia and/or electrolyte abnormalities, lethargy, hypoglycaemia, hypotension or shock, or failure to thrive (**may be presentation of CAH due to 21-hydroxylase deficiency**)
2. Any patient with known disorders of adrenal insufficiency exposed to stress e.g. illness
3. Patients on chronic corticosteroid treatment, especially if exposed to stress or during tapering phase
4. Patients with other autoimmune endocrine deficiencies
5. Critically ill patients with septic shock who are unresponsive to fluids and inotropic support (may have **bilateral adrenal haemorrhage**)
6. Patients with hyperpigmentation or vitiligo

Signs and Symptoms of Adrenal Crisis

- Vomiting and diarrhoea, sometimes with severe abdominal pain or unexplained fever, weight loss, and anorexia
- Serum electrolyte abnormalities: hyponatraemia +/- hyperkalaemia, metabolic acidosis, hypoglycaemia (especially in young children)
- Hypotension or shock, particularly if disproportionate to apparent underlying illness

Neurological symptoms/signs e.g. confusion, seizure, reduced GCS, are especially concerning.

Investigations

- Blood glucose (bedside and formal), will be low in cortisol deficiency
- EUC (hyperkalaemia and hyponatraemia, especially in primary adrenal insufficiency)
- Blood gas (acidosis indicates mineralocorticoid deficiency)
- Additional investigations if first presentation: cortisol, ACTH, 17-hydroxyprogesterone, aldosterone, plasma renin, urinary steroids/ sodium

Management

Children with known adrenal insufficiency should have a known sick day management plan – follow this!

Severely unwell/ in crisis

4 S's (slightly dodgy mnemonic):

Steroids	Stress glucocorticoids. IV/IM hydrocortisone, then hydrocortisone every 6 hours IV • High-dose hydrocortisone will have mineralocorticoid activity, so mineralocorticoid replacement (e.g. fludrocortisone) is not usually needed initially • Once stable, switch to oral hydrocortisone
Shock	10 mg/kg IV bolus 0.9% sodium chloride Replace remaining deficit + maintenance fluid requirements evenly over 24 hours with 0.9% NaCl and 5% glucose IV If mild/no dehydration, no bolus (obviously), give 1-1.5x maintenance fluid volume as 0.9% NaCl + 5% glucose IV administered over 24 hours Remember to check electrolytes and glucose (blood gas or formal glucose) regularly.
Sugars	Bolus 10% glucose 2-5 mL/kg IV, recheck glucose levels 15 minutes post-treatment and at 30 minutes. Ensure glucose > 4.0 mmol/L.
PotaSsium	Usually normalizes with fluid and hydrocortisone/fludrocortisone replacement. • If K > 5.5 mmol/L – ECG and cardiac monitoring • K > 7.0 and abnormal ECG – nebulized salbutamol and insulin/glucose infusion

Prevention of adrenal crisis

- Moderately unwell: increase fluid and carbohydrate intake, hydrocortisone 30 mg/m²/day
- Mildly unwell: increase dose of hydrocortisone to 20 mg/m²/day

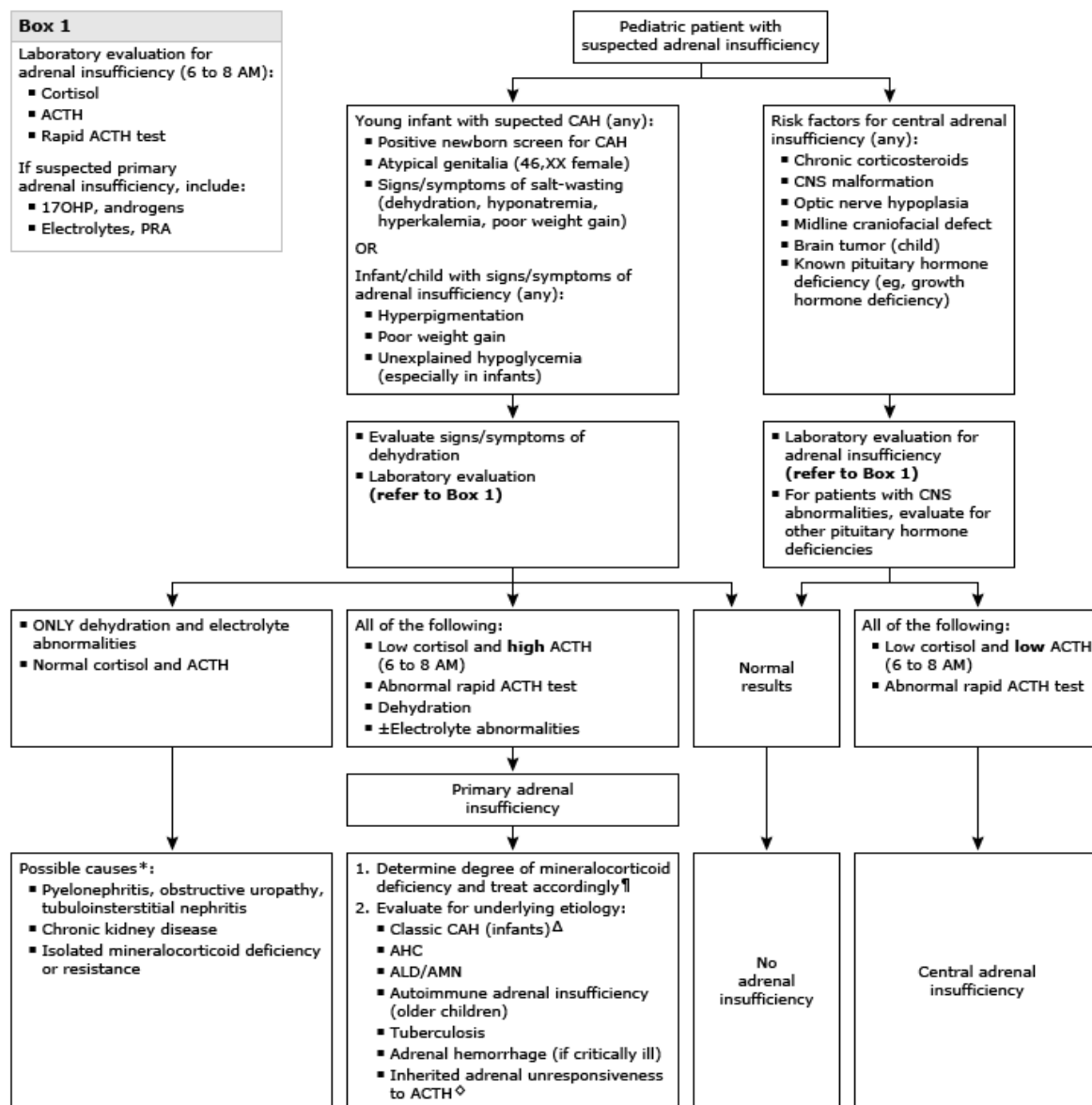
Perioperative management: children with adrenal insufficiency might need 'stress dosing' of IV hydrocortisone, typically at induction of anaesthesia. If the child requires emergency surgery, hydrocortisone is given stat.

Maintenance: oral hydrocortisone, oral fludrocortisone.

Note, while hydrocortisone has mineralocorticoid cover in the acute period, fludrocortisone (mineralocorticoid replacement) is needed chronically.

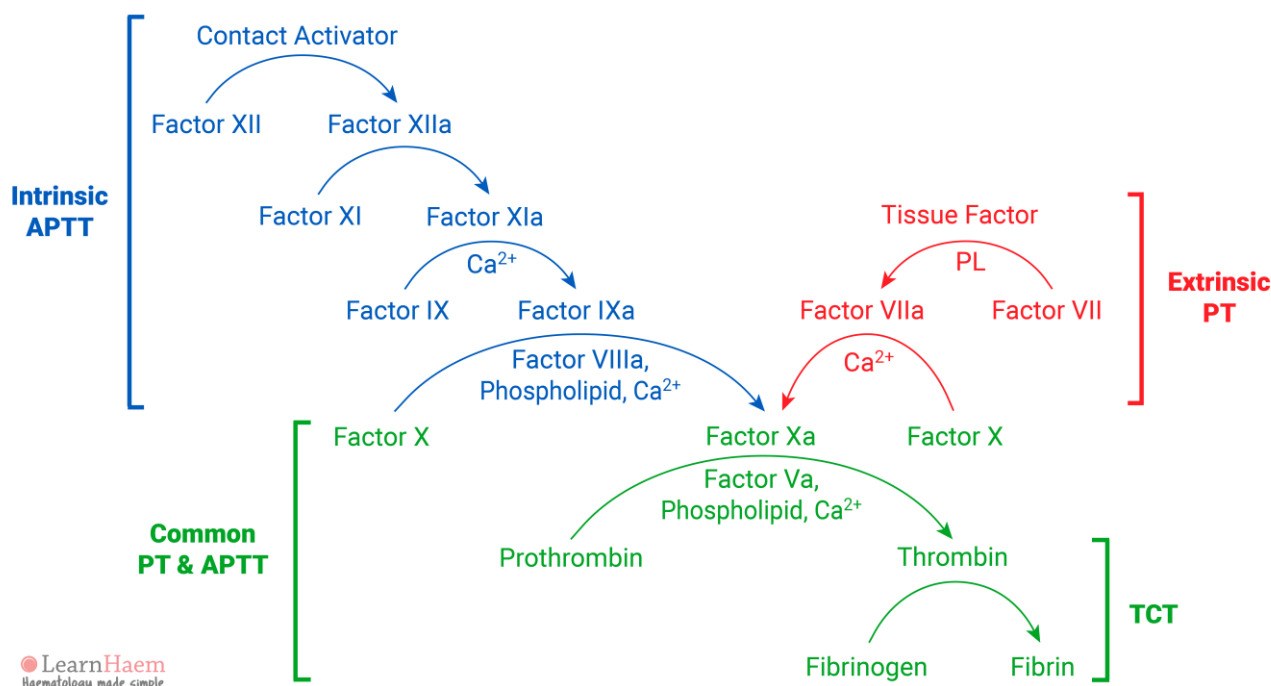
Supplemental Flowchart (UpToDate) – Diagnosis of adrenal insufficiency in children

Adrenal insufficiency in children: Clinical manifestations and diagnosis - UpToDate



13. Haematology

Coagulation Studies: Review



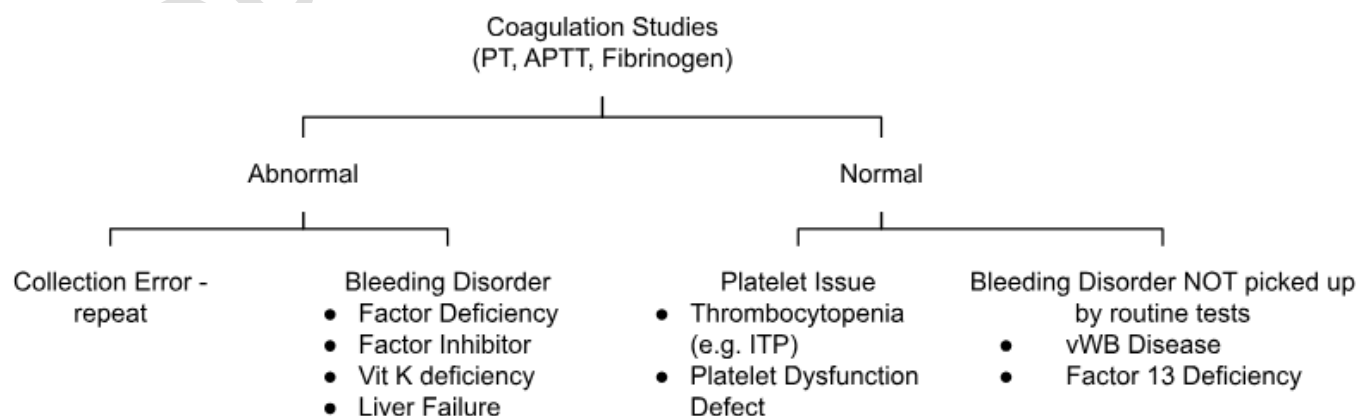
The standard coagulation studies are **PT (INR)** and **aPTT**.

aPTT	PT
Intrinsic pathway (table tennis)	Extrinsic pathway (tennis)
Also tests for common pathway – X, V, prothrombin, fibrinogen	
4 letters – 12, 11, 9, 8	2 letters – tissue factor (3), 7
Longer cascade – 45 seconds	Shorter cascade – 15 seconds
Heparin monitoring Haemophilis (VIII and IX)	Warfarin monitoring (via INR)
Activator is added	Heparinase added – less sensitive to heparin

Factor XIII cross-links fibrin to inhibitors of fibrinolysis (e.g. α_2 -antiplasmin), making the clot resistant to breakdown.

If coagulation studies are abnormal, a **mixing study** may help determine if there is a factor deficiency or inhibitor. A deficiency will be corrected with mixing; an inhibitor will not (antibodies will inhibit the exogenous factors).

There are also **specific clotting tests** e.g. vWF, or clotting factor assays, e.g. FVIII, FXIII.



Red Blood Cell Problems

Anaemia	Polycythaemia
Production problem: - Bone marrow - Failure (aplastic anaemia) with pancytopenia - Infiltration i.e. leukaemia - Isolated issue e.g. red cell aplasia (Blackfan Diamond Anaemia, transient erythroblastopenia of childhood), - Haemoglobinopathy i.e. thalassaemia, sickle cell disease - Deficiencies i.e. iron, B12, EPO (renal disease)	Most common is factitious (haemoconcentration due to dehydration) Production issue (rare in children) Polycythaemia rubra vera Myeloproliferative disease
Destruction/loss - Haemolysis - Bleeding - Splenic pooling	Secondary – Hypoxia - High altitude - Chronic heart or lung disease - Erythropoietin receptor mutations

Symptoms

Children who are chronically anaemic may be relatively well/ asymptomatic

- Pallor of lips, mucous membranes, conjunctivae
- Fatigue
- Failure to thrive
- Babies: feeding difficulties, hypersomnolence, poor suck
- Headaches
- Acute:** tachycardia, hypotension, lethargy, unsettled, SOB, chest pain
- Severe anaemia can lead to high output heart failure

Associated Features

Blood loss	- Signs of bleeding: bleeding, bruising, petechiae - Sources of bleeding: haematuria, melaena, epistaxis, menorrhagia - Cow's milk protein enteropathy: hypoalbuminaemia/ oedema - Thrombocytopenia, known bleeding disorder
Infection	Fever, viral infection, neutropenia (bone marrow failure, leukaemia)
Growth	Dietary history (selective eating/deficiencies), malabsorption (chronic diarrhoea, IBD)
Jaundice	Haemolysis
Red/brown urine	Haemoglobinuria seen in intravascular haemolysis i.e. G6PD deficiency, Malaria
Splenomegaly/ hepatomegaly	Haemoglobinopathies, viral infections, leukaemia

Causes of Anaemia by Age

Recall that there is a physiological Hb nadir at 6-8 weeks old. This is multifactorial: increased oxygen relative to the intrauterine environment ↓ EPO; increased haemolysis (transitioning from HbF to HbA); haemodilution due to a rapid expansion of blood volume.

Newborns/ early infancy	Older infants and toddlers	Older children and adolescents
Production Congenital anaemias Suppression from infections e.g. CMV, TORCH, drugs Haemolysis Alloimmune haemolysis e.g. Rh, ABO, or other RBC antibodies Inherited haemolytic anaemias e.g. hereditary spherocytosis, G6PD deficiency, PK deficiency Blood loss Foetomaternal haemorrhage, other bleeding, excessive blood tests Anaemia of prematurity (multifactorial)	Lack of production Iron deficiency anaemia, Vitamin B12 deficiency (poor diet, excess cow's milk, blood loss, malabsorption) Later presentation of congenital anaemia Transient erythroblastopenia of childhood (TEC) ¹ BM disorders Secondary to acute or chronic conditions Haemoglobinopathies Beta thalassaemia/ sickle cell disease present around 3-6 months of age Alpha thalassaemia intermedia Haemolysis Congenital haemolytic disorders: HS, G6PD deficiency, PK deficiency Autoimmune haemolytic anaemia: DAT/Coombs positive Blood loss	Acquired more common. Iron deficiency anaemia Dietary or malabsorption e.g. coeliac disease Blood loss (menorrhagia, IBD, Helicobacter pylori) Scurvy (severe Vitamin C deficiency) - Children with ASD and severe dietary restrictions are at risk - Vessel fragility and bleeding Anaemia in the context of acute or chronic conditions

1: TEC usually occurs secondary to viral infection and is typically self-resolving.

Initial Investigations

FBC and film

- Check other cell lines (platelets, WCC esp neutrophils)
- MCV (smaller in younger children, excluding neonates)
- **Film** is not always available immediately; scientist review and haematologist review

Reticulocyte count (compensatory)

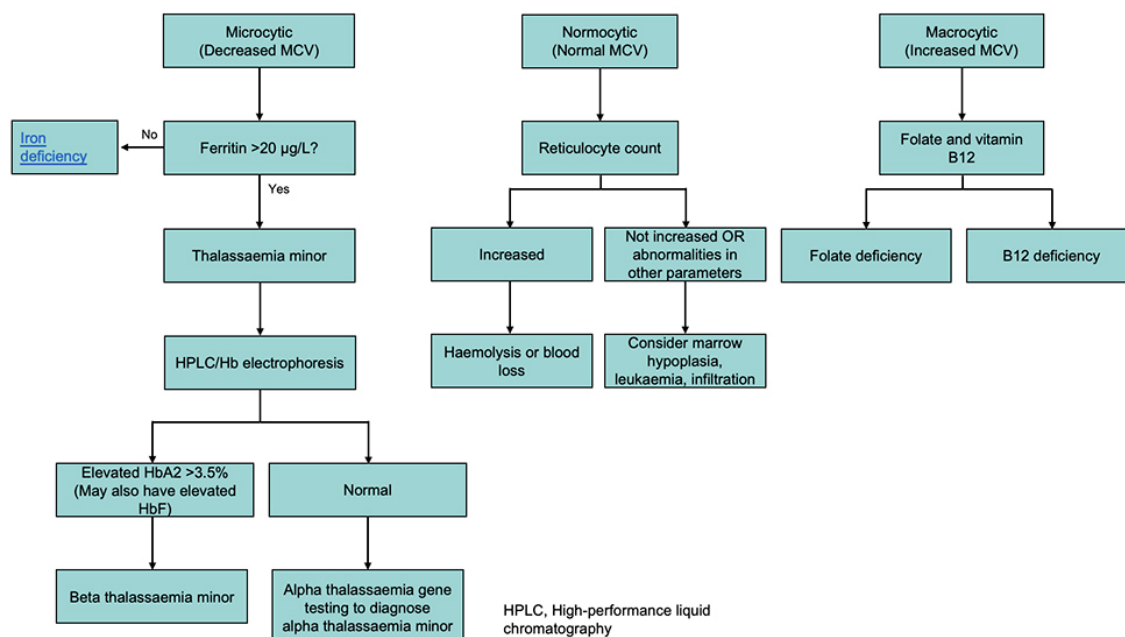
- Need to ask specifically

- Reticulocyte response may take 1-2 days
- Regenerative anaemia – high reticulocytes (polychromasia)
- Non-regenerative anaemia – low reticulocytes
- Bigger than RBCs and therefore may → **macrocytosis** (e.g. haemolysis, blood loss)

- **Ferritin (for iron stores)**. Beware false rise in context of inflammation (acute phase reactant)!

There are four critical parameters: MCV, reticulocytes, the “other” cells, film review.

Anaemia is initially classified according to MCV.



Select Conditions on Blood Film

Iron deficiency anaemia	Thalassaemia	G6PD Deficiency	Sickle Cell Disease
Hypochromic microcytic anaemia with teardrop and pencil cells.	Hypochromic microcytic anaemia, nucleated RBCs, polychromasia, tear drop and target cells. Poikilocytosis.	Bite cells and blister cells (oxidative haemolysis)	Sickle cells, Howell-Jolly bodies (RBCs with basophilic nuclear remnants, hyposplenism)

Additional Investigations in Anaemia

- Reticulocytes + film
- Haematinics (ferritin, B12, folate)
- Hb-EPG for haemoglobinopathy
- Red cell enzyme assays: G6PD assay, PK assay
- Flow cytometry for red cell EMA: Hereditary spherocytosis
- Red cell ADA for Diamond Blackfan Anaemia (adenosine deaminase)

- Bone marrow biopsy
 - Unexplained anaemia with low reticulocytes

For haemolysis:

- Bilirubin + LDH
- Haptoglobin (↓ but unreliable in kids)
- Coombs test/DAT (autoimmune haemolytic anaemia)

Red Cell Disorder	Classic Presentation
Iron Deficiency Anaemia	Pallor, lethargy, and sometimes pica in toddlers or adolescents with a history of dietary insufficiency or chronic blood loss (e.g., heavy menses). May have tachycardia and glossitis.
Sickle Cell Disease	Infant or young child with recurrent painful vaso-occlusive crises, dactylitis, chronic anaemia, and risk of infections; often has family history and ethnic predisposition (e.g., African, Mediterranean).
Thalassemia Major (Cooley's)	Infant with severe pallor, failure to thrive, hepatosplenomegaly, and frontal bossing due to marrow expansion; requires regular transfusions and shows characteristic microcytic hypochromic anaemia.
Hereditary Spherocytosis	Characterized by haemolytic anaemia, jaundice, splenomegaly, and often a positive family history; typically presents in childhood with anaemia and episodic jaundice triggered by infections or stress.
Glucose-6-Phosphate Dehydrogenase Deficiency (G6PD)	Acute haemolytic anaemia triggered by infections, certain drugs, or fava beans, often in males of Mediterranean, African, or Asian descent; episodic jaundice and dark urine.
Diamond-Blackfan Anaemia	Infant with severe macrocytic anaemia, congenital anomalies (such as craniofacial malformations or thumb abnormalities), and reticulocytopenia without splenomegaly.
Aplastic Anaemia	Pallor, fatigue, bleeding, and infections due to pancytopenia; no splenomegaly; often idiopathic or secondary to drugs, infections, or autoimmune causes in children.
Autoimmune Haemolytic Anaemia	Sudden onset pallor, jaundice, dark urine, and splenomegaly in older children; often follows infection or is associated with autoimmune disorders.
Transient Erythroblastopaenia of Childhood	Otherwise well child with recent viral infection that looks a bit pale. Isolated anaemia and low reticulocyte count. Recovers spontaneously in a few days.

Iron Deficiency Anaemia

Clinical Practice Guidelines : Iron deficiency

Iron deficiency is the most common cause of anaemia in children. Iron deficiency in infants and children is typically a nutritional disorder (e.g. excessive cow milk, insufficient iron-rich foods); malabsorption and blood loss become more of an issue in adolescents.

- **Clinical features:** impaired cognitive function (memory, learning, concentration), behavioural disturbance, fatigue, pica
- **Investigations:** serum ferritin; FBC needed to diagnose iron deficiency anaemia (microcytic, hypochromic)
- **Management:** dietary advice, oral iron supplementation. Only consider IV iron in select circumstances, e.g. oral iron contraindicated or not absorbed, severe ongoing blood loss, refractory to oral medications

Bleeding Disorders

Platelets	Coagulation proteins	Fragile vessels or vasculitis	Trauma
Thrombocytopenia Production (bone marrow) Destruction (ITP, TTP)	Factor deficiencies e.g. haemophilia, Von Willebrand's disease	HSP, Vitamin C deficiency Connective tissue disorders, rare genetic vessel disorders	NAI Factitious
Platelet function disorder			

1° Haemostasis	2° Haemostasis
Platelets, vWF form platelet plug	Coagulation factors form fibrin thrombus
Platelet number (thrombocytopenia) • Decreased production - bone marrow failure • Increased destruction - ITP or TTP, DIC Platelet Function Disorder • Inherited e.g. membrane defects • Acquired e.g. Myelodysplasia, Myeloproliferative disorder	Clotting factor deficiency • Haemophilia • Liver disease • Vitamin K deficiency Clotting factor increased destruction • DIC, inhibitors, transfusion, fibrinolysis
Test via: • Platelet Count - platelet number • Bleeding Time - platelet function • PLT Aggregometry	Test via: • PT: extrinsic pathway • aPTT: intrinsic pathway • TT • Coagulation factor assays
Mucocutaneous (superficial) bleeding	Anatomical (deep tissue) bleeding

- Cuts bleed Mucosal: Epistaxis, menorrhagia, gums Petechiae, purpura, bruises, ecchymosis	Haemarthrosis - Muscle hematoma - Brain haemorrhage
Immediate, mild bleeding after surgery	Delayed, severe bleeding after surgery

Bleeding History

A "lifecycle" approach can be helpful. We ought to establish whether the child was always "bleedy" e.g. haemophilia, or whether the bleeding is new e.g. ITP.

- Site (where), cause (spontaneous or reason) and duration
 - "Specific" challenges include epistaxis, joint bleeds, muscle bleeds, vaccinations, dental extractions, surgeries, injuries, menorrhagia, and GIT bleeding
- Non-blanching spots: petechiae (≤ 2 mm), purpura (> 2 mm)
- Bruises, haematomas, ecchymosis
- Active bleeding

Unusual bleeding

- Bruises in unusual locations (chest/back)
- Large bruises > 20 cent coin
- Spontaneous bruises
- Large numbers
- Consider age (neonate vs mobile toddler)
- Muscle bleed
- Joint bleed

Investigations

- FBC and blood film
- Coagulation studies (APTT, PT/INR)
- Other: fibrinogen, D-Dimers, TT, specific clotting tests e.g. vWF or clotting factor assays (e.g. FVIII) +/- mixing studies
- EUCs/LFTs

Remember that if the numbers don't make sense, it's likely a lab error (e.g. clotted sample, tube underfilled, wrong tube).

Classic Presentations

Diagnosis	Presentation
Immune thrombocytopenic purpura (ITP)	Autoimmune destruction of platelets; thrombocytopenia with size variation Sudden onset bruises and petechiae in a previously well child Normal coagulation studies
Haemophilia	Boy with long-standing excessive bruising since they were mobile with haemarthrosis; family history of "bleeding disorder"
Dysfibrinogenaemia	History of bleeding (easy bruising etc) and thrombosis (e.g. DVT, PE)
Factor XIII deficiency	Bleeding from umbilical stump and delayed bleeding
vWF disease, platelet function disorder	Mucocutaneous bleeding e.g. epistaxis, menorrhagia
Non-accidental injury	History not consistent with bruising

WBC Disorders

Remember that WCC varies with age. Always check whether just one cell line is affected or if it is a bicytopenia or pancytopenia.

Leukopenia

Most commonly neutropaenia or lymphopaenia.

	Neutropaenia	Lymphopaenia
Causes	Genetic: severe congenital neutropaenia, non-severe genetic neutropaenia (including Benign Ethnic Neutropaenia), metabolic Acquired: autoimmune neutropaenia (including alloimmune), bone marrow suppression/ infiltration (viral, drugs, aplastic anaemia, neutropaenia) Viruses are a common cause of transient neutropaenia	Viral infections (most commonly), immunodeficiencies, autoimmune disorders (e.g. SLE) Bone marrow suppression/ infiltration
Clinical Features	- Bacterial and fungal infections - Delayed healing of mucous membranes	Multiple, mild viral infections

	• Omphalitis, delayed umbilical cord separation • Recurrent mouth ulcers, recurrent or unusual bacterial infections e.g. boils, pimples, recurrent otitis, nailbed infections, abscesses (may be without pus) • Gingivitis, swelling of gums • Sepsis	Consider primary immunodeficiency or dysregulation, specially SLE
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Chronic Isolated Neutropaenia in Children

- Severe congenital neutropaenia is uncommon: autosomal dominant, severe, early-onset symptoms, needs G-CSF lifelong
- Cyclical neutropaenia: periodic mouth ulcers, infections and/or fevers, family history; strict pattern; needs investigation 2-3x per week for 6 weeks
- Benign chronic neutropaenia (acquired): thought to be autoimmune. Can be an incidental finding despite severe neutropaenia; spectrum of severity from asymptomatic to bacterial infections/sepsis. Usually resolving by 4-5 years, some may need G-CSF for some time.
- Benign ethnic neutropaenia: not a disease, just an artefact of our reference ranges being derived from Caucasian populations

Leukocytosis

- **Neutrophilia** is almost exclusively reactive e.g. bacterial and fungal infections or stress reactions. Neutrophilic leukaemia is very rare (nay, impossible) in children.
 - May have "toxic changes" e.g. granulation and vacuolation in serious bacterial infection, plus a left shift
- **Lymphocytosis**: commonly reactive to some viral infections (but lymphopaenia is more common), especially CMV, EBV, and pertussis. Immunological conditions e.g. atopia, immunodeficiencies, eczema can also cause.
 - **Beware!** Some analysers read leukaemic blasts as high lymphocytes, often accompanied by anaemia and thrombocytopaenia (requires film review).
 - **"Atypical" ("reactive")** lymphocytes can look similar to leukaemic blasts, but have more abundant cytoplasm, no nucleolus, broader morphologic diversity
- **Monocytosis**: usually reactive
- **Eosinophilia**: usually reactive (parasites, worms, drugs, atopia/eczema, asthma); primary hypereosinophilia is possible albeit very rare

Abnormal WBC Function

- Chronic granulomatous disease
- Neutrophil adhesion disorders
- Agammaglobulinaemia immunodeficiencies

The Lump

Reactive lymphadenopathy	Concerning lymph nodes
Small, soft, mobile cervical/inguinal nodes common May enlarge with infections, then recede May be tender Normal FBC, ultrasound (not always needed) shows reactive features	Large, widespread, hard and progressive nodes with no obvious infections May be associated with hepatosplenomegaly or abnormal FBC; bone pain/limp; unwell child with systemic/ constitutional features Any lymph node outside of the cervical or inguinal areas Accompanying mediastinal mass

Other causes of lumps include viral infections, unusual infections in regional drainage areas (e.g. bad/infected eczema, severe caries, head lice, infected earrings, chronic otitis), TB/atypical mycobacteria, immune

- A lymph node that is concerning (particularly for malignancy or TB) requires biopsy

White cell disorder tests

- FBC and blood film (may confirm but not exclude leukaemia)
- Infectious screening
- Bone marrow biopsy
 - If exclusion of leukaemia needed
 - Concerns about BM failure
 - Unexplained single or multiple cytopenia +/- elevated MCV
- Tissue biopsy (lumps / lymph nodes)
- Flow cytometry (less useful than adults)
- Immunology testing e.g. Ig, T/B lymphocyte subsets, Ab responses to vaccines, SLE/autoimmunity
- More specialized testing of immune cell function

14. Neurology

Seizures – Emergency Management

RCH [Clinical Practice Guidelines : Seizures acute management](#)

Seizures are common, occurring in **1 in 20 (5%) children**. Most are **febrile seizures** and are brief and self-limiting, ceasing within 5 minutes.

- **Key features of a febrile seizure:** 6 months to 6 years old; generalised tonic clonic seizure; brief (most < 1 minute); post-ictal phase; no increased risk of epilepsy or developmental issues.
- **Seizure red flags:** afebrile; prolonged (longer than 15 minutes); recurrent within 24 hours or same illness; focal; neural abnormality (new); highly febrile or young (meningitis); febrile + persistently altered neurology (encephalitis).
- **Potential underlying causes that might require specific management:** head trauma; stroke and ICH; drug/toxin overdose or envenomation; meningitis; electrolyte disturbance (hyponatraemia, hypocalcaemia, hypomagnesaemia); hypoglycaemia.
 - Look at the **skin** carefully: café au lait spots → neurofibromatosis; Sturge-Weber syndrome → port-wine stain birthmark on face + angiomas in brain.

Emergency Management of Seizures

A	B	C	D
Clear A irway, lie on side. Airway adjuncts (often not possible due to clenched teeth)	Give oxygen	IV access 20 mL/kg 0.9% saline if shock	Perform BSL: if hypoglycaemic (<3.5) give IV 10% dextrose 5 mL/kg or glucagon 0.5 mg (<5 years) or 1 mg (>5 yr). Follow with IV glucose or carbohydrate feed.

Status epilepticus

Seizure or clusters of seizures lasting 30+ minutes without regaining consciousness in between.

- Bimodal distribution: peaks in toddlers (<5 years and especially under 12 months) and elderly (>55 years)
- Most seizures self-terminate within 5 minutes; **observe if no airway compromise.**

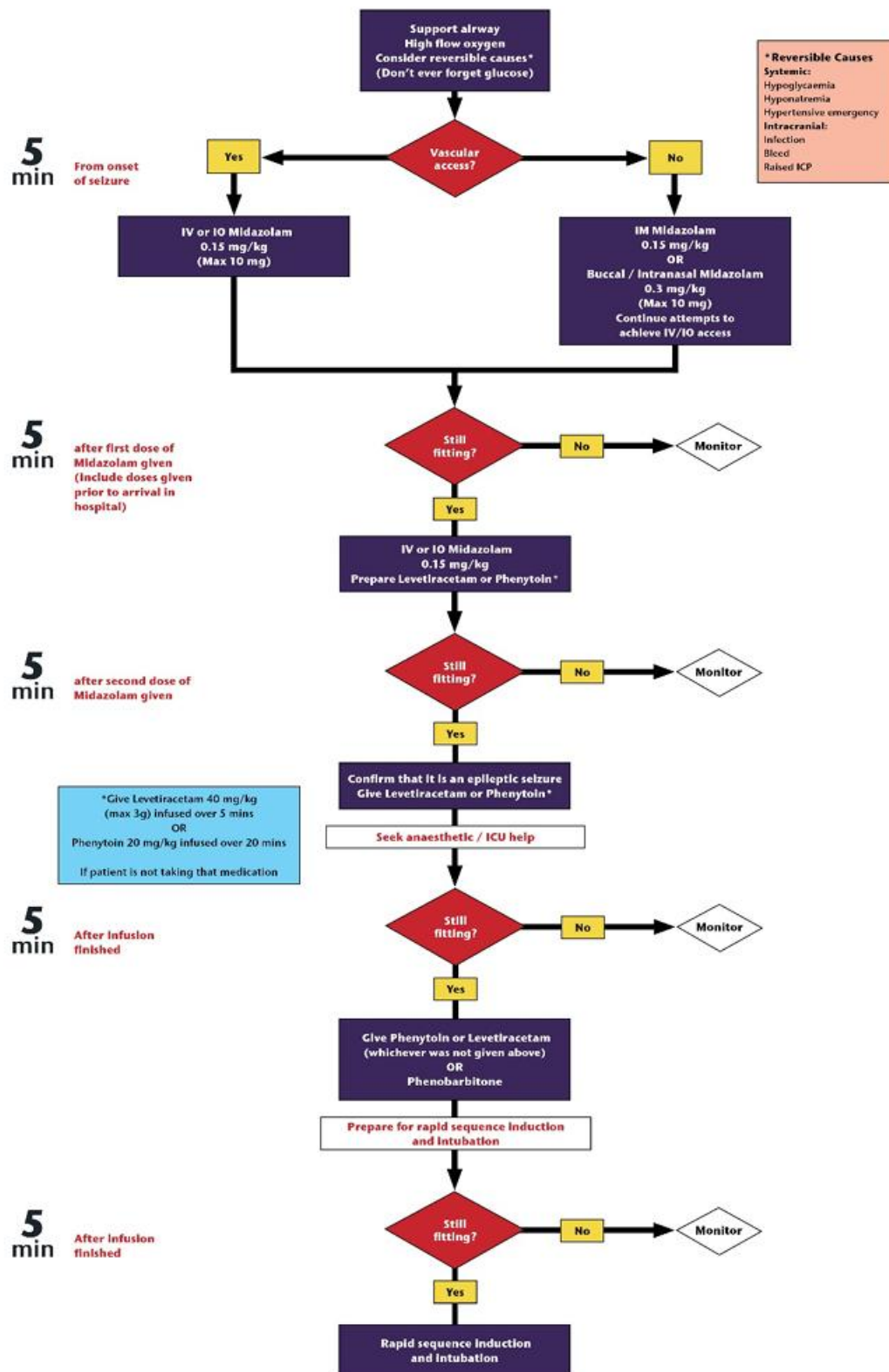
Investigation of SE:

- ABC (DEFG), Na, Ca, K, O₂ quickly
- FBC, EUC, LFT, CMP, formal glucose
- Blood/ urine culture if febrile
- Drug level / urine toxicology screen if relevant
- CSF (febrile, GCS > 8, stable, no raised ICP)
- EEG
- ECG to rule out arrhythmia if LOC
- Imaging (CT for trauma, MRI if focal neurological deficits or reduced consciousness)

Management of SE:

- If BSL < 3.5, give IV glucose **2 mL/kg 10% dextrose**
- Treat other reversible causes e.g. oxygen, calcium, sodium, etc.
- May consider antibiotics/ antivirals (acyclovir) if febrile
- Give **antiseizure medication** if seizure activity **≥ 5 minutes** – unlikely to terminate spontaneously.
 - 1st line: 2 doses of **benzodiazepine (IV midazolam)**, 5 minutes apart.
 - 2nd line: phenytoin (infusion over 20 minutes) or levetiracetam (or phenobarbitone in neonates).
 - If still fitting after 10 minutes, consider **RSI + intubation with thiopentone.**
- **No IV access:** start with buccal midazolam or PR diazepam, and establish IO access if still fitting after 2 doses.

Seizure Rule of 5's: **5%** of the population, medication if lasting **≥ 5 minutes**, 1st line Mx is 2 doses of benzodiazepine **5 minutes apart.**



Febrile Convulsions

Epileptic seizures precipitated by fever: $T > 38.5^{\circ}\text{C}$, seizure can be the first symptom in up to 42%.

- Febrile convulsions are **not due to CNS infection**, and **no previous afebrile seizure**. The most common aetiology is a **non-specific viral illness**.
- Aetiology is unknown but may be linked to a rapid rise in temperature (therefore antipyretics not helpful).
- Typical febrile convulsions:** brief, generalised and bilateral, clonic or tonic-clonic; occurs in a child between **6 months and 6 years** in the setting of an acute febrile illness; lasts <15 mins and does not recur within the same febrile illness.
 - Rare < 6 months old (check for meningitis)
 - 17-31% have a 1st degree relative with febrile convulsions

- **Complex febrile convulsions have 1+ of: focal** features at onset or during the seizure, **duration** > 15 minutes, **recurrence** within the same febrile illness (also incomplete recovery within one hour).
 - Often referred to as 'seizures associated with fever'.
 - Followed by **Todd's phenomenon** (transient focal neurological deficits, most often weakness or paralysis) after a seizure. Can last minutes to hours.
 - Children with all 3 components of a complex febrile seizure have a **50% chance of developing epilepsy**.
- **Febrile status epilepticus:** occurs in 5% with febrile seizures; very young (median age = 1.3 years), usually first seizure. Associated with lower temperature, longer duration of temperature before seizure, female, structural temporal lobe abnormalities, developmental delay, and first-degree FHx of febrile seizure.
- **Investigations:** BGL in all! EUC, CMP, and VBG if seizure requiring 2nd line agent, < 12 months, medical comorbidity, or not returned to neurological baseline.
 - A simple febrile convulsion rarely requires imaging – discharge once returned to neurological baseline, serious bacterial infection excluded or treated, and parents are educated regarding febrile convulsions.
 - Outpatient EEG if child < 12 months, initial focal component, or known genetic condition.
 - **Acute neurological deficit or encephalopathic after seizure** → needs **CT brain**.
- **Treatment of febrile convulsion:** treat the cause. Antipyretics do not help febrile convulsions (pathophysiology is linked to rapid increase in temperature, which antipyretics don't help).
 - If seizures are prolonged (>3 minutes), we may educate for buccal midazolam 0.3 mg/kg/dose
 - Higher-risk patients may need antiepileptic drugs
- **Prognosis: 30% risk of recurrent febrile seizure;** ¾ reoccurrences in first 12 months.
 - The younger the child, the greater the risk of recurrence.
 - The risk of epilepsy in a child with a simple febrile convulsion and no risk factors is **1%** - similar to the general population.
- **Patient education:** lie child on side during seizure; never place anything in child's mouth; call an ambulance if seizure lasts 5+ minutes.

Red flags for febrile convulsions:

Seizure after vaccination or prolonged hemiclonic seizure → consider **Dravet syndrome** (SCN1A variant; encodes Nav1.1 in inhibitory interneurons). Brain *extremely sensitive* to fever or mild immune stimulation.
 Incomplete recovery after seizure → CNS infection, inflammation, ischaemic stroke, or bleed.

Paroxysmal Disorders

Evaluation: semiology – history, home video, ancillary data e.g. EEG, MRI, genes, antibodies.

Terminology:

- Myoclonic jerk: brief tonic contraction lasting < 0.1s
- Spasm: sudden brief bilateral tonic contraction of axial and proximal limb muscles, abrupt on-off, lasts 0.2-2s
- Clonic: repetitive flexion/extension, rhythmically recurrent, range 1-3 Hz and up to 5 Hz
- Tonic: persistent contraction of limb muscles lasting 2-10s

Differentials by age:

Neonates (0 – 6 weeks)	Infants (2 mths – 2 yrs)	Childhood (2 – 12 yrs)	Adolescents (> 12 yrs)
Jittery (Ca, glucose, HIE) Startle reflex Benign sleep myoclonus	Breath holding spells GIT reflux Shuddering attacks Stereotypy (mannerisms) Rhythmic movement disorder (Sleep) Factitious disorder by proxy	Staring/ day dreaming Tics/ movement disorder Migraine equivalents Self-gratification Drugs (anti-emetics) Withholding behaviour Parasomnias (sleep)	Syncope/ cardiac Non-epileptic seizures

Other differentials include **Sandifer syndrome** which has abnormal posture in kids with reflux +/- hiatus hernia

Syncope: a history is often sufficient to differentiate malignant from benign types, but if the event is unclear, an ECG may help (e.g. to rule out long QT).

Benign syncope	Malignant syncope
Prolonged expiratory apnoea (blue breath-holding) Reflex anoxic (or asystolic) syncope (vagally-mediated) Pallid-type breath holding Convulsive Valsava	Hyperekplexia (stiff apnoea in neonate preceded by staccato cry). - Genetic mutation affecting glycine NT - Exaggerated startle response Cardiac: syncope on exertion, water immersion or strong emotions; SOB or chest pain; palpitations; FHx sudden unexplained death

Functional/Non-Epileptic Seizure

Precipitated by stressful events, can be induced by suggestion. Occur in wakefulness with witnesses, and lack stereotypical characteristics e.g. post-ictal confusion.

- 'Teddy bear' sign: sign of regressing and non-organic illness in older children
- Asynchronous, asymmetrical, waxing/ waning, accelerating/decelerating, pelvic thrusts
- Flailing, tremors, resisting passive eye opening
- Belle indifference (paradoxical lack of concern)
- May have incontinence, tongue biting

Benign Sleep Myoclonus

Onset from the first weeks of life. Rapid, random, bilateral/asynchronous myoclonic jerks (may be forceful and rhythmic) occurring during sleep (all stages).

- Does not stop on passive flexion (unlike jitter)
- Usually lasts for seconds (10-20s) but can go for 30 minutes
- Normal EEG, not seen during alert wakefulness
- Mostly disappear by late infancy (7 months)

Breath Holding Spells

6 months – 2 years old. Typically confused with a tonic seizure. Benign, without long-term impacts.

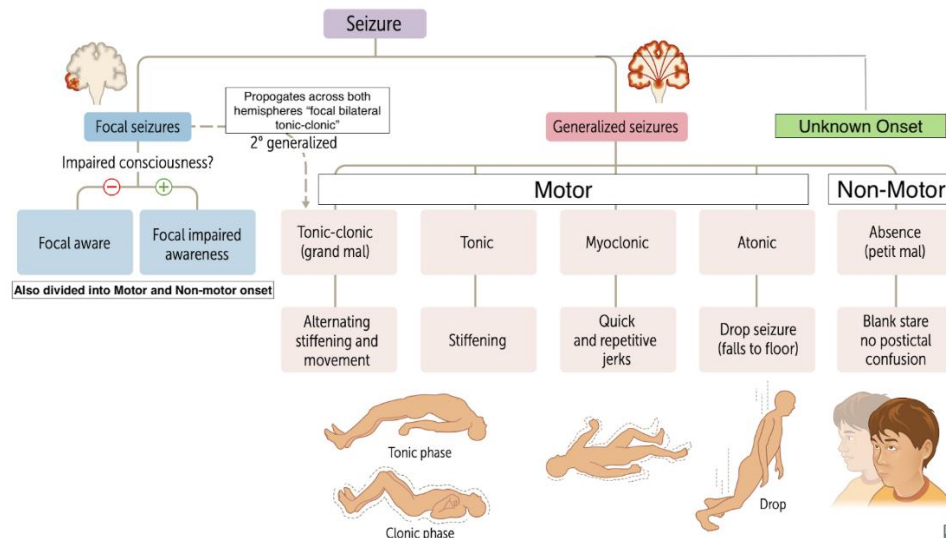
- Triggering event (fear, injury, frustration) → crying → expiratory breath holding → stiff, loss of awareness → clonic jerks
- Pathophysiology not well-understood, but **iron deficiency involved** – correction is helpful + counselling
- Check ECG to rule out arrhythmia
- Pallid type (reflex asystole):** parasympathetic dysregulation, pale and limp, with asystole

Tics

Sudden, rapid recurrent, non-rhythmic motor movements or vocalisations, often with premonitory urges. Simple tics affect 1 muscle group, complex if >1 goal directed (purposeless).

- <1 year, motor or vocal → **provisional tic disorder of childhood**
- >1 year → **persistent tic disorder**
- Tourette Syndrome:** motor + verbal tics in bouts, nearly daily for > 1 year, beginning before 18 years
- Tics may also be substance-induced or occur secondary to another medical condition
- Epidemiology:** 8 in 1000, M > F 4:1, typical age of onset 3 – 8 years, 50% have FHx, 90% have neurodevelopmental comorbidities (ADHD, OCD, ASD)
- Course:** waxing and waning, peaks prepubertally
- The most common tic is blinking; worsened with stress and anxiety, alleviated by distraction.

Epilepsy



Infantile Spasms

Classical syndrome: West syndrome (Infantile Spasms Syndrome)

- Characterised by a triad of **epileptic spasms**, **interictal EEG showing hypsarrhythmia (disorganised high-voltage background)**, and **regression**.
 - Not all with infantile spasms have this syndrome.
- Affects 1 in 3500 infants, onset 1st year of life
- Spasms last 1-2 seconds, repetitive, every 5 – 30 seconds, may have 100 episodes.
 - Spasms occur **when drowsy** (in waking or falling asleep)
- Aetiology:** 80% symptomatic (i.e. identifiable cause) e.g. HIE, neurocutaneous syndromes (TS, NF), brain malformations, metabolic (e.g. PKU, congenital glycosylation disorders), infectious, toxic
 - 15-20% are cryptogenic (idiopathic) – course is **more** favourable

Infantile spasms are considered a **neurological emergency**; prompt treatment is required to prevent irreversible effects on development.

- Urgent referral to paediatric neurologist
- UKISS protocol
 - Early hormonal treatment (oral prednisolone or ACTH) for initial control
 - Steroids suppress CRH (which ↑ neuroexcitability)
 - ACTH acts on melanocortin receptors in limbic regions → ↓ excitability
 - Vigabatrin for tuberous sclerosis

Tuberous sclerosis: rare genetic disorder characterised by growth of hamartomas (non-cancerous lesions) in multiple organs. Ungual fibroma, angiofibromas, hypomelanotic macules (ash leaf macules), calcification of cortical tubers, shagreen patches (connective tissue naevi).

Childhood Absence Epilepsy (CAE) and Juvenile Absence Epilepsy (JAE)

Feature	Childhood Absence Epilepsy (CAE)	Juvenile Absence Epilepsy (JAE)
Age of Onset	3–8 years (<10 years), girls > boys	>10 years
Development	Usually normal	May have mild cognitive issues
Aetiology	Presumed genetic (Chromosomes 5 and 8)	Genetic, related but distinct syndrome
Absence Seizure Features	Sudden cessation or marked decrease in consciousness with subtle automatisms (lip smacking, chewing, hand fiddling); brief (5-15s), abrupt onset/termination, multiple times/day.	Longer staring episodes, less frequent than CAE, often in clusters post-awakening
Seizure Triggers	Hyperventilation, flashing lights, sleep deprivation, stress, illness	Similar triggers
EEG	3 Hz spike-and-wave complexes, triggered by hyperventilation	3-4 Hz spike-and-wave, may be faster (4–6 Hz)

Associated Seizures	Typically absence only; if generalized tonic-clonic seizures (GTCS), treat with valproate	90% have absences with GTCS; can progress to Juvenile Myoclonic Epilepsy (JME)
Treatment	Ethosuximide if no GTCS; valproate if GTCS present	Valproate or lamotrigine preferred; ethosuximide not effective alone
Prognosis	65% remission; 40% without remission may develop JAE	Often lifelong; good control but less remission than CAE

Juvenile Myoclonic Epilepsy (JME)

- Most frequent, well-defined, idiopathic generalised epilepsy
- Triad: absences, myoclonic jerks, GTCS
- 12 – 18 years onset (mean 14 years); myoclonic jerks mainly on waking, 90% have associated GTCS, 15-30% have absence seizures.
- Triggers include sleep deprivation, arousals from sleep, photic stimulation, alcohol
- **Treatment lifelong:** valproate (lamotrigine or levetiracetam in women of childbearing age); 80-90% control
 - Seizures are worsened by carbamazepine and phenytoin

Self-Limited Epilepsy with Centrottemporal Spikes (SeLECTS)

- Previously known as **benign focal epilepsy of childhood (Rolandic Epilepsy)**
- 5 – 10 years old; M>F; strong genetic contribution; usually remits <16 years even in severe cases
- Most patients have < 10 seizures (1/4 only have 1 seizure)
- **History:** hemi-facial sensorimotor seizures (twitching of mouth, unilateral numbness), retained awareness (60%), oropharyngeal laryngeal ictal manifestation, arrest of speech, hypersalivation
 - **1/3 can have secondary GTCS;** some have nocturnal GTC with focal features
- Normal neurological exam, most neurodevelopmentally normal (though some evidence for comorbid ADHD and language difficulties)
- Most do not require anti-epileptic drugs. May consider if frequent seizures (carbamazepine, oxcarbazepine, valproate, levetiracetam)

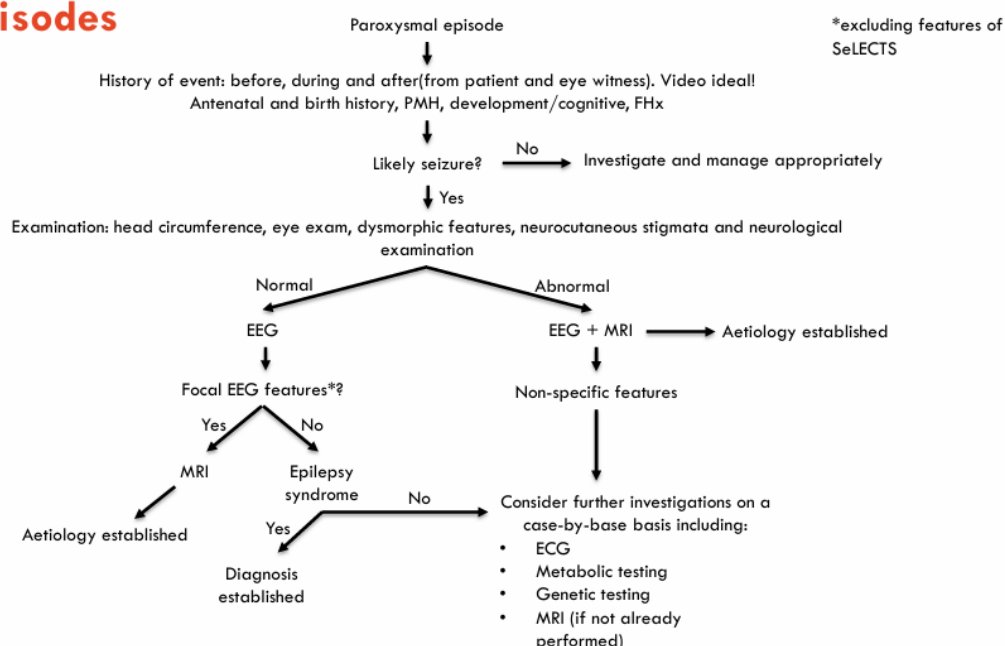
Classification of seizures in childhood including epilepsy and febrile seizures

Management of initial epileptic seizure or febrile seizure

Work-up of Paroxysmal Abnormal Episodes

After a full history and examination, perform **outpatient EEG**. If there are focal features that are out of keeping with SeLECTS, perform MRI for structural anomalies. If the EEG is non-specific, we may require further testing, e.g. lactate/ glucose/ urine metabolic screen; genetic testing (chromosome microarray then epilepsy panel).

Investigative approach to paroxysmal abnormal episodes



For all patients:
Manage comorbidities e.g. mental health disorders, ADHD, behavioural issues
Consider regular antiseizure medication if recurrent epileptic seizures

Motor Difficulties

Paediatric Motor Examination

Gait examination (>12 months)	Symmetry Stability: heel-to-toe, Romberg test Manoeuvres: toe walking, heel walking
180° manoeuvre	- Pull to sit (head lag may indicate hypotonia) Sitting (posture and head control) - Horizontal suspension (head control, leg positioning) Prone (at 3 months, should lift head and chest; by 6 months, head vertical with arms extended) Bearing weight (head control, leg positioning). Toes curled/ feet curved inwards may indicate spasticity.  

Other manoeuvres:

- Hopping: asymmetric weakness
- Gower's sign (rising from floor after supine): proximal weakness
- Squatting/rising from low chair: proximal weakness
- Fog manoeuvre (walking on outside of foot): dystonia

The "Floppy" Infant

Floppy strong	Floppy weak
Central hypotonia	Neuromuscular problem
Impaired alertness, seizures, poor feeding/ sucking, micro/macrocephaly, dysmorphic, nystagmus, retinal abnormalities	Alert baby that is 'not moving', characteristic findings (shape of chest, contractures)
No weakness, normal reflexes	Weakness and hypo/areflexia
e.g. HIE, meningitis, hyperbilirubinaemia, Down Syndrome	Less common but important e.g. Spinal Muscular Atrophy (SMA)

Recall that it is crucial to assess whether the abnormality is isolated or not (e.g. intellectual, social, speech difficulties).

Cerebral Palsy

A group of developmental disorders of movement and posture, causing activity restrictions or disability that are attributed to disturbances occurring in the foetal or infant brain. **Non-progressive disorder of motor control.**

- Motor impairment may be accompanied by a seizure disorder and by impairment of sensation, cognition, communication, and/or behaviour.
- Epidemiology:** 2.5 per 1000, most common cause of childhood disability affecting the industrialised world
- Aetiology:** brain injury occurring during prenatal, perinatal, or postnatal periods. 70-80% acquired prenatally, idiopathic.
- Neonatal risk factors:** premature < 32 weeks, BW < 2500g, IUGR, ICH, trauma
- 10-20% postnatally acquired, e.g. bacterial meningitis, viral encephalitis, hyperbilirubinaemia, MVA, NAI, falls

Classification of CP

- Topographic:** 39% hemiplegia, 38% diplegia, 23% quadriplegia.

Most kids with CP will walk: 60% independently ambulate, 10% walk with an aid, 30% use a wheelchair.

The **Gross Motor Function Classification Scale (GMFCS)** classifies gross motor function in CP on an ordinal scale I – V. It was initially designed for children aged 2 – 12; now expanded to age 18, with additional age-relevant descriptors.

GMFCS I	GMFCS II	GMFCS III	GMFCS IV	GMFCS V
Walk, climb stairs without railing, running, jumping, limited speed, jumping, coordination.	Walk, climb stairs with railing, difficulties long-distance/ uneven terrain, difficulty running/ jumping, may need mobility aid.	Walk with hand-held mobility device in most settings, may self-propel short distances; climb stairs with assistance	Powered mobility in most settings; may walk short distances	Transported in a manual wheelchair in most settings, limited antigravity movements/ control of legs and arms

- Motor types:** spastic is the most common form.

Spastic (80-90%)	Dyskinetic (6%)	Ataxic (5%)	Mixed
Stiff, tight	Dystonia, athetosis, chorea	Shaky movements	2+ motor types
Motor cortex damage	Basal ganglia damage	Cerebellum damage	Multiple sites affected

CP-Associated Problems: the most common overall is **pain** (especially due to spasticity). Half of patients with CP have intellectual disability.

Developmental	Functional	Physical
Intellectual disability (50%) Epilepsy (25%) Behaviour disorder (25%) Sleep disorder (20%)	Unable to walk (30%) Unable to talk (25%) Blind (10%) Deaf (4%)	Pain (75%) Hip displacement (30%) Bladder control problems (25%) Drooling (20%) Requiring tube feeding (7%)

Investigations

After fully characterising CP and associated conditions, obtain **neuroimaging (MRI preferred)**

- MRI has an 89% positive yield, and is often the only test needed
- If the MRI shows evidence of cerebral infarct, perform **thrombophilia screen**
- Normal MRI or unclear aetiology: metabolic and genetic testing, TORCH screen

Proximal vs Distal Weakness

Proximal weakness	Distal weakness
Usually muscle problem	Usually peripheral nerve problem
Duchenne muscular dystrophy (DMD) Steroid or other drug-induced myopathy Inflammatory and infectious myopathies Thyroid disease	Guillain-Barre Syndrome Hereditary motor-sensory neuropathy (Charcot-Marie Tooth) NMJ disorders Motor neuron

Clinical features of proximal weakness:

- Trendelenburg gait ('good side sags'), typically bilateral in neuromuscular conditions
- Gower's sign
- Difficulty heel walking
- Shoulder weakness e.g. can't remove jumper/shirt easily
- 180° exam (floppy, weak)



Demonstration of Gower's sign.

Duchenne Muscular Dystrophy (DMD)

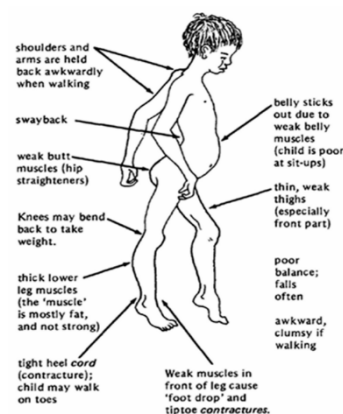
The most common neuromuscular condition in paediatrics.

- X-linked recessive condition, **1:3500 live male births**.
 - Note, 2-20% carriers (female) have weakness.
- 2/3 have FHx, 1/3 de novo (10 – 20% gonadal mosaics)
- Caused by **absent dystrophin**.
 - **Dystrophin**: large molecule, stabilises cell membrane. If defective, cell leaks.
 - **Becker Muscular Dystrophy (BMD)** is a milder form that presents in adolescents, caused by reduced dystrophin.

Onset: 3 – 6 years old

Clinical Features: Early clue – no jumping/ running. Diagnosed (latest) by age 5 years.

Classic Features	Other Findings
Gross motor delay (delayed walking) Generalised developmental delay Progressive weakness, mainly proximal (Gowers Sign) Incidental raised AST/ALT (muscular origin)	Pseudohypertrophy of calf muscles (muscle replaced by fat and fibrous tissue) Spinal deformity (later) Cardiomyopathy Mild-moderate intellectual disability



Investigations

Blood tests	Electromyography (EMG)	Molecular genetic testing	MRI muscle
CK ↑ 10 – 20x normal (CPK 50 - 300x) ↑ ALT/ AST (correlates w/ CK) ↑ LDH	Rarely used now Myopathic findings (low amplitude polyphasic motor potentials, abnormal spontaneous activity)	Gold-standard (multiplex ligation-dependent probe amplification, MLPA)	Not usually done for diagnosis, can monitor disease progression

Other tissues affected too e.g. intellectual development/neuronal tissue.

Management

Surveillance	Corticosteroids	Genetic
Orthopaedic (contractures, tendon release, hip dislocation, scoliosis) Cardiac (cardiomyopathy, conduction abnormalities, heart failure) Respiratory (neuromuscular sleep apnoea) General (constipation, school support, depression)	Unknown mechanism Offered to all DMD patients Improved muscle strength and prolonged time to loss of ambulation, improved cardiac function (less cardiomyopathy), less scoliosis	No gene therapy yet; being trialled on animal models. Gene counselling for parents, especially if considering having other children.

Prognosis: wheelchair-dependent at 12 years old. NIV has extended life expectancy to 30 years. The cause of death is usually either cardiac or respiratory failure.

- Once they stop walking, they can develop contractures – hip, knee, and ankle pain, and scoliosis.

Differential diagnoses of progressive proximal weakness (supplemental table):

Category	Example Conditions	Age/ Onset Pattern	Clinical Features	Ix	Notes
Muscular (myopathic)	Duchenne Muscular Dystrophy (DMD)	Boys, 2–5 yrs	Gower's sign, calf pseudo-hypertrophy, delayed motor milestones	↑CK (very high), genetic testing (dystrophin mutation)	X-linked recessive; progressive loss of ambulation ~12 yrs
	Becker Muscular Dystrophy	Later (5–15 yrs)	Similar to DMD but milder, slower progression	↑CK (moderate), dystrophin partially functional	X-linked; ambulant into adulthood
	Limb–Girdle Muscular Dystrophy	Variable childhood/ adolescence	Shoulder and hip weakness, normal intelligence	↑CK, genetic panel	Autosomal recessive or dominant; no calf hypertrophy
	Congenital Myopathies (e.g. nemaline, central core)	Neonatal–infant	Floppy infant, delayed motor development, facial weakness	CK normal or mildly ↑, muscle biopsy	Non-progressive or slowly progressive
	Inflammatory Myopathies (Polymyositis, Dermatomyositis)	School-age to adolescence	Subacute weakness, pain, rash (heliotrope, Gottron's papules)	↑CK, ESR/CRP, ANA, EMG, MRI	Often steroid-responsive; rash = dermatomyositis
Neuromuscular junction	Myasthenia Gravis (juvenile/congenital)	Any age	Fatigable weakness, ptosis, bulbar symptoms	AChR antibodies, edrophonium test, EMG	Improves with rest, no muscle atrophy
Motor neuron/anterior horn cell	Spinal Muscular Atrophy (SMA types I–III)	Type I: infancy; Type III: childhood	Symmetric proximal weakness, hypotonia,	Genetic test (SMN1 deletion), normal CK	Alert child, no sensory loss

			tongue fasciculations		
Peripheral nerve (neuropathic)	Charcot–Marie–Tooth (CMT)	Late childhood	Distal weakness → foot drop, pes cavus	Nerve conduction: slow velocity	Sensory loss + distal wasting (not purely proximal)
Metabolic / Endocrine / Toxic	Hypothyroid myopathy	Any age	Lethargy, weight gain, delayed relaxation of reflexes	↑TSH, ↓T4, mild ↑CK	Improves with thyroid replacement
	Mitochondrial myopathies	Childhood/adolescence	Exercise intolerance, multi-system (CNS, cardiac, ocular)	Lactate ↑, genetic testing, muscle biopsy	Often maternal inheritance
	Steroid-induced myopathy	After chronic steroid use	Painless proximal weakness	CK normal	No fasciculations or sensory changes; reversible
Functional Non-organic	Functional (psychogenic) weakness	Variable	Inconsistent exam findings, normal investigations	Normal	Rule of exclusion; often abrupt onset

Distal weakness by comparison might be reflected in poor toe walking!

Guillain-Barre Syndrome (GBS)

- Most common form of acute inflammatory peripheral neuropathy (AIPN)
- Autoimmune demyelination of peripheral nerves
- 2/3 have antecedent infection: EBV, CMV, Mycoplasma, Campylobacter, hepatitis, varicella

Clinical presentation: 'ascending symmetrical paralysis' that progresses over hours-days (nadir at 2 weeks)

- 50% have pain (may be only symptom)
- 15-30% have respiratory muscle involvement requiring ICU
- Ataxia
- Distal paraesthesia
- Urinary retention
- Cranial nerve involvement (IV, VI, VII), rarely bulbar
- Autonomic changes (bradycardia, postural hypotension, arrhythmias)
- **Miller Fisher variant:** ophthalmoparesis, ataxia, areflexia

Spinal cord	Peripheral nerve	Anterior horn cell	Neuromuscular junction	Muscle
Demyelinating disorder e.g. transverse myelitis Trauma Infarction Epidural abscess	Toxin CIDP	Poliomyelitis (rare) Enterovirus	Tick paralysis (springtime!) Myasthenia gravis (fatigability) Botulism	Myositis (inflammatory, infectious) Metabolic myopathies Periodic paralysis syndromes

Investigations

Note: the most important differential to rule out is **spinal cord compression (tumour, vascular)**. If the diagnosis of GBS is unclear, an urgent MRI spine is needed.

- Serology or culture (stool) for antecedent infection
- CSF (**albuminocytologic dissociation**): normal opening pressure, ↑ protein (> 0.4 g/L), normal cells
- Electromyogram (EMG): demyelination +/- axonal loss if severe, ↓ NCV
- MRI with gadolinium contrast: mainly to rule out SCC, but in GBS, may have enhancement of cauda equina and nerve roots.

Management

- **ABC**
 - Respiratory muscles
 - Autonomic instability
- Fluids for hypotension and orthostatic hypotension
 - May get paroxysmal hypertension which complicates treatment
- **Arrhythmias are the most life-threatening** but respond to standard treatment
- Evidence for its benefit is low-moderate quality; thus, only use if benefits outweigh risks
- Risks: viral hepatitis B/C transmission (low), aseptic meningitis, headache
- **Plasmapheresis** is the only therapy with proven benefit, but is seldom used in children.
- Do **not** use steroids for GBS! They do not work!

Mechanism of steroid inefficacy: They can inhibit macrophage infiltration into the CNS which is essential for clearing nerve debris.

Specific treatment

- **If ambulant/mild:** watch and wait
- **Non-ambulant:** IVIg, 2 g/kg over 3-5 days is standard

Inherited Neuropathies (Charcot-Marie-Tooth)

AKA Hereditary Motor and Sensory Neuropathy. The most common type is **CMT Type 1**, caused by a PMP22 gene duplication. These patients have slow nerve conduction velocities in their upper extremities with pathological evidence of demyelinating neuropathy.

Presentation

Recall the classic foot deformities e.g. pes cavus, hammer toe, champagne bottle legs

- **Classic:** begins walking on time, weakness or sensory loss in first 2 decades of life, progressive impairment
- **Infantile onset:** walk after 15 months old, loss of independent ambulation by 20 years old. Associated with areflexia and foot deformity, scoliosis, congenital hip dysplasia +/- facial weakness and tongue fasciculations.
- **Adult-onset:** can develop symptoms as late as 40 years old



Diagnostic pathway: NCS, needle EMG (differentiate myopathy vs neuropathy), nerve biopsy (superseded now by **genetic testing**).

Treatment: supportive – physio, orthotics, orthopaedic surgery, pain management, genetic counselling.

Headache

Can be primary (recurrent or episodic, intrinsic to the nervous system), or secondary (underlying disease process). Remember to **take BP in any child presenting with headache**.

How do childhood headaches differ from adult headaches? Male preponderance (60%), frontotemporal location, more bilateral, more nausea/vomiting, more aura, Hx seizures more common (5-12%), FHx migraine common (72%).

Acute severe headache

- Subarachnoid haemorrhage: rare in kids, order CT then LP for xanthochromia if negative
- Meningitis
- Acute raised ICP (beware **occipital headache!**)
- Hypertension

Red flags for raised ICP: recent onset of headache, increasing frequency/severity, change in character, localised **occipital headache**, headache wakes child at night, early morning headache, headache brought on by coughing or straining, higher centre changes, endocrine abnormalities (e.g. reduced growth), neurological signs.

- Always measure head circumference!

Investigation of headaches

- Only if suspicious of raised ICP, worsening, not relieved by analgesics, or associated with decreased growth
- Do **not** do EEG for headache alone!
- Neuroimaging: MRI if needed (*CT preferred acutely, but higher radiation risk*)

Primary Headaches

Tension-type headaches (commonest cause of recurrent headaches)

- 25% adolescents
- 'Band-like pain', bilateral, 4-6 hours; not worsened by activity, no photophobia or phonophobia
- **Aetiology:** Genetic heightened pain sensitivity +/- abnormal scalp muscle contraction
- **Triggers:** depression, school phobia, psychological ailments
- **Treatment:** NSAIDs/paracetamol acutely, TCA prophylaxis (low-dose amitriptyline); scalp massage can help; magnesium
 - **Random factoid:** do not give aspirin to children under 14 as it can lead to Reye Syndrome (severe cerebral swelling and liver failure)

Migraine

- Lasts 1-48 hours; unilateral or bilateral, pulsating, moderate-severe, aggravated by physical activity +/- nausea and/or vomiting, photophobia and/or phonophobia
- May include aura (precedes headache < 60 minutes, lasts 4 – 60 minutes)
- **Variants:** cyclical vomiting, hemiplegic migraine, basilar, vertiginous, ophthalmoplegic, acute confusional migraine, Alice in Wonderland syndrome, migraine stupor
- **Hemiplegic migraine:**
 - Unilateral motor weakness or paralysis occurring as part of aura; headache can start before, during, or after
 - 20-50% have family history (CACNA1A, ATP1A2, SCN1A, PRRT2 genes – involved in ion transport and neuronal signalling)
 - Usually transient
 - Infarct occasionally
- This child has ophthalmoplegic migraine: ptosis +/- fixed dilated pupil (paresis/paralysis of CN III)
- **Prognosis:** recurrence common in adult life, >50% have migraine age 50



Acute treatment of migraine	Interval (prophylactic) therapy
Quiet, cool dark room; sleep Paracetamol or NSAID, anti-emetic (ondansetron), sumatriptan nasal spray	Only if significant impact on daily life, acute treatment ineffective or overused, headaches intense and > 48 hr
Medication overuse is a problem. Use analgesics < 2-3x per week and triptans < 6x per month.	Beta blockers (e.g. propranolol), topiramate (more effective but can cause behavioural disturbance), sodium valproate, flunarizine (CCB), CGRP, amitriptyline

Prevention of headaches

Lifestyle measures are the most important, e.g. adequate sleep, hydration, balanced diet, exercise and relaxation. OTC supplements: riboflavin, magnesium, Coenzyme-Q10. Prescription meds for migraine (see above).

Common presentations of potentially serious problems +/- headache

Early morning headache with vomiting	Space-occupying lesion, raised ICP
New onset torticollis in a child	Space-occupying lesion (posterior fossa mass; CN XI)
"Bell's palsy" in a child who is too young, developmentally abnormal, encephalopathic, not resolving as it should	Leukaemia (deposits along CN VII), space-occupying lesion Always check FBC + film in kids with CN VII palsy
Afebrile focal seizure and decreased LOC	Non-accidental injury
CN dysfunction, cerebellar, and long tract signs	Brainstem lesion/ mass

Microcephaly and Macrocephaly

Occipitofrontal circumference (OFC): marker of growth and alterations of head contents (brain, CSF, blood, skull thickness); most relevant in first 2 years of life prior to fusion of cranial sutures. OFC can be unreliable in the newborn until day 3-4 of life due to moulding, cephalohaematoma, and caput succedaneum.

- Most informative plotted over time
- Specific charts for certain conditions e.g. prematurity, achondroplasia, NF1, Down syndrome

Aetiology of Microcephaly and Macrocephaly

	Microcephaly	Macrocephaly
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Primary – present at birth	Problems with brain forming • Microcephaly vera or MCPH • Chromosomal disorders • Neurulation defect (anencephaly, encephalocele) • Prosencephalisation defect • Migration defect	Syndromic - Non-syndromic • No developmental delay: familial macrocephaly • Developmental delay: many causes
Secondary – normal at birth	Postnatal systemic disease • Intrauterine disorder e.g. TORCH, toxins, vascular • Perinatal brain injuries e.g. HIE, ICH, stroke, meningoencephalitis • Postnatal systemic disease e.g. CKD, malnutrition, cardiopulmonary disease Craniosynostosis: sutures close too early	Too much brain (megalencephaly) e.g. familial, genetic, syndromic (achondroplasia), NF1 Too much fluid e.g. communicating hydrocephalus (meningitis, benign) or non-communicating (e.g. aqueductal stenosis) Too much skull e.g. osteogenesis imperfecta, rickets, anaemia (marrow expansion) Mass effect of SOL

Note: the classic condition causing normal head circumference in girls until 6-8 months, that then drops off, is Rett syndrome.

Microcephaly is overall less urgent than macrocephaly. **The only real red flag is a decelerating head circumference (suggests neurodegeneration).**

Macrocephaly red flags:

- Rapid ↑ OFC (> 2 cm/month)
- Change in consciousness
- Prolonged afebrile seizure (bleed from NAI?)
- Developmental arrest and/or regression
- Signs of raised ICP (papilloedema, new 6th nerve palsy)
- New focal neurological deficit
- Neurocutaneous stigmata

Investigations

Dictated by presentation (acute or insidious), rate of growth acceleration or deceleration, exam findings.

- Always consider parental head circumference
- US if < 6 months (fontanelles open)
- CT only in acute situations (radiation ++)
- MRI preferred for the brain parenchyma but requires GA after 3-4 months
- Bloods: toxic substances (microcephaly), genetic studies (CMA, Fragile X testing), metabolic panel, urine metabolites
- Do not perform LP without imaging first (risk of coning with raised ICP)

What Not to Miss in Neurology

Neurological Emergencies

- Children can also get strokes (beware of cardiac history, thrombophilia)
- Spinal cord compression: acute onset urinary/ bowel problems with lower limb symptoms; **urgent MRI spine**

Vitamin-Responsive Conditions

Acquired	Inborn errors of metabolism
Consider with maternal dietary restriction, or child dietary restriction (e.g. in children with autism) e.g. Vitamin B12 deficiency → spastic paraparesis; Vitamin D deficiency → seizures	Usually very early in the neonatal period – seizures, encephalopathy, low body tone e.g. Vitamin B6 responsive seizures Babies who don't respond to 2-3 seizure medications are often given pyridoxine for this rare possibility

Functional Neurological Disorders

- More common in teenage/adolescents, and in females, but not an absolute rule.
- Examples include functional seizures; tics, tremor, dystonia; gait problems; visual problems; dizziness
- Proactive diagnosis based on clinical examination (not just a diagnosis of exclusion): distractibility, variability of symptoms
- Mind-body rehabilitative approach

15. Oncology

1 in 600 children will develop cancer or leukaemia during childhood. Cancer is the second leading cause of death in children, after accidents.

Paediatric cancer is typically **not** carcinoma (cf. adults) and not associated with lifestyle factors (except for radiation). Rarely has a heritable basis (exceptions: Li Fraumeni syndrome, bilateral retinoblastoma).

Many cancers are highly chemo-sensitive with 75% cured permanently. Where adult cancer is a multi-step process, we believe paediatric cancer is a 1- or 2-step process.

Types of paediatric cancer

- Leukaemia (**most common**) (30%); 80% ALL, 20% AML (CML rare, CLL almost never).
- Brain tumours
- Neuroblastoma
- Lymphoma
- Sarcomas
- Wilms' tumour

Cardinal Symptoms of Cancer

- Persistent/ recurrent fever without a cause
- Persistent pain, especially at night, bone pain, or headache; out of proportion with findings
- Mass
- Hepatosplenomegaly or lymphadenopathy
- Purpura
- Anaemia or pallor
- Strabismus
- Neuromotor changes e.g. coordination, behaviour
- Weight loss

Acute Lymphoblastic Leukaemia (ALL)

Commonest childhood malignancy, usually arises from a lymphoid progenitor. Presents with anaemia, thrombocytopaenia, bruising, fever, bone pain, lymphadenopathy, hepatomegaly, splenomegaly, malaise, anorexia.

- Symptoms primarily arise from bone marrow infiltration
- WCC may be normal or high +/- circulating blasts (depending on spillage from bone marrow)
- Diagnosed via **bone marrow biopsy**, LP for CNS disease
- **Poor prognostic factors:** <2 years old or >10 years old, high WCC, Philadelphia t(9;22), t(4;11), CNS disease

DDx: ITP, neutropaenias, anaemias, infectious mononucleosis (hepatosplenomegaly and lymphadenopathy), arthritis (e.g. systemic JIA).

Treatment: induction chemotherapy (1 month), 5-6 months heavy chemotherapy, then 1.5 years "gentle" oral chemotherapy. CNS disease may require IT chemo or cranial RT.

- 75% cured, 25% go into remission (relapses hard to cure, need BM transplant)

Acute Myeloid Leukaemia (AML)

More serious than ALL; mostly in hospital (e.g. neutropaenia, sore mouth, TPN). Different appearance on bone marrow aspirate. Requires **intensive chemotherapy**; common complications include infection and hyperviscosity.

Brain Tumours

Children tend to have more posterior fossa tumours, e.g. medulloblastoma. Other subtypes include ependymoma, glioma (high- and low-grade; brainstem).

May present with **hydrocephalus** (if blocking mesencephalic aqueduct), signs of raised ICP (e.g. headache worse on waking, vomiting), focal neurological signs.

Treatment may include surgery, radiation therapy, and chemotherapy.

- Radiation carries several risks; avoid in kids < 5.
- Risks include neurocognitive effects, hypopituitarism, and secondary cancer

Neuroblastoma

Tumour of **neural crest cells** i.e. adrenal medulla and sympathetic ganglia. Seen in pre-school children. Most are abdominal, malignant, and require intensive chemotherapy, surgery, +/- bone marrow transplant.

- >12 months old carries worst prognosis: flank mass, unhappy, bony metastases (e.g. 'black eyes', pain, limp, lumps), anaemia
- Better prognosis < 12 months old

Other presentations of neuroblastoma: neonate with hepatomegaly, neck mass, paraspinal chest mass, spinal cord compression, adrenal mass on antenatal ultrasound.

A simple bedside test for neuroblastoma is **urine catecholamines** – specifically, **homovanillic acid (HMA)** and **vanillylmandelic acid (VMA)**.

Wilms' Tumour

Kidney tumour in young child 1-7 years old. Large, painless flank mass, haematuria. Metastasises to the lungs/ liver. Generally good prognosis with surgery, chemo, RT.

- Kids with **bilateral Wilms' tumour** often have a predisposing genetic condition.

Hodgkin's Disease

Typically teenager. Presents with a neck or mediastinal mass, lymphadenopathy, constitutional symptoms (fever, weight loss, night sweats). Generally good outcomes with chemotherapy and RT.

- A **mediastinal mass** may be an emergency as it can result in SVCO and airway compression.
- Side-effects of treatment include secondary cancer (breast cancer due to chest irradiation), fertility problems

Non-Hodgkin's Lymphoma

Various cell types of origin

- T-lymphoblastic: older child, mediastinal mass, pleural effusion
- Burkitt's (B cell): abdominal disease, intussusception (in 6-7 years old), abdominal mass, marrow, CSF disease; rapid-growing
- Large cell NHL

Bone sarcomas

Typically adolescents, corresponding to growth spurt. Have a 70% survival rate if non-metastatic.

- Ewing sarcoma in diaphyseal long bones and the axial skeleton (e.g. pelvis, ribs), due to t(11;22) translocation (leading to EWS-FLI1 fusion transcript)
- Osteosarcoma occurs at the growing ends of long bones (distal femur, proximal tibia, and proximal humerus most common).

Soft tissue sarcomas

Rhabdomyosarcoma is the most common, and is sensitive to chemotherapy and RT. It may present in the genitourinary tract, extremities, parameningeal, head/neck, and orbit.

The really bad diseases: metastatic sarcoma, stage IV neuroblastoma (over age 1 year), high grade glioma, ALL with bad chromosomes (e.g. Ph+), anything recurrent.

Chemotherapy is generally given as multi-agent protocols with 6-12 months of cycles of therapy (the exception is ALL, which requires 2 years of treatment). It has massive toxicities and can disrupt normal life. Adverse effects include alopecia, myelosuppression, immune suppression, sepsis, opportunistic infection, and mucositis.

Late effects of cancer treatment

1 in 1000 adults are a survivor of childhood cancer. Late effects include infertility, secondary neoplasia (e.g. solid tumours from RT, AML from alkylating agents), cardiac damage (anthracyclines), hypopituitarism and neurocognitive damage (cranial RT), ototoxicity, nephrotoxicity, and others.

Death and dying: many families choose to have their child die at home. Death and dying have a profound impact on the child's family. Supports include paediatric palliative care, Bear Cottage, and bereavement support.

Oncological Emergencies

Febrile Neutropaenia

Infants and Children: Initial Management of Fever/Suspected Sepsis in Oncology /Transplant Patients

Fever is a **medical emergency** in neutropaenic patients. Fever may be the only sign of severe underlying infection, since the signs and symptoms of inflammation are typically attenuated.

- Recall that WCC is lowest **7-10 days after chemotherapy**

- Because septicaemia usually occurs between cycles, we typically aim to treat aggressively with antibiotics and commence the next cycle as scheduled.
- **Mucositis** commonly co-exists with fever and neutropaenia.
- Most children have a CVAD in situ which is a potential source of infection.

Fever in paediatric oncology may be secondary to drugs e.g. cytarabine and bleomycin, blood products, or viruses. However, as it is impossible to rule out bacterial infection, prompt antibiotics are needed.

Management: Immediate measures include ABCDE, blood cultures (including from CVAD), and broad-spectrum antibiotics **within 60 minutes of presentation**.

- Other bloods: blood culture, lactate, FBC, EUC, LFT, BGL (sepsis can → hypoglycaemia in infants), urine MCS, CXR if symptomatic only
- Mortality in febrile neutropaenia is typically attributed to gram-negative infections, especially *Pseudomonas*, *E. coli*, and *Klebsiella*
- Therefore, the recommended antibiotic is typically an anti-pseudomonal beta-lactam (**piperacillin + tazobactam**) with **gentamicin** for gram negatives.
- **Vancomycin** is added in severe sepsis/shock or in signs of infected vascular devices to account for the possibility of MRSA.
- Do not wait for laboratory test results before starting antibiotics.
- Children might acutely deteriorate after administering antibiotics due to septic shower/endotoxin release – **closely monitor!**

Spinal Cord Compression

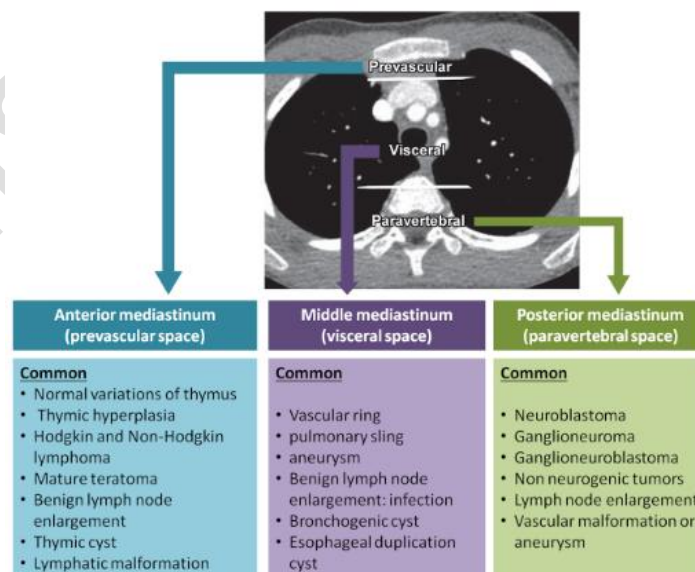
Medical emergency! Occurs in **3-5% oncology patients** and is the **commonest cause of paraparesis in children 3 months – 17 years**.

- **Red flags:** developmental regression, back pain, abnormal gait, lower limb asymmetry, limb weakness, sensory deficit.
- Investigated with MRI spine
- Needs urgent treatment to avoid irreversible spinal cord damage – multidisciplinary e.g. chemo, surgery.

Mediastinal Masses and Superior Vena Cava Obstruction

Resource: Starship NZ [Mediastinal Mass diagnosis and management](#)

An anterior mediastinal mass in a paediatric patient is considered an oncologic emergency. These patients are at increased risk of cardiorespiratory arrest that can be exacerbated under GA or sedation.



- Over 80% of mediastinal masses are malignant; typically haematological malignancies
- **Investigations:** Chest X-Ray (frontal and lateral), CT chest with contrast, echocardiography. *Do not force the child to lay flat – CT may be performed prone, lateral, or semi-upright.*
- **Diagnostic work-up:** FBC + flow cytometry, biopsy of mediastinal mass or pathological lymph nodes, aspirate of pleural effusion (if preset), bone marrow aspirate/biopsy and LP by experienced clinician

- **Management:** ABCs, systemic steroids and treatment specific to diagnosis

Interestingly, the child may have a history of LRTI or asthma that has partially improved with steroids. This is because steroids are given to treat lymphoma.

Raised Intracranial Pressure

Red flags for raised ICP:

- Persistent vomiting
- Worse in the morning
- Worse with coughing
- Personality change (head banging)
- Hypertension
- Papilloedema
- Gait changes
- Neurological signs
- Visual disturbance
- Drowsiness or reduced LOC

Management

- MRI
- Elevate head
- Dexamethasone
- IV fluids 75% maintenance
- Aim for higher Na
- Consider hypertonic saline/mannitol
- Intubate to control CO₂
- Urgent neurosurgery

16. Adolescent Health

Adolescence is the most important developmental stage in life after early childhood. The gains in childhood are not maintained in adolescence, representing a major loss on health investment with new morbidities and mortalities. The main drivers of adolescent morbidity include mental health, injury, chronic illness, and risk-taking behaviours.

Isolation and marginalization bring the risk of poor health outcomes; consider especially the risk of poorer health outcomes in rural and remote youth and Indigenous youth.

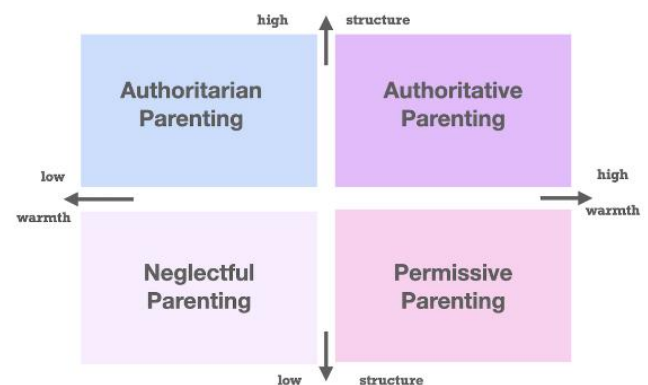
Resilience

An individual's ability to face and rise above adversity, and to learn from that experience of adversity.

- Multiple networks and connections, 'one adult who cares'
- Academic and social competence and a positive school ethos
- Outside activities with multiple friendship networks
- Internal locus of control

Parenting Style

- The ideal is **authoritarian**: warm, involved, responsive to child's wellbeing; firm and consistent but responsive to developmental change; fosters psychological autonomy
- The chaotic parent may be one with mental health or substance use concerns



Supporting Parents in the Health System

- Avoid labelling as 'hopeless'
- Be vigilant about home safety, DV, and abuse
- Acknowledge parent's own experience
- Understand current social circumstances and demands
- Support changes, one goal at a time
- Praise effort
- Help adjust to transition of care in chronic illness and disability

Developmental Tasks of Adolescence

Approximately 2nd decade of life (10 – 19 years). Involves rapid physical, cognitive, emotional, and social changes.

- Physical/biological: puberty, sexual maturation, body image development
- Cognitive: concrete → abstract thinking, develop problem-solving, future planning
- Emotional: identity formation, independence, managing emotions
- Social: peer relationships, intimacy/ romantic relationships, separation from parents, experimentation with roles
- Moral/ethical: development of personal values, autonomy, responsibility

HEEADSS Assessment

Home

Education, eating, exercise

Activities and peers

Drugs including alcohol and tobacco

Sexuality

Suicidality and depression

Sleep, safety, spirituality, social media

The Adolescent Brain and Risk-Taking

- Adolescence is a period of dramatic neuronal overgrowth and subsequent pruning.
- The prefrontal cortex (PFC) is the last region of the brain to mature.
- The developing brain is susceptible to the addictive effects of alcohol and other drugs.
- Chronic stress and its effect on the HPA axis may interfere with PFC development
- Those with intellectual disability may take longer to develop maximal cognitive maturity

Early adolescence (10 – 14 y)	Middle adolescence (14 – 18 y)	Late adolescence (18 – 21 y)
Puberty heightens emotional arousability, sensation-seeking, reward orientation Maturation of cognitive processes Improved reasoning, information processing, expertise	Increased vulnerability to risk-taking and problems in regulation of affect and behaviour Become capable of planned, abstract, multidimensional thought; maximal cognitive abilities	Maturation of frontal lobes → competent regulatory function

Consent to Medical Treatment

In Australia, anyone below age 18 is a minor. An adult (18 years and over) is presumed competent to consent to medical treatment.

The Mature Minor (Gillick Competency): someone under age 18 who is determined competent to consent to medical treatment.

Legal competence to consent to treatment accounts for a minor's

- Maturity
- Intelligence
- Experiences
- Ability to manage influences on decision-making
- Understanding
- Ability to weigh risks and benefits and longer-term impacts
- Gravity/seriousness of the treatment

The laws vary by state. Below are some points from the NSW Health Consent to Medical and Healthcare Treatment Manual. Note, a **child** is defined as someone 15 or younger; a **young person** is someone aged 16 or 17.

- Medical practitioners can carry out **emergency/life-saving treatment** without the consent of the Minor or their parents.
- If the parents object to a potentially foreseen emergency treatment (e.g. blood transfusion in a Minor requiring heart surgery) it might be necessary to obtain a **court order** for further advice.
- A Minor with capacity may refuse treatment, however a court can override this decision to avoid serious consequences for the Minor

Level of maturity and understanding	Recommendation for obtaining consent
Immature and insufficient	Consent from a parent or guardian must be obtained
Intermediate understanding	Consent from the young person may be sufficient, however, consent of a parent or guardian should also be obtained, unless the young person objects
Mature understanding	Consent of the young person is usually sufficient

If a Minor or their parent refuses consent, or there is conflict between the parent(s) and the Minor:

- Explain there is no suitable alternative treatment
- Obtain specialist opinion on the Minor's capacity (if applicable)
- If dispute about the treatment plan, seek second medical opinion
- Attempt to reach agreement by counselling and repeat discussion with the family
- Explain to parents how treatment delays may affect the patient
- If the Minor is 'at risk of significant harm' due to parent's refusal to consent, may consider a Mandatory Report
- Court order (last-resort) to authorize treatment
- Document everything!

Special Medical Treatment (<16 years old)

- Includes those which lead to permanent infertility, involve administering a drug of addiction > 10 days, experimental procedure
- Consent from Guardianship Division of NCAT required – neither the parent nor the Mature Minor can consent

Laws around Gender-Affirming Medical Treatment (GAMT) are also evolving

- Parental consent is required regardless of whether the child is Gillick competent (consent of *both parents* or *all persons with parental responsibility*)
- Accessing GAMT also requires assessment by a multidisciplinary team and a diagnosis of gender dysphoria (DSM-5-TR)

Medicare for Adolescents

- From 14, only the adolescent can access online Medicare claims history
- From age 15, adolescents can have their own Medicare card
- Children remain on the Medicare Safety Net until age 25 if they are a full-time student

Adolescent Sexual Health

Sexual health: a state of physical, emotional, mental, and social wellbeing related to sexuality.

Age of Consent and Related Sexual Offences (NSW)

- It is a crime for a person to do a sexual act with another person who is under 14 years old
- The age of consent is 16; someone who is ≥ 16 years old can legally have sexual intercourse or participate in other sexual acts with another person who is 16 years or older as long as both freely consent
- In NSW, consent to a sexual activity with a person who is under 16 can be a defence to sexual crimes if both people are 14 or over, and the age difference is ≤ 2 years

Adolescent Sexual Behaviour

- Age of first intercourse has decreased slightly, from age 19 in cohorts born in 1940s to age 17 in cohorts born from late 1990s.
- Older adolescents have more **unprotected sexual intercourse** than younger adolescents, possibly linked to relationship status.
- Proportion of adolescents reporting they “feel good” after intercourse has declined since the 90s, now stable at around 50%.
- Sexually-oriented social media behaviour is also common

Due to contextual changes, there is an increasing gap between first sexual experience and monogamy. However, while **STIs are on the rise**, **teen pregnancy/births have been stable since the 90s**. This probably reflects improved sexual education.

Risky Sexual Behaviour in Adolescents

Driven by developmental and psycho-social vulnerabilities and propensity for risk taking. This can be exacerbated by conditions that increase impulsivity, e.g. ADHD.

- Behaviour that leads to unplanned pregnancy/ STI
- Sexual behaviour that impacts mental health and wellbeing, including sexual behaviour without pleasure.

Note, there is a sketchy (and probably bidirectional) relationship between exposure to sexualized material and sexual behaviour.

Alcohol and Other Drugs

Adolescence is a peak age of initiation of substance use. This can be a manifestation of risk-taking.

- The top 3 substances used in adolescence: alcohol, tobacco, cannabis
- Concerning trends in e-Cigarettes, normalization of MDMA, sedatives
- Smaller percentages of adolescents use amphetamine-type substances and sedatives, but their harms are often greater
- Substance use has short- and long-term impacts (bio-psycho-social)

In 2015, alcohol and illicit drug use were the leading cause of total burden of disease in males aged 15-24 and the second and third leading causes for females.

- Males experience nearly 2x the burden of disease compared to females
- Tobacco smoking, and the use of alcohol and illicit drugs is declining
- E-Cigarettes are on the rise (95.7% increase between 2016 and 2019); those who use e-cigarettes are 3x more likely to take up traditional cigarettes

Trends in alcohol consumption

- Age of initiation is increasing
- Consumption of alcohol at **risky levels** remains high
- Adolescents are more likely to binge drink, be alcohol dependent, have withdrawal, mix alcohol with other substances, suffer harms

- Most important risk factors: peer alcohol use, normalization of alcohol, behavioural problems, parental alcohol supply
- Early onset alcohol use increases the risk of problematic use later in life, alcohol dependence in adulthood (x4), mental health issues, other substance use, memory and concentration

Cannabis

- Most common illicit drug in Australia; the most common principal drug of concern for those seeking AOD treatment
- Early-onset cannabis increases the risk of impaired attention, IQ, memory, concentration, anxiety/ depression/ psychosis
- Common withdrawal symptoms: anger, aggression, irritability, anxiety/nervousness, anorexia, weight loss, sleeping difficulties (including vivid dreams), restlessness, depressed mood

Screening for Substance Use: CRAFFT

Have you ever ridden in a car driven by someone under the influence of drugs or alcohol?	Have you ever used drugs or alcohol to relax or fit in?	Have you ever used drugs or alcohol while alone ?	Have you ever forgotten anything while using drugs or alcohol?	Have your friends or family told you to cut down?	Have you gotten into trouble using drugs or alcohol?
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AUDIT (10 questions) or **AUDIT-C** (3 questions) may be used to screen for alcohol dependence specifically.

Brief Interventions (5 A's): Ask, Assess (stage of change, dependence), Advise (to quit), Assist (e.g. NRT), Arrange (follow-up)

Comprehensive Adolescent Drug and Alcohol Assessment

- Engagement and discuss confidentiality
- HEEADDSSS Assessment
- Substance Use Assessment (CRAFFT, AUDIT-C)
- Risks, harms, and mental health comorbidities
- Neurocognitive (and FASD) assessment
- Identify risks and address child protection issues

- Trauma-informed framework

Treatment Options

- Psychoeducation
- Brief interventions
- Motivational Enhancement Therapy and Motivational Counselling
- CBT +/- MET
- Multi-dimensional family therapy
- Symptomatic treatment

Chronic Illness and Disability

Chronic illness: permanent condition with residual disability, involving a long period of supervision, observation, and/or rehabilitation.

- 10-15% of young people have a chronic or disabling condition
- This is on the rise, a combination of increased survival rates, a true increase in prevalence (e.g. DM, asthma, ALL, IBD), and conditions appearing earlier.

Leading causes of chronic illness: 10-14 y/o → asthma (followed by ADHD); 15-19 y/o → anxiety and depression

- Technically, overweight and obesity are the most common.

Adolescents with chronic illness are **more likely** to engage in **risk behaviours**.

Adolescent Sleep Disorders

Changes to sleep-wake regulation: ↓ sleep duration, ↓ slow-wave sleep, change in circadian rhythm

- Circadian rhythm shifts to eveningness (vesperal) coinciding with sexual maturity
- 53% adolescents have problematic levels of fatigue, 30% have delayed sleep onset

Factors contributing to poor sleep

- School start times
- Excess after school activities
- Electronic media
- Caffeinated substances
- Parents sleep patterns/ home environment
- Genetics

Consequences of poor sleep: mental health (bidirectional), academic performance, risk-taking behaviour, weight gain, driving accidents

Assessing sleep in adolescents:

- Sleep history and exam
- Formal scales: Adolescent Sleep Hygiene Scale, Paediatric daytime sleepiness scales, Flinders fatigue scale, Composite morningness eveningness scale
- Sleep diaries
- Actigraphy (estimates sleep-wake and activity)
- Polysomnogram

Management

- Sleep hygiene
- Parent set bedtime
- Mindfulness +/- CBT-I
- Address underlying sleep/mood disorders

Specific Sleep Disorders

Narcolepsy	
Definition	Lifelong neurological disorder of REM sleep
Clinical Features	Attacks of irresistible daytime sleepiness +/- cataplexy (sudden loss of muscle tone due to emotional stimuli, with intact consciousness/ respiratory/ cardiovascular systems), hypnagogic hallucinations, sleep paralysis, fragmented night sleep
Subtypes	Narcolepsy with cataplexy (50%) less common in paediatrics Narcolepsy without cataplexy
Epidemiology	M:F 1.8:1, most commonly in adolescents
Associated Problems	Behavioural, mood disorders, QOL
Management	Scheduled naps Treatment of daytime sleepiness e.g. dexamphetamine, methylphenidate, modafinil Treatment of cataplexy: clomipramine (TCA), sodium oxybate (GABA-B agonist, not readily available in Australia)

Delayed Sleep Phase Disorder	
Definition	Major sleep episode delayed by 1-2 hours of desired bedtime
Clinical Features	Normal sleep quality and quantity, sleep later on weekend, pattern present > 1 month Coexisting ADHD and depression common
Diagnosis	History and exam, actigraphy
Management	Sleep hygiene e.g. early morning light, advance wake time by 15 minutes every few days, don't revert weekends and holidays Melatonin 3-6 mg nocte Hospital admission for sleep rescheduling

Transition

The purposeful, planned movement of healthcare from paediatric to adult services.

- The largest numerical transitioning group is **developmental disorders** and **cerebral palsy** (though these patients may not necessarily need regular adult specialty care)

Trapeze is the SCHN transition service. It provides support to young people aged 14-25 years old, living with a complex chronic illness currently treated at SCHN, and who have an unclear health transition pathway.

Keys to a good transition:

- Start discussing transition early
- Ensure increasing confidence in self-management
- Re-educate on the chronic condition
- Shared paediatric and adult clinics
- Specific transition support services
- Proactive approach towards no-show patients

More formally, the ACI Key Principles of Care for Young People Transitioning to Adult Health Services are:

- A systematic and formal transition process

- Early preparation (education and empowerment commencing ~14 years old)
- Identification of a transition coordinator/ facilitator (from the young person's paediatric and adult specialty team)
- Good communication
- Individualised transition plan
- Empower, encourage, and enable young people to self-manage
- Follow up and evaluation

Ineffective or delayed transition often means poor health outcomes, e.g. loss of illness control, ED presentation with life-threatening conditions (**DKA is a common one**), deterioration of physical or mental health, secondary complications expedited, no primary care.

Louisa Leone Notes

17. Dentistry

Dental Anatomy

We are not just tonsil people!

Nomenclature: the face is divided into quadrants (RUQ, LUQ, RLQ, LLQ are 1-4 respectively). Then, teeth are numbered 1 onwards from medially to laterally.

- For example, **11** is the right central incisor

Gingivae blend into mucosa at the mucogingival junction. Gingivae are pink, while the mucosa is red.

- Oral herpes can cause **gingivostomatitis**
- Aphthous ulcers affect the **mucosa** not gingival tissues



Saliva is important for oral health

- There are minor salivary glands inside the lower lip. Mucocoeles (mucin retention cysts) can occur after lip-biting.
- Mumps can cause parotid gland swelling and produce a halo around the opening of the parotid gland in the buccal mucosa.

Check the uvula, tonsils, and posterior pharyngeal wall. Check the **hard palate**.

- Damage to the hard palate can be a sign of child sexual abuse.

Occlusion refers to the way the upper and lower teeth relate; implications in trauma.

- Remember a jaw injury can mean a head injury!

Perioral tissues

Feel for crepitus, discontinuities (on the external border of the mandible) and assess for mucosal tears.

Analgesia/anaesthesia: nitrous oxide is a good adjunct; local anaesthetic is often given as an infiltration (via 30G needle).

Oro-Facial Infections

Remember to put odontogenic infections in your differentials for **face swelling**. The child might have a history of dental caries, but not necessarily.

- One red flag for a dental cause is repeat antibiotics for facial infection

Assorted Conditions

- Eruption cyst: bluish cyst above where tooth is about to erupt, can cause feeding problems, aspirate to relieve
- Natal tooth: loose, no root structure, can lead to tongue ulceration
- Pacifier use can lead to an anterior open bite (resolves after pacifier stopped)
- Ankyloglossia: might need frenectomy
- **Dental caries:** increased risk if poor saliva quality/flow; prolonged night feeding
 - Part of the "tsunami of chronic disease": 75% of caries are found in 25% of children
 - High caries rates run in families, though typically environmental
 - Risk categories: special needs/late order offspring, mother with high caries rate, visible plaque, inappropriate prolonged feed, low SES and mother's education level
 - Preventative measures: clean for 2 minutes, 2x per day, use fluoride toothpaste, give teeth 2 hour rest between meals, limit sugar, drink enough water (to promote saliva production/flow)
- Stained teeth: purely aesthetic, supplemental iron can cause





Normal dental development

Primary (deciduous, milk)

- Develop in utero, and erupt between 6 – 36 months
- 20 primary teeth
- Development can give insight into intrauterine conditions

Permanent teeth

- Development commences from 3-4 months in utero, calcify from birth (so indicate postpartum health)
- Development complete at age 3, eruption occurs from 6 – 18 years old

Question if arrival of teeth is not bilateral, e.g. pathology in maxilla or mandible.

Oro-Facial Trauma

- Peak age 2-4 years, M:F 2:1
- Keep **non-accidental injury** on your radar
- Injuries include **degloving** (mucosa removed off alveolus), luxation, fracture, avulsion (knocked out), laceration, fracture of maxilla/ mandible/ nasomaxillary complex
- Most common tooth affected: anterior maxillary tooth (central incisor)
- Treat every head injury as a potential tooth injury, and vice versa

Avulsed primary teeth	Avulsed permanent teeth
Do not reposition	Reposition urgently
If borderline age (e.g. 5.5 years) and unsure if tooth is primary or permanent, place tooth in milk or saline	Temporising measure: place in cold milk, saline, or Hank's balanced salt solution

- Chin-point trauma: suspect a submandibular fracture
 - Always check **under the tongue** – a **sublingual haemorrhage** is a mandibular fracture until proven otherwise!

Medical-Dental Interface

Lots of medical conditions can present with dental problems. These were the ones given in the online video.

- **Ectodermal dysplasia**: alveolar underdevelopment, missing teeth, bubbly saliva, febrile convulsions
- **Single upper central incisor** in holoprosencephaly
- **Cleidocranial dysplasia** causing supernumerary teeth
- **Osteogenesis imperfecta**: blue sclerae, translucent-looking teeth
- **Primary immunodeficiency** can cause damage to alveolus/oral mucosa
- **Papillon-Lefevre Syndrome (autosomal recessive disorder of neutrophil function)**: skin lesions (palmar-plantar hyperkeratosis), hyperhidrosis, severe periodontal destruction.
- **Gastrointestinal reflux disease** can cause erosion of teeth, morning cough, halitosis, retrosternal pain
- **Calcium scale deposits** from poor dental hygiene pose an aspiration risk
- **Velocardiofacial syndrome**: bifid uvula, bulbous tip to nose, hypernasality of speech, soft palate not working, intellectual disability
- **Haemophilia** can cause prolonged bleeding from minor dental trauma
- **Radiation to face/head** can affect tooth development; also consider osteoradionecrosis
- **Foreign bodies**
- **Self-inflicted injury e.g. tongue biting**
- **Post-meningococcal sequela**: disruption of odontogenesis, discolouration of permanent teeth, or delayed tooth eruption

18. Emergency Medicine/Surgery

18.1 Recognising and Managing the Sick Child

Clinical Practice Guidelines : Recognition of the seriously unwell neonate and young infant

- Infections are the commonest cause of illness in neonates (<28 days corrected) and young infants (<3 mo.).
- UTI is the most common bacterial infection.
- Fever is classified in a neonate as $\geq 38^{\circ}\text{C}$ and warrants initial investigation and empiric IV antibiotics.
- Parental concern is everything!
- **Sepsis**: life-threatening organ dysfunction caused by dysregulated host response to infection.
 - **Septic shock** is sepsis + haemodynamic instability.

Airway	Breathing	Circulation/Hydration	Disability/Neurological
Position: tripoding Sounds: inspiratory stridor (upper airway), biphasic stridor (subglottic obstruction), expiratory wheeze (lower airway), grunting (airspace disease; self-PEEP); any talking or babbling? Effort: seesawing chest movement Secretion: drooling, pooling of saliva	Respiration: RR (Age-dependent), WOB, silent chest (exhaustion), SaO_2 Oxygenation : hypoxia, cyanosis. Abdominal pain might occur secondary to respiratory distress.	Colour and skin temperature. • Mottling • Central cyanosis: inside mouth (<i>lips are peripheral!</i>) Perfusion : central capillary refill $\geq 3\text{s}$, tachycardia (especially $> 160\text{ bpm}$), hypotension (late). • Sternum or forehead more reliable than peripheries HR (age-dependent) Hydration : feeding, mucous membranes, skin turgor, urine output	Consciousness: altered, semi-conscious, comatose, lethargy, irritability, cry, AVPU*, pupils, motor response, modified GCS if ≤ 4 years old (<i>verbal – crying and vocalisation</i>) Seizures Neurological signs: bulging fontanelle (9 – 12 months), neck stiffness, focal neurology, poor tone Swelling of limb or joint, or not weight-bearing/ not using an extremity.

*An AVPU at **pain** level is guaranteed GCS 9 or below ... start thinking about intubation!

Exposure for hypothermia or hyperthermia, rash, petechial bruising, injuries, fracture.

Beware the kid with non-blanching rash! An acutely ill, toxic child with a non-blanching rash has meningococcal septicaemia until proven otherwise.

- Henoch Schonlein Purpura is another cause of non-blanching rash. This is a small vessel vasculitis that is often preceded by an URTI or other mild illness 1-3 weeks prior. Therefore, it should be pretty easy to distinguish from meningococcal septicaemia, because the kid will not be acutely unwell.
- Unlike meningococcal septicaemia, the HSP rash is also palpable – a hallmark of vasculitis.

Causes to consider in an unwell neonate and young infant (from RCH Guideline)

	Condition	Salient Features
Infective: bacterial	UTI/pyelonephritis	Fever, vomiting, poor feeding
	Bacteraemia/ sepsis	Fever, tachycardia, tachypnoea, increased work of breathing
	Pneumonia	Irritable, neck stiffness, bulging fontanelle
	Meningitis	Reduced movement of limb +/- swelling
	Bone or joint	Erythema, swelling, tenderness, or lesions
Infective: viral	Skin	
	Bronchiolitis	Tachypnoea, increased work of breathing, wheeze or crackles on auscultation
	Primary HSV in first 1 month of life	Skin vesicles (not in 1/3 of neonates, and can be afebrile), <u>seizures</u> , <u>encephalopathy</u>
	Influenza and other respiratory viruses	Fever, poor feeding, lethargic, nasal congestion

	Enterovirus or parechovirus	Fever, rash, poor feeding, irritable, possible seizures, persistent or marked tachycardia (<u>myocardial involvement</u>)
Surgical	Malrotation with volvulus	Bile-stained vomit, abdominal distension, <u>75% occur in neonatal period</u>
	Pyloric stenosis	<u>3-6 weeks old</u> ; progressive, non-bilious and projectile vomiting, pyloric olive-shaped mass, hypochloreaemic hypokalaemic metabolic alkalosis
	Incarcerated hernia	Irreducible inguinal swelling
	Hirschsprung's Disease and meconium ileus	<u>Neonate <48 hours</u> (or infant with chronic constipation if short segment disease); abdominal distension with absent or infrequent bowel motions
	Necrotising enterocolitis (NEC)	Preterm neonate 1-3 weeks after birth; abdominal distension, tenderness, vomiting, blood in stool, discolouration or erythema of abdominal wall
	Intussusception	<u>6-18 month old (peak 9 months)</u> ; intermittent severe abdominal pain, vomiting, pallor, lethargy and rectal bleeding (red currant stool; a late sign).
Cardiac	Congenital cardiac disease	Cyanosis, murmur (not always), diaphoresis (sweating) with feeding, cardiac failure (tachypnoea, hepatomegaly, hypoperfusion), poor or absent peripheral pulses.
	Supraventricular tachycardia (SVT) and other arrhythmias	Persistent marked tachycardia, pallor, and poor feeding.
Respiratory	Meconium aspiration	Meconium-stained liquor, tachypnoea, increased WOB, cyanosis
	TTN and RDS	<u>Neonate</u> ; tachypnoea, increased WOB, possible cyanosis, radiological findings.
	Pneumothorax	Tachypnoea, hyper-resonance, decreased breath sounds
Endocrine and metabolic	Congenital adrenal hyperplasia	Ambiguous genitalia, hypotension, dehydration, hyponatraemia, hyperkalaemia, hypoglycaemia
	Hypoglycaemia	Altered level of consciousness, hypotonia, seizures, jaundice, organomegaly, dysmorphism
	Inborn errors of metabolism	Hypoglycaemia, metabolic acidosis
Non-accidental injury (NAI)	Child abuse	Bruising, unexplained injury, signs of intracranial injury, signs of neglect
Other	Acute bilirubin encephalopathy	Jaundice
	Brief resolved unexplained event (BRUE)	
	Toxin	
	Electrolyte abnormalities	e.g. iatrogenic from diluted formula water ingestion

Cardinal ED Presentations: [Breathing Difficulty](#), Fever, [Vomiting](#)/Diarrhoea/Abdominal Pain, [Seizures](#), [Injury](#).

Also see the section on [Urinary Tract Infection](#).

Meningitis and Encephalitis (Bonus)

- Bacterial meningitis is a medical emergency that requires empiric antibiotics without delay. Maintain a high index of suspicion in any unwell child, especially if there is altered mental state or no clear focus of infection.
- Consider HSV encephalitis in any child with encephalopathy, and in neonates. Treat with aciclovir.
- In immunocompromised, consider cryptococcus and mycobacteria.

It can be difficult to distinguish viral from bacterial meningitis clinically, but the aetiology differs.

	Meningitis	Encephalitis
History	Fever Infant: non-specific, <u>irritability</u>, lethargy or drowsiness, poor feeding, apnoea, vomiting and diarrhoea, temperature instability Child: headache, neck pain, <u>photophobia</u>, nausea, <u>altered conscious</u>.	Fever <u>Altered mental status</u> (unusual behaviour, confusion, personality change, emotional lability) Seizures (common) Abnormal movements Headache Vomiting Lethargy Prodromal illness

Associated Features	Preceding URTI Seizures Recent antibiotics Immunisation history, travel history Predisposing medical condition e.g. CNS anatomical abnormality or shunt, immunosuppression, immunodeficiency	Features of prodromal illness Exposure Hx: travel, location (e.g. NT, far north Australia), mosquito/ tick/ insect bites, animal contact Maternal HSV Immunisation history Immunosuppression or immunodeficiency Consider other causes e.g. acute disseminated encephalomyelitis (ADEM), toxins or metabolic
Examination	Full or bulging fontanelle High-pitched cry Fever or hypothermia Hyper or hypotonia Neck stiffness (may be absent in infants) Focal neurological signs Purpuric rash Pain and involuntary effort to reduce meningeal "stretch" e.g. Kernig and Brudzinski signs	Altered conscious Confusion Fever Bradycardia and hypertension (raised ICP) Focal neurological signs Abnormal movements e.g. chorea Vesicular rash, corneal ulcer/ keratitis (HSV)

Investigations

- **Urgent lumbar puncture:** urgent CSF microscopy and biochemistry with simultaneous blood glucose.
 - **Defer if:** focal neurological signs, ongoing seizures, markedly reduced GCS, cardiovascular compromise or coagulopathy.
 - Do not delay antimicrobials if LP will take 30+ minutes.

Parameter	Bacterial Meningitis	Viral Meningitis	Tuberculous Meningitis
Predominant cell	Polymorphonuclear cells	Lymphocytes	Lymphocytes predominate
Cell count/mm³	1000-10 000	<300	50-500
Protein	0.5 – 2 g/L (usually > 1 g/L) (increased)	0.5 – 1 g/L	0.5 – 4 g/L (markedly increased)
Glucose	< ½ blood glucose	> ½ blood glucose	< ½ blood glucose

Note: **red and white cell ratios** can be used to compare the CSF result with peripheral blood. Some clinicians suggest RBC:WBC 500:1 but it can be unreliable. It is hard to interpret blood-stained CSF; err on the side of caution and treat as meningitis until senior review.

- **Bloods:** blood culture, FBC (may be normal), BGL, VBG (lactate), EUC (hyponatraemia), coagulation studies (if shock or coagulopathy suspected), LFTs, metabolic and toxicology testing (if non-infective cause suspected); serum bacterial and viral PCR
- **Neuroimaging** if encephalopathy, signs of raised ICP, or focal neurology. **EEG** might be useful in suspected encephalitis.
 - Normal CT brain does not exclude raised ICP and should not influence the decision to perform LP.
 - MRI can be helpful but often requires general anaesthetic.

Treatment: empiric treatment should cover meningitis; cover for encephalitis if clinical suspicion and/or child is obtunded. Give antibiotics within **30 minutes of the decision to treat** and consider **steroids if 2+ months and meningitic**.

- The evidence for steroids is sketchy. Not recommended in neonates (can affect neurodevelopment). Can help reduce the risk of hearing loss following Hib meningitis and possibly pneumococcal meningitis.
- Give first dose of IV dexamethasone just before or with first dose of antibiotics; if after Abx, should be within 4 hours.

Age group	Common organisms	Empiric antibiotic	Dexamethasone
Meningitis			
0-2 months	GBS, E. coli, Listeria monocytogenes (rare)	Benzylpenicillin or ampicillin IV <u>and</u> Cefotaxime or ceftriaxone IV (<i>not in neonates <41/40 gestation</i>)	Not advised
2+ months	N. meningitidis, S. pneumoniae, Hib	Ceftriaxone IV or cefotaxime IV Add vancomycin if gram-positive cocci on gram stain	0.15 mg/kg (max 10 mg) IV 6 hourly for 4 days
Any age	Streptococcus pneumoniae	Add vancomycin if suspected	

Encephalitis: as above, plus			
	HSV	Aciclovir IV*	Not advised
	Mycoplasma pneumoniae, EBV, CMV, HHV6, influenza, arboviruses	Consider azithromycin	

Basically, *N. meningitidis* in over 2 months. Consider neonatal sepsis causes in <2 months.

Benzylpenicillin covers GBS, *Listeria*, *N. meningitidis* and *S. pneumoniae*; **Cefotaxime/ceftriaxone** cover penicillin-resistant *S. pneumoniae*, *HiB*, gram negatives.

***Aciclovir** is not routinely prescribed for infants 3 months or older. It is indicated in children 1-3 months old with a strong suspicion of Herpes simplex virus as the causative organism.

Ongoing management:

- Treat seizures immediately
- Fluid management: resuscitation is **10 mL/kg 0.9% sodium chloride IV bolus**
 - Conservative end of normal because *children with CNS infections are at higher risk of cerebral oedema*.
 - You must ensure normal BP even with intracranial HTN.
- 6-12-hourly monitoring of: hydration (weight, Na and acid-base status), and clinical assessment of neurological state
- Monitor weight, head circumference under 2 years, vital signs (HR, BP), EUC, blood glucose

Recall also that ***Neisseria meningitidis*** and ***Haemophilus influenzae*** are notifiable diseases.

Complications:

- Multi-organ involvement, venous sinus thrombosis, hydrocephalus, seizures/epilepsy
- **Bacterial meningitis is fatal in 5-40% of children.**
- Hearing and neurodevelopmental impairment or permanent focal neurological deficit.
 - **Hearing impairment (direct infection and post-infectious ossification)** is most common with *S. pneumoniae* (*can occur from 48 hours*): needs inpatient audiology testing and early ENT/audiology referral
- **Persistent fever 4-6+ days:** nosocomial infection, subdural effusion or empyema, cerebral or parameningeal abscess, inadequate treatment.

Follow-up: formal audiology assessment 6-8 weeks post-discharge; monitor neurodevelopmental progress; investigate for complement deficiency if 1+ episode of meningococcal disease.

Complement is essential for killing *Neisseria*, particularly the membrane attack complex (C5-C9).

Vomiting

- Most commonly gastroenteritis: fever, vomiting, diarrhoea.
- **Aetiology of gastroenteritis:**
 - **Viral most common (70%):** rotaviruses, enteric adenoviruses, small round viruses, caliciviruses, astroviruses, enteroviruses.
 - **Bacteria (15%):** *Campylobacter jejuni*, *Salmonella* spp, atypical enteropathogenic *E. coli*, enteropathogenic *E. coli*, *Shigella* spp, *Yersinia enterocolitica*, Shiga toxin producing *E. coli*, *Salmonella typhi* and paratyphi, Cholera, *Clostridium difficile*
 - **Parasites (15%):** *Cryptosporidium*, *Giardia lamblia*, *Entamoeba histolytica*, *Strongyloides stercoralis*.
- **Red flags:** bilious, fever > 39 °C or persistent < 36 °C, significant abdominal pain, babies and kids < 2 years old, colicky or crescendo-type abdominal pain (*note: appendicitis progresses quickly*)
 - **Abdominal pain, vomiting, and rapid/irregular breathing in an older child → DKA!**

Investigations

- Unwell neonates and young infants: BGL, VBG, FBC, EUC, LFT, CRP, blood culture, urine MCS
- Lumbar puncture in neonates, *consider* for young infants.

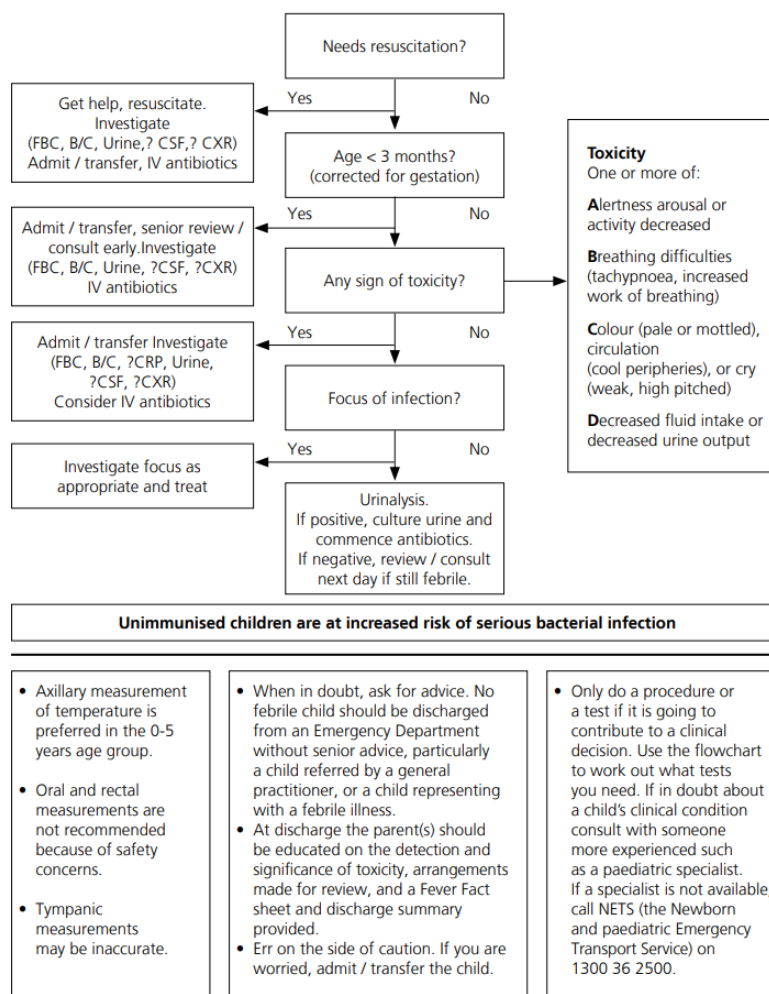
Preferred Methods of Urine Collection in Febrile Child	
<i>Toilet trained</i>	<i>Urgent or prior to toilet training</i>
Clean catch (<i>can be slow</i>)	In-out catheter urine (preferred invasive method); suprapubic aspirate if < 1 year

- Investigations targeted to cause: NH3 level (inborn errors of metabolism, acute liver failure → encephalopathy); viral swabs e.g. HSV PCR; ECG
- Imaging depending on cause: CXR, abdominal imaging, neuroimaging.

Treatment: early referral to paediatric, surgical and/or sub-specialist team as indicated plus involvement of senior clinician.

- Resuscitation: DRS ABCD
- Seriously unwell neonates and young infants: prompt management of sepsis with early empiric antibiotics (IV/ IM/ IO); aciclovir if indicated; adequate analgesia and sedation.
- Judicious use of fluids
- Treatment targeted to cause
- NG tube on free drainage for suspected bowel obstruction
- In neonates with suspected *duct-dependent congenital cardiac condition*, early discussion with cardiology/ ICU/Retrieval Team.

Flowchart for child < 5 years old with fever (>38°C axillary)



Source: NSW Health [POLICY FRAMEWORK](#)

Most Common Sources of Paediatric Infection

Age Group	Infection Type	Common Pathogens	Key Notes
Neonate (<28 days)	Sepsis, meningitis, pneumonia	Group B streptococcus, E. coli, Listeria monocytogenes	GBS in term infants; E. coli in preterm; consider HSV if vesicular rash or encephalitis
	Conjunctivitis	Neisseria gonorrhoeae, Chlamydia trachomatis	Acquired during vaginal delivery; N. gonorrhoeae → purulent in 2-5 days; Chlamydia later (5 – 14 days)

Infants (1 – 12 months)	<i>Meningitis</i>	S. pneumoniae; N. meningitidis; H. influenzae type B	After 3 months, GBS and E. coli less common; fully immunised infants less likely to have Hib
	<i>Pneumonia</i>	RSV, rhinovirus, HMPV, adenovirus, S. pneumoniae	Viral pneumonia predominates under 2 years; pneumococcal or pertussis if not immunised
	<i>UTI</i>	E. coli	Especially in uncircumcised males or structural urinary anomalies
	<i>Gastroenteritis</i>	Rotavirus, norovirus, salmonella, E. coli	Viral more common overall, bacterial in severe or bloody diarrhoea
Toddler/ preschool (1 – 5 years)	<i>Otitis media/ sinusitis</i>	S. pneumoniae, H. influenzae, Moraxella catarrhalis	Often follows viral URTI
	<i>Pneumonia</i>	RSV, HMPV, S. pneumoniae, Mycoplasma pneumoniae (>2 years)	Mixed viral-bacterial infections common
	<i>Meningitis</i>	S. pneumoniae, N. meningitidis	Vaccine-preventable in most developed settings
	<i>Skin/ soft tissue infections</i>	S. aureus, S. pyogenes (GAS)	Impetigo, cellulitis, abscesses
School-aged child (5 – 12 years)	<i>Pharyngitis</i>	Group A Streptococcus	Must test for GAS to guide antibiotic use
	<i>Pneumonia</i>	S. pneumoniae, Mycoplasma pneumoniae (<i>walking pneumonia</i>), Chlamydophila pneumoniae	
	<i>Gastroenteritis</i>	Rotavirus, norovirus, salmonella, campylobacter	
	<i>UTI</i>	E. coli, Enterococcus	More common in girls
Adolescent (13 – 18 years)	<i>Meningitis</i>	N. meningitidis; S. pneumoniae	Teen outbreaks; meningococcal vaccine schedule applies
	<i>Pneumonia</i>	Mycoplasma pneumoniae, Chlamydia pneumoniae	Atypical, gradual onset; contagious in close contact settings
	<i>Sexually transmitted infection</i>	Chlamydia trachomatis, Neisseria gonorrhoeae, HSV	Consider in all sexually active adolescents with discharge or pelvic pain

18.1.1 The Febrile Infant

- **Definition of fever:** body temperature $>38.0^{\circ}\text{C}$
 - Normal temperature is $36.5 - 37.5^{\circ}\text{C}$ by axillary measurement.
- The best site to measure a kid's body temperature is typically **axillary**.
 - Technically, rectal temperature is the most accurate, but understandably it is not commonly performed (one exception is if a child is hypothermic from drowning).
- **Aetiology:** viruses most common ($>90\%$); need to consider serious bacterial infections (SBIs).
- The most useful indicator of severity is **the appearance of the child** ... not the degree of fever, rapidity of onset, response to antipyretics, or the presence of febrile seizures.
- Any child with fever for **5+ days must** be assessed for **Kawasaki Disease**, or, if history of COVID-19, **Paediatric Inflammatory Multisystem Syndrome (PIMS-TS)**.

At-risk groups: prematurity, immunosuppression/oncology, central line or other foreign device (e.g. shunt), chronic lung disease, congenital heart disease, previous SBI, Indigenous Australian/Maori, multiple presentations.

History

- Localising symptoms e.g. cough, headache, photophobia, diarrhoea, vomiting, abdominal pain, MSK pain, rash
- Travel
- Sick contacts
- Immunisation (beware children < 6 months age or with incomplete immunisation)
 - **Post-vaccination fever** is common. It typically onsets within 24 hours of immunisation and lasts up to 2-3 days. However, in an unwell child, **fever should not be attributed to vaccination alone**.

Random factoid: *teething* does not cause fever.

Examination: assess for signs of shock/toxicity and also for a clinical focus of infection.

- **Red flags for sepsis/toxicity (bacterial endo/ exotoxin):** ABCD (see below); drowsy, lethargic or irritable, pale, mottled, or tachycardic.

Signs of Toxicity			
<u>A</u> lertness, arousal, or activity decreased	<u>B</u> reathing difficulties	<u>C</u> olour (pale or mottled); circulation (cool peripheries, central cyanosis); cry (weak or high-pitched)	<u>D</u> ecreased fluid intake or urine output.

Investigations

Any **seriously unwell child** should be managed as suspected **sepsis**, regardless of the degree of fever. **Do not accept apparent AOM or URTI** as the source of infection in young infants or unwell children – they still need assessment for possible SBI.

- Children with a definite focus of infection should only have investigations specific to that focus unless very young, or toxic.
 - Blood culture should be performed whenever a blood count is performed on children with toxicity and/or a focus of infection. WCC and CRP can guide antibiotics.
 - CXR if signs of respiratory illness, otherwise defer.
 - Lumbar puncture in young infant, toxic child, irritable, or complex febrile convulsions; however, do not delay resuscitation and antibiotics.
 - **Contraindications:** impaired consciousness, focal neurological signs, impaired coagulation, haemodynamic instability.
 - If LP contraindicated, commence treatment for meningitis/encephalitis and perform LP when the child is stable.
 - Urine culture in all febrile children < 3 months old and all toxic children (clean catch recommended; if urgent, suprapubic aspirate). If non-toxic, febrile, and > 3 months, urinalysis is acceptable.
 - **UTI is the most common SBI.** Always perform **urine collection and testing** if there is no clinically obvious focus for fever.

Infants ≤ 28 days corrected	Infants 4 – 12 weeks	Infants > 3 months
Full septic workup: FBC, CRP, blood culture, urine culture (clean catch; if urgent, suprapubic aspirate), LP +/- CXR	Sepsis workup and empiric antibiotics if suspicion of sepsis +/- delayed LP pending blood results.	Consider clinical appearance and history for threshold. Hypothermia and temperature.

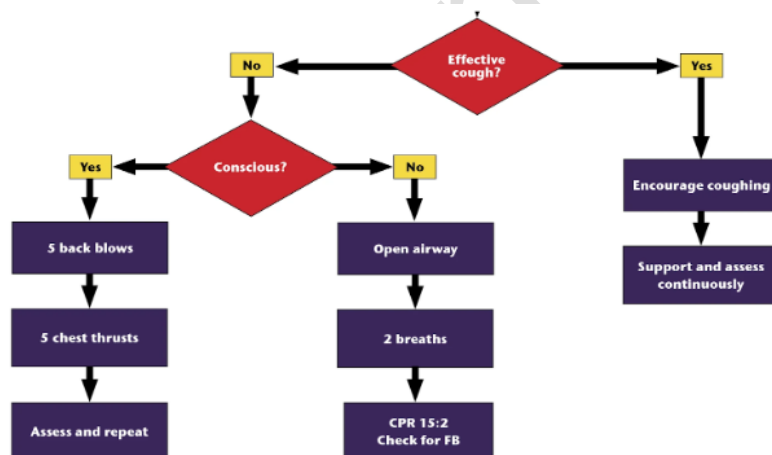
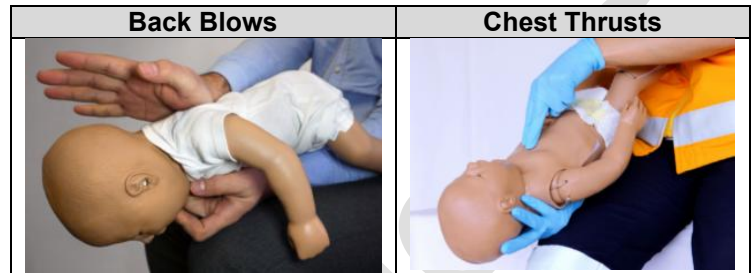
- Antipyretics can provide comfort: **no ibuprofen for < 6 months old!**
- Tepid sponging and other physical methods of reducing temperature are not recommended; unwrapping an overdressed child is appropriate.
- **Follow-up:** if discharged from ED with fever, follow up the next day to assess progress. Parents should be encouraged to look for toxicity every 4-6 hours and seek clinical review if the kid becomes toxic or unwell.

18.2 Foreign Body Aspiration

- Usually **under 4 years old**; nuts, raw carrot, apple, seeds, popcorn, coins, balloons, and pieces of toys. Most commonly in **right main bronchus**.
 - Choking accounts for 80% of deaths under age 1 year.
- Clinical features depend on location. A child with proximal impaction **may present with cardiorespiratory arrest**.
- The diagnostic triad of **cough, wheeze, and unilaterally diminished breath sounds** is present in <50%. Up to 25% may not have a witnessed choking episode.

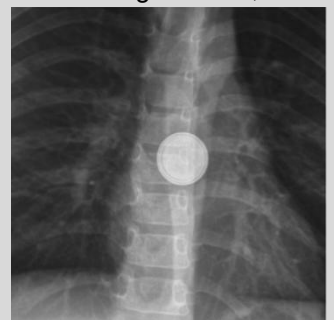
Impaction in larynx or trachea	Impaction in main bronchus (or lower)
Sudden and catastrophic; coughing, choking, vomiting; severe respiratory distress; stridor; cyanosis; altered mental state; drooling and voice changes.	Witnessed choking, coughing, or wheezing while eating or playing; tachypnoea and respiratory distress; persistent wheeze (+/- focal) or cough; fever; haemoptysis; shortness of breath; recurrent or persistent consolidation. May present as focal pneumonia, abscess, bronchiectasis.

- **Investigations:** consider CXR +/- inspiratory and expiratory views (air trapping); bronchoscopy.
 - Many inhaled foreign bodies are radiolucent. **Only 10% are radio-opaque, and 30-40% CXR are completely normal.**
 - Any kid with **reasonable suspicion** of airway foreign body requires exploratory rigid **bronchoscopy!**
- URGENT intervention: evidence of actual or potential airway obstruction, actual or imminent oesophageal perforation, battery ingestion, unable to manage secretions, organic material (e.g. peanuts or peas) which can swell with moisture → asphyxiation.
- **Management:**
 - Encourage coughing and assess for deterioration. Give oxygen and consider adrenaline nebulisers and steroids to reduce airway swelling/irritation **before considering bronchoscopy.**
 - Ineffective cough or total/near-total airway obstruction
 - Call for anaesthetics and/or ENT to assist with securing the airway.
 - 5 back blows, 5 chest thrusts, remove foreign body under vision (may use laryngoscopy + Magill forceps if obstruction persists > 1 minute and FB visualised).
 - Unresponsive → **CPR**; prior to ventilation, direct laryngoscopy + remove FB with Magill forceps if visualised.
 - Persistent FB above vocal cords → needle or surgical cricothyroidotomy; below → right mainstem bronchus intubation.
- After immediate management, patient requires transfer to theatre for complete removal of FB.



Foreign Body Ingestion

- Highest risk 6 months to 3 years old, plus children with developmental concerns or psychiatric problems.
- Ingested foreign bodies are twice as common as inhaled foreign bodies but only 10-20% require removal.
- Food bolus impaction is rare but may occur in kids with oesophageal strictures or eosinophilic oesophagitis.
- **High-risk FB: button batteries, magnet + metal object or >1 magnet**, sharp objects in oesophagus (e.g. fish bones, pins, needles), lead based, large (>6 cm long or >2 cm wide), expandable foreign bodies, multi-component objects, lodged in oropharynx.
- **Most common site of impaction:** just below the cricopharyngeus muscle, the "upper oesophageal sphincter".
- **Clinical features:** dysphagia (solids and liquids depending on severity and level of obstruction); regurgitation, vomiting. Drooling is a sign of complete obstruction.
- **X-ray (neck, chest abdomen)** if high-risk or unknown FB, high-risk or unwell child.
- **Acutely symptomatic** children (drooling, neck, or chest pain) require urgent endoscopy.
- **Button battery ingestion:** double-ring or halo sign on CXR. Button battery in oesophagus requires a time-critical endoscopy (within 2 hours).
 - Button batteries can cause full-thickness burns in 4 hours.



- Administer honey to children >1 year with an oesophageal button battery within 12 hours of ingestion while awaiting theatre. **10 mL every 10 minutes up to 6 doses.**
- Not all foreign bodies need to be removed (unless they are batteries!). Once an object passes through the oesophagus, it is most likely to pass PR via the digestive tract.

[Clinical Practice Guidelines : Foreign body ingestion](#)

Fish Bones: most commonly lodge in the palatine tonsils, vallecula, or pyriform fossa. Causes localised discomfort.

- These kids require laryngobronchoscopy or CT scan.

Foreign body aspiration: foreign body lodged on R side with corresponding hyperinflation (ball-valve effect). The foreign body partially obstructs expiratory airflow but allows some air to enter during inspiration, causing air trapping and overinflation of that lung or lobe. The opposite lung appears relatively less inflated due to compensatory mechanisms or poor ventilation.



18.3 Anaphylaxis

[Clinical Practice Guidelines: Anaphylaxis](#)

- Severe allergic reaction, usually IgE-mediated. Lifetime prevalence is **1-3%**.
- Clinical diagnosis made in the setting of **acute onset** of:
 - Skin features (urticaria, flushing, and/or angioedema) plus involvement of one other system:
 - Respiratory (e.g. cough, wheeze, stridor, chest pain, dyspnoea)
 - Cardiovascular (pale and floppy, palpitations, tachy/bradycardia, hypotension, arrest)
 - Severe GI symptoms (**especially after exposure to non-food allergens**)
 - Note, abdominal pain after an insect bite is anaphylaxis until proven otherwise!
 - OR hypotension, bronchospasm, or upper airway obstruction where anaphylaxis is possible.
- Most anaphylaxis occurs within **30 minutes of exposure** but can occur **up to 4 hours later**.

Foods	Peanut, tree nut, cow milk, eggs, soy, shellfish, fish, wheat
Bites/ stings	Bee, wasp, jack jumper ants, ticks
Medications	Beta-lactams, NSAIDs
Other	Exercise, idiopathic, rubber latex (bottle nipples, pacifiers, toys)

To have anaphylaxis to foods, you need ingestion.

Cofactors: physical activity and exercise, NSAIDs, alcohol, sleep deprivation (↓ reaction threshold or ↑ severity).

Treatment

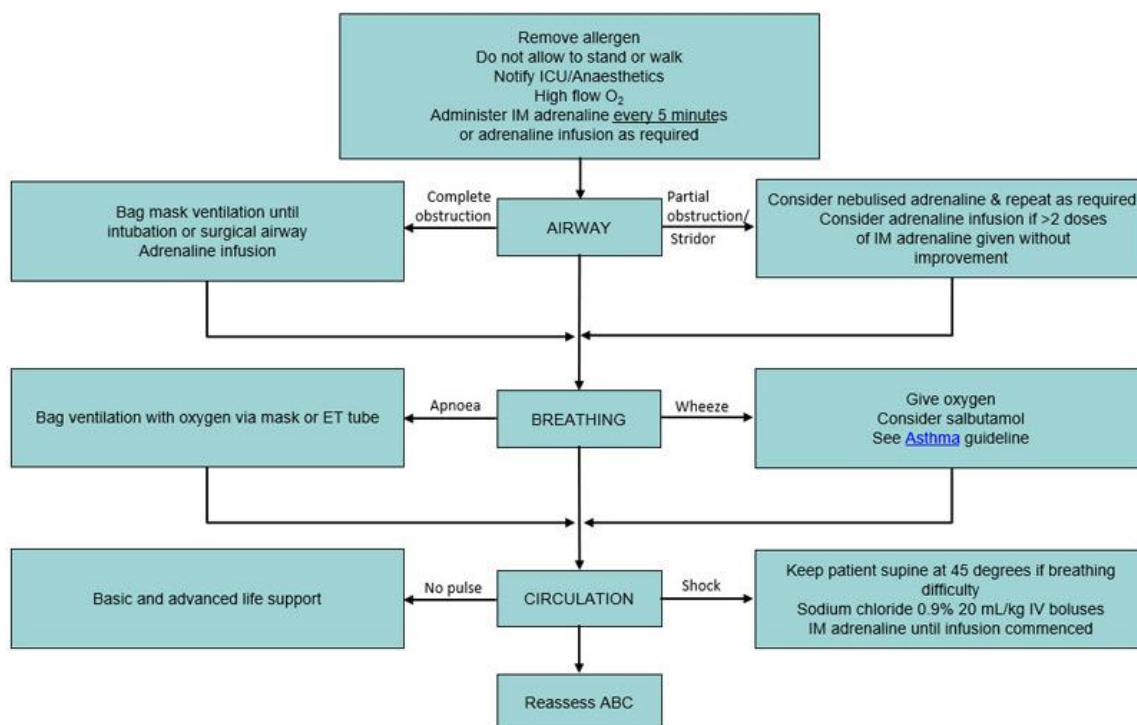
- Remove allergen if still present e.g. insect stinger, food debris in mouth.
- Lay the child flat; **do not allow them to stand or walk**. Treat supine or lying on side; if vomiting child is sat upright, monitor for hypotension.
- **10 µg/kg or 0.01 mL/kg of 1:1000 IM adrenaline (max 0.5 mL)** into lateral thigh; repeat after 5 minutes if the child is not improving.
 - **This is the same dose as in arrest but different concentration (1:1000 IM vs 1:10 000 IV).**
- No IV adrenaline unless arrest is imminent.
- A new intranasal adrenaline **Neffy** is game-changing: 1 mg dose for 15 – 30 kg, 2 mg for 30+ kg.

Adrenaline Doses			
Age (years)	Weight (kg)	Volume 1:1000 adrenaline (mL)	Adrenaline autoinjectors
<1	<7.5	0.1	Not available
<1 – 2	7.5 – 10	0.1	150 mcg device (EpiPen Jr, green) 
2 – 3	15	0.15	
4 – 5	18	0.18	
5 – 6	20	0.2	300 mcg (EpiPen, yellow) 
7 – 10	30	0.3	
10 – 12	40	0.4	
> 12 and adults	> 50	0.5 mL	300 mcg or 500 mcg device

So, children < 5 years old and 7.5 – 20 kg can get **EpiPen Jr**; children >5 or > 20 kg (think, school-age) get **EpiPen!**

Give IM adrenaline **every 5 minutes** until IV access is established.

- Nebulised or MDI salbutamol if respiratory distress with wheezing.
- Consider nebulised adrenaline if upper airway obstruction/ bronchospasm present.
- If hypotensive, **20 mL/kg 0.9% NaCl resuscitation for shock.**



Make sure that any kid with anaphylaxis is observed for at least 4 hours to manage deterioration.

18.4 Paediatric Basic Life Support

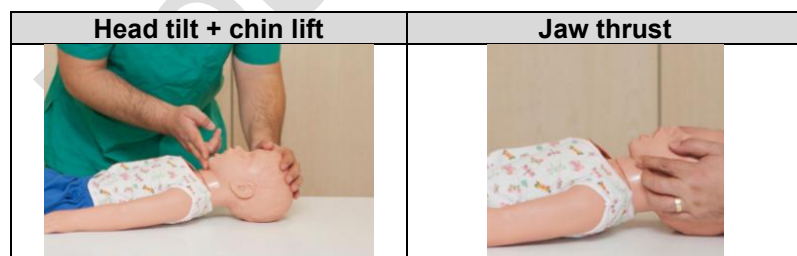
Danger: slipping, fire, contamination/infection, wires, sharps, unsteady objects, people.

Response: Call name, 'Are you okay?', move arm, assess pain response (trapezius squeeze)

Send for help: 2222 (NSW Health)

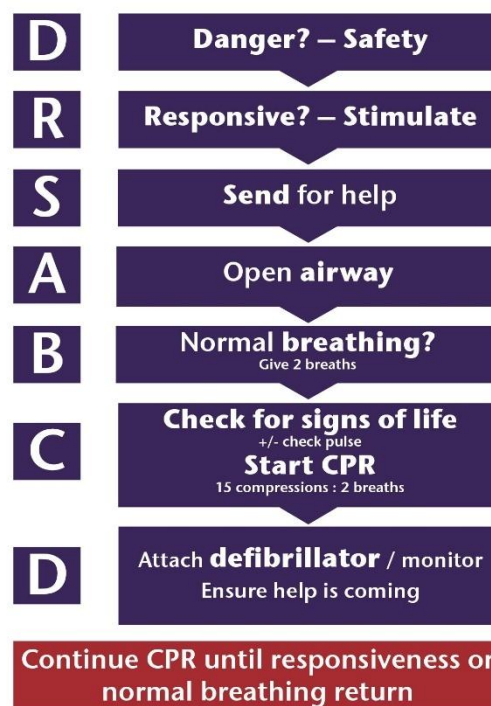
Airway:

- Consider suctioning any foreign body.
- Infant < 1 year: neutral position (towel under shoulder; large occiput → neck flexion when prone).
- Child > 1 year: sniffing position.
- Basic manoeuvres: head tilt + chin lift; jaw thrust



- Airway adjuncts:
 - Guedel: middle incisor to angle of mandible ("hard to hard"); insert facing down (laryngoscope or tongue depressor for view).
 - Nasopharyngeal tube: lateral nare to tragus of ear ("soft to soft"); insert straight with lubricant; **contraindicated in base of skull fracture.**

Breathing: look, listen, feel. If not, give 2 rescue breaths with the bag valve mask.



Circulation: check for signs of life +/- check pulse (brachial or femoral).

- Pulses can be unreliable; compress CPR if unresponsive.
- **CPR:** if one responder, 30:2; in hospital and multiple responders, 15:2.
- **Delivery:** lower half sternum, rate 100 – 120 compressions/minute, 1/3 depth of chest.
- **Techniques:**

Encircling (infant)	Two fingers (infant/ small child)	One hand (small/ medium child)	Two hands (larger child)
			

Next Steps:

- Continue CPR and ensure help is coming.
- Attach defibrillator (COACHED): determine shockable vs. non-shockable rhythm.
 - Shockable rhythm: DC shock 4J/kg biphasic.
- IO access or IV if < 90 seconds. Bloods from IV (not IO) only if quick and not delaying medications.
- **Adrenaline: 10 mcg/kg IV or IO (0.1 mL/kg of 1:10 000).**
- Consider causes (4 H's and 4 T's).

18.5 Managing Fluids in Young Children and Infants

Clinical Practice Guidelines : Intravenous fluids

- Use the enteral route whenever possible.
- The preferred fluid is usually **sodium chloride 0.9%** (with **5% glucose +/- potassium** for maintenance)
- Most sick children will retain water and need **less than full maintenance fluids**
- **Serial weights** are the best measure of acute changes in fluid status.

Red flags: abnormal sodium levels, increased ADH secretion (especially acute CNS and pulmonary conditions), short gut or GI pathology, fluid resuscitation > 20 mL/kg required

Examination

- Hydration status, and signs of fluid overload: fontanelles, sunken eyes, central capillary refill, temperature (peripheral to central)
- Weight at least daily (more often if severe dehydration or ongoing losses)

Risk factors for dehydration: young ($\uparrow$ SA:V ratio), low body weight, large stool/stoma losses in past 24 hours, vomited 2+ times, cannot tolerate enteral feeds, no feeding, renal/ cardiac/ endocrine disease (e.g. CAH, DM), inborn errors of metabolism.

Assessment of the severity of dehydration

Degree of Dehydration	% body weight	Clinical signs
Mild	3%	Reduced urine output
Moderate	5%	Tachycardia, reduced skin turgor
	7%	Marked reduction in skin turgor, reduced fontanelle tension, lassitude and/or restlessness
Severe	10%	Shock, marked tachycardia, prolonged capillary refill +/- hypotension

Investigations

The 3 core tests in every kid receiving IV fluids are VBG, EUC, and glucose.

- Check **electrolytes and glucose** before starting IV fluids and at least daily if receiving > 50% maintenance.
 - More often if unwell, large fluid losses, or abnormal electrolytes.
- Document inputs, outputs, and repeated weights
 - **Remember to perform a bare weight on admission!**
- In anything more than mild dehydration: VBG, EUCL BGL, FBC, targeted investigations e.g.
 - Stool microscopy, culture, virology if gastroenteritis (debatable since usually self-resolving ... probably only if very sick)
 - Urinalysis and culture if suspecting UTI
 - Blood culture if T > 38.5 C
 - AXR, CXR, US, air enema (intussusception)

Treatment

Resuscitation fluid (for shock): 10-20 mL/kg bolus sodium chloride 0.9%. In the case of haemorrhage (E.g. trauma) blood may be given if immediately available (usually O negative).

- Do not include in subsequent calculations
- Remember, **no resuscitation with glucose** (it enters the cell!)
- Once the fluid bolus is administered, reassess perfusion (e.g. capillary refill).
- After 2 20 mL/kg boluses, we might think about **inotropic support (ICU level)**

Rehydration (treat deficit):

- Mild-moderate: enteral (oral or NG) rehydration preferable.
- Severe or cannot tolerate enteral intake: IV fluid rehydration

Total fluid requirement = Maintenance + Replacement of deficit + Replacement of ongoing losses

Calculating fluid deficit:

$$\text{Deficit (mL)} = [\text{Premorbid weight (kg)} - \text{current weight (kg)}] \times 1000$$

If a pre-morbid weight is not available, use:

$$\text{Deficit (mL)} = \text{weight (kg)} \times \% \text{ dehydration} \times 10$$

Sub in the % dehydration as a number between 1 and 100; not as a decimal.

The deficit should be replaced **over 24 – 48 hours**:

- Usually prescribe rehydration fluids calculated at the **5% level of dehydration over 24 hours**.
 - The exception is **DKA where kids often have a greater total body water deficit than their initial clinical assessment; always seek senior advice!**
- For children with a greater deficit, replace the rest over the next 24 hours.

Ongoing Fluid Losses:

- Replace **ongoing losses** based on the previous hour (if significant) or 4-hour period (e.g. you may replace a 200 mL loss over 4 hours with 50 mL/hr over 24 hours).
- GIT losses are typically replaced with **sodium chloride 0.9% + potassium chloride 20 mmol/L**

Oral Rehydration Fluids

- Gastrolyte (Hydralyte and Pedialyte also available in Australia). Contains: glucose (90 mmol/L), sodium (60 mmol/L), Cl (60 mmol/L), K (20 mmol/L), base (citrate) (10 mmol/L).
- For example, during **gastroenteritis**, aim for oral rehydration at **5 mL/kg/hr** (or **0.5 mL/kg every 5 minutes via syringe**)
 - Offer 2-hourly breastfeeds if children are breastfeeding!

Maintenance Fluids

Recommendations for Fluid Maintenance AND Replacement	
< 28 days	0.45% NaCl + 10% glucose +/- KCl*
> 28 days	0.9% NaCl + 5% glucose +/- KCl

* Alternatives: Plasma-Lyte 148 with glucose 5%; Hartmann's with glucose 5%

- Holiday-Segar method (100:50:20 rule, or 4:2:1 rule)
- Full maintenance fluids would be appropriate for a well child e.g. fasting for elective surgery. See below for **fluid restriction in unwell children**.

Body weight	Fluid requirement mL/day	Fluid requirement mL/hr
First 10 kg	100 mL/kg	4 mL/kg/hr
Second 10 kg	50 mL/kg	2 mL/kg/hr
Subsequent kg	20 mL/kg	1 mL/kg/hr

2400 mL/day (100 mL/hr) is the normal maximum amount.

Fluid requirements may be modified in certain conditions:

- Increased: fever, hyperventilation, high ambient temperature, extreme activity, abnormal fluid loss
 - A typical example is **increasing maintenance fluids by 10-20% when fever is present**
- Decreased: hypothermia, very high humidity, oliguria/anuria, extreme inactivity, fluid retention

Maintaining Electrolytes and Glucose

Sodium	Potassium	Glucose
First week: 1 to 1.5 mmol/kg/day 2-3 months: 3 mmol/kg/day 2 years: 2 mmol/kg/day	2 mmol/kg/day	Neonate: 6-8 mg/kg/minute Older: 4-5 mg/kg/minute

In summary:

1. Calculate bolus, maintenance, rehydration, and ongoing loss volumes separately.
2. Calculate fluids needed for the next 24 hour period: if rehydration is needed, assume 5% dehydration
3. Review after 6-8 hours.

Fluid Restriction: 2/3 maintenance rates should be used in most unwell children unless they are dehydrated.

- This is because unwell children are likely to secrete excess ADH.
- Children at risk of even more ADH requiring further fluid restriction: acute CNS conditions, pulmonary conditions (including those on mechanical ventilation), post-operatively and in trauma.

Glucose:

Treat hypoglycaemia specifically if BSL < 2.6 or BSL < 3 + symptoms. If **severely unwell** or not tolerating enteral intake, administer **IV 2 mL/kg 10% dextrose**.

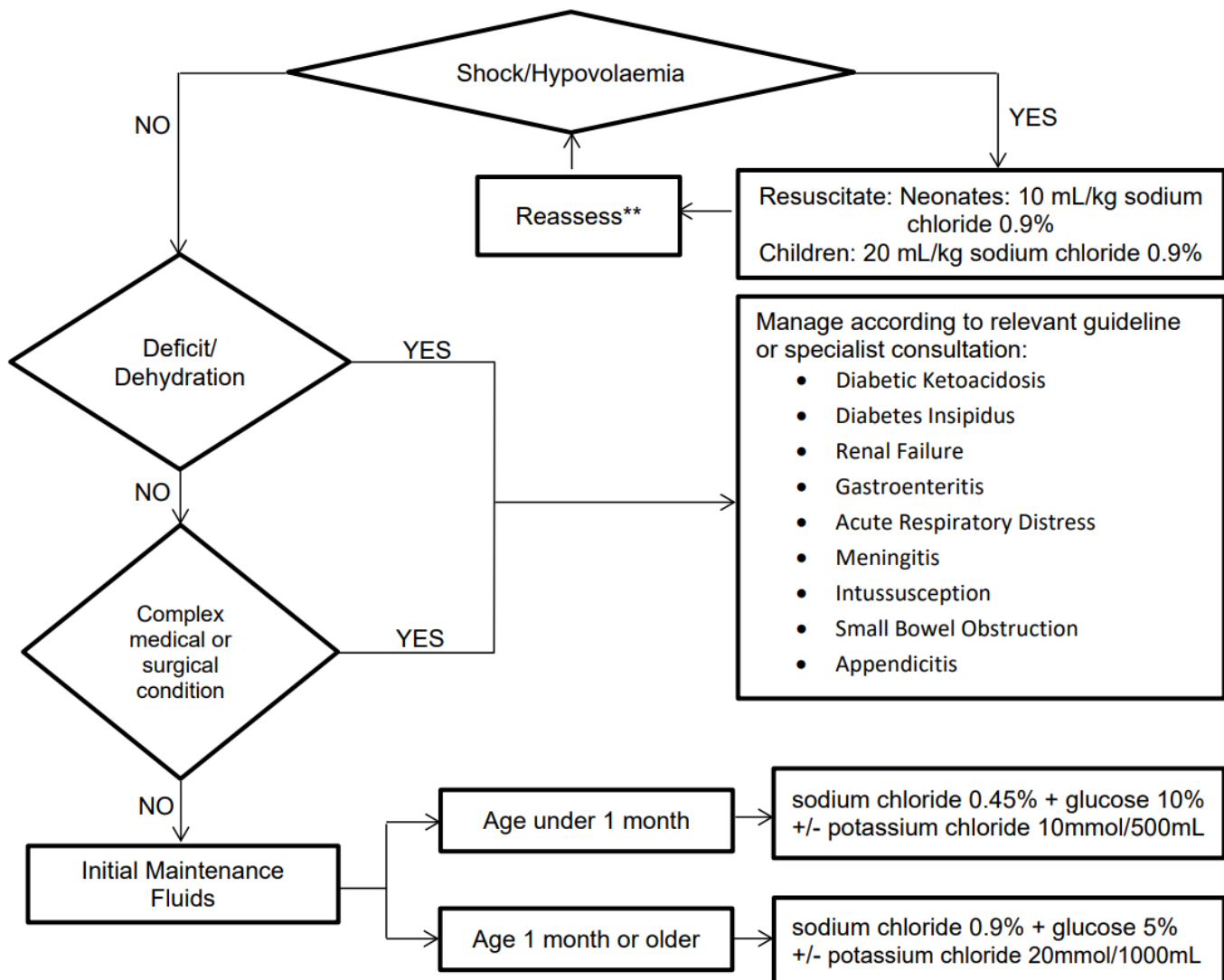
< 1 year old	Glucogel 40% 1 mL/kg
1-5 year old	75 mL juice
5+	150 mL juice

More on Potassium:

- Only add potassium **once normal baseline electrolytes and renal function** have been confirmed.
- SCHN advises 2 mmol/kg/day up to 4 mmol/kg/day
 - Standard is **20 mmol/L** of KCl added to a bag of 0.9% NaCl + 5% dextrose

Management of Hypokalaemia			
Degree	Serum K (mmol/L)	Essential minimum monitoring	Potassium prescription
Critical	Less than 2.0	Continuous ECG, hourly observations, strict fluid balance, hourly serum K levels, monitor urea and creatinine	Consult ICU urgently IV emergency replacement 0.5 mmol/kg/hr (max 20 mmol/hr) for 2 hours , then 0.1-0.3 mmol/kg/hr
Severe	2.0 to 2.4	Continuous ECG, 2-hourly observations, strict fluid balance (4-hourly urine output), K levels 2-6 hourly	Oral: 1 – 1.5 mmol/kg/dose up to 4x a day IV: 0.1 – 0.25 mmol/kg/hr (up to 20 mmol/hr)
Moderate	2.5 to 2.9	2 hourly observations, strict fluid balance (4-hourly urine output), K levels between 8 and 24 hours after initiating treatment Any patient on > 0.25 mmol/kg/hr (or > 10 mmol/hr if over 40 kg) needs ECG .	Oral: 0.5 – 1.5 mmol/kg/dose, up to 4 times daily IV: 0.1 – 0.25 mmol/kg/hr (up to 10 mmol/hr)

Mild	3.0 to 3.5	4 hourly observations, strict fluid balance, biochemistry as clinically indicated	Oral normal diet of 0.5 – 1 mmol/kg/dose up to qid IV maintenance: 20 – 40 mmol/L potassium solution; run infusion at rate that meets patient's fluid needs
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From: [IV Fluids and Electrolyte Therapy](#)

18.6 Emergency Drug Dosing

From the card that came with our Westmead ID's; it was recommended we look at it.

Indication	Drug	Dose	Indication	Drug	Dose
Anaphylaxis	Adrenaline 1:1000 IM	10 mcg/kg or 0.01 mL/kg (max 0.5 mL)	Pain	Fentanyl	1.5 mcg/kg (max 100 mcg) intranasal
Cardiac Arrest	Adrenaline 1:10000 IV	10 mcg/kg or 0.1 mL/kg		Morphine	0.05 – 0.1 mg/kg IV
	Amiodarone	5 mg/kg IV (max 300 mg)		Oxycodone	0.1 mg/kg oral
	DC shock	4 J/kg ¹	Seizure	Midazolam	0.15 mg/kg (max 10 mg) IV or IM
Croup²	Adrenaline 1:1000	5 mL nebulised			0.3 mg/kg (max 10 mg) intranasal or buccal

Hypoglycaemia	10% dextrose	2 mL/kg IV bolus		Phenytoin	20 mg/kg IV infusion
Shock	0.9% sodium chloride	10 – 20 mL/kg			

1: This makes sense because the adult shock is **200 J biphasic** and would presumably be calculated on a 50 kg weight.

2: This is only for severe croup (stridor at rest, distress, agitation/drowsiness, marked work of breathing); oral dexamethasone is otherwise first-line

Louisa Leone Notes

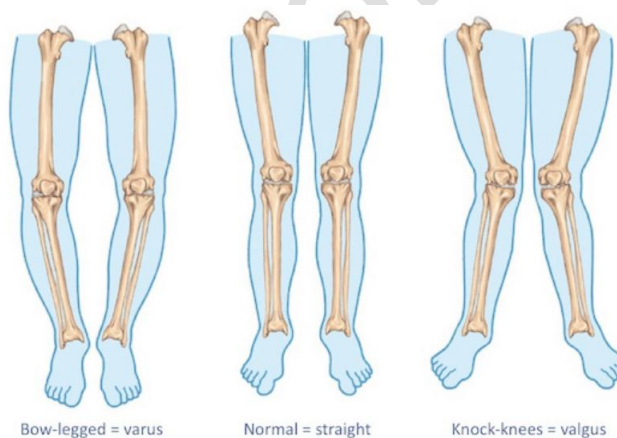
19. Fractures and Orthopaedics

Orthopaedic Examination and Findings

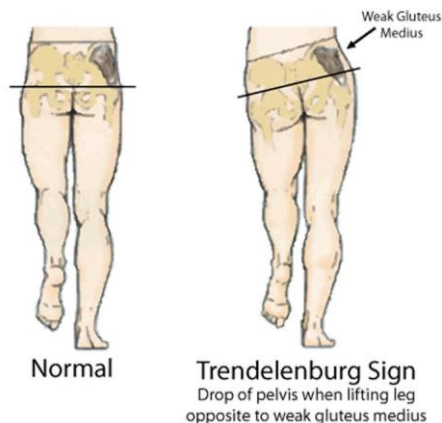
Table adapted from PassMed, but the topic came up in our handbook.

Description	Onset	Signs	Notes
Flat feet (pes planus)	All ages	Absent medial arch on standing	Typically resolves between 4-8 years old Reassure
In toeing	1 st year		Various causes, e.g. metatarsus adductus, internal tibial torsion, femoral anteversion adductus, internal tibial torsion, femoral anteversion Usually resolves spontaneously
Out toeing	All ages		Common in early infancy, usually resolves by 2 y/o Usually due to external tibial torsion Intervention might be needed if doesn't resolve → risk of patellofemoral pain
Bow legs (genu varum)	1-2 years	Increased intercondylar distance	Typically resolves by age 4-5 years
Knock knees (genu valgum)	3-4 years	Increased intermalleolar distance	Typically resolves spontaneously

Varus (bow legs) vs Valgus (knock knees)



Trendelenburg gait: unilateral weakness in hip abductors. The sound side “sags”, and the torso swings toward the affected side to compensate for the weakness in hip abduction.



The Limping or Non-Weight-Bearing Child

Clinical Practice Guidelines : The limping or non-weight bearing child

Most children presenting with a limp do not require investigations. However, acute inability to walk or weight bear is a red flag. **Septic arthritis** is an orthopaedic emergency; consider in all children with severe localised joint pain and fever.

- Pain after a viral infection in a young child would be classic of transient synovitis or acute myositis

Red Flags

History: duration > 7 days, severe localised joint pain (septic?), change to urinary or bowel habits, complete inability to weight bear, nocturnal symptoms, systemic symptoms, constitutional symptoms (consider malignancy/ haematological cause).

Examination: generalised wasting, fever, petechiae/ purpurae/ ecchymosis

Investigations

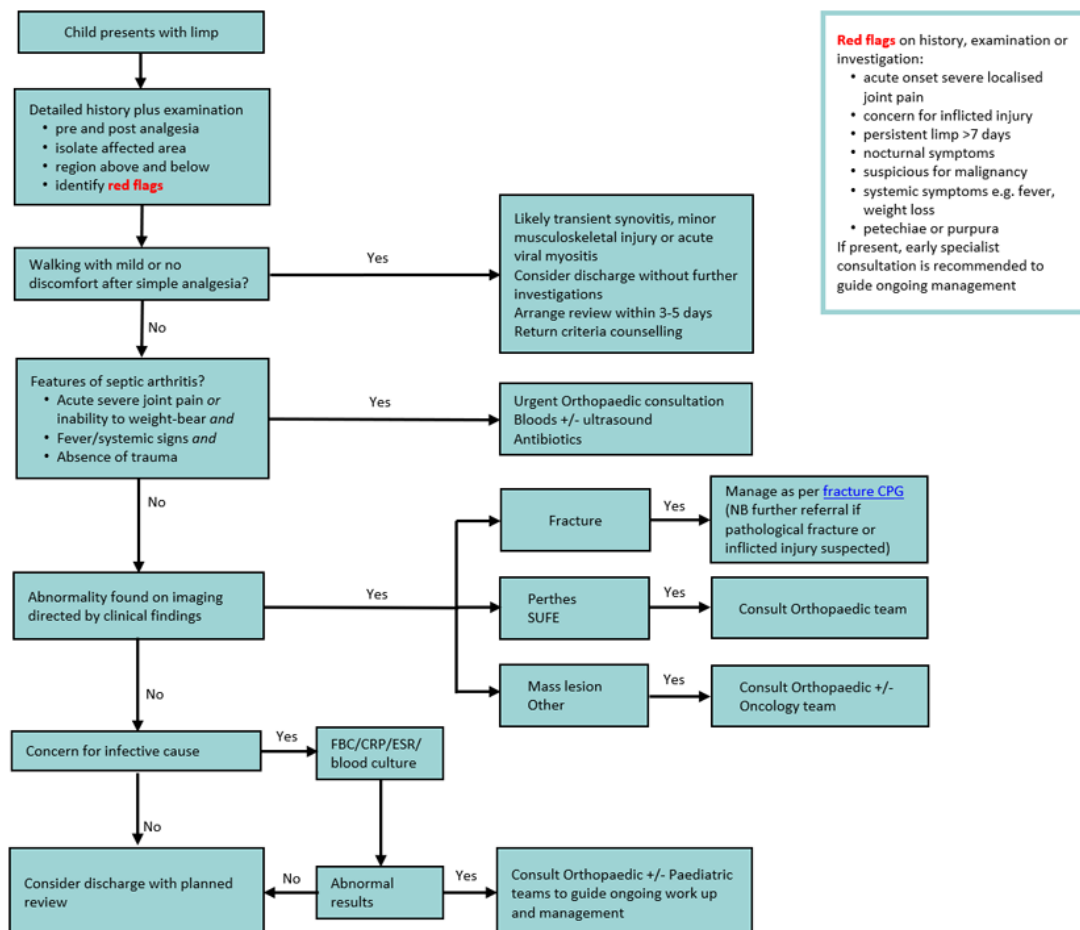
If there are no red flags, the child is ambulating with little/no discomfort with simple analgesia, no investigations are needed.

Imaging: X-ray (normal X-ray cannot exclude septic arthritis or early OM); hip ultrasound (effusion, cannot differentiate septic arthritis vs. hip synovitis); bone scan or MRI if limping > 7 days.

Laboratory: inflammatory markers (platelets, WCC, CRP +/- ESR) for acute infection/inflammation, blood cultures, FBC and blood film if concern for haematological malignancy.

Treatment

- Simple analgesia: paracetamol +/- NSAID.
- Child-directed limitation of activity.
- Management directed to cause e.g. fracture
- Septic arthritis: obtain aspirate if the child is stable, antibiotics, and orthopaedic consult



Developmental Dysplasia of the Hip

Nearly always present at birth. There is a spectrum ranging from subluxable (e.g. shallow acetabulum) to fully dislocated.

Epidemiology: 4 in 1000 live births; F:M 6:1 (the only hip condition that has a *female preponderance*).

Risk factors: family history of DDH, breech, first born, oligohydramnios (foetal packing), race (e.g. Canadian Inuit population), swaddling.

- **Remember the 5 F's of DDH:** female, first born, family history, feeet down (breech), fluid (oligohydramnios)
- Swaddling is bad for the hips; one reason Asian babies tend to have good hips is because mothers carry their babies on their back.

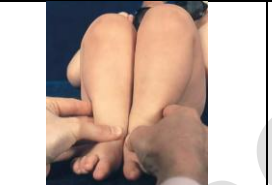
Clinical Features

Ordinarily, the infant's knee should be able to touch the bed during abduction.

Barlow's test: provocation test; hip slightly adducted, knee flexed; positive Barlow's = dislocates posteriorly.

Ortolani test (ortolini): abduct hip and apply inward pressure; positive Ortolani test = hip relocates

Galeazzi sign positive: difference in height of knees when hips and knees flexed to 90 degrees.

Asymmetrical creases	Limited abduction in flexion	Shortening of affected limb	Barlow's test	Ortolani test
				

Plus associated features of foetal packing e.g. plagiocephaly, torticollis, and postural foot deformities.

Imaging

Indications: family history of DDH, breech, abnormal signs

<6 months: ultrasound (hip joint is largely cartilaginous)

>6 months: X-ray showing delayed ossification (smaller femoral head ossification centre)

- The image shows left-sided DDH
- Note that the ossification centre should become visible at 4-6 months old

Treatment

Closed reduction or brace: Pavlik Harness for infants, abduction brace for older children. These splint the hips in a flexed and abducted position.



Surgical options are required in older infants:

- Closed or open reduction under GA and brace in spica cast (open reduction more likely in >12 months)
- >18 months: open reduction under GA, pelvic osteotomy (reshapes acetabulum as it has insufficient growth potential), spica

Hip Pain

By age of presentation:

2 – 6 years old	4 – 10 years old	10 – 14 years old
Transient synovitis	Avascular necrosis of the femoral head (Legg-Calve-Perthes Disease)	Slipped capital femoral epiphysis

Young child with acute hip pain and ↓ ROM; preceding URTI	Progressive hip pain and limp, ↓ ROM, no EMS, positive Trendelenburg	Overweight male with ↓ ROM and externally rotated hip
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Legg-Calve-Perthes Disease

When your hip is necrosed and your leg falls off, your leg and calf go to Perth.

Aetiology: disturbed blood supply (medial circumflex artery) to femoral head

Specific type of AVN of the femoral head in children; usually boys aged 4 – 10. Due to idiopathic interruption of blood supply to the capital femoral epiphysis.

- Incidence 1 in 1000, bilateral in 10% (often not simultaneous)



Presentation

- Progressive hip pain and limp with **positive Trendelenburg test**
- Recall that **pain originating in the hip might refer to the knee!**
- No EMS but ↓ ROM in hip with muscle wasting and limited abduction
- May be **initially misdiagnosed with transient synovitis** – suspect if persistent symptoms.

Investigate with X-ray. Will show fragmentation and collapse of the femoral head (condensation of the femoral epiphysis → sclerosis) +/- bony cysts in the metaphysis.

Prognosis and Management

- Older children have less remodelling and therefore a poorer prognosis.
 - <8 years old / the younger the child, the better the possible outcome.
 - **Waldenstrom classification:** condensation (infarction), fragmentation, reossification, remodelling
 - **Herring Lateral Pillar classification** on severity of Perthe's disease: >50% collapse of femoral head, poorer outcome
- Usually conservative treatment (self-resolving over 4-5 years) but operation (osteotomy) required if subluxation
 - Treatment may include anti-inflammatory medication, limiting physical activity, casting and bracing, or surgical intervention depending on severity.
- 50% have osteoarthritis by age 50 and will need total hip replacement
 - The femoral head can grow back larger (Coxa magna) and does not fit perfectly into the acetabulum

This girl has right-sided Perthes disease, with muscle wasting, limited hip abduction, and a positive Trendelenburg.



Slipped Capital Femoral Epiphysis (SCFE)

Surgical emergency! Usually peripubescent, slightly overweight male aged 10-14 years old.

- The femoral head slips off the neck through the growth plate of the femur.
- 20-50% bilateral

Presentation: M:F 3:2; presents during growth spurt i.e. 12-15 years in males, and 10-13 years in females.

- Acute < 2 weeks, chronic > 2 weeks; can also be acute on chronic.
- Stable = able to walk, unstable = unable to walk (orthopaedic **emergency!**)

Clinical features: pain in the hip/knee/thigh, external rotation deformity, out-toeing gait

- Recall that “knee pain” is a red herring

Investigation: X-Ray AP and frog lateral view

- An AP x-ray can miss early slips; a **frog leg view** assesses the hip in a **flexed and abducted position**
- The femoral head looks perfectly intact, and has ‘slipped’ off the femoral neck like a scoop of ice cream off a cone.



Management: screw fixation of the femoral capital epiphysis (percutaneous) including prophylactic screw fixation of the contralateral side (if the angle between the physis and femoral neck $> 14.5^\circ$, indicates $\uparrow$ risk of SCFE)

Outcome: exactly like Perthes disease (except all of these kids need surgery), 50% will need hip replacement after 50 due to OA

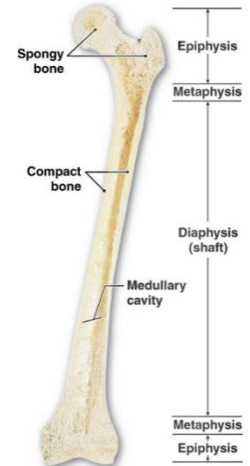
- Note, one risk is avascular necrosis

Fractures

Adult vs Paediatric Skeleton

Fracture Education : Biomechanical differences between adult and child

- Thicker periosteum (appositional growth) – children's bone typically heals faster than adults.
 - Fractures, even if complete, usually have intact periosteum on at least one side (which acts as a hinge)
- The physis produces a relative weak point, and ligaments are stronger than the growth plate.
 - Children are more likely to incur fractures than dislocations or sprains.
- Young bones are more porous → will bend before they break, but fail in compression (leading to **buckle injury**, particularly at the metaphysis).
- Long bones can bend to 45° before disrupting the cortex, however may undergo plastic bowing (especially the **fibula** and **ulna**) if they do not return to their initial position.
- Remodelling can occur due to remaining growth, which can allow non-anatomic reduction to be acceptable in some circumstances.
- Children can also experience fractures of the **physis or epiphysis** which are distinct from adult fracture patterns



History: age, mechanism, time of injury, witnesses, treatments prior to presentation.

Examination:

Look	Feel	Move
Compare to opposite side Limb position Swelling and deformity Skin abrasions or lacerations	Palpation of bony tenderness Joint effusion Vascular status (pulse, capillary refill) Neurological status (sensation)	Limit to essential activities for assessing neurological status and tendon injury

e.g. **Kiloh-Nevin sign** in anterior interosseous nerve syndrome due to forearm fracture – inability to perform “OK” sign due to weakness in flexor pollicis longus and flexor digitorum profundus.

Investigations

Imaging modality of choice: plain radiograph

Rule of 2's: 2 views, 2 joints (above and below), 2 times (repeat at 7-10 days if clinical suspicion persists)

- Further X-rays: oblique or tangential, mortise view of ankle

Advanced imaging: ultrasound for fluid, CT for bony lesions and occult fractures (rare due to radiation risk), MRI for occult fractures or soft tissue lesions, bone scan for osteomyelitis or occult fractures.

Describing the X-Ray Features of a Fracture

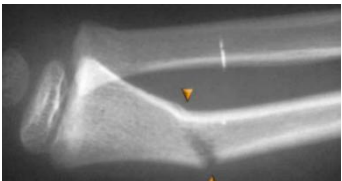
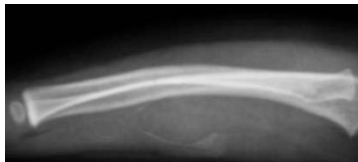
1. **Location:** which bone, side, location (proximal, diaphysis, distal), part (epiphysis, metaphysis, neck, diaphysis), intra-articular
2. **Open or closed**
 - Open (aka compound) if the tissue is not intact; there may be a foreign body visible; beware of subcutaneous emphysema.
3. **Fracture pattern** (see below)
4. **Dislocation**
 - Subluxation (articular surfaces partially in contact), dislocation (fully dislocated), diastasis (e.g. pubic symphysis; moved apart).
5. **Displacement:** including translation, angulation, and rotation. Always describe how the distal segment has moved.

Paediatric Fracture Patterns

Complete:

Transverse	Spiral	Oblique	Butterfly
			
Transverse fracture of the mid-tibia	Femoral diaphyseal spiral fracture	Oblique fracture of the mid tibia	Butterfly tibial fracture (with comminuted oblique fibular fracture)

Incomplete:

Greenstick Fractures	Buckle (Torus) Fracture	Plastic Deformation
Incomplete fracture due to incomplete ossification. Considered a stable fracture as the bone is 'bent'; Only one side of the cortex is broken. 	More likely in children due to less strength against compression Bone bulges and compresses on one side without breaking completely. 	Permanent bending or bowing of a bone, caused by low-energy, slow bending forces that don't cause a complete break or a distinct fracture line 

Epiphyseal Fractures

These are intra-articular and include avulsion (of ligamentous attachments), osteochondral fractures (through bone and articular cartilage), and compression fractures.

Physeal Fractures

Break in the growth plate of a long bone. The most common is Type II. Type III – V fractures have the highest rates of physeal arrest.

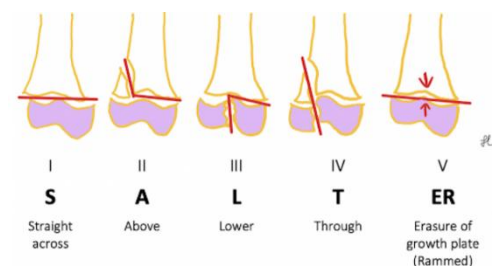
Straight across

Above the growth plate (metaphysis)

Lower than growth plate (epiphysis)

Through growth plate (epiphyseal diaphysis and metaphysis)

ERasure of growth plate (rammed)



Management of Paediatric Fractures

Emergency Treatment

If a deformity is obvious, **do not delay reduction** to allow imaging if (i) the deformity is causing **pressure to the skin**, or (ii) any **neurovascular compromise** is present (e.g. pulseless, pale limb, weakness, altered sensation).

Initial Treatment

ABCDs especially if high energy mechanism of injury (e.g. open fracture).

- Analgesia
- IV fluids
- Reduce and splint injured limb (slings, plaster, traction)
- Elevate
- Ice
- Regular neurovascular observations (Avoid compartment syndrome)

Maintaining Traction

Slings		Plaster	Traction
Broad arm supports elbow, e.g. clavicle or scapula fracture 	Collar and cuff allows elbow to hang e.g. humeral neck and shaft or undisplaced supracondylar humerus fracture 	Back-slab can be above or below the joint Below elbow/knee for distal injury only e.g. Salter Harris type distal radius/tibia #, torus #, wrist/hand/foot # Above elbow/knee for all other injuries (safest when in doubt) <u>Do not trap fingers; stop cast short of any joints</u>	By orthopaedics, e.g. Thomas splint or Gallows traction for proximal femur / femoral shaft fractures in children and babies, respectively Not for SCFE!

Keys to successful plaster: **attention to detail**; maintain reduction until dry, counteract deforming forces, broad plaster roll. Refer any open injuries/ those with NV compromise early.

Management of Open Fractures

Often associated with high-energy mechanisms of injury and concomitant injuries to C-spine, chest, abdomen, pelvis, and other limbs.

- ABCDs
- Photograph wound for later reference
- Remove large contaminants
- Copious wash with warm normal saline
- Reduction (may resolve neurovascular deficits)
- Sterile dressing
- Splint
- **Early IV antibiotics critical**
- Tetanus booster or immunoglobulin as necessary

After emergency treatment, full **history**, **examination** (including neurovascular status – you can still get a compartment syndrome!), **secondary survey**, **X-ray of injured limb** and **trauma series** if appropriate.

- Early referral to Orthopaedic team
- **Surgical debridement** is urgent, not emergent (should occur as soon as practical within 24 hours, ideally within 6 hours)

Implications of residual growth for paediatric fractures

Remaining growth may have positive or negative outcomes in different situations.

Remodelling: non-anatomic reduction can be acceptable in some circumstances e.g. if >2 years of growth remain, fracture is within metaphysis, and the deformity is in the plane of movement of the joint.

- Remodelling will **not** restore articular congruency, so all intra-articular fractures (e.g. **Salter Harris III and IV**) require anatomic reduction.
- It also will not accommodate the initial deformity if fractures are **within the diaphysis**, there is **gross angulation/ shortening/ rotation**, or deformity is **perpendicular** to the plane of movement of the joint (e.g. varus or valgus deformity with distal femoral fracture)

Overgrowth

- Fracture → blood flow, when coupled with loss of periosteal “tether” → ↑ longitudinal growth and limb length.
- Most often **post-femoral fracture** (1-2 cm overgrowth)
- If a limb length difference exists, it may be dealt with by **epiphysiodesis** (growth plate destruction on the long leg), exquisitely timed to allow the short leg enough time to ‘catch up’.

Complete physeal arrest

- Permanent growth plate damage leading to premature closure of the physis
- Most commonly following **Salter Harris V** injury, and more likely in kids approaching skeletal maturity (i.e. **adolescents**). However, limb length discrepancy most significant in younger children.
- This time, **epiphysiodesis** can be performed to stop growth in the **unaffected side**; limb lengthening procedures are another option.

Partial physeal arrest

- Permanent but localised growth plate damage → 'bony bar' which tethers causing angular growth
- Commonly Salter Harris IV and III injuries
- Management includes 'guided growth' (e.g. '8-plate' which tethers physis on one side, slowing growth), resection of the bony bar (epiphysiodesis), epiphysiodesis, or osteotomy following skeletal maturity (bone resection).



20. Rheumatology

The pGALS Musculoskeletal Screen

Screening questions:

- Do you (or does your child) have any pain or stiffness in your (their) joints, muscles, or back?
- Do you (or does your child) have any difficulty getting yourself (him/herself) dressed without any help?
- Do you (or does your child) have any problems going up and down stairs?

Figure	Screening manoeuvres	What is being assessed?
	Observe the child standing (front, back, sides)	Posture and habitus Skin rashes, e.g. psoriasis Deformity, e.g. leg length inequality, leg alignment (valgus or varus at the knee or ankle), scoliosis, joint swelling, muscle wasting, flat feet
	Observe the child walking, including on heels ('duck walk') and tip toes	Ankles, subtalar, midtarsal, and small joints of feet and toes Foot posture (note if presence of normal longitudinal arches of feet when on tiptoes)
	"Hold your hands out straight in front of you"	Forward flexion of shoulders Elbow extension Wrist extension Extension of small joints of fingers
	Turn your hands over and make a fist	Wrist supination Elbow supination Flexion of small joints of fingers
	"Pinch your index finger and thumb together"	Manual dexterity Coordination of small joints of index finger and thumb, and functional key grip
	"Touch the tips of your fingers"	Manual dexterity Coordination of small joints of fingers and thumbs
	Squeeze the metacarpophalangeal joints for tenderness	Metacarpophalangeal joints
	Put your hands together palm to palm (praying hands) Put your hands together back to back	Extension of small joints of fingers Wrist extension Elbow flexion

	Reach up, touch the sky, and look at the ceiling	Elbow extension Wrist extension Shoulder abduction Neck extension
	Put your hands behind your neck	Shoulder abduction External rotation of shoulders Elbow flexion
	Try and touch your shoulder with your ear	Cervical spine lateral flexion
	Open wide and put three (child's own) fingers in your mouth	Temporomandibular joints (and check for deviation of joint movement)
	Feel for effusion at the knee (patella tap, or cross-fluctuation) with child supine	Knee effusion
	Active movement of knees (flexion and extension) and feel for crepitus	Knee flexion and extension
	Passive movement of hip (knee flexed to 90 degrees) and internal rotation of hip	Hip flexion and internal rotation
	Bend forwards and touch your toes	Forward flexion of thoracolumbar spine (and check for scoliosis and kyphosis)

Adapted from UpToDate

Musculoskeletal Symptoms

Arthritis: joint swelling/effusion, OR 2 of tenderness/pain on movement, limited movement, increased warmth.

- May be associated with systemic autoimmune diseases e.g. SLE, vasculitis, dermatomyositis, scleroderma
- The clue is that there would be other systemic features e.g. skin changes

Septic Arthritis

Acute onset of joint pain, swelling, tenderness, unable to walk (if lower limb) + fever

- Haematogenous spread from primary infection (UTI, respiratory, skin), commonly *staph aureus*

Investigations

- Joint aspiration is definitive (WCC, gram stain, culture): CRP, synovial fluid WCC > 50,000
- Blood cultures, urine, sputum, wound cultures, etc.

Management: steroid and local anaesthetic injection; IV antibiotics; surgical emergency (arthroscopic washout)

Osteomyelitis

Routes of infection include direct implantation (e.g. open fracture or surgery), haematogenous, contiguous spread.

- Most common organism: staphylococcus aureus

Presentation

- Fever
- Pinpoint bone tenderness, effusion (serous) if near joint
- Less acute than septic arthritis

Investigation: X-rays may be normal early, so instead use bone scan or MRI

Treatment: IV antibiotics +/- surgery to remove affected bone

Post-infectious/reactive arthritis

Immunological reaction 1-3 weeks after infection at another site. Common causes:

- Bacterial: Group A streptococcus, enteric (e.g. salmonella, yersinia, campylobacter)
- Viral: rubella, measles, EBV, CMV, "transient synovitis"

Reactive arthritis is oligoarticular (≤ 4 joints) and typically involves large joints in the lower limbs (knee, ankle, hip, sacroiliac joint). May be associated with enthesitis (plantar fasciitis, Achilles tendonitis), conjunctivitis, etc depending on the infective trigger.

Transient Synovitis

Preschool-aged child with sudden onset pain in hip or knee, limp, and morning stiffness. Often follows a viral URTI but has a benign course. The child is constitutionally well. Typical history – no additional investigations required. Excellent prognosis, managed with **rest, ibuprofen** and full recovery in 2 weeks.

Reactive arthritis bears some similarity to transient synovitis, so we will compare them in a table.

Feature	Transient Synovitis	Post-Infective Arthritis
Aetiology	Non-infectious inflammation, often post-viral	Immune-mediated inflammation after systemic infection (bacterial or viral)
Common Age Group	Preschool or very early school age (2-5 years)	Children and adolescents, variable age
Sex distribution	M:F 3:2	M:F 2-3:1
Preceding Illness	Recent upper respiratory viral infection common	Preceded by documented infection (e.g., Salmonella, Chlamydia, streptococcal)
Onset	Acute, sudden hip pain and limp, but no refusal to weight bear	Acute or subacute joint pain and swelling
Joints Affected	Mainly hip (unilateral)	Mono- or oligoarthritis, commonly lower extremity joints (hip, knee, ankle)
Constitutional Sx	Usually absent or mild	May have fever and systemic signs
Physical Exam	Limp, limited hip range of motion due to pain	Joint swelling, tenderness, limited motion
Duration	Self-limiting, resolves in 1-2 weeks with rest and NSAIDs	Can last longer, weeks to months, variable recovery
Joint Fluid	Sterile effusion, no infection	Sterile inflammatory synovial fluid (aseptic)
Treatment	Supportive: rest, NSAIDs	NSAIDs, sometimes more prolonged treatment
Prognosis	Excellent, no long-term joint damage	Usually good, but risk of longer inflammation
Recurrence	Possible but uncommon	Can recur, dependent on infection control and immune response

Malignancy

Suspect in: fever, weight loss, pain out of proportion to physical findings, nocturnal pain, hepatosplenomegaly, lymphadenopathy, anaemia.

- **Frank polyarticular arthritis** is also a red flag and can indicate marrow infiltration (by leukaemic cells) or paraneoplastic effects.

May be systemic (leukaemia, lymphoma, neuroblastoma), or local (sarcoma, or benign conditions like osteoid osteoma or eosinophilic granuloma).

X-ray may show metaphyseal lucencies due to impaired bone formation.

Mechanical Conditions

Overuse syndromes: little leaguer's elbow, swimmer's shoulder etc

Avascular necrosis (hip): Legg-Perthe's

Slipped capital femoral epiphysis: surgical emergency!

Also see section in Orthopaedics on [Hip Pain](#).

Growing Pains

Age 3-10 years. Usually occurs at night, bilateral, limited to calf, thigh, and shins. No relationship to growth and normal examination/investigations. Requires leg massage and analgesia often.

- Children often do not remember episodes of pain the next morning
- Remember **unilateral nocturnal pain** is a red flag for malignancy!

Apophysitis (Growth-Related Disease)

Osgood-Schlatter Disease (knee)	Sinding-Larsen-Johansson Syndrome	Sever's disease (calcaneal apophysitis)
Tibial tubercle (patellar tendon insertion)	Inferior pole of the patella	Achilles tendon insertion
		

Children will complain of more mechanical-type pain (no early morning stiffness!); insidious, usually age 10-14 (Sever's disease appears a bit younger, age 8-12). Tenderness and possible swelling; activity-related (due to tendon pull at growth centre).

Pain Syndromes

- **Chronic regional pain syndrome**
 - Allodynia and hyperalgesia, swelling
 - Altered skin colour, temperature, texture, sweating
 - Motor/ muscle changes e.g. stiffness, tremors, dystonia, muscle wasting
- **Generalised pain syndromes** (fibromyalgia) characterised by widespread MSK pain, fatigue, sleep and/or cognitive disturbance without underlying tissue inflammation/damage
- History of mild trauma, inflammation, and disuse
- Consider in those with pain **out of proportion to** clinical signs
- Investigation not suggestive of serious disease
- Painful areas light up on bone scan (increased bone turnover and localised hyperaemia)
- Early recognition and intervention are critical
- Physio, psychological support, occasional medications

Hypermobility

Joint swelling is usually absent or transient. Hypermobility may be limited or generalised. Mild hypermobility is normal in younger children; many are asymptomatic. Older children may experience recurrent mechanical joint and muscle

pain, recurrent dislocations, usually worse with exercise. Hypermobility is a feature of some connective tissue disorders e.g. Marfan's or Ehlers-Danlos syndrome.

MSK Diagnostic Keys

Childhood arthritis will cause limp and morning stiffness, but not refusal to bear weight.

Refusal to bear weight	Night pain
Septic arthritis/ osteomyelitis Malignancy Reactive arthritis Pain syndromes/ conversion syndrome/ CRPS	Malignancy Osteoid osteoma (benign, responds to NSAIDs)

Juvenile Idiopathic Arthritis (JIA)

Incidence of arthritis: 2 in 1000

JIA is the most common cause of chronic arthritis in children. Persistent joint swelling (>6 weeks duration) in <16 years old in the absence of defined causes of arthritis. It is a clinical diagnosis based on history and examination.

- Classification** depends on whether it is oligoarthritis (≤ 4 joints) or polyarthritis (>4 joints); systemic (Still's), enthesitis-related, or psoriatic arthritis.

Oligoarticular (50%)	Polyarticular (30%)	Systemic JIA (Still's) (10%)	Enthesitis-related (ERA)	Psoriatic arthropathy
- Age < 5 - F > M - Large joints, knee commonest 80% ANA + - RF – Complications - Knee flexion contracture - Quadriceps atrophy - Leg-length discrepancy (affected leg shorter due to early fusion) - Cartilage damage - Ophthal review for uveitis in ANA+ (asymptomatic until significant damage)	RF-negative - Young - F>M - Small and large joints, neck, TMJ 50% ANA+ 10% have uveitis RF+ ('early onset RA') - Older girls - Symmetrical, small and large joints - Rheumatoid nodules over pressure points - ANA+ Worse prognosis, needs aggressive Rx	Mimic of malignancy with: - Quotidian fever that returns to below baseline - Evanescient salmon-coloured rash correlating with fever (trunk, thighs, upper arms) - Arthritis – early morning stiffness, pain, limping, swelling - Hepatosplenomegaly - Lymphadenopathy - Serositis e.g. pericarditis - Anaemia, high ESR, CRP, thrombocytosis (cf leukaemia)	Considered juvenile onset seronegative spondyloarthropathy e.g. ankylosing spondylitis - Older M>F - FHx - Peripheral and axial - ANA – - RF – - HLA-B27+ - Enthesitis (same location as apophysitis, but improves with activity)	Also seronegative Psoriasis Dactylitis "Pencil in cup" erosions Note: spondyloarthropathies may also have <u>extra-articular manifestations</u> Skin: psoriasis Eye: iritis, conjunctivitis (sore and red, unlike uveitis) GI: IBD, mouth ulcers

Treatment: high-dose corticosteroids, immunosuppression

Rates of Remission

Oligoarticular JIA has the highest rate of remission: 50-60%

This is followed by systemic arthritis (50% remission)

Polyarticular arthritis: 40% remission if RF-, no remission in RF+

Enthesitis-related arthritis: 20% remission

Psoriatic arthritis: 0% remission

Vasculitis

All adult vasculitides can occur in kids, with the notable exception of **Giant Cell Arteritis (GCA; a disease of the elderly)**.

Large vessel	Medium vessel	Small vessel
Tayakasu arteritis (aorta) GCA	Kawasaki disease (paediatric-specific) Polyarteritis nodosa	ANCA positive vasculitis (hard to diagnose, rare) Henoch-Schonlein Purpura (HSP)

Henoch-Schonlein Purpura (HSP)

Clinical Practice Guidelines : Henoch-Schönlein purpura

Most common vasculitis of childhood; **leukocytoclastic vasculitis (neutrophil infiltration and release of contents)**

- **Typical age:** 2 – 8 years old
- Now called IgA vasculitis, as it is driven by deposition of abnormal immune complexes, specifically galactose-deficient IgA1.
- **Aetiology:** idiopathic; various infectious and chemical triggers
- 50% have **recent URTI**; commonly viral or Group A streptococcus infection



Clinical features:

HSP tetrad: palpable purpura without thrombocytopaenia and coagulopathy, arthralgia and/or arthritis, abdominal pain, kidney disease.

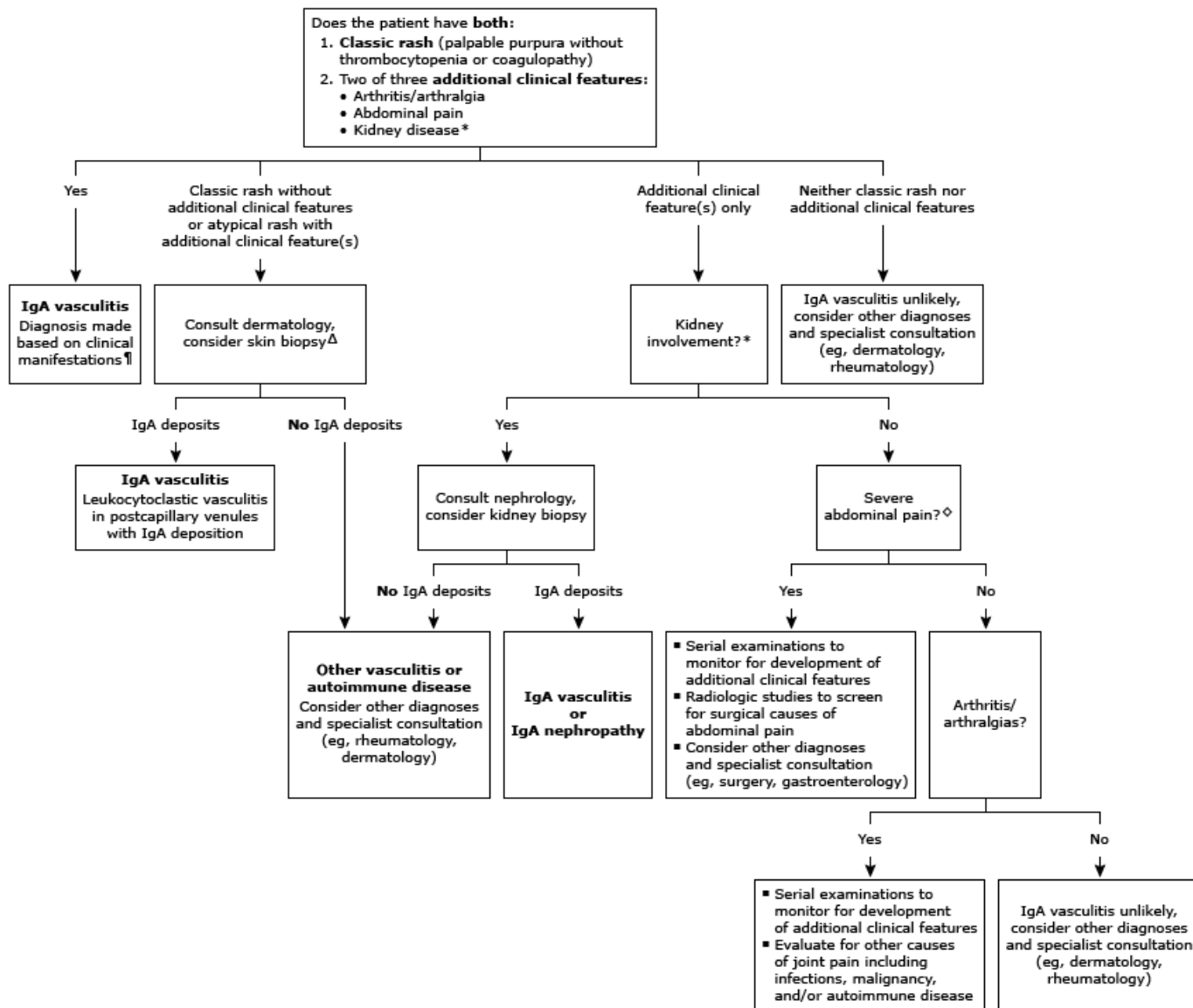
In more detail:

- Hypertension
- Palpable purpura, petechiae, ecchymoses
 - Symmetrical
 - May be preceded by urticarial, erythematous, maculopapular, or bullous skin lesions
 - Gravity/pressure-dependent areas e.g. buttocks and lower limbs in ambulatory children
- Painful, non-pitting subcutaneous oedema e.g. periorbital and/or dependent areas
- Arthralgia +/- arthritis (large joints of lower limbs, no significant effusion or warmth)
- Abdominal: diffuse abdominal pain, tenderness, signs of bowel obstruction or peritonism
 - Most common complication is **intussusception** though GI haemorrhage, bowel ischaemia/ necrosis, protein-losing enteropathy, pancreatitis can also occur
- Genital: testicular pain +/- orchitis +/- necrosis +/- cord haematoma
- Respiratory (rare): respiratory distress due to diffuse alveolar haemorrhage
- Neurological (rare): AMS, focal neurological signs (Consider ICH)

For diagnosis, children must have at least 2 of the 4 criteria:

- Palpable purpura
- Age < 20 years at disease onset
- Bowel angina (abnormal pain after meals or bowel ischaemia, usually with bloody diarrhoea)
- Granulocytes in walls of arterioles or venules

Can affect kidneys, joints, and gut. Recall that HSP is a “systemic” version of IgA nephropathy



Investigations: urinalysis and blood pressure (for renal failure; though, rare occurrence)

- Urinalysis may show haematuria (microscopic or macroscopic), proteinuria
- If hypertension, macroscopic haematuria, or significant proteinuria: formal urine microscopy, UPCR, EUC, albumin

Children may require other investigations to rule out differentials if the diagnosis is unclear e.g. FBC, EUC, albumin, blood/urine cultures, abdominal imaging, ASOT and anti-DNAse B, SLE investigations (ANA, dsDNA, ANCA, C3/C4)

Management: usually supportive (bed rest, elevation, simple analgesia)

- Consider steroids for moderate-severe abdominal and joint pain (oral prednisolone or IV methylprednisolone); though steroids **do not impact the long-term rate of renal complications**
- Review **BP/urinalysis** regularly since renal involvement can be delayed

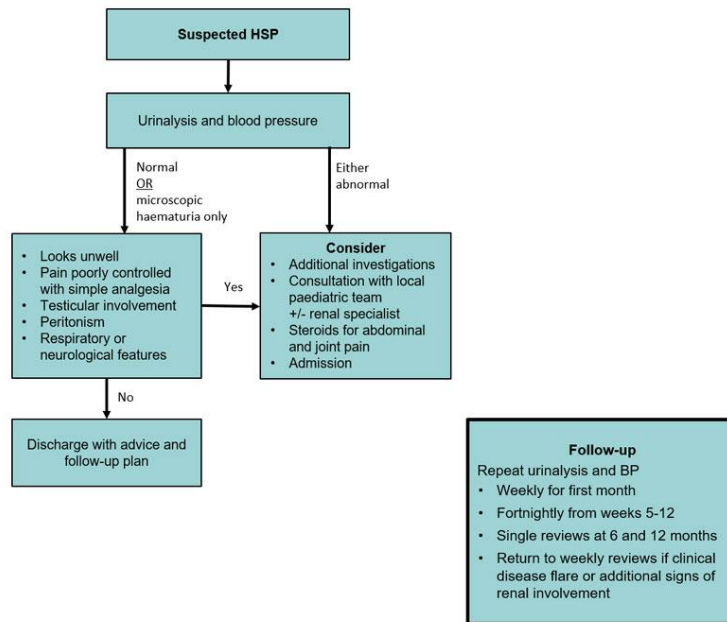
Prognosis: good, usually self-limiting

Unusual manifestations of HSP

Appendicitis-like picture: look for lower limb rash in any patient with abdominal pain

Serositis of tunica vaginitis (acute scrotum): again, lower limb rash suggests vasculitic cause

Scalp swelling is another unusual manifestation.



Kawasaki Disease

Covered in [ENT: Throat](#).

But, it's kinda important, so let's recap it here. **Recall the Kawasaki Rule of 5 – 5 days of fever, with 5 features, under 5 years old (80% cases).**

- Self-limited vasculitis of unknown aetiology; **second commonest after HSP**
- Male preponderance; seasonal peaks (Winter) link to potential viral trigger
- Most common cause of acquired heart disease in children in the developed world

Clinical Features

There is no particular order, and the manifestations may present at different times.

Good mnemonic: CRASH and BURN – Conjunctivitis, Rash, Adenopathy, Strawberry tongue, Hand & feet swelling, Burn (5 days of fever)

Criteria

Red, injected conjunctiva with limbal sparing (bilateral nonexudative)	Cracked red lips (mucositis)	Lymphadenopathy (usually nontender)	Rash (can be polymorphous)	Red palms and soles
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Prolonged fever + 4 of the above is definite Kawasaki; incomplete Kawasaki if <4 features but high clinical suspicion.

Other features:

- Fever: high unremitting fevers are the hallmark, minimally responsive to antipyretic agents, remains above 38.5 °C
- Arthritis is the most common MSK manifestation of KD but interestingly is not part of the diagnostic criteria.
- “Strawberry tongue” is due to sloughing of filiform papillae and denuding of inflamed glossal tissue; the bumps are the remaining fungiform papillae.
- **Extremity changes** are present in most patients in the acute and convalescent phase of KD
 - **Acute:** indurated oedema of the dorsum of hands and feet, and diffuse erythema of palms and soles
 - **Convalescent:** sheet-like desquamation that begins in the periungual region of the hands and feet

Treatment: IVIg, aspirin (for thrombocytosis); reduces complication rate from 25% to 5%

Complications

- Important long-term sequelae: **coronary artery aneurysm**
- **Complications:** primarily cardiovascular → may also have depressed myocardial contractility and heart failure, myocardial infarction, coronary artery rupture, arrhythmia and peripheral arterial occlusion.

- Non-cardiac complications are rare: macrophage activation syndrome, altered renal function, acute abdominal catastrophe, SNHL.

Dermatomyositis

Small vessel autoimmune vasculopathy. Though associated with malignancy in adults, tends to be an isolated immune condition in children.

- Scaly erythematous rash over extensors (Gottron lesions)
- Progressive proximal muscle weakness e.g. Gower sign, difficulty walking up stairs
- Nail fold capillary dilation
- Calcinosis cutis in soft tissues
- Heliotrope rash of eyelids (eyes usually spared in other conditions)
- Arthritis, dysphagia, ulceration



Treatment: corticosteroids and systemic immunosuppression

Systemic Lupus Erythematosus (SLE)

Multisystem chronic autoimmune condition due to multiple autoantibodies. Relapsing and remitting pattern. Asian and African American more common. F >>> M (4-5:1); average age at diagnosis is 11-14 years old

Presentation (RASH OR PAIN)

- Rash (malar, butterfly, photosensitive)
- Arthritis and arthralgia (deforming, **non-erosive**, Jaccoud arthritis with swan-neck deformity and ulnar deviation)
- Serositis (pleuritis, pericarditis, effusions)
- Haematologic (cytopenia), vasculitis (especially in cold months)
- Oro/nasopharyngeal ulcers e.g. hard palate
- Renal disease (most common cause of death in SLE)
- Photosensitivity
- ANA+
- Immuno dysfunction of dsDNA, Sm
- Neurologic (central and peripheral – seizures, psychosis)



Investigations

- dsDNA most specific for SLE, follows disease activity
- Pancytopenia
- Low C3 and C4
- Urinalysis suggesting nephritis
- X-ray: no bony erosions
- Antiphospholipid antibodies are a common association (can get cerebral venous sinus thrombosis)

Management: NSAIDs, steroids, immunosuppressive agents

Scleroderma

Localised scleroderma is more common in children. If there are systemic changes, it is **systemic sclerosis**.

Presentation

- Thick, swollen fingertips
- Hemiatrophy (growth distortion of face)
- Raynaud's
- Joint pain
- Shiny, thickened patches of skin
- Flexion contractures



21. Genetics

Accurate diagnosis relies on head-to-toe systematic observation, objective data and measurements (e.g. head circumference), and genomic data.

- We can do **reverse phenotyping**, starting with genomic information then matching onto phenotype.

Approach: context of family, systematic, document distinctive features

First-tier testing for intellectual disability:

- Clinical assessment and history
- Routine bloods including creatine kinase (CK) and thyroid function tests
- Fragile X DNA testing
- Urine metabolic screen
- Chromosome microarray (CMA): chromosomal abnormalities, copy number variants (microdeletions/microduplications) **not detected on genomic testing.**
 - 15% yield

Types of Genomic Testing

Sanger sequencing assesses one gene, e.g. FGFR3 for achondroplasia. Gene panel testing assesses a few genes linked to particular conditions e.g. Noonan syndrome or hearing loss. Whole exome sequencing assesses all genes.

Exome sequencing has become Medicare-funded for children < 10 years old with intellectual disability, developmental delay, and/or congenital anomalies, **after normal CMA, Fragile X, and UMS.**

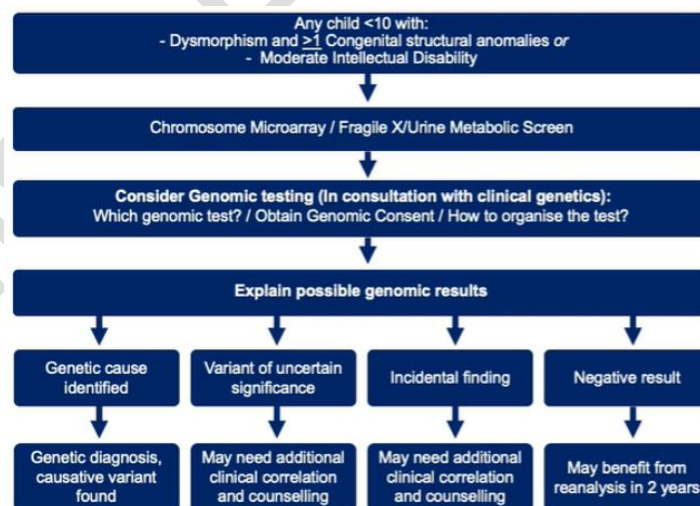
- Yield 36-41% for genetic anomaly
- Trio (child and both parents) has highest yield and also reduces likelihood of VOUS

Results of Genomic Testing

Genetic cause identified (40%), variant of uncertain significance (VOUS) (20%), or negative result (40%). Note, **incidental findings** are rare, and most labs do not actively screen for incidental findings.

- Those with a negative result may benefit from re-analysis in 2 years

Reasons for a negative exome: tiny deletion, mitochondrial pathogenic variant, methylation abnormalities (e.g. Angelman syndrome), expansion disorder (e.g. myotonic dystrophy, Fragile X), deep intronic variants.



Congenital Malformations

Major: produce significant long-term disability or death, occur in 15-20% stillborn babies (1/3 due to chromosomal anomalies), 2-3% life-born babies.

- Apert syndrome: syndactyly of the hands and feet, associated with **craniosyntosis**

Minor: unusual morphological features of no serious medical or cosmetic consequence, but dose-dependently increase the risk of major malformations.

- Examples: single transverse palmar crease, cleft lip, bifid uvula, familial clinodactyly, haemangioma

- Beware minor anomalies that may be associated with major ones, e.g. a **haemangioma over the lumbosacral spine** may be associated with **spinal dysraphism (neural tube defects of the spine)**.

Dysmorphology: multiple minor anomalies, often no clinical consequence in themselves, but may provide clues to generalised pattern of altered morphogenesis, timing of problem in development, buildup of storage materials (e.g. mucopolysaccharidoses).

Malformation: primary error in morphogenesis

- Agenesis: failure of organ to form
- Aplasia: absence of cell proliferation
- Hypoplasia: insufficient cell proliferation
- Hyperplasia: excessive cell proliferation
- Dysplasia: abnormal cellular organisation within tissue, leading to structural changes e.g. syndactyly

Disruption: destruction of previously normal tissue, e.g. amniotic banding (obstruction of distal fingertips)

- Altered shape/configuration, division of parts not usually divided, fusion of parts not usually fused, amputation of parts
- Vascular occlusion is a common mechanism

Deformation: distortion by physical force of an otherwise normal structure

- Usually extrinsic (e.g. co-twin, fibroids, oligohydramnios) but may be intrinsic (e.g. oedema)
- After organogenesis
- Loss of symmetry, altered alignment, abnormal positioning, distorted configuration

Syndrome: group of features that occur together, implies a common specific cause

- Example: **CHARGE Syndrome (caused by mutation in CHD7 gene)**. Coloboma, Heart defects, Atresia choanae, Retardation of growth/ development, Genital anomalies, Ear abnormalities.

Association: non-random concurrence of two or more features occurring together more frequently than expected by chance, no known cause

- Example: **VACTERL Association**. Vertebral defects, Anal atresia, Cardiac defects, Tracheo-Esophageal fistula, Renal anomalies, Limb abnormalities.

Complex: composite of manifestations

Spectrum: entities with multiple features, especially where the prominent features are variable in expression

Sequence: pattern of anomalies arising from single primary anomaly or mechanical factor

Developmental field: finite areas of embryonic tissue that develop into multiple, related morphologic structures; helps explain why certain malformations co-occur. May be **monotopic (single)** or **polytopic (multiple)**.

Common Genetic Syndromes

1:1000 – 1:5000

Whole Chromosomal Aneuploidy

- Trisomy 13 (Patau Syndrome)
- Trisomy 18 (Edwards Syndrome)
- Trisomy 21 (Down Syndrome)
- Sex chromosome aneuploidies
 - Turner (phenotype - female): 45 X
 - Klinefelter (phenotype - male): 47 XXY
 - Triple X, 46 XXX

Historically, T13 and T18 were considered lethal in utero, but with more aggressive management (e.g. of cardiac complications), prognosis has extended a couple years.

Klinefelter Syndrome

- Prevalence: 1 in 500
- Most common sex chromosome aneuploidy; very variable and undiagnosed
- **Physical features:** tall stature, long extremities, gynaecomastia; hypergonadotrophic hypogonadism after puberty, micro-orchidism, most adult men are infertile



22q11.2 Deletion Syndrome (Velocardiofacial Syndrome +/- Di George Syndrome)

- The most common microdeletion syndrome associated with developmental delay +/- intellectual disability
- 3 megabases recurrent microdeletion, detectable on chromosomal microarray
- Cardiac surveillance indicated due to incidence of cardiac anomalies (Tetralogy of Fallot, VSD, persistent truncus, interrupted aortic arch)
- Other (**mostly around the neck/chest**)
 - Palate anomalies (no adenoidectomy due to velopharyngeal insufficiency!), cleft palate +/- lip
 - Hypocalcaemia (hypoplastic parathyroid gland)
 - Immune dysfunction (T cells) (hypoplastic thymus)
 - C spine instability



Fragile X Syndrome

- Incidence:** 1 in 7000 males, 1 in 11 000 females.
- Most common cause of X-linked intellectual disability
- Triplet repeat expansion (CGG) in **FMR1**; unstable beyond 45 repeats, full mutation range consists of 200+ CGG repeats
- Males are affected more severely than most females.
 - Almost all males have intellectual deficit, IQ ~ 40.
 - 25% females have IQ < 70, typically shyness, social anxiety, mild learning problems
- Physical features: protruding ears, long narrow face, prominent jaw/chin/forehead, macrocephaly, macro-orchidism
- Not detectable on chromosome microarray (need PCR/ Western Blot)
- Rule of thumb:** any child with global developmental delay requires Fragile X screen!



Beckwith-Wiedemann Syndrome

- Polyhydramnios, congenital hyperinsulinism (with neonatal hypoglycaemia), macrosomia (large tongue), earlobe malformations (anterior linear earlobe creases, posterior helical ear pits), postnatal overgrowth, hemihyperplasia, facial naevus flammeus, umbilical hernia/ omphalocele,
- There is a 7-10% risk of embryonal tumours, that often do well with early detection.
 - Wilms tumour, hepatoblastoma.
 - Patients require 3-monthly abdominal ultrasound until 8 years old
- Development is generally normal
- Not detectable on chromosomal microarray; methylation studies.

Macroglossia	Posterior helical pits	Umbilical hernia	Anterior linear earlobe creases	Hemihyperplasia and facial nevus flammeus
				

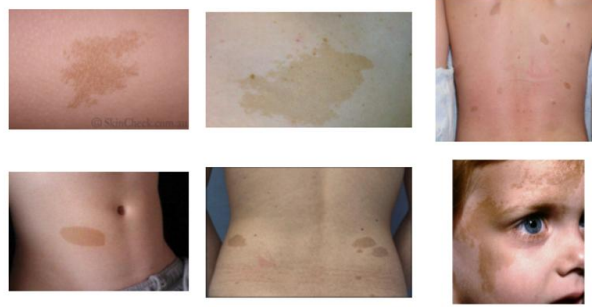
Noonan Syndrome

- Incidence: 1 in 1000 to 1 in 2500
- Autosomal dominant, genetically heterogeneous
 - Might incidentally diagnose parent with child
- Facial features (change with age):** relative macrocephaly, broad forehead, ptosis/ eyelid fullness, downward-slanting palpebral fissures, flat nasal bridge, short broad neck, pectus deformity
- Associations**
 - Congenital heart disease** (50 – 80%): **pulmonary stenosis**, hypertrophic cardiomyopathy (can be neonatally lethal; some MEK inhibitors are good for neonatal HCM).
 - Short stature, hearing loss, variable developmental delay, cryptorchidism.



<https://research.nhgri.nih.gov/atlas/condition/noonan->

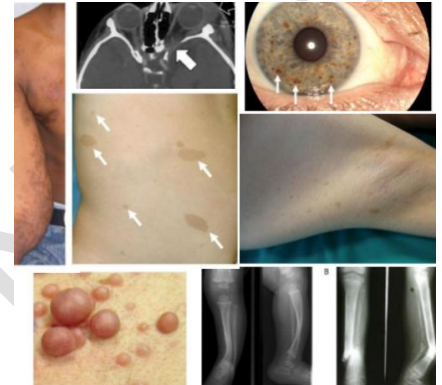
Pulmonary stenosis is Noonan until proven otherwise!



Atypical café-au-lait macules (top); typical café-au-lait macules bottom.

Neurofibromatosis Type 1 (NF1)

- Autosomal dominant, incidence 1:2000, ½ de novo, ½ inherited
- Mostly **clinical** diagnosis! NF1 can be diagnosed if 2+
 - 6+ café-au-lait macules, >5 mm in pre-pubertal children, > 15 mm in post-pubertal
 - Freckling in axilla (armpit) or groin
 - 2+ neurofibroma tumours
 - 2+ Lisch nodules (on iris)
 - Optic pathway glioma
 - Osseous lesions (e.g. pseudoarthrosis)
 - Pathogenic NF1 gene variant
 - Parent with NF1
- Presents with: learning difficulties (mostly mild), short stature with relative macrocephaly, scoliosis, hypertension
- NF1 is a **tumour predisposition syndrome** – benign and malignant
 - Annual ophthalmology surveillance for optic pathway glioma until 8
 - Plexiform neurofibromas (10-15% transition to malignant peripheral nerve sheath tumour)
 - Breast MRI for women from 35 years old
 - Early or prompt investigation of **HELP** symptoms: hardening or enlarging lump, leg/ arm weakness, pain



Marfan Syndrome

- Autosomal dominant connective tissue disorder, 1:500, FBN1 mutation
- Aortic root dilatation is a major source of mortality
- Ectopia lentis
- Tall, thin stature, long bone overgrowth, pectus deformities, scoliosis, joint laxity
- **Facial features:** dolichocephaly (long, thin head), malar hypoplasia (flat cheek bones), enophthalmos (deep set eyes), high-arch palate



Prader-Willi Syndrome

- Severe hypotonia and feeding difficulties in the newborn, followed by hyperphagia, central obesity, and short stature
- Cognitive delay, IQ typically 60s to 70s
- Caused by absence of the **paternally** active gene expression in the Prader-Willi/ Angelman syndrome critical region on 15q11-q13
 - "Imprinting disorder": genetic material from one parent silenced
 - Interestingly **Angelman syndrome** involves a deletion in **maternal genes** of the same chromosomal region – developmental delay, speech impairment, happy demeanour



22. Paediatric Surgery

General Paediatric Surgery

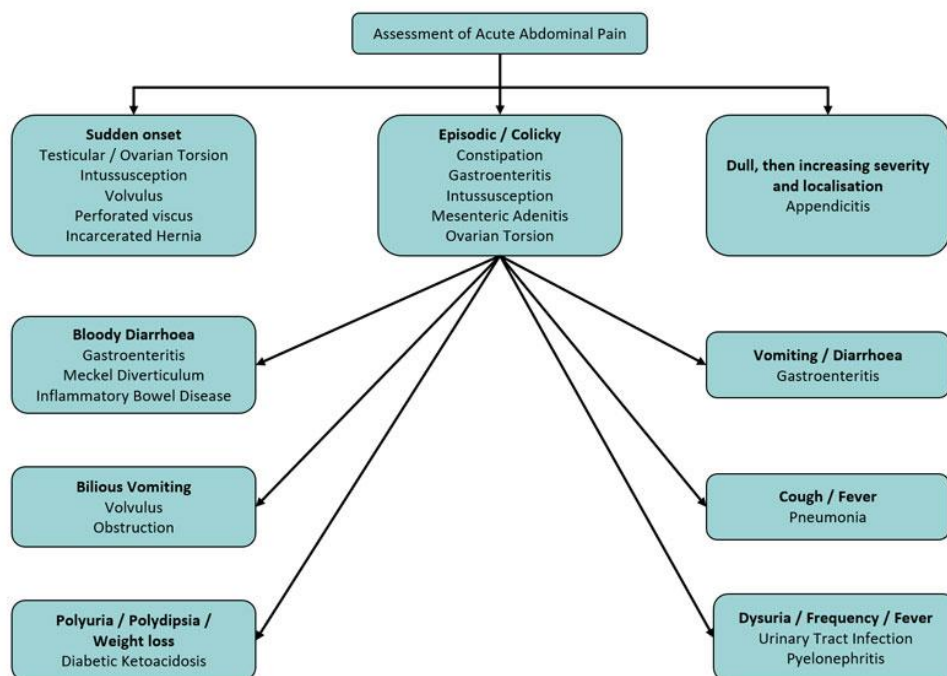
Approach to Acute Abdominal Pain

Clinical Practice Guidelines : Abdominal pain - acute

- Recall that abdominal pain can be medical or surgical.
- Analgesia should be used during examination, and does not mask causes of pain.

History

- SOCRATES for pain
- Significant if it results in **functional impairment** or **wakes a child from sleep**
- Systemic symptoms: fever, lethargy, irritability, anorexia, **weight loss**
- Dietary history
- Stool e.g. diarrhoea, constipation, **blood**, mucus
- Vomiting e.g. **bilious**, blood
- Urinary symptoms, e.g. dysuria, polyuria, polydipsia
- Menstrual and sexual history in adolescents
- Psychosocial history
- Underlying medical conditions



Examination

- General appearance and posture, e.g. rigidity, abdominal masses, palpable faeces
- Vital signs, hydration
- Abdominal examination
 - Scars
 - Palpate for focal vs generalised tenderness, percussion tenderness (+/- rebound tenderness), guarding or
- Examine genitalia
- Rectal or vaginal exam rarely needed

Note: a boy with testicular torsion might complain only of abdominal pain (embarrassed to reveal scrotal symptoms). It is thus essential to examine external genitalia in any boy complaining of abdominal pain.

Investigations

Not routine; targeted towards likely differential.

- Urinalysis +/- culture +/- β -hCG
- LFTs for hepatitis
- Lipase for pancreatitis
- VBG for lactate/acidosis
- BGL for DKA
- LFT, lipase, and EUC for abdominal injury
- FBC and CRP for inflammatory/ infectious cause

Imaging

- AXR if obstruction suspected
- CXR if pneumonia suspected

- US judiciously
 - No role in testicular torsion, but good for ovarian torsion
 - Good for intussusception
 - May identify renal calculi

Management: fluid resuscitation, adequate analgesia (including IV morphine or intranasal fentanyl for severe pain). Early surgical referral and fasting if surgical pathology suspected. NGT for bowel obstruction.

Red flags for a surgical problem: bilious vomiting, bloody stool, localised tenderness, distension and guarding, mass, inguinoscrotal swelling.

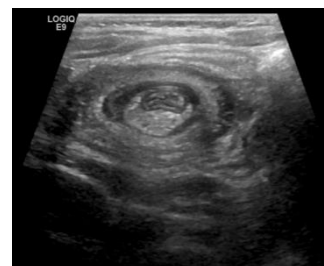
- By comparison, medical causes should **not** cause guarding/rigidity, with the exception of **DKA!**

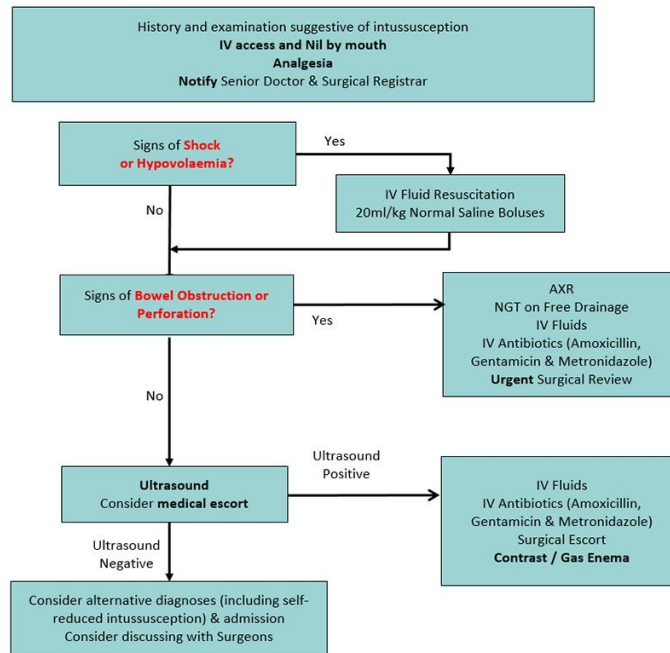
Appendicitis

- **Risk factors:** most common in 2nd decade, slight male preponderance
- **Classic triad in a minority:** right lower quadrant pain, fever, and vomiting.
- **Diagnosis** can be made on history and examination in 70%.
- The classic story is periumbilical abdominal pain that migrates to the RIF after 36 – 48 hours
 - Initial inflammation stimulates visceral afferent pain fibres (T10 dermatome), producing umbilical pain.
 - The appendix then irritates the parietal peritoneum → localised pain, usually in the RIF.
- **Relevant investigations:** FBC (↑ WCC, neutrophilia), CRP, EUC if significant nausea/vomiting
- **Imaging:** not needed for diagnosis, but helps determine the anatomical location of the appendix, the extent of inflammation, and whether operative management is required.
 - **Typical US:** non-compressible appendix (> 6 mm diameter), appendicolith, wall thickening ≥ 3 mm + hyperaemia
 - **Secondary sonographic signs:** free fluid in RIF
- **The Paediatric Appendicitis Score (PAS)** can be used to risk stratify patients aged 3 – 18 years with less than 4 days of abdominal pain. Essentially, high-risk PAS → surgical consult, equivocal → imaging, low-risk → consider other differentials.
- **Management:** appendectomy is the most common management strategy.
 - If the patient has an appendiceal mass, we may elect to commence antibiotics first and delay appendectomy by several months (**interval appendectomy**) to allow inflammation to settle.
 - **Active observation** is a new concept and involves antibiotics and fluid therapy without planned interval appendectomy. It may reduce costs and the need for surgery.

Intussusception

- Invagination (telescoping) of a proximal segment of bowel into the distal bowel lumen. The commonest site is a **segment of ileum moving into the colon through the ileo-caecal valve**.
- Typically presents **3 months to 6 years old**. Outside this age range, consider a **pathological lead point**.
- **Aetiology:** 90% are idiopathic.
- **Risk factors:** recent intussusception, potential lead point (e.g. Meckel's diverticulum, Henoch Schonlein Purpura, lymphoma, polyps), recent bowel surgery, recent rotavirus vaccination
- **Triad:** intermittent abdominal pain, palpable abdominal mass, red currant jelly stools in **1/3 only!**
 - "Sausage-shaped mass" in the right abdomen
 - Red currant jelly stools: late sign, suggesting bowel ischaemia and infarction
- **Investigations**
 - Ultrasound has high sensitivity and specificity when performed by an experienced clinician. "**Target Sign**" or "**Crescent in a donut**".
 - AXR if you suspect obstruction or perforation
 - Blood tests: blood glucose, VBG/FBC/EUC if unwell, Group & Hold pre-operatively
- **Management:** resuscitation and analgesia first.
 - If there is no evidence for perforation, contrast/gas enema is preferred (diagnostic and therapeutic).
 - If sick or perforated, surgery required.





Vomiting

Like abdominal pain, vomiting can be medical or surgical.

When is vomiting significant?

- Bilious vomiting is due to a gastrointestinal obstruction until proven otherwise. This includes green bilious vomiting and yellow bile-stained vomitus.
- Persistent vomiting
- Projectile vomiting (obstructive)
- Blood staining
- Weight loss or failure to thrive
- Unwell child
- Associated with a mass e.g. groin or abdominal mass

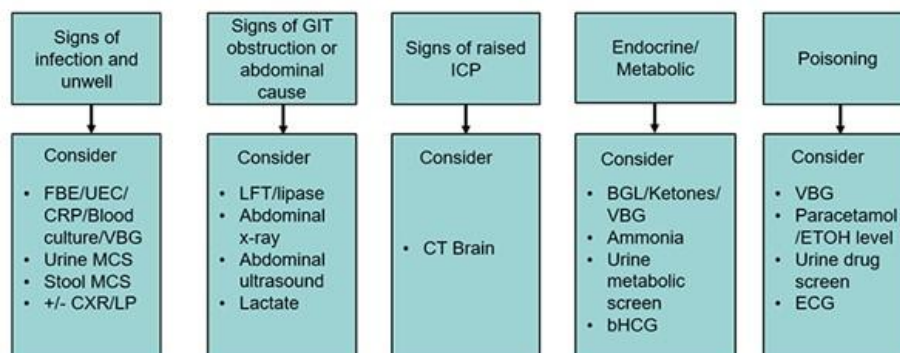
Examination

- Abdominal distension – acute abdomen or bowel obstruction
- Tenderness
- 'Tinkling' high-pitched or absent bowel sounds in obstruction
- **Always** check the **scrotum!**
- Abnormal neurology or bulging fontanelle for raised ICP
- Assess fluid status

Neonates & infants	Children	Adolescents
- Infection – particularly gastroenteritis; consider UTI, pneumonia and other - Raised intracranial pressure - Gastrointestinal obstruction - Poisoning - Metabolic/endocrine disease - Volvulus - Testicular/ovarian torsion	- Gastroesophageal reflux disease - Pyloric stenosis - Intussusception - Food protein induced enterocolitis - Diabetic ketoacidosis - Appendicitis	- Eating disorder - Drug use eg alcohol - Pregnancy

Investigations

Consider BGL and ketones if BGL < 3 or > 11 mmol/L. Note, vomiting can lead to **ketotic hypoglycaemia** especially in younger children. Like abdominal pain, most vomiting kids don't need extra investigations. Investigations would be targeted to cause.



Pyloric Stenosis

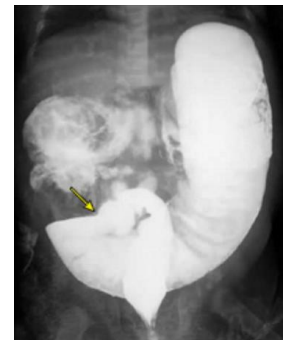
- Commonest cause of non-bilious surgical vomiting
- **Epidemiology and Aetiology:** 1 in 450, predominantly male, 20% have positive family history
- **Presentation:** projectile vomit after 2 weeks of age with olive-shaped RUQ mass. The baby will be hungry and thriving well.
- **Diagnosis:** abdominal ultrasound; remember **pi is 3.14** → **pyloric stenosis suggested by thickness > 3 mm and length > 4 mm in term baby**
- **Management:** pyloromyotomy

Malrotation Volvulus

- **Presentation:** term baby thriving well with acute episode of bilious vomiting in first week of life
- **Plain AXR and contrast study** may help locate the position of the duodenojejunal junction (should be left of midline).

Hirschsprung Disease AKA Congenital Aganglionic Megacolon

- Developmental disorder of the enteric nervous system characterised by absence of ganglionic cells in the myenteric and submucosal plexus of the intestine in continuity from the anus to a variable distance proximally
- **Incidence:** 1:5000
- **Associated syndromes:** T21, neural crestopathies, multiple endocrine neoplasia, Type 2 congenital central hypoventilation syndrome, neurofibromatosis. Also associated with neuroblastoma.
- **Clinical presentation:** **50-90% present in the neonatal period.** suspect in a baby who **has not passed meconium in 24 hours**; abdominal distension, bile-stained vomiting
 - **May develop enterocolitis and perforation**
 - It can also present in older children with chronic constipation, abdominal distension, and failure to thrive.
- Usually at rectosigmoid level.
 - The **aganglionic distal segment** lacks enteric ganglion cells → contracted, spastic, narrow.
 - The proximal bowel is normal, and becomes dilated due to intestinal contents and gas.
- **Histological diagnosis** via rectal suction biopsy showing an absence of ganglion cells in the bowel mucosa (**negative calretinin stain**).
- **Surgical management:** remove aganglionic bowel and bring normally innervated bowel to the anus whilst preserving sphincter function. There are various procedures; clinician preference.
 - Infants who present require surgery in the first 2 weeks of life.
- Of course, **long segment Hirschsprung's** or **whole aganglionic bowel** is incredibly challenging → short bowel, intestinal failure, long-term TPN, and resulting complications.



Additional Congenital Surgical Problems

Congenital Abdominal Wall Defects

	Gastroschisis	Exomphalos/ Omphalocele
Definition	Herniated 'free floating' abdominal contents	Herniated abdominal contents contained within sac
Incidence (live births)	1:4000	1:4-6000
Aetiology	Defect in abdominal wall formation	Failure of viscera to return to abdominal cavity following physiological herniation
Associated anomalies	Uncommon, possibly intestinal atresia and undescended testes	Common, may be syndromic e.g. Beckwith-Weidemann, T13/18/21, Pentalogy of Cantrell
Mode of delivery	Normal vaginal delivery in neonatal surgical centre	C-section
Consequences	Malrotated, susceptible to volvulus	
Management	Resuscitation of baby, NG suction, reduction of bowel into abdominal cavity	Small defects: primary closure Large defects: scarification (e.g. betadine, chlorhexidine, alcohol)
Prognosis	Excellent depending on postnatal condition of bowel	Depends on associated anomalies

Oesophageal Atresia and Tracheo-Oesophageal Fistula

- Incidence: 1:2500 – 3000 live births
- 50 – 70% have associated anomalies: **50% VACTERL/CHARGE syndrome, 10% chromosomal.**

- Classification: Gross System. Based on whether a **trachea-oesophageal fistula** is present, and its location.
 - The most common type is **Gross C** which is an **oesophageal atresia with distal TOF**.



Gross 1953	A	B	C	D	E	F
Incidence (%)	7	1	86	2	4	-

- Diagnosis: typically postnatal with difficulty feeding, salivation, regurgitation and/or aspiration. NGT usually fails to progress beyond T2-4.
- Management: bronchoscopy to assess fistula and other anomalies; thoracotomy or thoracoscopic repair. A lengthening procedure (e.g. **Focker technique**) may be used to bridge the distance between the proximal and distal oesophagus.
- Prognosis is good, but again depends on associated anomalies.
- Post-operative sequelae might be gastrointestinal or respiratory
 - GI: dysphagia and choking (oesophageal strictures, dysfunctional oesophageal activity), GORD, TOF recurrence, oesophagitis, chronic inflammation
 - Respiratory: recurrent chest infections, chronic respiratory Sx (associated tracheomalacia or laryngeal clefts); restrictive and obstructive ventilatory defects

Congenital Diaphragmatic Hernia

- **Incidence:** 1:2-5000
- Developmental discontinuity of the diaphragm, allowing abdominal viscera to herniate into the chest.
- 50% mortality (including in utero death), 50% have associated anomalies. May again be associated with various syndromes e.g. T13/18/21, Beckwith-Wiedemann, Frey syndrome
- CDH is typically **left-sided**. Two types:
 - Posterolateral Bochdalek Hernia (Bochdalek at the back!): most common
 - Anterior Morgagni Hernia: usually in older kids
- CDH can also lead to **pulmonary hypoplasia** and **pulmonary hypertension** since the herniated viscera take up space in the hemithorax.
- **Prognosis:** like *all other conditions*, the **biggest prognostic factor** is the degree and severity of other anomalies (especially cardiac).
- **Management:** physiological emergency – remember that pulmonary hypoplasia and pulmonary hypertension are an ICU not surgical problem!
 - Surgical repair can be done open, laparoscopic, or thoracoscopic during the postnatal “honeymoon” period (where the neonate is clinically stable) or after stabilisation on ventilation/ECMO
 - Resurgence of fetoscopic surgery

Necrotising Enterocolitis

- Multifactorial exaggerated inflammatory response due to an initiating event (typically **physiological stress**) in the context of an immature intestine and immune system.
- **Incidence:** 1:1000 live births, **90% pre-term**.
- **Severity:** graded by the Modified Bell's Classification System

Stage	Stage I (Suspected NEC)	Stage II (Confirmed NEC)	Stage III (Progressive NEC)
Clinical Signs	Lethargy, temperature instability, apnoea, bradycardia, emesis, abdominal distension, haematochezia	+ metabolic acidosis, thrombocytopaenia, abdominal tenderness, absent bowel sounds	+ respiratory or circulatory arrest
Radiographic	Intestinal dilatation	+ Portal venous gas, pneumatosis intestinalis	+ Ascites or pneumoperitoneum
Treatment	Oral feeding cessation Parenteral nutrition	As in Stage I, plus correction of metabolic acidosis	Surgery

	NGT suction Antibiotics		
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- **AXR:** gaseous distension of the bowel, pneumatosis intestinalis, portal venous gas, pneumoperitoneum
- **Laboratory findings:** leukocytosis, thrombocytopaenia, metabolic acidosis, rising CRP
- **Management:** the only absolute indication for surgery is perforation. Otherwise consider bowel rest, gastric decompression, IV fluids, IV antibiotics, and observation.
- The biggest prognostic factor is the degree of prematurity.

Inguino-Scrotal Disorders in Children

Acute Scrotal Pain or Swelling

Clinical Practice Guidelines : Acute scrotal pain or swelling. This RCH guideline is excellent. The tables below are directly from them.

Remember to check the scrotum in any boy presenting with abdominal pain!

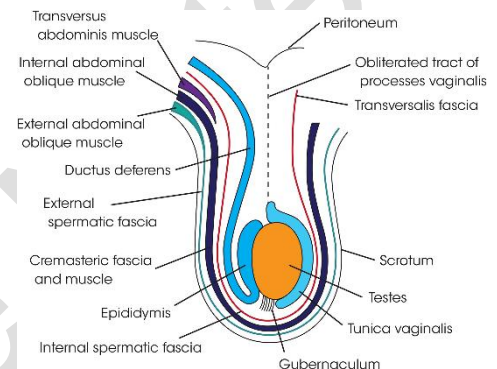
Testicular torsion is a surgical emergency. It is also a clinical diagnosis. Do not ultrasound if the clinical picture is consistent with torsion. Urgent surgical referral.

The most common cause of acute scrotal pain and/or swelling are **torsion of the testicular appendage (appendix testis), epididymitis, and testicular torsion.**

- Epididymitis is less common nowadays with the mumps vaccine.

Investigations

- Blood tests, ultrasound, and Doppler ultrasound are not useful in the acute setting.
- Only use **ultrasound** if the diagnosis is unclear and testicular torsion and irreducible hernia have been confidently excluded.
- For suspected epididymo-orchitis: urine MCS (ideally first pass), chlamydia/gonorrhoea PCR if clinically suspected, if mumps suspected RT-PCR and/or IgM



Scrotal pain +/- swelling

	Testicular Torsion	Irreducible Hernia*	Torsion of Testicular Appendage	Epididymo-Orchitis	Trauma e.g. Testicular or epididymal rupture
	 		 		
Typical age group	Pubertal (and rarely neonates)	Infants	Pre-pubertal (7-12 years)	<2 years and post-pubertal	
Pain	Severe Usually sudden onset May radiate to iliac fossa or thigh May be painless in neonates	Irritable	Usually sudden onset Usually minimal at rest	Sudden or subacute onset May improve with elevation	May be delayed
Swelling	Yes	Yes, may extend to scrotum	Yes	Yes	May be delayed
Fever	-	-	-	Yes	-

Nausea and vomiting	Common (90%)	Common	-	-	-
Dysuria or discharge	-	-	-	Common	-
Gait	Impaired	-	-	-	-
Position of testis	High-riding or horizontal	-	Normal	-	-
Palpation	Tender Thickened spermatic cord	Firm and tender Irreducible swelling, "can't get above"	Focal tenderness of upper pole	Tender posterolateral testis	Tender
Oedema crosses midline	-	-	-	Possible	Possible
Discolouration	Red/blue Dark in neonate	-	Blue dot sign	Red	Bruising (consider NAI)
Cremasteric reflex	Usually absent	Present	Present	Present	Present
Reactive hydrocoele	Possible	-	-	Possible	Possible
Treatment	Urgent surgical review (orchidectomy if testis necrotic) Fasting or clear fluids until surgical review Adequate analgesia	Urgent surgical review Fasting or clear fluids until review Adequate analgesia Consider NGT on free drainage if bowel obstruction suspected	Surgical exploration if unable to exclude testicular torsion Supportive treatment, analgesia and rest Pain should resolve in 2-10 days	Antibiotics: IV if unwell or infant, oral if well Second episode: renal tract ultrasound and urological review Slow to resolve – may have weeks of subsiding scrotal discomfort and swelling	Surgical review unless testis is clearly normal and without significant tenderness

*Inguinal hernia may be painless if it is still reducible. Note that the risk of incarceration is highest in the first 6 months of life, so the hernia needs surgical repair as soon as practically possible.

Non-Painful Scrotal Swelling

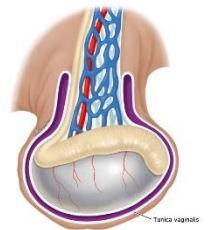
	Hydrocoele	Varicocoele	Idiopathic scrotal oedema	Tumour/ leukaemia
				
Typical age group	Infants	Peri-pubertal	3 – 7 years	1 – 8 years
Fever				Possible
Palpation	Soft Non-tender Fluctuant	"Bag of worms" Occasionally tender	Non-tender May have low-grade discomfort	Hard Non-tender May be painful if rapidly growing
Swelling pattern	Scrotal Can be communicating (increases with Valsalva or standing) or non-	Predominantly left-sided (L testicular vein comes off L renal vein) Worse on prolonged standing	Can extend across midline and into perineum, groin, penis	Unilateral or bilateral

	communicating (more tense, does not change)			
Discolouration	No	No	Bland, purplish	No
Transilluminable	Brightly*	No	No	No
Reactive hydrocoele		No	No	Possible
Management	Spontaneous resolution in first year, 90% by 2 years Consider outpatient surgical referral for repair if present after 2 years old	Surgery if signs and symptoms related to impaired testis growth (laparoscopic ligation of testicular veins)	Can occur in setting of systemic disease e.g. nephrotic syndrome May trial antihistamines, steroids If idiopathic, resolves spontaneously in 1-5 days	

*Transillumination can be unreliable in neonates because the skin is fairly translucent.

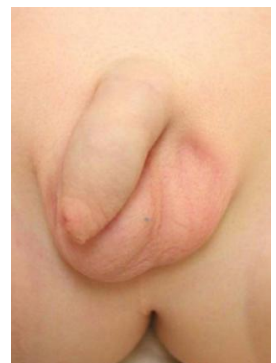
Testicular Torsion

- In **neonates**, testicular torsion is often “extravaginal”, meaning the twist involves the entire testis, epididymis, and tunica vaginalis twisting on the spermatic cord.
- It occurs because, in newborns, the attachments that normally anchor the testis to the scrotum are not fully developed.
- It thus makes sense that **prematurity** is a risk factor for neonatal testicular torsion, and this is linked to **cryptorchidism** (undescended testes are more mobile).
- The “**mini-puberty**” at 1-5 weeks of life can also heighten the risk of torsion, since the temporary increase in testicular volume → increased mobility or tension in testicular attachments.
- For the same reason, **the risk of testicular torsion is increased during puberty**.
- However, during puberty, the most common form is **intravaginal torsion**, where the testis twists within the tunica vaginalis. This is usually due to a congenital malformation e.g. **bell-clapper testis**.



Cryptorchidism

- Incidence:** 3-5% at birth. Often descend spontaneously, about 1-2% remain undescended by 3 months.
- Risk factors:** prematurity (up to 30-45%) and low birth weight, family history, use of assisted reproductive technologies, maternal factors during pregnancy (e.g. maternal diabetes, smoking, obesity, pre-eclampsia)
- May be **congenital undescended testes** or **acquired**.
 - Retractile testes: due to hyperactive cremasteric reflex; testes retract but can be milked into scrotum and stay in scrotum for substantial period.
- The testis may be arrested along the normal line of descent from the kidney, or ectopic. It may or may not be palpable in the inguinal canal.
- Risks of cryptorchidism: testicular torsion, hernia, reduced fertility, risk of testicular cancer (*increased risk remains with surgical correction*)
- Orchidopexy** is required if testes remain undescended by 6 months; should be performed **12 months**.
 - The main risks are an unsuccessful procedure and testicular atrophy.**



Penile Disorders

- Parents may be concerned about penis size (measure stretched length and girth and chart).
- At birth, the normal foreskin (prepuce) is attached to the glans and has a tight opening at the distal end.
- A non-retractable foreskin is a normal variant and needs no intervention.
 - Only 30-40% of males have a fully retractile foreskin at 4 years old, 99% retract by age 17.
- The foreskin should only be retracted for cleaning once it becomes freely retractable naturally.
- Smegma:** desquamated epithelial cells and sebaceous matter between the glans penis and foreskin. May collect in small white/yellow “rice grain” lumps – normal, no intervention needed.
- Foreskin attachments:** if separation of the foreskin is uneven, it may lead to a day or two of soreness and dysuria. This does not need intervention.

Phimosis

- Scarring of preputial ring preventing retraction.
- **Features:** ring of scar tissue at foreskin opening, foreskin not retractable at conclusion of puberty, persistent ballooning of foreskin on urination with pinhole opening
 - **Kids may report difficulty aiming the urine stream properly which can lead to “missing the toilet” or urine spraying.**
- **Aetiology:** most commonly due to repeated attempts to forcibly retract the foreskin before it has become naturally retractable.
 - Another cause is Balanitis Xerotica Obliterans (BXO), an aggressive scarring condition that typically occurs >8 years old.
 - The image to the right is typical of BXO: tight, white signature ring involving meatus; inflammation of proximal foreskin; difficult circumcision due to fibrosis
- **Treatment:** trial topical steroids (0.05% betamethasone cream 2-3 times daily) for 2-4 weeks. If good response, continue for 6-12 weeks.
 - Steroids have a 50% success rate.
 - **If poor response to steroids or unable to pass urine → refer to paediatric surgery for circumcision/meatotomy!**



Balanitis/Balanoposthitis

- Bacterial infection of the skin and glans penis.
- Clinical features: preschool age, often in nappies, low-grade fever (37 – 38 C), dysuria, dark red discolouration of foreskin, purulent meatal discharge.
- If fever is present, urine culture should be performed to exclude concomitant UTI.
- **Initial treatment:**
 - Mild balanitis is very common in uncircumcised boys still in nappies, manage with hygiene
 - Moderate-severe: topical antimicrobials, (oral if systemic features), hygiene, simple analgesia, encourage to pass urine by placing in warm bath.
- Once the acute episode has resolved, the patient should receive evaluation by a paediatric surgeon for any underlying abnormality of the penis. If the boy has a pathological phimosis, further episodes of balanitis can be avoided via circumcision.



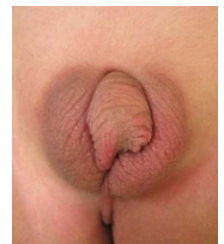
Hypospadias

- Common, 1:200 – 1:300 males
- **Risk factors:** family history (10 – 15%), assisted reproductive technology
- **Hallmarks:** abnormal location of urethral meatus (ventrally positioned), downward curvature of penis (chordee), abnormal foreskin development (ventrally deficient).
 - **Meatus classified as distal/anterior (at coronal sulcus or distally), middle, or proximal.**
 - The meatus has a splayed appearance.
 - Anterior hypospadias is most common (70%)
- **Predictors of surgical repair:** severity of chordee, location of urethral meatus
- **Main complication of hypospadias repair:** urethrocutaneous fistula. Also recurrent chordee.



Ambiguous External Genitalia

- The example given was a **female patient with congenital adrenal hyperplasia (CAH)**. The patient classically presents with a salt-losing crisis in the first 5 days of life.
- **Investigations:** biochemical, genetic testing, laparoscopic assessment of internal gonads/genitalia
- Not a surgical emergency, but rather a medical and social one.



Chromosomal Makeup	Common Causes of Ambiguous Genitalia	Key Features / Examples
46,XX (Genetic Female)	Congenital Adrenal Hyperplasia (CAH) most common	Excess androgen → virilization: enlarged clitoris, fused labia
	Maternal androgen exposure	Virilized female genitalia

	Aromatase deficiency	Virilization due to impaired estrogen synthesis
46,XY (Genetic Male)	Androgen Insensitivity Syndrome (AIS)	Female or ambiguous genitalia despite XY, poor androgen effect
	5-alpha-reductase deficiency	Undervirilized genitalia; micropenis, hypospadias
	Gonadal dysgenesis	Streak gonads, ambiguous genitalia
	Leydig cell hypoplasia / testosterone synthesis defects	Undervirilization
Sex chromosome abnormalities (e.g., mosaics, Turner, Klinefelter)	Mixed gonadal dysgenesis, ovotesticular DSD	Variable ambiguous genitalia depending on mosaicism
Ovotesticular DSD	Presence of both ovarian and testicular tissue	Ambiguous genitalia with mixed features

DSD = Disorder of Sexual Development

We must carefully work up any child with severe hypospadias, or hypospadias associated with undescended testes, microphallus, or bifid scrotum.

- Check for proteinuria, and if positive, refer to renal and genetics team, renal USS – **syndromes involving genital and renal systems, e.g. Denys Drash Syndrome**
- Blood tests for FSH, LH, testosterone, AMH, Inhibin B, chromosome microarray, and store DNA.
- Testosterone supplementation may be provided to boys with a small phallus prior to surgical intervention.
 - The recommended timing of surgery is 6-18 months old.

Acute and Chronic Neck Lumps

Neck Lumps Resource: [RACGP - AJGP \(Australian Journal of General Practice\)](#)

Note, the categorisation below is according to the online videos – but some of these lumps can be both midline and lateral.

Midline Neck Lump			
Lump	Features	Image	Management
Note: Most cysts feel firm, however, an infected cyst or lymphatic malformation may feel fluctuant. In an infected cyst , the overlying skin will be erythematous and tender, and you might see an abscess or draining sinus.			
Thyroglossal duct cyst Most common midline congenital neck mass (70-75% midline lumps)!	Mobile, soft, painless, cystic Superior to thyroid cartilage; attached to hyoid bone Move up with swallowing or tongue protrusion Transilluminates (filled with viscous content) Feels firm		Neck US may be warranted if Dx unclear, especially if there is concern about the thyroid or ectopic thyroid tissue suspected. Excise completely (can recur) – Sistrunk's procedure (middle 1/3 hyoid bone + thyroglossal tract removed). 10% risk of recurrence and small risk of malignant transformation.
Ectopic thyroid	Painless, slow-growing; often lingual or sublingual; may present with mass effect (dysphagia, dysphonia, snoring, dyspnoea) or features of hypothyroidism e.g. poor growth.		TFTs (↑ TSH and subclinical/overt hypothyroidism); neck US (absent eutopic thyroid); 99m-Tc scan definitive Further assessment from paediatric surgeon and endocrinologist

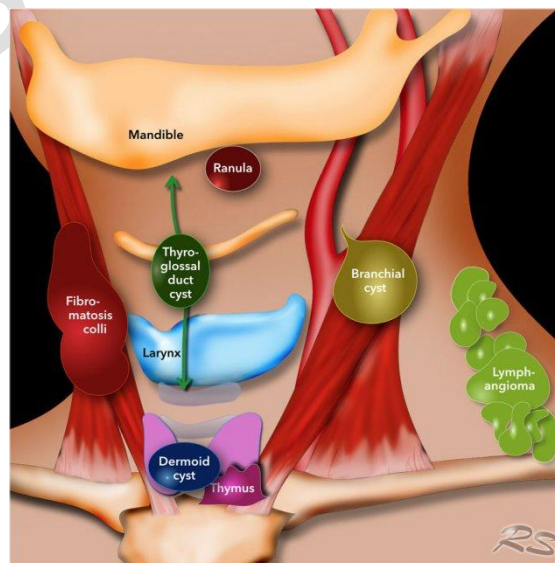
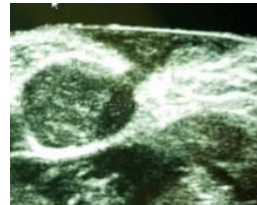
Midline dermoid cyst	Midline, non-tender, soft, fluctuant; tethered to overlying skin, does not move with tongue protrusion or swallowing. Looks yellowish; suprasternal notch or submental area.		Can communicate intracranially, so investigate carefully before surgery.
Midline cervical lymph node	Small, tender, mobile lump; may be multiple.		
Plunging ranula	Fluctuant, bluish mass in the floor of mouth/ upper neck. Pseudocyst associated with sublingual glands and submandibular ducts; 'plunges' through mylohyoid. Inspect inside of the mouth.		Intra-oral excision to include sublingual gland of origin.
Lateral Neck Lumps			
Acquired: inflammatory (acute lymphadenitis, atypical mycobacterial infection, lymph node abscess); malignancy (primary neoplasia e.g. lymphoma, secondary tumour e.g. from neuroblastoma), submandibular gland calculus, parotid gland (sialectasis), thyroid gland goitre.			
Congenital: branchial cleft (cyst, sinus, or fistula), lymphatic malformation, epidermal cyst.			
Reactive lymphadenopathy	Mobile, tender (reactive) Recent viral infection		Expectant management
Cervical lymphadenitis	Reactive lymphadenopathy complicated by secondary infection Erythematous and warm to touch → superimposed bacterial infection (usually Staphylococcus aureus or Streptococcus pyogenes) Accompanied by fever, malaise, local infective signs e.g. sore throat		Oral or IV antibiotics for bacterial infection If abscess forms, may need surgical drainage
Lateral cervical lymphangioma (cystic hygroma); lymphatic malformation	Soft, cystic Overlying skin not inflamed May be diagnosed antenatally		Sclerotherapy e.g. bleomycin or OK 432 (streptococcal toxin) Early referral to multidisciplinary vascular malformation clinic Can be a medical emergency if increases rapidly in size due to infection or haemorrhage!
Branchial remnant.cyst.	Fluctuant, anterior to sternocleidomastoid. Failure of pharyngeal clefts to involute Aetiology uncertain: possible necrosed reactive cervical LN (post RTI). Previously thought congenital remnant.		If asymptomatic, can manage conservatively Surgical excision to confirm histopathological diagnosis and excise.

Branchial fistula	Can discharge; requires excision Punctum at skin may conceal larger remnant below.		Surgery (circumferential incision around punctum; 'stepladder' approach).
Sternomastoid 'tumour'	Firm, fusiform mass in SCM muscle in infants; assoc with torticollis		
Dermoid Cyst	As above; most common location is around orbit (junction between frontonasal process and maxillary process)		
Malignancy	Hard, rapidly growing, non-tender		Refer to oncology unit and surgeon for biopsy and further management

- Note: enlarged lymph nodes (>3cm in diameter) that are present for more than 4 weeks need further investigation, including a biopsy.
- Long-standing torticollis (e.g. due to sternocleidomastoid tumours, deep neck space infections) can cause hemi-hypoplasia of the face secondary to limited neck movement.

Atypical mycobacterial disease

- Chronic painless unilateral submandibular or cervical lymphadenitis.
 - The most common cause of a 'cold' cervical lymph node abscess**
- Overlying skin can become violaceous and may fistulise; central crusting.
- Superficial cold abscess, break in deep fascia, with necrotic node.
- Diagnosis:** positive Mantoux test + cervical pathology.
- Management:** complete surgical excision; prolonged antibiotic therapy if early.



Burns

A major burn causes major physical and psychological injury; management needs a multidisciplinary approach. A **severe burn** is any burn that affects $\geq 10\%$ **body surface area (BSA)**.

Incidence: 1% in Australia and New Zealand per year. Incidence has plateaued. Paediatric burns have 2 peaks.

- <4 years old: scalds/contact burns, >80% are <5% BSA, treated as outpatients and day-stay surgery
- Adolescents: flame/contact burns, due to adventurous and nefarious outlook
- **75% of burns occur at home**

Water temperature: at 60 °C it takes 1 second for hot water to cause a full-thickness burn, at 50 °C, it takes 5 minutes.

The **NSW Statewide Burn Injury Service** consists of 3 Burns Units (2 adult, 1 paediatric). Minor burns are referred to the LHD. LHDs confer with Burns Units through digital photo consultation and refer according to NSW SBIS guidelines.

Guideline for burn transfer:

Retrieval	Referral
Intubated Inhalation injuries Head/neck burns >10% BSA in children Significant comorbidities or pre-existing medical disorder Associated trauma Circumferential burn to limbs or chest Significant electrical (e.g. lightning) or chemical burn (e.g. HF)	>5% BSA Burns to hands, feet, genitalia, perineum or major joints Chemical or electrical burns Pre-existing medical conditions Suspected NAI Infant/neonate

Jackson Burn Wound Model

3 zones:

1. Necrosis
2. Stasis (oedema)
3. Hyperaemia (capillary dilatation, can involve entire body)

The goal of resuscitation is to **protect the zone of stasis**.

Pathophysiology of Burns

Recall that small burns cause a local inflammatory reaction, while large burns cause systemic disruption.

- Hormonal: $\uparrow$ catabolic hormones (glucagon, cortisol, catecholamines); $\downarrow$ anabolic hormones (T4, GH)
- $\downarrow$ Immune function (T-cells, phagocytosis, IgG)
- Inflammatory mediators $\rightarrow \uparrow$ capillary permeability $\rightarrow$ oliguria, renal failure, GI ileus/bacterial dislocation, pulmonary oedema and ARDS, cerebral oedema

Effect on Airway

- Upper airway obstruction: mucosal swelling from heat; soft tissue swelling in neck burns
- Lower airway: inhalational burns and chemical burns to tracheobronchial tree (**vesicles, soot**)
 - The most important management is **bronchial tree ventilatory support**; bronchoscopy and lavage are not usually advised.
- Systemic intoxication e.g. CO, HCN



Circulation

- Oedema
- Capillary dilation + fluid loss $\rightarrow$ 'burn shock'
- Circumferential full thickness burns can cause compartment syndrome and distal ischaemia (may require **escharotomy**)
- Can similarly lead to **abdominal compartment syndrome** with rising intra-abdominal pressure > 20 mmHg $\rightarrow$ escharotomy and laparotomy

Burns can also lead to a systemic inflammatory response and infection. In fact, **major burns** almost inevitably have **invasive sepsis** necessitating antibiotics.

- IV antibiotics are needed to treat this invasive sepsis but interestingly there is no role for antibiotic prophylaxis.

Assessment of the Burn

Depth of Burn

Recall the dermal capillary plexus provides blood supply to the skin, allowing for regeneration.

Epidermal (like a mild sunburn): very painful, surface of skin red but not broken, no blister, regeneration from the intact basal layer.

- May be caused by flash flame, injury, sunburn, or scald injury.
- Red colour, dry surface, may peel, good capillary return
- Generally takes **1 week** to heal



Dermal:

Superficial dermal	Mid dermal	Deep dermal
Blister, pink, blanching	Red, blotchy, +/- blanching (some Hb extravasation)	Non-blanching (complete Hb extravasation)
Epidermal regeneration from dermal papillae, hair follicles, sweat glands	Regeneration from hair follicles and sweat glands, can have scarring	No spontaneous healing, heal by contraction and scarring
Healing <14 days	Healing 14-21 days	
		

Full-thickness burn: outer skin and some underlying tissue dead



Assessment of Burn Depth

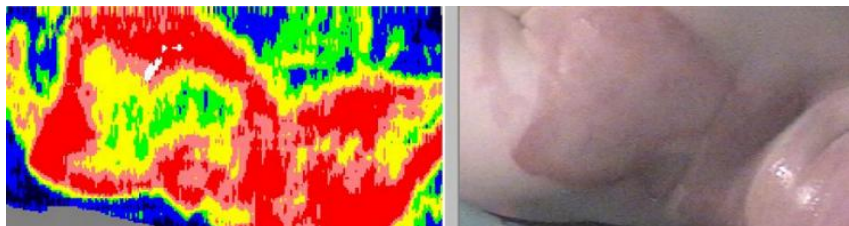
Burns are dynamic. Clinical assessment is unreliable over the first 5 days.

- They are also **heterogeneous** i.e. depth varies within a given area
- **Laser Doppler Imaging** is a useful adjunct to clinical evaluation and is validated to assess depth between **48 – 72 hours of the burn**. It measures the reflection of US waves off RBCs in the dermal capillary plexus.

Unless the burn is obviously superficial or deep, it is worth waiting a few days for the burn to evolve, either towards healing or towards requirement for skin grafting.

- This is particularly the case for **mid-dermal burns**, where most burns that heal within 10-14 days will not scar.

This laser Doppler image shows an area of diminished flow in the superficial dermal capillary plexus (green) consistent with a mid-dermal burn.



Burn Management

Primary Survey (ABCDE): **A**irway (+ C-spine), **B**reathing, **C**irculation, **D**isability (neurological assessment), **E**nvironment (expose and assess area of burn), **F**luid (using modified formula)

First Aid

If not given at the scene, should be given within 3 hours of the burn.

- Remove the patient from the source of heat/ extinguish flames
- Remove clothes
- Apply **cold running water (15 °C) for 20 minutes (ideal)** or immerse in river/ sea/ pool or rotating cold packs
- "Cool the burn, warm the child"
- **Hydrogels e.g. Burn Aid** are useful for analgesia/cooling but are dangerous if used for a long time in major burns as they can cause hypothermia.

Burns > 10% TBSA will induce a systemic inflammatory response (SIRS). Therefore, fluid resuscitation is mandatory, and the IV route is ideal (though IO and oral can be used).

Burns to **faces and heads** are significant and will involve marked swelling over 24-48 hours. Early eye review and consideration of intubation.

- Also look at the scalp, and consider shaving the hair back.

You will also initially do a wound swab, but **do not administer antibiotics unless there is evidence of proven infection.**

Assessing Total Body Surface Area of Burn

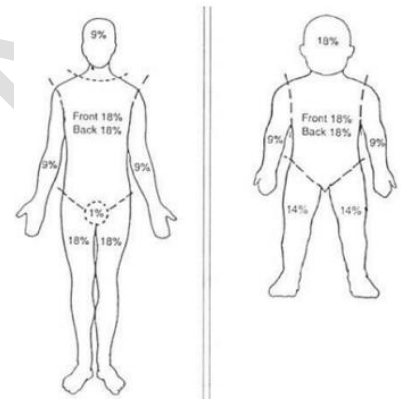
Recall that you only count dermal burns (i.e. exclude superficial burns). The Rule of 9's (adult) can be used for any children aged 10 and above.

Young kids have a proportionally large head. Up until **age 1**:

- The **head** is **18% of BSA**; **subtract 1% each year until it reaches 9% (adult proportions) from 10 y/o.**
- The **legs** are 14% each; add ½% each year until it reaches 18% (adult proportions)

The simplest way to think about it is that the head starts at 18%, then for every year over 1 year old, 1% is subtracted from the head and divided between the lower limbs.

The palms are 1% TBSA.



Fluid resuscitation

Using 2 large-bore IVC or IO, with catheter in situ
0.9% sodium chloride or Hartmann's solution

Parkland Formula (for any burn >10% TBSA): **3 mL/kg/% burn over the first 24 hours**

- Give ½ in the first 8 hours when most fluid loss occurs)
- Second ½ in subsequent 16 hours

Maintenance fluid (0.9% NaCl + 5% glucose): use 100-50-20 rule for 24h fluid requirements.

Note that the formulae are only guides, and fluid resuscitation should be monitored via urine output (through catheter). Those with oliguria or anuria may also need a central venous catheter.

Once the Parkland formula is calculated, and fluid resuscitation commences, monitor and maintain urine output at **0.5 – 1 mL/kg/hour.**

Pain Relief

Severe pain, hospital presentation: IV opioids (titrated to effect) +/- anxiolytic (midazolam, ketamine, or N2O)

80% of patients attend as outpatients. Assess the level of pain, type of burn, and pre-existing conditions that may influence the child's anxiety. Set the parents' expectations i.e. burn dressing is not a painless procedure.

- Pre-dressing preparation with Child Life Therapist
 - Hypnosis
 - Distraction therapy
 - Reduces opioid requirement

- Conscious sedation and analgesia, e.g. oral narcotic (or intranasal fentanyl), anxiolytic, N2O, ketamine (**classic combo: oral morphine, oral paracetamol, oral midazolam**)
- Ultimately, the choice of management depends on the expertise of the staff

Wound Management

"The solution to pollution is dilution!"

- Contaminated: irrigate with saline/water, wash with soapy chlorhexidine
- Remove exudate and dead skin. You need to **touch the wound** when cleaning it.
 - If the patient already has a dressing that has dried onto the wound, you might need to soak it off
- Temporary dressings for transfer: <8 hrs → glad wrap, >8 hrs → consider antibacterial dressing (Bactigras)
 - Glad wrap is a good option because it maintains visibility and prevents fluid loss; avoid for circumferential use
 - Bactigras is a dressing with soft paraffin and chlorhexidine 0.5%. It gives antimicrobial coverage for Staphylococcus and promotes moist wound healing.

Dressings

Superficial (epidermis intact)	Moisturise
Dermal	Silver-containing antimicrobial dressing (Acticoat 3 or 7 day)* • De-roof any blisters if analgesia is good; aspirate if not • Debriding the blister will allow you to fully assess the burn wound Mepilex Ag+ is a good alternative for mid-dermal burns. It is painfree to apply, absorbs exudate, and provides a moist environment for wound healing.
Eschar	Acticoat 7 + early split skin graft (SSG)

*Acticoat hurts to apply. You wet the dressing, apply it, then add a clear film to retain the moisture + secondary dressing
 Recall that for mid-deep dermal burns, 10-14 days of conservative management are appropriate, allowing you to work out how the wound is progressing. If proceeding with surgery, it should occur at **~Day 14**.



If the patient has a proven sepsis, you might use **flamazine (silver sulphadiazine 1%)**. This is a cream; may discolour the wound (leading to pseudoeschar), reapply every 24 hours.

Paraffin can be applied to the face, for superficial burns that are hard to dress. **Chloromycetin** is used around the eyes.

Grafting

- Required when the burn has no further propensity to heal, involving removal of dead eschar via **tangential excision** or **Versajet**.
- Once the base is viable, a **split skin graft** is applied. This involves taking a very thin shaving of normal, healthy skin using a dermatome.
- In some areas, e.g. palms, and ears, if the surface is small enough, a full-thickness graft is needed. This involves excising skin from the donor site down to subcutaneous tissue.

Methods of Skin Expansion

- When there is not enough donor skin
- Mesh (note – can cause a persistent pattern), sandwich autograft with overlying homograft, meek micrografts (5mm graft "stamps"), bioengineered skin substitute (epidermal cells grown in culture)

One complication of burn healing is **hypertrophic scarring**, the likelihood/severity of which is influenced by the genetic makeup of the child and the time it takes for the burn to heal (less time for healing = less time for scarring). Early wound closure is essential for primary prevention.

- Hypertrophic scar: physiological response to injury, rapid growth phase then regression; increase in collagen formation and scar is elevated
- Compare to **keloid** which is an inherited metabolic alteration to collagen → scar grows beyond wound borders
- Other factors influencing scarring: age (younger), skin type (increased pigment), infection, area of high mobility and concave areas (hard to compress)
- Problem areas: flexor surfaces of joints, hands/feet, axilla, face (mouth/eyes), neck, elbows and knees

Time to healing	Likelihood of scar
<14 days	No scar; occasional loss of pigment
14-21 days	May or may not scar
>21 days or grafted	Will definitely scar

Management of Burn Scars

We can think of burn scars as an acute inflammatory process. There are 2 simple remedies

1. Apply pressure to the scar (decreases blood flow)
2. Silicone gel (rehydration effect)

Splinting will prevent contracture (worn continuously for 3 months then weaned to 12 months). Steroids, CO2 laser, microneedling, and other strategies can be used to diminish the hypertrophic scar.

Burn Rehabilitation

Maintain joint mobility, prevent contractures, initiate scar management

- Exercise maintains joint mobility, prevents skin/joint contractures; perform little and often during the day; exercise when splints are off
- Splinting (begins immediately, adjusted regularly, continued until risk of contracture diminished)
- Pressure controls hypertrophic scarring, itch and hypersensitivity; worn 23/24 for 12/12
- Contact media (silicon products) minimise scarring and improves cosmetic appearance

Treatment may be ceased when the scar is **mature** and **optimal ROM is achieved**

- Scars are typically soft, flat, pale, and white/brown/skin colour
- Burn scars generally take 12-18 months to mature; surgical scars take around 6 months
- Ongoing reconstructive surgery and scar management might be needed

Itch in Burns

Can be hard to control. Antihistamines, moisturisers, special additives.

Psychological Support

Burns have enormous psychological impacts. The best predictor of **positive emotional adjustment** for children with burns is **parental adjustment** – so there is enormous imperative for parents to actively engage in their own psychological healing process.

- Mental health risk; size and visibility of burn does not always predict outcome, though those with facial disfigurement may be at greatest risk

Trauma

Injury is the leading cause of death in children over 1 in the developed world. Children are developmentally predisposed to injury.

- Smaller body mass, with more force per unit area.
- Head proportionately larger to body.
- Compliant skeleton i.e. force transmitted to soft tissue
- Energy transmitted to body with less fat, less connective tissue, and close organs.
- High SA:V ratio, increasing thermal loss
- At risk of road trauma due to inquisitive, unpredictable behaviour
- Infants and toddlers are more dependent on carers
- Harder to see as pedestrians because of size
- Adolescents make poor risk assessments and have a feeling of invincibility
- Extra stress from separation anxiety

Table of Injury Patterns

Note: up to 90% of paediatric trauma involves the head (kids have a proportionally bigger head, so it is a greater target in trauma).

MVA	Pedestrian	Bicycle	Fall
Restrained: chest/ abdo, lumbar spine Unrestrained: multiple, head and neck, facial	Low speed: leg fracture High speed: multiple, head and neck, leg fracture	No helmet: head and neck, facial, arm fracture Helmet: arm fracture	Low: arm fracture Medium: head and neck, arm and leg fracture High: multiple

Outcomes for the injured child are best when managed in a **paediatric centre**.

- Management is very similar to adults and relies on **(C)ABCDE**
- One difference is that **solid organ injuries** e.g. hepatic or splenic lacerations are mostly managed conservatively

Technically, **tourniquet life-threatening sources of haemorrhage first**.

'Lethal 6' Injuries in Trauma: Airway obstruction, tension pneumothorax, open pneumothorax, massive haemothorax, flail chest, cardiac tamponade.

Airway (and C-spine) and Breathing

- Children have a larger tongue, floppier epiglottis, larger head and occiput than adults. They also have a limited ability to increase their tidal volume.
- **Neutral airway position** is favourable: depression of the head or a towel under the shoulders can help.
- Head tilt and chin lift is contraindicated in suspected C-spine injury. If the patient is immobilised in a collar, jaw thrust is the only safe manoeuvre.
- Suction catheter must be inserted under **vision** (lest you may cause trauma/bleeding or worsen foreign bodies).
- All major trauma patients benefit from **high-flow oxygen via non-rebreather mask**.

For breathing, we assess **effort, efficacy, and effects**.

Effort	Efficacy	Effects
RR, chest wall expansion, subcostal/ intercostal recessions, tracheal tug.	Air entry (equal bilateral) Added sounds	Oxygen saturation, skin colour, heart rate, conscious level

Reduced air entry/chest movement – **percussion** may differentiate between a pneumothorax (hyperresonant) and haemothorax (dull).

Children have more compliant ribs that are less likely to fracture but are more likely to transfer force to the viscera. Therefore, children have a **higher incidence of pulmonary contusion or other visceral injuries in the absence of chest wall damage or rib fracture**.

Circulation

- Assess circulation (central capillary refill, pulse, BP)
- Identify sources of bleeding (**PLACES**): Pelvis (and retroperitoneum), Limbs, Abdomen, Chest, External, Scalp
- Resuscitation
 - Recall, hypotension indicates >40% blood loss in a child

- Permissive hypotension is **seldom** used. The one exception is **single penetrating injury (e.g. stab wound)** provided there is no evidence of head or spinal injury.
- **Haemostatic resuscitation** ("replace blood with blood"): limited isotonic fluids, early blood products, activate MTP if needed, medications as a last-resort (e.g. TXA, FVIIa, prothrombinex), consider IR/surgery.
- **TEG/ROTEM** is also being increasingly used albeit less common than in adults

Recommendations for resuscitation fluid are to give **up to 40 mL/kg of crystalloid fluid (0.9% NaCl or Hartmann's)** in aliquots of either **10 or 20 mL/kg**. Frequently reassess for response. Remember a kid's circulating volume is only 60-80 mL/kg, so we are effectively replacing 50% of their circulating volume with fluid. By this stage, or **even earlier, replace losses like-for-like (O⁻, ABO compatible, or cross-matched blood)**.

- Cross-matched blood is ideal but takes up to 1 hour to obtain; ABO compatible blood is acceptable but still a touch delayed (10 – 15 minutes). Most ED's have immediate access to O⁻ blood.

Disability

- AVPU initially
- Formal GCS eventually needed
- Change from baseline (parents important)
- Modify neurological status to a child's age and developmental stage

Exposure

- Children are more susceptible to hypothermia than adults (↑ SA: Mass ratio)
- Hypoglycaemia, metabolic acidosis

Imaging in Paediatric Trauma

The **PANSCAN** is not useful in paediatrics. Recall the **ALARA** principle for minimising radiation exposure (As Low as Reasonably Achievable).

- The traditional trauma series includes **X-Ray C-spine, CXR, PXR**.
- May not be necessary if GCS > 14, full clinical examination, and no evidence of major injuries/ no high-risk mechanism.
- If a child has localised symptoms only, targeted imaging of the isolated site is more appropriate than a full series.

Interestingly, the **FAST Scan** is not routinely recommended in paediatric blunt trauma. Intra-abdominal injury may not cause much free fluid, so a negative FAST does not rule out serious injury.

- It has more of a role in **unstable** patients; a positive FAST effectively rules bleeding in.
- This is important when deciding whether to transfer to theatre for laparotomy (you don't want to operate on an unstable patient if you're not sure the abdomen is even their source of bleeding...)

A **CT abdomen with double contrast** is the gold-standard for identifying abdominal or pelvic injuries. However, it should not be used in patients who are haemodynamically unstable. This is why it's often called the 'Donut of Death!'

An unstable trauma patient unresponsive to fluid resuscitation and with a positive FAST needs to be transferred to theatre for laparotomy. Fluids should continue but should not delay transfer. Note, inotropes are useless if the 'pipes are empty'.

Injury Patterns

Beware of **non-accidental injury**. Screen whether the carer recognises the injury, seeks aid promptly, tells a consistent story, and identifies an event consistent with the magnitude of the injury.

Head	Chest	Abdomen/Pelvis
Most common cause of death Commonest mechanism: falls Scalp laceration can cause significant bleeding Skull fractures carry ↑ risk of intracranial injury Monitor fontanelle in kids < 18 months (when still open)	Intrathoracic injury can occur without external signs of trauma Most common paediatric chest injury: pulmonary contusion , followed by intrapleural injuries First-line Ix: CXR and/or USS	Solid organ injuries are the most common; managed based on physiology (i.e. conservative if stable, resuscitate +/- surgery if unstable) Hollow viscus injuries are always a surgical problem Handlebar injuries (from falling off bicycle) are common since children have a thin abdominal wall – can be

		accompanied by solid organ and hollow viscous injuries Ix: <u>bloods</u> (FBC, lipase/amylase, LFT, crossmatch), <u>urinalysis</u>, <u>CT with contrast</u>, <u>X-ray</u> (if suspect intestinal perforation), <u>U/S</u> for solid organ injury +/- free fluid (but CT is gold-standard)
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Chest injury is the only type of injury for which CT is not first-line. All patients undergoing CT should have an orogastric or nasogastric tube inserted prior to the investigation, to reduce the risk of vomiting.

PREDICT Algorithm for Paediatric Head Trauma

GCS, clinical features (e.g. altered mental status, headache, vomiting, neurological Sx) and the presence/absence of skull fracture are used for risk stratification. High-risk kids get a head CT, low-risk kids can get discharged without further investigations.

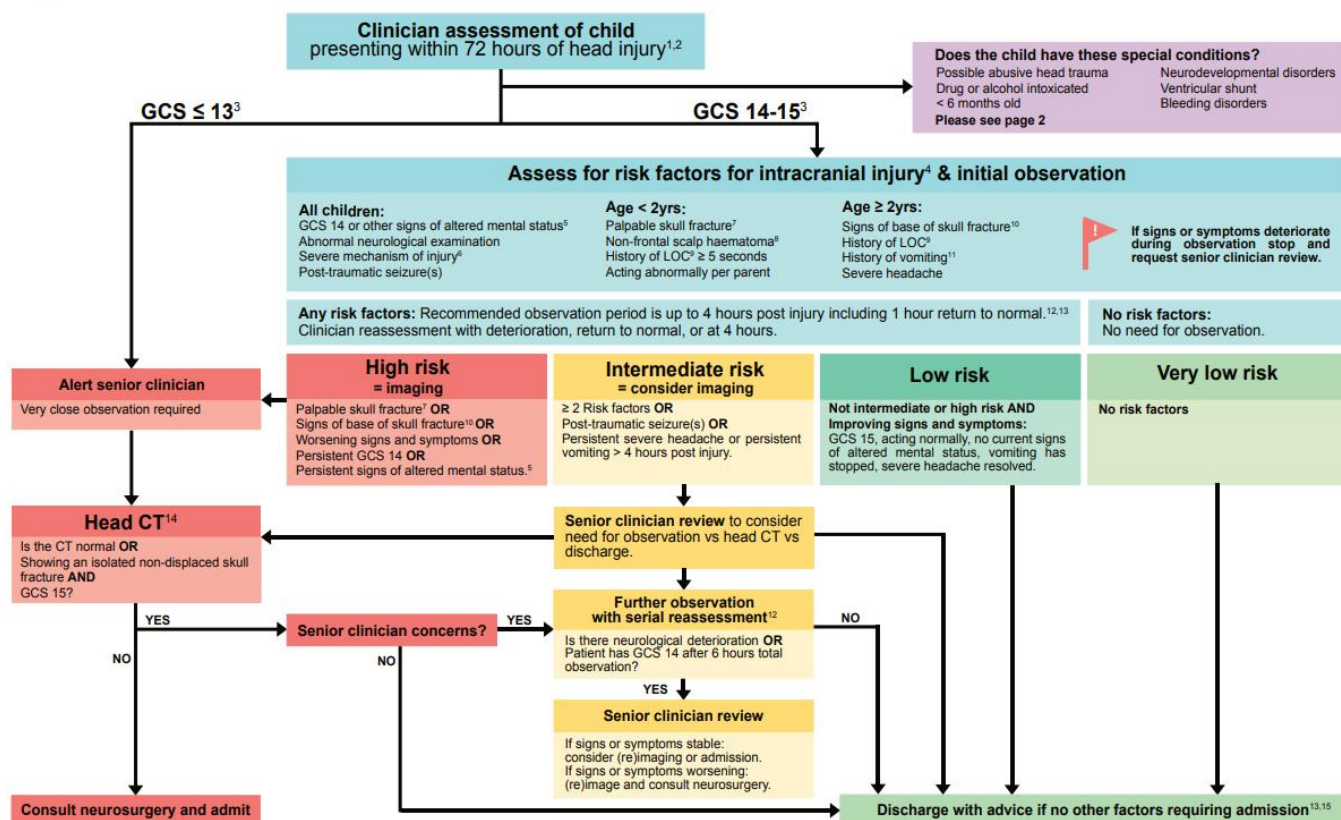
Kids with any risk factors for intracranial injury should be observed up to 4 hours post-injury. They should be normal for at least 1 hour before discharge.



From: PREDICT GUIDELINE FOR CHILDREN WITH MILD TO MODERATE HEAD INJURIES (see www.predict.org.au) Version 1.0 [150121]

Algorithm: Imaging & Observation Decision-Making for Children with Head Injuries

Further details and footnotes are important to interpretation of the algorithm. Please see page 2.



Recall the signs of base of skull fracture: haemotympanum, 'raccoon' eyes, CSF otorrhoea or rhinorrhoea, Battle's sign (bruising over the mastoid process).

Pain Management in Childhood

Pain is an unpleasant sensory and emotional experience associated with, or resembling that associated with, actual or potential tissue damage.

Pain descriptors:

Allodynia: pain due to a stimulus that is usually not noxious
 Hyperalgesia: increased response to a normally painful stimulus
 Dysaesthesia: spontaneous or evoked unpleasant abnormal sensation

Chronic pain:

Pain lasting longer than the usual expected time after illness or injury (3 months); affects 1 in 4 kids. The pathophysiology is nervous system hypersensitivity.

Types of pain:

Neuropathic pain	Lesion or disease of the somatosensory nervous system. Signs and symptoms: burning, shooting, pins and needles, sharp, allodynia, hyperalgesia, dysaesthesia, hyperaesthesia.
Nociceptive pain	Actual or threatened damage to non-neural tissue which activates nociceptors. Normal somatosensory nervous system
Nociplastic pain (central sensitisation)	Altered nociception despite no clear evidence of actual or threatened tissue damage activating peripheral nociceptors or disease/lesion of the somatosensory system.

Pain Assessment:

0 – 2 years old or developmental delay FLACC Scale	2 – 4 years old Faces Scale	5 – 7 years old Faces Scale or Verbal Pain Intensity Scale	Older Numeric pain ladder or Visual analogue scale
FLACC Scale 1Face No particular expression or smile 2Legs Normal position or relaxed 3Activity Lying quietly, normal position, moves easily 4Cry No crying (suckle or soothe) 5Consolability Content, relaxed 0 Occasional grimace or frown, withdrawn, distressed 1 Uneasy, restless, tense 2 Frequent to constant frown, clenched jaw, squinting, grimacing 0 No Hurt 2 Hurts Little Bit 4 Hurts Little More 6 Hurts Even More 8 Hurts Whole Lot 10 Hurts Worst Verbal Pain Intensity Scale No Pain Mild Pain Moderate Pain Severe Pain Very Severe Pain Worst Possible Pain 0-10 Numeric Pain Intensity Scale No Pain 0 1 2 3 4 5 6 7 8 9 10 Moderate Pain Worst Possible Pain Visual Analogue Scale No Pain Worst Possible Pain			

Routes of Analgesic Administration

Oral	Painless and convenient. Palatability can be a problem – difficult if nausea.
Rectal	Well-tolerated in young infants, good if NBM. Remember to get parental consent.
Transdermal	Bigger role in palliative care.
Intravenous	Rapid acting, easily titratable. Preferred for opioid infusions. Useful if NBM.
Intranasal	Rapid-acting.

Principles of Analgesia in Children

“By the clock”: give pain medications at regular intervals.

“By the mouth”: whenever possible, use oral medications

“By the ladder”: start with non-opioids for mild pain, advance to weak opioids for moderate pain, and strong opioids for severe pain.

Analgesic Drugs



Drug	Dose	Adverse Effects/CI	Caution
Paracetamol	>3 months PO/IV 15 mg/kg/dose q6 hourly Maximum 4 doses/24 hours or 1g per dose	Hepatotoxicity	Review dose every 48 hours in hospitalised patients
NSAIDs	Ibuprofen: > 3 months PO 10 mg/kg/dose Q8h Naproxen: > 1 month 5 mg/kg/dose q12hrly	Gastrointestinal ulceration Thrombocytopaenia Clotting anomalies Severe asthma	

	Diclofenac: >10kg PO 1 mg/kg/dose BD-TDS	Renal impairment	
Opioids	Morphine: poor oral bioavailability >1 y/o PO 0.2 – 0.5 mg/kg/dose Q4 PRN < 1 y/o PO 0.1 – 0.2 mg/kg/dose Q4 PRN IV/SC 0.05 mg/kg (>12 months up to 0.2 mg/kg max 5 – 10 mg)	GI: constipation, N&V Autonomic: urinary retention, postural hypotension, xerostomia CNS: drowsiness, cognitive impairment, hallucinations, respiratory depression, myclonus Cutaneous: itching	Reversal: IV naloxone 5 mcg/kg over 1-2 mins Antihistamines can help itch

There are **adjuvant analgesics** which are particularly helpful for neuropathic pain e.g. antidepressants (such as TCA amitriptyline), anticonvulsants (gabapentin, pregabalin), ketamine (good for post-operative pain). Cannabinoids have a role in the research sphere.

Non-pharmacological management: Medicines, Movement & The Mind.

Procedural Pain Management

- **Medical trauma:** exposure to a single or series of medical event(s) related to injury/ illness, painful or invasive medical intervention, and/or other hospital experiences that may be perceived by the child and/or caregiver as **threatening, overwhelming, or frightening**.
- **Being kinds to kids:** 5 P's

Presence	Preparation	Positioning	Positive Reinforcement	Play
Staff: medical, nursing, allied staff, Child Life Therapist (for play and developmentally appropriate support)		Position of comfort Young child: skin to skin "Hugging hold" for IV cannula, bloods, vaccination On side for LP Swaddled/ wrapped (0 – 3 months)	One calm voice Be sensitive to child's cues Reinforce coping strategies and use positive language Encourage verbalisation of pain or distress Validate Provide a tangible reward	Active distraction is better than passive Offer the child the choice to watch the procedure if they want (empowering) Create a visual shield for child

"One Voice" Approach

One voice should be heard during procedures.

Need for parent/carer involvement.

Educate parent/carer before procedure.

Validate child with words

Offer position of comfort

Individualise approach

Choose appropriate distraction techniques

Eliminate unnecessary people from the room

Procedural medicine

- Emla/topical anaesthetic agent given 1 hour before procedure
- Sucrose (28 days to 3 months old) or non-nutritive sucking: provide before procedure then as needed if distressed during procedure
- Oral premedication: analgesia, midazolam, ketamine
- Nitrous oxide (for suturing in ED): traditionally anaesthetist only, but nurses being trained

23. Indigenous Health and Social/Ethical Issues

Aboriginal and Torres Strait Islander Health

There are currently about 700 000 Aboriginal people in Australia. 25% live remotely, and 9% live very remotely.

- Though there is more funding per person for Aboriginal vs non-Aboriginal Australians, it is not enough to close the gap.
- Specific Aboriginal health initiatives, e.g. Aboriginal Community Controlled Health Organisations, Aboriginal Child Health Checks, closing the gap initiatives etc.
- **Barriers in accessing health care:** availability + affordability + acceptability of services

Immunisation differences (recall 5): BCG, Hepatitis A, Influenza, Meningococcal B, pneumococcal.

- The flu shot is funded for Aboriginal children from 6 months old.

High-Risk Conditions

Condition	Background	Prevalence	Outcomes
Iron deficiency	Multifactorial aetiology e.g. diet, worms	20 – 60% in remote settings	Common cause of cognitive impairment
Otitis media	Earlier, more frequent, last longer, more complications (perforations, CSOM)	AOM: 94% in remote area CSOM: 25 to 40%	Hearing loss, cognitive and behavioural problems
MRSA	Skin infection; may be secondary to scabies	54% of Aboriginal people	Requires antibiotics: clindamycin for adults, cotrimoxazole for children
Rheumatic Heart Disease	Group A β -haemolytic strep, acute rheumatic fever $\rightarrow$ RHD	ARF: 3 per 1000 children 5 – 14 years ($> 100\times$ general pop)	Prevented with throat swab and empiric IMI penicillin for sore throats Notifiable disease IMI benzathine penicillin (Penicillin G) monthly for tertiary prevention
Chronic kidney disease	$\downarrow$ nephron number in utero $\uparrow$ rates of proteinuria commencing from adulthood Associated with PSGN	CKD accounted for 45% of hospitalisations of Aboriginal people in 2013	Treat scabies and skin infections Routine screening not indicated except in diabetes
Foetal alcohol spectrum disorder	Indigenous women vs. non-Indigenous women have more abstinence but more dangerous consumption	Unknown	Prevented on supply and demand side; harm minimisation

Other high risk conditions include eyes, teeth, skin, chronic suppurative lung disease, developmental delay, learning difficulties, and mental health concerns.

Fortunately, many health outcomes are improving for Indigenous Australians: mortality, low birthweight, smoking during pregnancy, episodes of primary healthcare, Y12 or equivalent educational attainment, overcrowding/homelessness, home ownership.

- **Life expectancy** is 76 years for females and 72 years for males; infant mortality is 6 per 1000.
- **Indigenous immunization rates (at 5 years old)** are also higher than in non-Indigenous children. Though delays may occur more frequently.

Children and Adolescents at Risk

All doctors are mandatory reporters.

Types of abuse: physical abuse, sexual abuse, neglect (general, medical, educational), psychological or emotional harm, exposure to domestic and family violence, medical child abuse (factitious disorder).

How to report:

First, determine the risk of harm using your clinical acumen. You only need a reasonable level of suspicion to file a report; no complete investigation.

- The **Mandatory Reporter's Guide** helps with decision-making around the child's risk and initial actions.
- You should also discuss with a senior colleague or the Child Protection Unit (CPU).
 - The CPU is different to the FaCS (Department of Family and Community Services). It is staffed by child protection paediatricians/forensic paediatricians who can conduct forensic medical assessment of children for suspected maltreatment.
- There are **3 possible outcomes**:

"Risk of Significant Harm" Report to Child Protection Helpline	Call to Child Wellbeing Unit (CWU)	Continue to Monitor and Support
Made to the Department of Family and Community Services (FaCS) helpline. Call 132 111 any day, any time. You need sufficient evidence e.g. child's details, family background, why you're concerned (incident details). E-reporting can be used for non-imminent cases e.g. chronic neglect.	During business hours: 1300 480 420 Any report that does not meet "significant risk" but is still concerning - cumulative risk of harm If level of concern is sufficient, CWU can escalate to Helpline	No reporting necessary Keep working with family Consider community supports e.g. NGO If situation changes, do MRG again.

Chapter 16A of the Children and Young Persons (Care and Protection) Act 1998 allows certain government and non-government agencies (e.g. clinicians) to exchange information related to the safety, welfare, and wellbeing of children and young people.

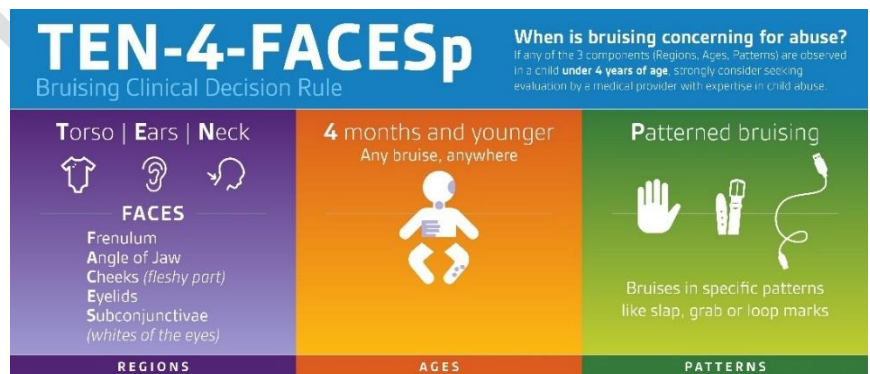
Bruising in Children

When is bruising concerning for abuse? Recall **TEN-4-FACESp**.

Though only clinically validated in kids under 4 years old, it still applies later.

Location

- Torso
- Ears
- Neck
- Frenulum (sentinel injury)
- Cheeks (fleshy part)
- Eyelids
- Subconjunctivae (whites of eyes)
 - May indicate **suffocation** which → triad of bilateral subconjunctival haemorrhage, oral injury, and facial bruising.
 - This includes gripping the baby, and placing it face-down.



Injuries to the face or mouth are often linked to parents wanting the child to stop crying.

Age: any child <4 months old with bruises is alarming.

Pattern: e.g. fingers, belt, hot appliance.

Section 173: legislation that DCJ can use in the event that parents do not agree to voluntary admission and medical assessment.

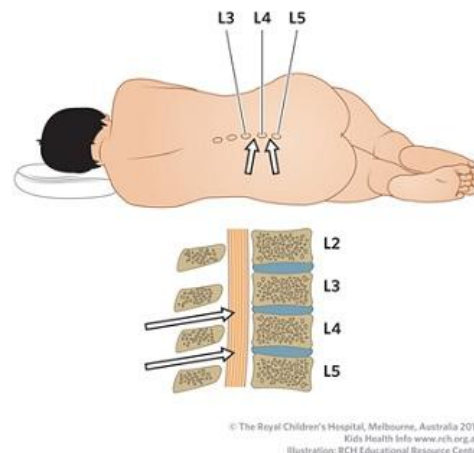
Child Sexual Assault: most children will *not* have injuries. Many forms of sexual assault e.g. grooming, inappropriate sexual touching, exposing child to inappropriate sexual material, penetrative sexual assault. If you suspect CSA, refer to the NSW Health Sexual Assault Service.

24. Miscellaneous

Procedural Skills

Lumbar Puncture

- Consent, and full explanation with fact sheet.
- Local anaesthetic patch on lower back, unless extreme urgency. We may use local anaesthetic injection, but this can obscure landmarks.
 - Inhaled nitrous oxide or a mild sedative (e.g. oral midazolam) may be suitable in an older child.
 - Consider oral sucrose in a neonate.
- Positioning: on side with legs curled up under chin.
- Clean skin with iodine or chlorhexidine; sterile drape.
- Needle inserted into the spinal space between L3/L4 or L4/L5. This is located by palpating the posterior-superior iliac crest and drawing an imaginary line between the two crests that intersect with L4.
- Insert the needle bevel facing up and aim towards the umbilicus, advancing slowly. You may hear a “pop” as the needle enters the subarachnoid space.
- Collect CSF in 3-4 separate tubes with 10 drops in each tube.
- Replace stylet and needle removed. Dressing over site.
- Complications: the two commonest complications are traumatic tap and unsuccessful procedure. Other complications include dermoid (from introduction of skin tissue during procedure), headache (difficult to assess in infants), and coning (less risk with open anterior fontanelle).



Refer to the section on **Meningitis** for interpretation of CSF.

Common CSF appearances at LP

CSF	Diagnosis
Clear and colourless	Normal CSF, viral meningitis, tuberculous meningitis
Turbid	Bacterial meningitis
Slightly turbid	Tuberculous meningitis
Blood-stained	Traumatic tap, subarachnoid haemorrhage

Specific indications to delay LP:

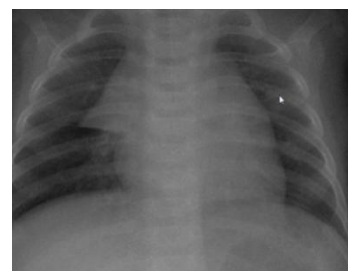
Broad Categories	Specific Indications to Delay LP
Local site for LP	Skin infection or anatomic abnormality at LP site
Patient instability	Respiratory or cardiovascular compromise Continuing seizure activity
Suspicion of space-occupying lesion or raised ICP	Focal seizures, focal neurological signs, reduced consciousness, fixed dilated or unequal pupils, absent vestibulo-ocular reflex ('dolls eye movement'), papilloedema, Cushing triad (HTN, bradycardia, irregular respirations)
Haematological	Coagulopathy

Medical Imaging in Paediatrics

Chest X-Ray

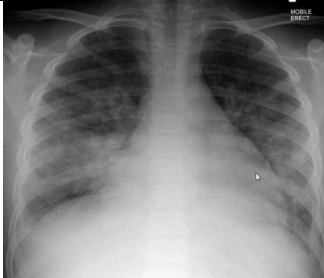
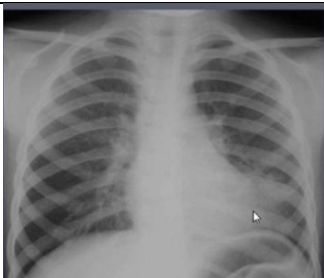
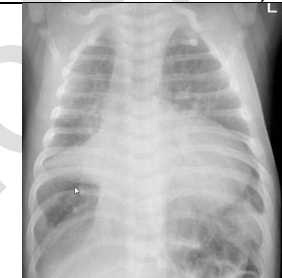
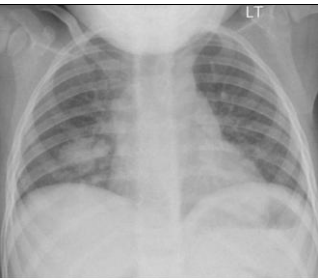
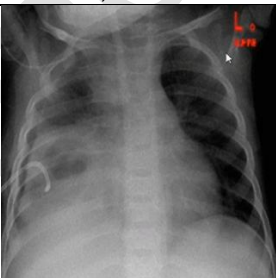
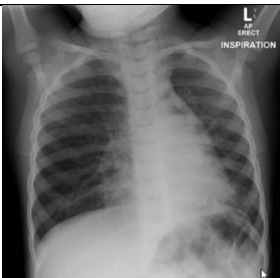
One big way the paediatric CXR differs from adults is that kids have a **thymus** ('sail sign'). Notice a triangular opacity in the superior mediastinum; sharp border on the right, scalloped border on the left.

- The thymus becomes smaller after age 2.



Lung Pathologies

Diffuse Disease

Primarily interstitial, perihilar	Airspace Process	Pulmonary Fibrosis (disorganised, non-branching)	Cystic fibrosis with bronchial thickening
			
Focal Disease			
Atelectasis + LLL consolidation	LLL pneumonia	Pleural effusion (layers lateral to lung)	Collapse (sharply demarcated, shift towards consolidation)
			
Round pneumonia ¹	Complicated pneumonia ²	Pneumonia, cavitation, chest drain, and PTX	Foreign body aspiration ³
			

1: Round pneumonia is unique to infants and arises because the interconnections between alveoli are less well-developed than in adults.

2: Left lower lobe consolidation, air-fluid level, gas lucency (hydropneumothorax, necrotising pneumonia, pleural effusion, bronchopleural fistula).

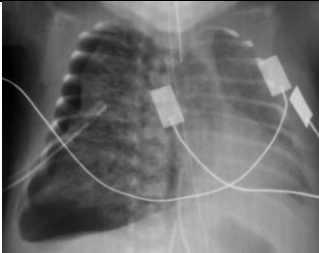
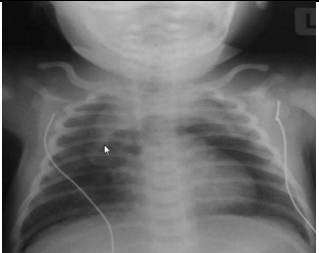
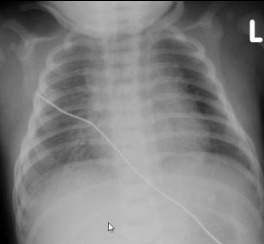
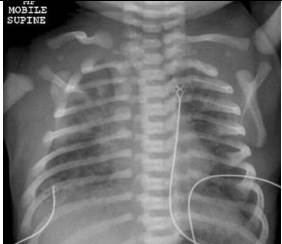
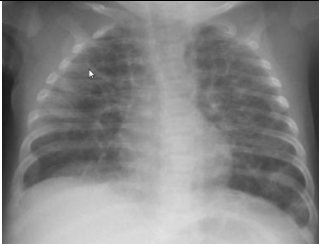
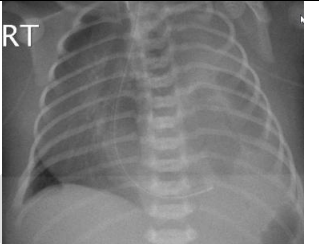
3: Kids get air trapping distal to the FB (increased lucency in one hemithorax); older people get collapse. The volume of the hyperinflated lung will change little between an inspiratory and expiratory film. In the CXR provided, the foreign body is in the right main bronchus.

The silhouette sign can help us localise a pathology on CXR. Recall, if the **diaphragm border** is obscured → lower lobe, **heart border** → middle lobe (which would be the **lingula** on the left side), upper zone for upper lobe.

Neonatal Lung Disease

Note: you can work out if a CXR was done on a term or preterm neonate by looking for the ossification centre of the humeral head. It presents at 38/40.

RDS: Diffuse granular consolidation, hypo-inflated, pre-term baby	RDS with right-sided pneumothorax (demarcated pleural line)	Complication of PPV: pulmonary interstitial emphysema	Pneumomediastinum ('angel wing')
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TTN: term infant, first few hrs of life, resolves by 48h ¹	Meconium aspiration (diffuse, coarse densities)	Chronic neonatal lung disease (coarse interstitial markings, distorted architecture)	Diaphragmatic hernia (usually L sided with pulm hypoplasia); posterior defect
			

1: An older infant beyond the first few days of life with respiratory distress might have *congenital heart disease*.

Abdominal Imaging

Abdominal X-Ray: 5 main indications – for free gas (perforation), evaluation of abdominal dilatation (obstruction), suspected renal tract calcification, foreign body ingestion, and lines/tubes.

- Basically, anything bowel-related where US is useless!

Normal		Abnormal	
Neonates (mosaic pattern)	Older children	Gasless abdomen (ascites)	Dilated small bowel loops
			

Signs of Free Intraperitoneal Gas

Remember, in a supine AXR, you won't see free air under the diaphragm.

Football Sign: outlining of falciform ligament (left)

Rigler's sign: gas on both sides of the bowel wall (right)

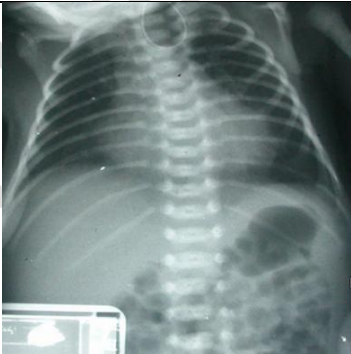
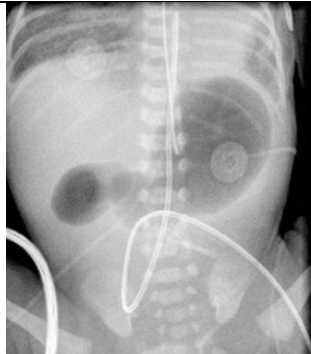


Indications for AXR in Neonates

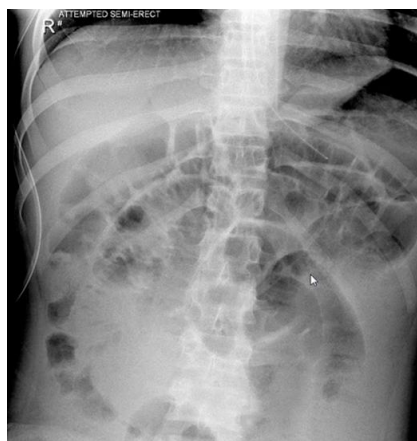
Catheter position, necrotizing enterocolitis, congenital anomalies (malrotation, atresia, webs, anorectal), meconium ileus, Hirschsprung's disease.

Catheter Position:

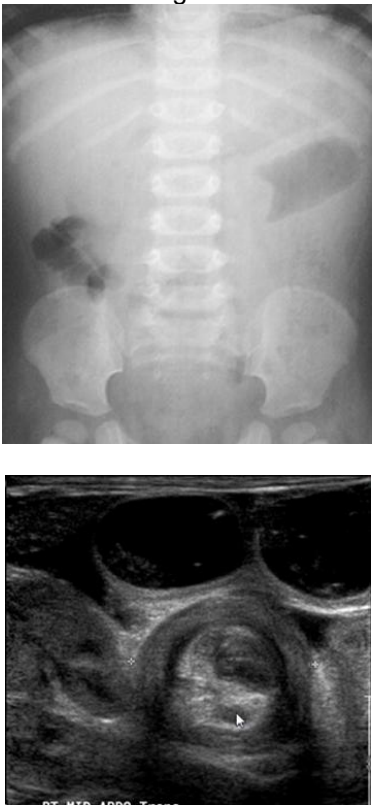
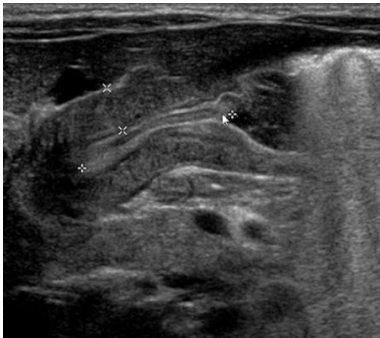
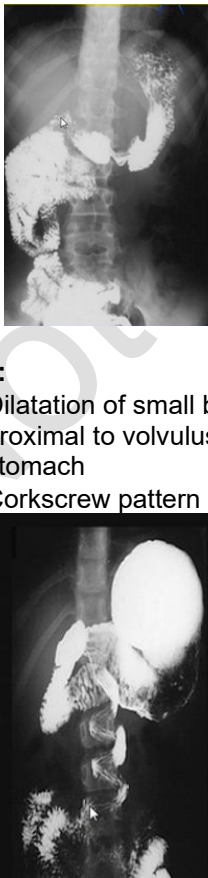
Catheter/Tube	Typical Position
Umbilical venous catheter (UVC)	Umbilicus, via ductus venosus into hepatic veins and IVC. T8-T9, at IVC/ near right atrium border
Umbilical arterial catheter (UAC)	High position (T6-T9) or low position (L3-L4) below renal arteries
Endotracheal tube (ETT)	Level of T3-T4 vertebrae, 1-2 cm above carina
Nasogastric tube (NGT)	Tip in stomach fundus, below diaphragm and preferably below the gastroesophageal junction
Peripherally Inserted Central Catheter (PICC)	Lower third of SVC for upper limb PICC; high IVC for lower limb PICC.

Necrotising enterocolitis	Oesophageal atresia	Duodenal atresia
<u>mural gas</u> ; may extend to branching lucencies in the portal venous system.	NG tube coiled in oesophagus; may have tracheal fistula (gas in bowel loops suggestive)	Double bubble sign
		

Intestinal obstruction: may occur secondary to appendicitis, adhesions, intussusception, incarcerated hernia, malrotation, or Meckel's. The image below shows a small bowel obstruction with density in the right iliac fossa (**highly suspicious of appendicitis**).



Other pathologies in infants

Intussusception	Hypertrophic Pyloric Stenosis	Malrotation/ Midgut Volvulus
Soft tissue density obscuring transverse colon gas. 	Thickness > 3 mm and length > 14 mm abnormal (remember, pi is 3.14) 	Malrotation is a congenital condition that predisposes to volvulus: • Narrow mesenteric fixation • Abnormal positioning of caecum and small bowel  Volvulus: • Dilatation of small bowel proximal to volvulus + stomach • Corkscrew pattern